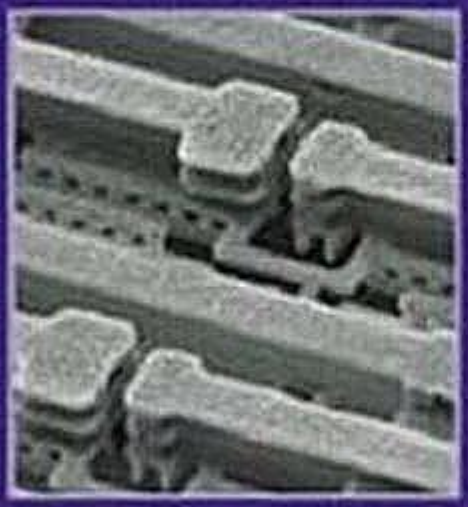
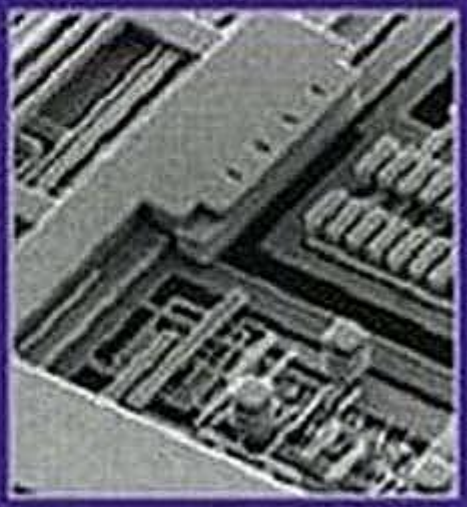
Second Edition

Principles of Electronic Materials and Devices





S.O. Kasap

Solutions Manual

to accompany

Principles of Electronic Materials and Devices

Second Edition

S.O. Kasap
University of Saskatchewan



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Chapter 1

1.1 The covalent bond

Consider the H_2 molecule in a simple way as two touching H atoms as depicted in Figure 1Q1-1. Does this arrangement have a lower energy than two separated H atoms? Suppose that electrons totally correlate their motions so that they move to avoid each other as in the snapshot in Figure 1Q1-1. The radius r_o of the hydrogen atom is 0.0529 nm. The electrostatic potential energy *PE* of two charges Q_1 and Q_2 separated by a distance r is given by $Q_1Q_2/(4\pi\varepsilon_o r)$.

Using the Virial Theorem stated in Example 1.1 (in textbook) consider the following:

a. Calculate the total electrostatic potential energy (*PE*) of all the charges when they are arranged as shown in Figure 1Q1-1. In evaluating the *PE* of the whole collection of charges you must consider all pairs of charges and, at the same time, avoid double counting of interactions between the same pair of charges. The total *PE* is the sum of the following: electron 1 interacting with the proton at a distance r_o on the left, proton at r_o on the right, and electron 2 at a distance $2r_o$ + electron 2 interacting with a proton at r_o and another proton at $3r_o$ + two protons, separated by $2r_o$, interacting with each other. Is this configuration energetically favorable?

b. Given that in the isolated H-atom the *PE* is $2 \times (-13.6$ eV), calculate the change in *PE* with respect to two isolated H-atoms. Using the Virial theorem, find the change in the total energy and hence the covalent bond energy. How does this compare with the experimental value of 4.51 eV?

Solution

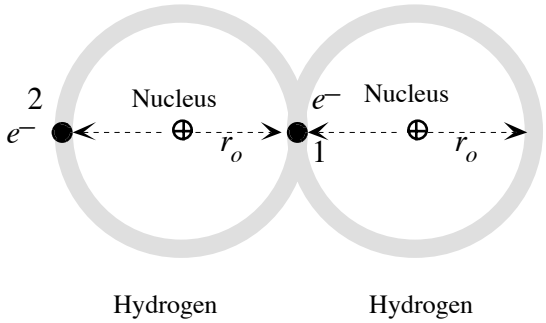


Figure 1Q1-1 A simplified view of the covalent bond in H_2: a snap shot at one instant in time. The electrons correlate their motions and avoid each other as much as possible.

a Consider the *PE* of the whole arrangement of charges shown in the figure. In evaluating the *PE* of all the charges, we must avoid double counting of interactions between the same pair of charges. The total *PE* is the sum of the following:

Electron 1 interacting with the proton at a distance r_o on the left, with the proton at r_o on the right and with electron 2 at a distance $2r_o$

\+ Electron 2 on the far left interacting with a proton at r_o and another proton at $3r_o$

\+ Two protons, separated by $2r_o$, interacting with each other

$$PE = -\frac{e^2}{4\pi\varepsilon_o r_o} - \frac{e^2}{4\pi\varepsilon_o r_o} + \frac{e^2}{4\pi\varepsilon_o (2r_o)} - \frac{e^2}{4\pi\varepsilon_o r_o} - \frac{e^2}{4\pi\varepsilon_o 3r_o} + \frac{e^2}{4\pi\varepsilon_o 2r_o}$$

substituting and calculating we find

$$\boldsymbol{PE} = -1.0176 \times 10^{-17} \text{ J or } \mathbf{-63.52\ eV}$$

The negative *PE* for this particular arrangement indicates that this arrangement of charges is indeed energetically favorable compared with all the charges infinitely separated (*PE* is then zero).

b The potential energy of an isolated H-atom is -2×13.6 eV or 27.2 eV. The difference between the *PE* of the H_2 molecule and two isolated H-atoms is,

$$\Delta PE = -(63.52 \text{ eV}) - 2(-27.2) \text{ eV} = -9.12 \text{ eV}$$

We can write the last expression above as changes in the total energy as

$$\boldsymbol{\Delta \overline{E}} = \tfrac{1}{2}\Delta \overline{PE} = \tfrac{1}{2}(-9.12 \text{ eV}) = \mathbf{-4.56\ eV}$$

This is the change in the total energy which is negative. The H_2 molecule has lower energy than two H-atoms by 4.56 eV which is the bonding energy. This is very close to the experimental value of 4.51 eV. (Note: We used an r_o value from quantum mechanics - so the calculation was not totally classical).

1.2 Ionic bonding and NaCl

The interaction energy between Na^+ and Cl^- ions in the NaCl crystal can be written as

$$E(r) = -\frac{4.03 \times 10^{-28}}{r} + \frac{6.97 \times 10^{-96}}{r^8}$$

where the energy is given in joules per ion pair, and the interionic separation *r* is in meters. Calculate the binding energy and the equilibrium ionic separation in the crystal; include the energy involved in electron transfer from Cl^- to Na^+. Also estimate the elastic modulus *Y* of NaCl given that

$$Y \approx \frac{1}{6r_o}\left[\frac{d^2E}{dr^2}\right]_{r=r_o}$$

Solution

The *PE* curve for NaCl is given by

$$E(r) = \frac{-4.03 \times 10^{-28}}{r} + \frac{6.97 \times 10^{-96}}{r^8}$$

where *E* is the potential energy and *r* represents interionic separation.

We can differentiate this and set it to zero to find the minimum *PE*, and consequently the minimum (equilibrium) separation (r_o).

$$\frac{dE(r_o)}{dr_o} = 0$$

$\therefore$ $$-55.76\frac{1}{10^{96} r_o^{\,9}} + 4.03\frac{1}{10^{28} r_o^{\,2}} = 0$$

isolating r_o: $\boldsymbol{r_o = 2.81 \times 10^{-10}\ \text{m}}$ or **2.81 Å**

The apparent "ionic binding energy" (E_b) for the ions alone, in eV, is:

$$E_b = -\frac{E(r_o)}{q} = -\left(\frac{-4.03\times10^{-28}}{r_o} + \frac{6.97\times10^{-96}}{r_o^{\,8}}\right)\left(\frac{1}{q}\right)$$

$\therefore$ $$E_b = -\left(\frac{-4.03\times10^{-28}}{(2.81\times10^{-10}\ \text{m})} + \frac{6.97\times10^{-96}}{(2.81\times10^{-10}\ \text{m})^8}\right)\left(\frac{1}{1.602\times10^{-19}\ \text{J/eV}}\right)$$

$\therefore$ $\boldsymbol{E_b = 7.83\ \text{eV}}$

Note however that this is the energy required to separate the Na^+ and Cl^- ions in the crystal and then to take the ions to infinity, that is to break up the crystal into its ions. The **actual bond energy** involves taking the NaCl crystal into its constituent *neutral* Na and Cl atoms. We have to transfer the electron from Cl^- to Na^+. The energy for this transfer, according to Figure 1Q2-1, is -1.5 eV (negative represents energy release). Thus the **bond energy** is 7.83 - 1.5 = **6.33 eV** as in Figure 1Q2-1.

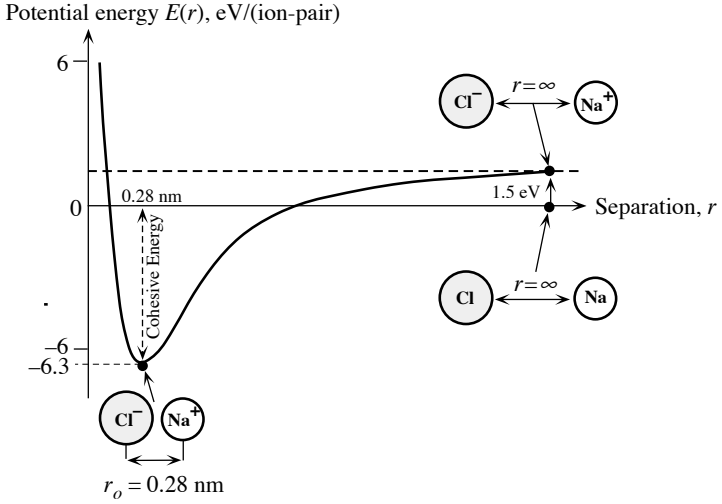


Figure 1Q2-1 Sketch of the potential energy per ion pair in solid NaCl. Zero energy corresponds to neutral Na and Cl atoms infinitely separated.

If r is defined as a variable representing interionic separation, then Young's modulus is given by:

$$Y = \frac{1}{6r}\left[\frac{d}{dr}\left(\frac{dE(r)}{dr}\right)\right]$$

$\therefore$ $$Y = \frac{1}{6}\left(\frac{-8.06\dfrac{1}{10^{28} r^3} + 501.84\dfrac{1}{10^{96} r^{10}}}{r}\right)$$

$\therefore$ $$Y = 8.364\times10^{-95}\frac{1}{r^{11}} - 1.3433\times10^{-28}\frac{1}{r^4}$$

Substituting the value for equilibrium separation (r_o) into this equation (2.81×10^{-10} m),

$$\boldsymbol{Y = 7.54 \times 10^{10}\ Pa = 75\ GPa}$$

This value is somewhat larger than about 40 GPa in Table 1.2 (in the textbook), but not too far out.

*1.3 van der Waals bonding

Below 24.5 K, Ne is a crystalline solid with an FCC structure. The interatomic interaction energy per atom can be written as

$$E(r) = -2\varepsilon\left[14.45\left(\frac{\sigma}{r}\right)^6 - 12.13\left(\frac{\sigma}{r}\right)^{12}\right] \qquad \text{(eV/atom)}$$

where ε and σ are constants that depend on the polarizability, the mean dipole moment, and the extent of overlap of core electrons. For crystalline Ne, $\varepsilon = 3.121 \times 10^{-3}$ eV and $\sigma = 0.274$ nm.

a. Show that the equilibrium separation between the atoms in an inert gas crystal is given by $r_o = (1.090)\sigma$. What is the equilibrium interatomic separation in the Ne crystal?

b. Find the bonding energy per atom in solid Ne.

c. Calculate the density of solid Ne (atomic mass = 20.18 g/mol).

Solution

a Let E = potential energy and x = distance variable. The energy E is given by

$$E(x) = -2\varepsilon\left[14.45\left(\frac{\sigma}{x}\right)^6 - 12.13\left(\frac{\sigma}{x}\right)^{12}\right]$$

The force F on each atom is given by

$$F(x) = -\frac{dE(x)}{dx} = 2\varepsilon\left[145.56\frac{\sigma\left(\frac{\sigma}{x}\right)^{11}}{x^2} - 86.7\frac{\sigma\left(\frac{\sigma}{x}\right)^{5}}{x^2}\right]$$

$\therefore$ $$F(x) = 2\varepsilon\left[145.56\frac{\sigma^{12}}{x^{13}} - 86.7\frac{\sigma^6}{x^7}\right]$$

When the atoms are in equilibrium, this net force must be zero. Using r_o to denote equilibrium separation,

$$F(r_o) = 0$$

$\therefore$ $$2\varepsilon\left[145.56\frac{\sigma^{12}}{r_o^{13}} - 86.7\frac{\sigma^6}{r_o^7}\right] = 0$$

$\therefore$ $$145.56\frac{\sigma^{12}}{r_o^{13}} = 86.7\frac{\sigma^6}{r_o^7}$$

$$\therefore \qquad \frac{r_o^{13}}{r_o^7} = \left(\frac{145.56}{86.7}\right)\frac{\sigma^{12}}{\sigma^6}$$

$$\therefore \qquad \boldsymbol{r_o = 1.090\sigma}$$

For the Ne crystal, $\sigma = 2.74 \times 10^{-10}$ m and $\varepsilon = 0.003121$ eV. Therefore,

$$\boldsymbol{r_o} = 1.090(2.74 \times 10^{-10} \text{ m}) = \mathbf{2.99 \times 10^{-10}\ m\ for\ Ne.}$$

b Calculate energy per atom at equilibrium:

$$E(r_o) = -2\varepsilon\left[14.45\left(\frac{\sigma}{r_o}\right)^6 - 12.13\left(\frac{\sigma}{r_o}\right)^{12}\right]$$

$$\therefore \qquad E(r_o) = -2(0.003121 \text{ eV})(1.602\times10^{-19} \text{ J/eV})\left[14.45\left(\frac{2.74\times10^{-10}\text{ m}}{2.99\times10^{-10}\text{ m}}\right)^6 - 12.13\left(\frac{2.74\times10^{-10}\text{ m}}{2.99\times10^{-10}\text{ m}}\right)^{12}\right]$$

$$\therefore \qquad \boldsymbol{E(r_o) = -4.30 \times 10^{-21}\ J} \text{ or } \mathbf{-0.0269\ eV}$$

Therefore the bonding energy in solid Ne is 0.027 eV per atom.

c To calculate the density, remember that the unit cell is FCC, and density = (mass of atoms in the unit cell) / (volume of unit cell). There are 4 atoms per FCC unit cell, and the atomic mass of Ne is 20.18 g/mol.

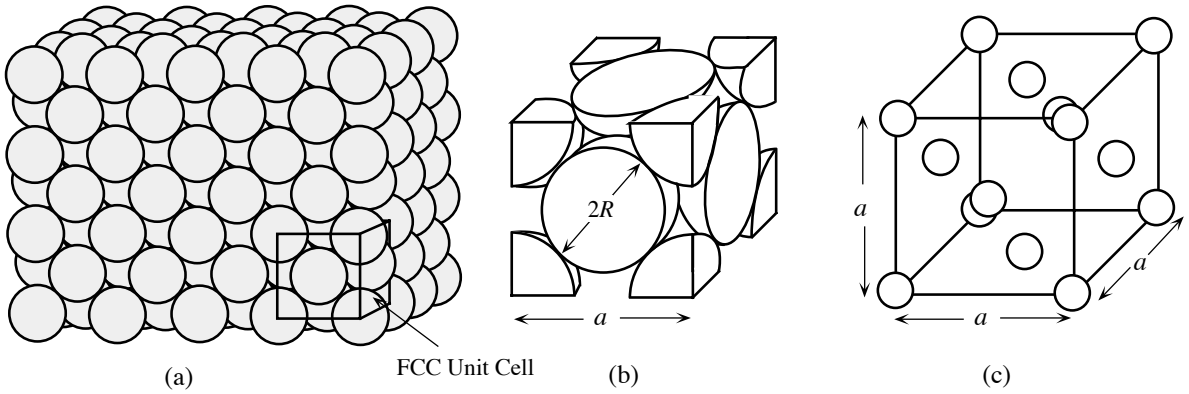


Figure 1Q3-1

(a) The crystal structure of copper which is face-centered-cubic (FCC). The atoms are positioned at well-defined sites arranged periodically, and there is a long-range order in the crystal.

(b) An FCC unit cell with close-packed spheres.

(c) Reduced-sphere representation of the FCC unit cell.

Examples: Ag, Al, Au, Ca, Cu, γ-Fe (>912 °C), Ni, Pd, Pt, Rh.

Since it is an FCC crystal structure, let a = lattice parameter (side of cubic cell) and R = radius of atom. The shortest interatomic separation is $r_o = 2R$ (atoms in contact means nucleus to nucleus separation is $2R$ (see Figure 1Q3-1).

$$R = r_o/2$$

and $$2a^2 = (4R)^2$$

$$\therefore \qquad a = 2\sqrt{2}R = 2\sqrt{2}\left(\frac{r_o}{2}\right) = \sqrt{2}\left(2.99\times10^{-10}\ \text{m}\right)$$

$$\therefore \qquad a = 4.228\times10^{-10}\ \text{m}$$

Therefore, the volume (V) of the unit cell is:

$$V = a^3 = (4.228\times10^{-10}\ \text{m})^3 = 7.558\times10^{-29}\ \text{m}^3$$

The mass (m) of 1 Ne atom in grams is the atomic mass (M_{at}) divided by N_A, because N_A number of atoms have a mass of M_{at}.

$$m = M_{at} / N_A$$

$$\therefore \qquad m = \frac{(20.18\ \text{g/mol})(0.001\ \text{kg/g})}{6.022\times10^{23}\ \text{mol}^{-1}} = 3.351\times10^{-26}\ \text{kg}$$

There are 4 atoms per unit cell in the FCC cell. The density (ρ) can then be found by:

$$\rho = (4m) / V = [4\times(3.351\times10^{-26}\ \text{kg})] / (7.558\times10^{-29}\ \text{m}^3)$$

$$\therefore \qquad \rho = \mathbf{1774\ kg/m^3}$$

In g/cm^3 this density is:

$$\rho = \frac{1774\ \text{kg/m}^3}{(100\ \text{cm/m})^3}\times(1000\ \text{g/kg}) = \mathbf{1.77\ g/cm^3}$$

The density of solid Ne is 1.77 g cm^{-3}.

1.4 Kinetic Molecular Theory

Calculate the effective (rms) speeds of the He and Ne atoms in the He-Ne gas laser tube at room temperature (300 K).

Solution

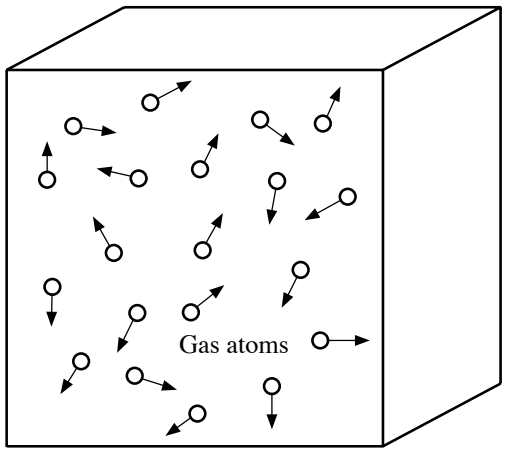


Figure 1Q4-1 The gas molecules in the container are in random motion.

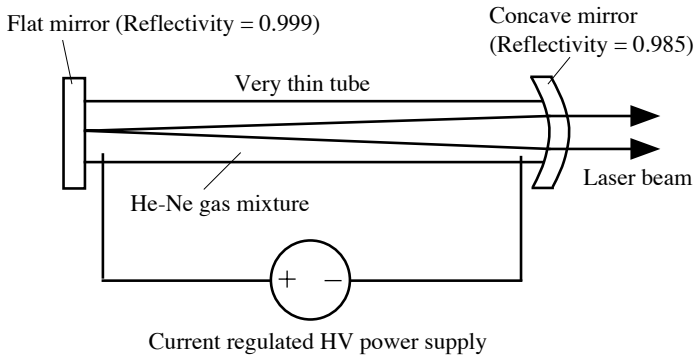


Figure 1Q4-2 The He-Ne gas laser.

To find the root mean square velocity (v_{rms}) of He atoms at T = 300 K:

The atomic mass of He is (from Periodic Table) M_{at} = 4.0 g/mol. Remember that 1 mole has a mass of M_{at} grams. Then one He atom has a mass (m) in kg given by:

$$m = \frac{M_{at}}{N_A} = \frac{4.0 \text{ g/mol}}{6.022\times10^{23} \text{ mol}^{-1}} \times (0.001 \text{ kg/g}) = 6.642\times10^{-27} \text{ kg}$$

From kinetic theory:

$$\frac{1}{2}m(v_{rms})^2 = \frac{3}{2}kT$$

$\therefore$ $$v_{rms} = \sqrt{3\frac{kT}{m}} = \sqrt{3\frac{(1.381\times10^{-23} \text{ J K}^{-1})(300 \text{ K})}{(6.642\times10^{-27} \text{ kg})}}$$

$\therefore$ $$\boldsymbol{v_{rms} = 1368 \text{ m/s}}$$

The root mean square velocity (v_{rms}) of Ne atoms at T = 300 K can be found using the same method as above, changing the atomic mass to that of Ne, M_{at} = 20.18 g/mol. After calculations, the mass of one Ne atom is found to be 3.351×10^{-26} kg, and the root mean square velocity (v_{rms}) of Ne is found to be $\boldsymbol{v_{rms} = 609 \text{ m/s}}$.

*1.5 Vacuum deposition

Consider air as composed of nitrogen molecules N_2.

a. What is the concentration n (number of molecules per unit volume) of N_2 molecules at 1 atm and 27 °C?

b. Estimate the mean separation between the N_2 molecules.

c. Assume each molecule has a finite size that can be represented by a sphere of radius r. Also assume that ℓ is the **mean free path,** defined as the mean distance a molecule travels before colliding with another molecule, as illustrated in Figure 1Q5-1a. If we consider the motion of one N_2 molecule, with all the others stationary, it is apparent that if the path of the traveling molecule crosses the cross-sectional area $S = \pi(2r)^2$, there will be a collision. Since ℓ is the mean distance between collisions, there must be at least one stationary molecule within the volume $S\ell$, as shown in Figure 1Q5-1a. Since n is the concentration, we must have $n(S\ell) = 1$ or $\ell = 1/(\pi 4r^2 n)$. However, this must be corrected for the fact that all the molecules are in motion, which only introduces a numerical factor, so that

$$\ell = \frac{1}{2^{1/2} 4\pi r^2 n}$$

Assuming a radius r of 0.1 nm, calculate the mean free path of N_2 molecules between collisions at 27 °C and 1 atm.

d. Assume that an Au film is to be deposited onto the surface of an Si chip to form metallic interconnections between various devices. The deposition process is generally carried out in a vacuum chamber and involves the condensation of Au atoms from the vapor phase onto the chip surface. In one procedure, a gold wire is wrapped around a tungsten filament, which is heated by passing a large current through the filament (analogous to the heating of the filament in a light bulb) as depicted in Figure 1Q5-1b. The Au wire melts and wets the filament, but as the temperature of the filament increases, the gold evaporates to form a vapor. Au atoms from this vapor then condense onto the chip surface, to solidify and form the metallic connections. Suppose that the source (filament)-to-substrate (chip) distance L is 10 cm. Unless the mean free path of air molecules is much longer than L, collisions between the metal atoms and air molecules will prevent the deposition of the Au onto the chip surface. Taking the mean free path ℓ to be $100L$, what should be the pressure inside the vacuum system? (Assume the same r for Au atoms).

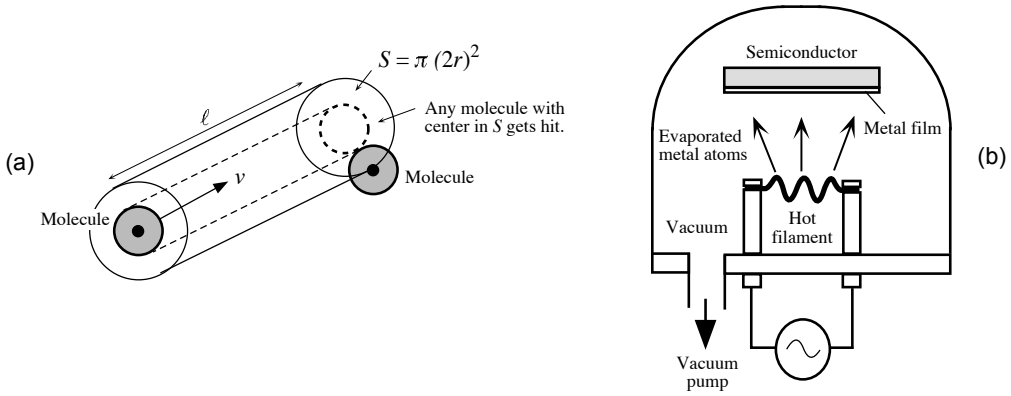


Figure 1Q5-1

(a) A molecule moving with a velocity v travels a mean distance ℓ between collisions. Since the collision cross-sectional area is S, in the volume $S\ell$ there must be at least one molecule. Consequently, $n(S\ell) = 1$.

(b) Vacuum deposition of metal electrodes by thermal evaporation.

Solution

Assume $T = 300$ K throughout. The radius of the nitrogen molecule (given approximately) is $r = 0.1 \times 10^{-9}$ m. Also, we know that pressure $P = 1$ atm $= 1.013 \times 10^5$ Pa.

Let N = total number of molecules , V = volume and k = Boltzmann's constant. Then:

$$PV = NkT$$

(Note: this equation can be derived from the more familiar form of $PV = \eta RT$, where η is the total number of moles, which is equal to N/N_A, and R is the gas constant, which is equal to $k \times N_A$.)

The concentration n (number of molecules per unit volume) is defined as:

$$n = N/V$$

Substituting this into the previous equation, the following equation is obtained:

$$P = nkT$$

a What is n at 1 atm and $T = 27$ °C + 273 = 300 K?

$$n = P/(kT)$$

$$\therefore \quad n = \frac{1.013\times10^{5}\ \text{Pa}}{\left(1.381\times10^{-23}\ \text{J K}^{-1}\right)(300\ \text{K})}$$

$$\therefore \quad \boldsymbol{n = 2.45\times10^{25}\ \text{molecules per m}^3}$$

b What is the mean separation between the molecules?

Consider a crystal of the material which is a cube of unit volume, each side of unit length as shown in Figure 1Q5-2. To each atom we can attribute *a portion of the whole volume* which for simplicity is a cube of side *d*. Thus, each atom is considered to occupy a volume of d^3.

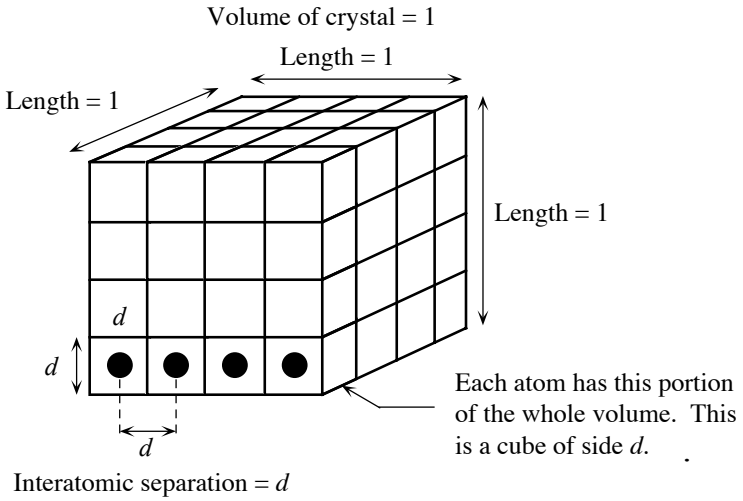


Figure 1Q5-2 Relationship between interatomic separation and the number of atoms per unit volume.

The actual or true volume of the atom does not matter. All we need to know is how much volume an atom has around it given all the atoms are identical and that adding all the atomic volumes must give the whole volume of the crystal.

Suppose that there are *n* atoms in this crystal. Then *n* is the atomic concentration, number of atoms per unit volume. Clearly, *n* atoms make up the crystal so that

$$n\,d^{3} = \text{Crystal volume} = 1$$

Remember that this is only an approximation. The separation between any two atoms is *d*. Thus,

$$d = \frac{1}{n^{\frac{1}{3}}} = \frac{1}{\sqrt[3]{\left(2.45\times10^{25}\ \text{m}^{-3}\right)}}$$

$$\therefore \quad \boldsymbol{d = 3.44\times10^{-9}\ \text{m}}\ \text{or}\ \boldsymbol{3.4\ \text{nm}}$$

c Assuming a radius, *r*, of 0.1 nm, what is the mean free path, ℓ, between collisions?

$$\ell = \frac{1}{\sqrt{2}\cdot 4\pi r^{2} n} = \frac{1}{\sqrt{2}\cdot 4\pi\left(0.1\times10^{-9}\ \text{m}\right)^{2}\left(2.45\times10^{25}\ \text{m}^{-3}\right)}$$

$$\therefore \quad \boldsymbol{\ell = 2.30\times10^{-7}\ \text{m}}\ \text{or}\ \boldsymbol{230\ \text{nm}}$$

d We need the new mean free path, $\ell' = 100L$, or 0.1 m × 100 (*L* is the source-to-substrate distance)

$$\ell = 10\ \text{m}$$

The new ℓ' corresponds to a new concentration n' of nitrogen molecules.

$$\ell' = \frac{1}{\sqrt{2} \cdot 4\pi r^2 n'}$$

$$\therefore \quad n' = \frac{1}{8}\frac{\sqrt{2}}{\pi r^2 \ell'} = \frac{1}{8}\frac{\sqrt{2}}{\pi\left(0.1\times10^{-9}\ \text{m}\right)^2(10\ \text{m})} = 5.627 \times 10^{17}\ \text{m}^{-3}$$

This new concentration of nitrogen molecules requires a new pressure, P':

$$\boldsymbol{P' = n'kT} = (5.627 \times 10^{17}\ \text{m}^{-3})(1.381 \times 10^{-23}\ \text{J K}^{-1})(300\ \text{K}) = \mathbf{0.00233\ Pa}$$

In atmospheres this is:

$$\boldsymbol{P'} = \frac{0.00233\ \text{Pa}}{1.013\times10^5\ \text{Pa/atm}} = \mathbf{2.30\times10^{-8}\ atm}$$

In units of torr this is:

$$\boldsymbol{P'} = \left(2.30\times10^{-8}\ \text{atm}\right)(760\ \text{torr/atm}) = \mathbf{1.75\times10^{-5}\ torr}$$

There is an important assumption made, namely that the cross sectional area of the Au atom is about the same as that of N_2 so that the expression for the mean free path need not be modified to account for different sizes of Au atoms and N_2 molecules. The calculation gives a magnitude that is quite close to those used in practice , e.g. a pressure of 10^{-5} torr.

1.6 Heat capacity

a. Calculate the heat capacity per mole and per gram of N_2 gas, neglecting the vibrations of the molecule. How does this compare with the experimental value of 0.743 J g^{-1} K^{-1}?

b. Calculate the heat capacity per mole and per gram of CO_2 gas, neglecting the vibrations of the molecule. How does this compare with the experimental value of 0.648 J K^{-1} g^{-1}? Assume that CO_2 molecule is linear (O-C-O), so that it has two rotational degrees of freedom.

c. Based on the Dulong-Petit rule, calculate the heat capacity per mole and per gram of solid silver. How does this compare with the experimental value of 0.235 J K^{-1} g^{-1}?

d. Based on the Dulong-Petit rule, calculate the heat capacity per mole and per gram of the silicon crystal. How does this compare with the experimental value of 0.71 J K^{-1} g^{-1}?

Solution

a N_2 has 5 degrees of freedom: 3 translational and 2 rotational. Its molar mass is $M_{at} = 2 \times 14.01$ g/mol = 28.02 g/mol.

Let C_m = heat capacity per mole, C_s = specific heat capacity (heat capacity per gram), and R = gas constant, then:

$$\boldsymbol{C_m} = \frac{5}{2}R = \frac{5}{2}\left(8.315\ \text{J K}^{-1}\ \text{mol}^{-1}\right) = \mathbf{20.8\ J\ K^{-1}\ mol^{-1}}$$

$$\therefore \quad \boldsymbol{C_s} = C_m / M_{at} = (20.8\ \text{J K}^{-1}\ \text{mol}^{-1})/(28.02\ \text{g/mol}) = \mathbf{0.742\ J\ K^{-1}\ g^{-1}}$$

This is close to the experimental value.

b CO_2 has the linear structure O=C=O. Rotations about the molecular axis have negligible rotational energy as the moment of inertia about this axis is negligible. There are therefore 2 rotational degrees of

freedom. In total there are 5 degrees of freedom: 3 translational and 2 rotational. Its molar mass is

$$M_{at} = 12.01 + 2 \times 16 = 44.01 \text{ g/mol.}$$

$$\boldsymbol{C_m} = \frac{5}{2}R = \frac{5}{2}\left(8.315 \text{ J K}^{-1} \text{ mol}^{-1}\right) = \mathbf{20.8 \text{ J K}^{-1} \text{ mol}^{-1}}$$

$\therefore$ $\boldsymbol{C_s} = C_m / M_{at} = (20.8 \text{ J K}^{-1} \text{ mol}^{-1})/(44.01 \text{ g/mol}) = \mathbf{0.47 \text{ J K}^{-1} \text{ g}^{-1}}$

This is smaller than the experimental value 0.648 J K^{-1} g^{-1}, because vibrational energy was neglected in the 5 degrees of freedom assigned to the CO_2 molecule.

c For solid silver, there are 6 degrees of freedom: 3 vibrational *KE* and 3 elastic *PE* terms. Its molar mass is M_{at} = 107.87 g/mol.

$$\boldsymbol{C_m} = \frac{6}{2}R = \frac{6}{2}\left(8.315 \text{ J K}^{-1} \text{ mol}^{-1}\right) = \mathbf{24.9 \text{ J K}^{-1} \text{ mol}^{-1}}$$

$\therefore$ $\boldsymbol{C_s} = C_m / M_{at} = (24.9 \text{ J K}^{-1} \text{ mol}^{-1})/(107.87 \text{ g/mol}) = \mathbf{0.231 \text{ J K}^{-1} \text{ g}^{-1}}$

This is very close to the experimental value.

d For a solid, heat capacity per mole is 3*R*. The molar mass of Si is M_{at} = 28.09 g/mol.

$$\boldsymbol{C_m} = \frac{6}{2}R = \frac{6}{2}\left(8.315 \text{ J K}^{-1} \text{ mol}^{-1}\right) = \mathbf{24.9 \text{ J K}^{-1} \text{ mol}^{-1}}$$

$\therefore$ $\boldsymbol{C_s} = C_m / M_{at} = (24.9 \text{ J K}^{-1} \text{ mol}^{-1})/(28.09 \text{ g/mol}) = \mathbf{0.886 \text{ J K}^{-1} \text{ g}^{-1}}$

The experimental value is substantially less and is due to the failure of classical physics. One has to consider the quantum nature of the atomic vibrations and also the distribution of vibrational energy among the atoms. The student is referred to modern physics texts (under heat capacity in the Einstein model and the Debye model of lattice vibrations).

1.7 Thermal expansion

a. If λ is the thermal expansion coefficient, show that the thermal expansion coefficient for an area is 2λ. Consider an aluminum square sheet of area 1 cm^2. If the thermal expansion coefficient of Al at room temperature (25 °C) is about 24×10^{-6} K^{-1}, at what temperature is the percentage change in the area +1%?

b. The density of silicon at 25 °C is 2.329 g cm^{-3}. Estimate the density of silicon at 1000 °C given that the thermal expansion coefficient for Si over this temperature range has a mean value of about 3.5×10^{-6} K^{-1}.

c. The thermal expansion coefficient of Si depends on temperature as,

$$\lambda = 3.725 \times 10^{-6}\left\{1 - \exp\left[-0.00588(T - 124)\right]\right\} + 5.548 \times 10^{-10}\, T$$

where *T* is in Kelvins. The change $\delta\rho$ in the density due to a change δT in temperature is given by

$$\delta\rho = -\rho_0 \alpha_V \delta T = -3\rho_0 \lambda \delta T$$

By integrating this equation , calculate the density of Si at 1000 °C and compare with the result in *b*.

Solution

a Consider an rectangular area with sides x_o and y_o. Then at temperature T_0,

$$A_0 = x_0 y_0$$

and at temperature *T*,

$$A = \left[x_0(1+\lambda\Delta T)\right]\left[y_0(1+\lambda\Delta T)\right] = x_0 y_0(1+\lambda\Delta T)^2$$

that is

$$A = x_0 y_o\left[1+2\lambda\Delta T+(\lambda\Delta T)^2\right].$$

We can now use that $A_0 = x_0 y_0$ and neglect the term $(\lambda\Delta T)^2$ because it is very small in comparison with the linear term $\lambda\Delta T$ (λ<<1) to obtain

$$A = A_0(1+2\lambda\Delta T) = A_0(1+\alpha_A\Delta T)$$

So the area expansion coefficient is

$$\alpha_A = 2\lambda$$

The area of the aluminum sheet at any temperature is given by

$$A = A_0\left[1+2\lambda(T-T_0)\right]$$

where A_o is the area at the reference temperature T_0 . Solving for *T* we receive

$$T = T_0 + \frac{1}{2}\frac{A/A_0 - 1}{\lambda} = 25\ ^o\text{C} + \frac{1}{2}\frac{(1.01)-1}{24\times10^{-6}\ ^o\text{C}^{-1}} = \mathbf{233.3\ °C}.$$

b. As it is shown in Example 1.6 (in textbook) the volume expansion with temperature leads to density reduction

$$\rho = \frac{\rho_0}{1+3\lambda(T-T_0)} \approx \rho_0\left[1-3\lambda(T-T_0)\right] =$$

$$= \left(2.329\ \text{g cm}^{-3}\right)\left[1-3\left(3.5\times10^{-6}\ ^o\text{C}^{-1}\right)\left(1000\ ^o\text{C}-25\ ^o\text{C}\right)\right] = \mathbf{2.305\ g\ cm^{-3}}$$

c. By integrating the equation we receive

$$\int_{\rho_o}^{\rho} d\rho = -3\rho_0\int_{T_0}^{T}\lambda(T)dT$$

which gives the following relation

$$\rho = \rho_0\left(1-3\int_{T_o}^{T}\lambda dT\right)$$

Substituting 300 K for T_0, 1273 K for *T* and λ with the given expression, and carrying out the integration, which is straightforward, we receive

$$\rho(1273\ K) = \mathbf{2.302\ g\ cm^{-3}}$$

which is very close to the value in **b**.

*1.8 Thermal fluctuations

The cross section of a typical moving-coil ammeter is shown in Figure 1Q8-1. The rectangular moving coil (extended into the paper) is placed in the air gap between the poles of a permanent magnet and a

fixed, cylindrical iron core. The iron core ensures a strong magnetic field through the coil. The coil is hinged to rotate in the air gap and around the iron core. When a current I is passed through the coil, it experiences a torque and tries to rotate, but the rotation is restricted by the spring. A needle attached to coil indicates the amount of rotation. The deflection θ of the needle is proportional to the torque τ, by virtue of Hooke's Law

$$\theta = K\tau$$

where, from electromagnetic theory, the torque is proportional to the product of the current I and the magnetic field B. Thus, θ measures the current I.

A given ammeter has a coil inductance of 0.1 mH and a coil resistance of 100 Ω. The spring constant K is 10^{11} rad N^{-1} m^{-1} and the length of the needle is 10 cm.

a. What is the rms fluctuation in the position of the ammeter needle?

b. Assume the ammeter is placed in a closed circuit where the dc current is zero ($I = 0$), as shown in Figure 1Q8-1. Sketch the instantaneous current $i(t)$ versus time. What is the minimum current this ammeter can measure (noise in the current)? How can this be improved?

Solution

We can assume a room temperature of $T = 300$ K, and we are given the ammeter's characteristics.

a The spring constant (K) in this case relates θ and τ (torque) by the equation:

$$\theta = K\tau$$

Spring constant: $K = 10^{11}$ N^{-1} m^{-1} (Remember that rads are dimensionless units, and do not need to be carried through calculations)

Length of pointer: $d = 0.1$ m

Inductance of coil: $L = 0.0001$ H

θ is the rms (root mean square) value of the angle fluctuations. It must be in radians. The average energy fluctuations in the spring must be of the order of $(1/2)kT$.

$$\frac{1}{2}\frac{1}{K}\theta^2 = \frac{1}{2}kT$$

$$\therefore \quad \theta = \sqrt{TKk} = \sqrt{(300\ \text{K})\left(10^{11}\ \text{N}^{-1}\ \text{m}^{-1}\right)\left(1.381\times10^{-23}\ \text{J K}^{-1}\right)}$$

$$\therefore \quad \boldsymbol{\theta = 2.04 \times 10^{-5}\ \text{rad}}$$

As the needle rotates randomly left and right about its equilibrium position, its tip is displaced randomly. These random fluctuations in the displacement have an rms value of (where d is the length of the needle):

$$x_{rms} = \theta d = (2.04 \times 10^{-5}\ \text{rad})(0.10\ \text{m/rad}) = \mathbf{2.04 \times 10^{-6}\ m}\ \text{or}\ \mathbf{2\ \mu m}$$

Certainly this extent is visible under a microscope, so if you tried to measure the position of the needle accurately by focusing a microscope on it to obtain the current accurately, you'd be wasting time.

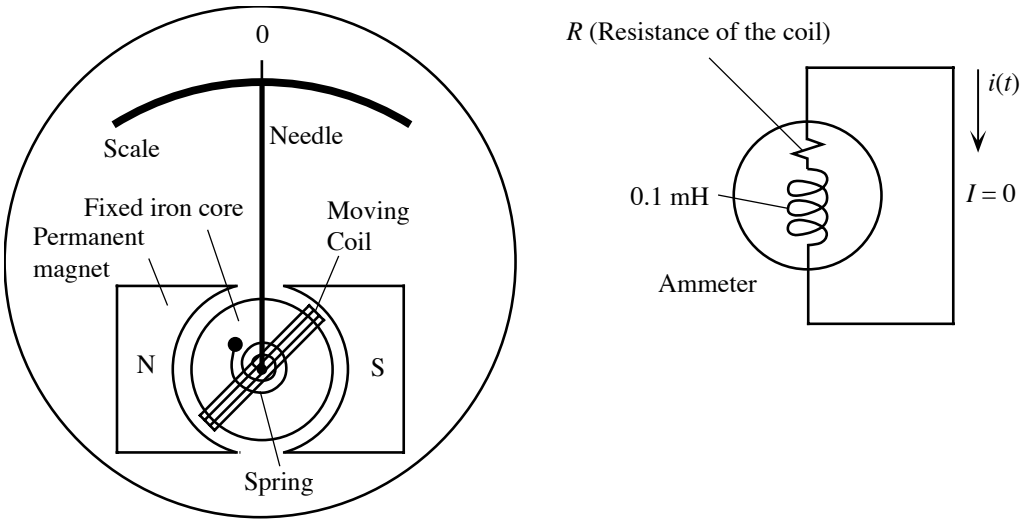


Figure 1Q8-1 Left: The ammeter. Right: The ammeter short circuited. It should be reading a zero current at all times (short circuit). But is it?

b Suppose that the ammeter is short circuited as in Figure 1Q8-1 and the current through the ammeter is monitored by an invisible, non-intrusive, second noiseless ammeter. Let i be the instantaneous current in the circuit. Its rms value is then i_{rms}. The average energy fluctuation due to inductance in the coil must be in the order of (1/2)kT:

$$\frac{1}{2}Li_{rms}^{\ 2} = \frac{1}{2}kT$$

$$\therefore \qquad i_{rms} = \sqrt{\frac{Tk}{L}} = \sqrt{\frac{(300\ \text{K})\left(1.381\times10^{-23}\ \text{J K}^{-1}\right)}{(0.0001\ \text{H})}}$$

$$\therefore \qquad \boldsymbol{i_{rms} = 6.44 \times 10^{-9}\ \text{A}} \text{ or } \mathbf{6\ nA}$$

Noise in the current measurements is 6 nA. If you were to examine the current in the circuit you would see the following:

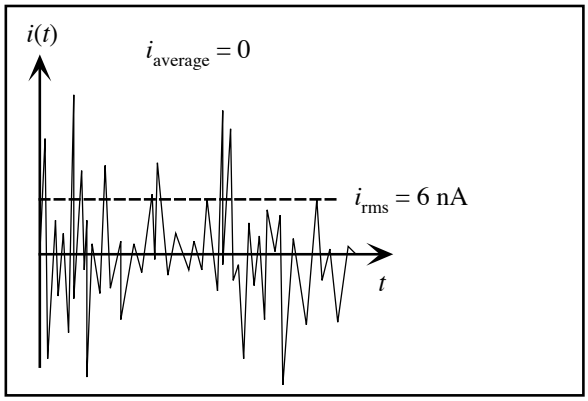


Figure 1Q8-2 Noise fluctuations of current in ammeter.

Assumption: When we considered the random fluctuations, we assumed that the mechanical motion containing the spring (energy storage element) was decoupled from the electrical circuit containing the inductance (energy storage element). An exact analysis is quite difficult but this first order treatment provides rough values on the limits of the system. We should calculate how much current, denoted i_n, is needed to provide a deflection that is equal to x_{rms}. For this approach we have to know more about the electromagnetic operation of the ammeter coil. For a more rigorous calculation see *Electricity and Magnetism*, Third Edition, B. I. Bleaney and B. Bleaney (Oxford University Press, Oxford, 1976), Ch. 23.

1.9 Electrical noise

Consider an amplifier with a bandwidth B of 5 kHz, corresponding to a typical speech bandwidth. Assume the input resistance of the amplifier is 1 MΩ. What is the rms noise voltage at the input? What will happen if the bandwidth is doubled to 10 kHz? What is your conclusion?

Solution

Bandwidth: $B = 5 \times 10^3$ Hz

Input resistance: $R_{in} = 10^6\ \Omega$

Assuming room temperature: $T = 300$ K

The noise voltage (or the rms voltage, v_{rms}) across the input is given by:

$$v_{rms} = \sqrt{4kTR_{in}B} = \sqrt{4\left(1.381\times10^{-23}\ \text{J K}^{-1}\right)(300\ \text{K})\left(10^6\ \Omega\right)\left(5\times10^3\ \text{Hz}\right)}$$

$\therefore$ $$\boldsymbol{v_{rms} = 9.10 \times 10^{-6}\ \text{V}}\ \text{or}\ \boldsymbol{9.1\ \mu\text{V}}$$

If the bandwidth is doubled: $B' = 2 \times 5 \times 10^3 = 10 \times 10^3$ Hz

$\therefore$ $$\boldsymbol{v'_{rms} = \sqrt{4kTR_{in}B'} = 1.29\times10^{-5}\ \text{V}}\ \text{or}\ \boldsymbol{12.9\ \mu\text{V}}$$

The larger the bandwidth, the greater the noise voltage. 5-10 kHz is a typical speech bandwidth and the input signal into the amplifier must be much greater than ~13 μV for amplification without noise added from the amplifier itself.

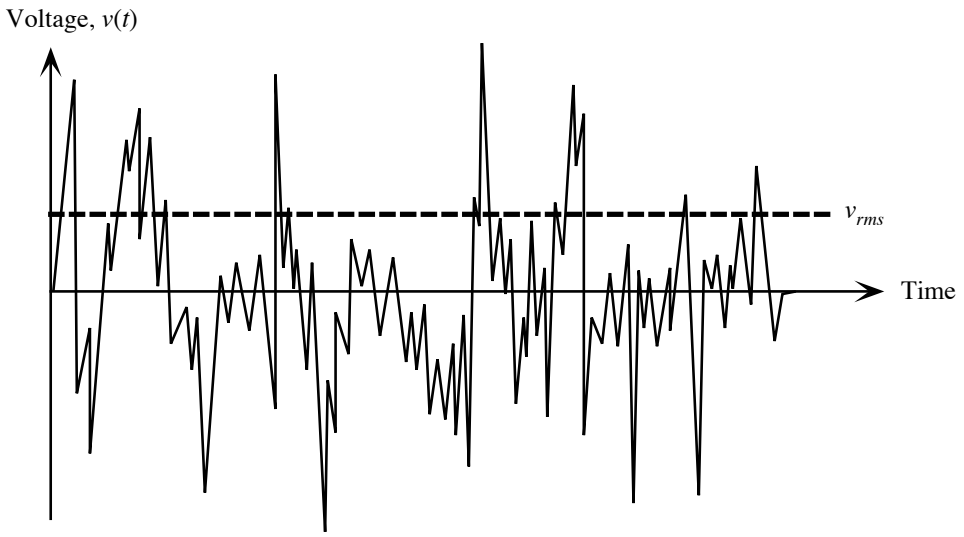


Figure 1Q9-1 Random motion of conduction electrons in a conductor results in electrical noise.

1.10 Thermal activation

A certain chemical oxidation process (e.g., SiO_2) has an activation energy of 2 eV atom^{-1}.

a. Consider the material exposed to pure oxygen gas at a pressure of 1 atm. at 27 °C. Estimate how many oxygen molecules per unit volume will have energies in excess of 2 eV? (Consider numerical integration of Equation 1.17, listed in textbook)

b. If the temperature is 900 °C, estimate the number of oxygen molecules with energies more than 2 eV. What happens to this concentration if the pressure is doubled?

Solution 1: Method of Estimation

The activation energy (E_A) of one atom is 2 eV/atom, or:

$$E_A = (2 \text{ eV/atom})(1.602 \times 10^{-19} \text{ J/eV}) = 3.204 \times 10^{-19} \text{ J/atom}$$

a We are given pressure $P = 1$ atm $= 1.013 \times 10^5$ Pa and temperature $T = 300$ K. If we consider a portion of oxygen gas of volume $V = 1$ m^3 (the unit volume), the number of molecules present in the gas (N) will be equal to the concentration of molecules in the gas (n), i.e.: $N = n$. And since we know $\eta = N/N_A$, where η is the total number of moles and N_A is Avogadro's number, we can make the following substitution into the equation $PV = \eta RT$:

$$PV = \frac{n}{N_A} RT$$

isolating n,
$$n = \frac{N_A PV}{RT} = \frac{\left(6.022 \times 10^{23} \text{ mol}^{-1}\right)\left(1.013 \times 10^5 \text{ Pa}\right)\left(1 \text{ m}^3\right)}{\left(8.315 \text{ m}^3 \text{ Pa K}^{-1} \text{ mol}^{-1}\right)(300 \text{ K})}\left(\frac{1}{\text{m}^3}\right)$$

$\therefore$
$$n = 2.445 \times 10^{25} \text{ m}^{-3}$$

Therefore there are 2.445×10^{25} oxygen molecules per unit volume.

For an estimation of the concentration of molecules with energy above 2 eV, we can use the following approximation (remember to convert E_A into Joules). If n_A is the concentration of molecules with $E > E_A$, then:

$$\frac{n_A}{n} = \exp\left(-\frac{E_A}{kT}\right)$$

$\therefore$
$$n_A = n \exp\left(-\frac{E_A}{kT}\right) = \left(2.445 \times 10^{25} \text{ m}^{-3}\right) \exp\left[-\frac{\left(3.204 \times 10^{-19} \text{ J}\right)}{\left(1.381 \times 10^{-23} \text{ J K}^{-1}\right)(300 \text{ K})}\right]$$

$\therefore$
$$\boldsymbol{n_A = 6.34 \times 10^{-9} \text{ m}^{-3}}$$

However, this answer is only in the right order of magnitude. For a better calculation we need to use a numerical integration of $n(E)$ from E_A to ∞.

b At $T = 900$ °C + 273 = 1173 K and $P = 1$ atm, the same method as above can be used to find the concentration of molecules with energy greater than E_A. After calculations, the following numbers will be obtained:

$$n = 6.254 \times 10^{24} \text{ m}^{-3}$$

$$\boldsymbol{n_A = 1.61 \times 10^{16} \text{ m}^{-3}}$$

This corresponds to an increase by a factor of 10^{25} compared to a temperature of $T = 300$ K.

Doubling the pressure doubles n and hence doubles n_A. In the oxidation of Si wafers, high pressures lead to more rapid oxidation rates and a shorter time for the oxidation process.

Solution 2: Method of Numerical Integration

To find the number of molecules with energies greater than $E_A = 2$ eV more accurately, numerical integration must be used. Suppose that N is the total number of molecules. Let

$$y = n_E/N$$

where n_E is the number of molecules per unit energy, so that $n_E\, dE$ is the number of molecules in the energy range dE.

Define a new variable x for the sake of convenience:

$$x = \frac{E}{kT}$$

$\therefore$ $$E = Tkx$$

where E is the energy of the molecules. Using partial differentiation:

$$dE = Tk\ dx$$

The energy distribution function is given by:

$$y = \frac{2}{\pi^{\frac{1}{2}}}\left(\frac{1}{kT}\right)^{\frac{3}{2}} E^{\frac{1}{2}} \exp\left(-\frac{E}{kT}\right)$$

substitute: $$y = 2\frac{\left(\frac{1}{kT}\right)^{\frac{3}{2}} \exp(-x)}{\sqrt{\pi}}\sqrt{Tkx}$$

simplify: $$y = 2\frac{\exp(-x)\sqrt{x}}{\sqrt{\pi}Tk}$$

Integration of y with respect to E can be changed to that with respect to x as follows:

$$\int y dE = 2\frac{\int \exp(-x)\sqrt{x}dE}{\sqrt{\pi}Tk}$$

$\therefore$ $$\int y dE = 2\frac{\int \exp(-x)\sqrt{x}dx}{\sqrt{\pi}} \qquad \text{(substitute } dE = Tk\ dx\text{)}$$

We need the lower limit x_A for x corresponding to E_A:

$$x_A = \frac{E_A}{kT} = \frac{3.204\times10^{-19}\ \text{J}}{\left(1.381\times10^{-23}\ \text{J K}^{-1}\right)(300\ \text{K})} = 77.36$$

This is the activation energy E_A in terms of x.

We also need the upper limit, x_b which should be ∞, but we will take it to be a multiple of x_A:

$$x_b = 2 \times x_A = 154.72$$

We do this because the numerical integration is difficult with very small numbers, e.g. $\exp(-x_A) = 2.522 \times 10^{-34}$.

The fraction, F, of molecules with energies greater than E_A ($= x_A$) is:

$$F = \int_{E_A}^{\infty} y dE = \int_{x_A}^{x_B}\left(2\frac{\exp(-x)\sqrt{x}}{\sqrt{\pi}}\right)dx = 2.519\times10^{-33}$$

At $T = 300$ K, $P = 1$ atm $= 1.013 \times 10^5$ Pa, and $V = 1$ m^3, the number of molecules per unit volume is n:

$$PV = \frac{n}{N_A}RT$$

$\therefore$ $$n = \frac{N_A PV}{RT} = 2.445 \times 10^{25} \ \text{m}^{-3}$$

The concentration of molecules with energy greater than E_A (n_A) can be found using the fraction F :

$$\boldsymbol{n_A} = nF = (2.445 \times 10^{25}\ \text{m}^{-3})(2.519 \times 10^{-33}) = \mathbf{6.16 \times 10^{-8}\ m^{-3}}$$

If we estimate n_A by multiplying n by the Boltzmann factor as previously:

$$n_A = n \exp\left(-\frac{E_A}{kT}\right) = 6.34 \times 10^{-9} \ \text{m}^{-3}$$

The estimate is out by a factor of about 10.

b The concentration of molecules with energy greater than E_A (n_E) can be found at T = 900 °C + 273 = 1173 K using the same method as in part **a**. After calculations, the following values will be obtained:

$$F = 1.3131 \times 10^{-8}$$

$$n = 6.254 \times 10^{24}\ \text{m}^{-3}$$

$$\boldsymbol{n_A = 8.21 \times 10^{16}\ m^{-3}}$$

If we compare this value to the one obtained previously through estimation (1.61×10^{16} m^{-3}), we see the estimate is out by a factor of about 5. As stated previously, doubling the pressure doubles n and hence doubles n_A. In the oxidation of Si wafers, high pressures lead to more rapid oxidation rates and a shorter time for the oxidation process.

1.11 Diffusion in Si

The diffusion coefficient of boron (B) atoms in a single crystal of Si has been measured to be 1.5×10^{-18} m^2 s^{-1} at 1000 °C and 1.1×10^{-16} m^2 s^{-1} at 1200 °C.

a. What is the activation energy for the diffusion of B, in eV/atom?

b. What is the preexponential constant D_o?

c. What is the rms distance (in micrometers) diffused in 1 hour by the B atom in the Si crystal at 1200 °C and 1000 °C?

d. The diffusion coefficient of B in polycrystalline Si has an activation energy of 2.4-2.5 eV atom^{-1} and D_o = (1.5-6) $\times$ 10^{-7} m^2 s^{-1}. What constitutes the diffusion difference between the single crystal sample and the polycrystalline sample?

Solution

Given diffusion coefficients at two temperatures:

T_1 = 1200 °C + 273 = 1473 K $\qquad$ $D_1 = 1.1 \times 10^{-16}$ m^2/s

T_2 = 1000 °C + 273 = 1273 K $\qquad$ $D_2 = 1.5 \times 10^{-18}$ m^2/s

a The diffusion coefficients (D_1 and D_2) at certain temperatures (T_1 and T_2) are given by:

$$D_1 = D_o \exp\left(-\frac{E_A q}{kT_1}\right) \qquad\qquad D_2 = D_o \exp\left(-\frac{E_A q}{kT_2}\right)$$

where E_A is the activation energy in eV/atom. Since we know the following:

$$\frac{\exp(-u)}{\exp(-w)} = \exp(w - u)$$

we can take the ratio of the diffusion coefficients to express them in terms of the activation energy (E_A):

$$\frac{D_1}{D_2} = \frac{D_o \exp\left(-\frac{E_A q}{kT_1}\right)}{D_o \exp\left(-\frac{E_A q}{kT_2}\right)} = \exp\left(\frac{E_A q}{kT_2} - \frac{E_A q}{kT_1}\right) = \exp\left(\frac{q[T_1 - T_2]E_A}{T_1 T_2 k}\right)$$

$\therefore$ $$E_A = \frac{T_1 T_2 k \ln\left(\frac{D_1}{D_2}\right)}{q(T_1 - T_2)}$$

$\therefore$ $$E_A = \frac{(1473\ \text{K})(1273\ \text{K})\left(1.381\times10^{-23}\ \text{J K}^{-1}\right)\ln\left(\frac{1.1\times10^{-16}\ \text{m}^2/\text{s}}{1.5\times10^{-18}\ \text{m}^2/\text{s}}\right)}{\left(1.602\times10^{-19}\ \text{J/eV}\right)(1473\ \text{K} - 1273\ \text{K})}$$

$\therefore$ $\boldsymbol{E_A = 3.47\ \text{eV/atom}}$

b To find D_o, use one of the equations for the diffusion coefficients:

$$D_1 = D_o \exp\left(-\frac{E_A q}{kT_1}\right)$$

$\therefore$ $$D_o = \frac{D_1}{\exp\left(-\frac{E_A q}{kT_1}\right)} = \frac{\left(1.1\times10^{-16}\ \text{m}^2/\text{s}\right)}{\exp\left[-\frac{(3.47\ \text{eV})\left(1.602\times10^{-19}\ \text{J/eV}\right)}{\left(1.381\times10^{-23}\ \text{J K}^{-1}\right)(1473\ \text{K})}\right]}$$

$\therefore$ $\boldsymbol{D_0 = 8.12\times10^{-5}\ \text{m}^2/\text{s}}$

c Given: time (t) = (1 hr) × (3600 s/hr) = 3600 s

At 1000 °C, rms diffusion distance ($L_{1000\ °C}$) in time t is given by:

$$L_{1000\ °C} = \sqrt{2(D_2 t)} = \sqrt{2\left(1.5\times10^{-18}\ \text{m}^2/\text{s}\right)(3600\ \text{s})}$$

$\therefore$ $\boldsymbol{L_{1000\ °C} = 1.04\times10^{-7}\ \text{m}}$ or $\boldsymbol{0.104\ \mu\text{m}}$

At 1200 °C:

$$L_{1200\ °C} = = \sqrt{2(D_1 t)} = \sqrt{2\left(1.1\times10^{-16}\ \text{m}^2/\text{s}\right)(3600\ \text{s})}$$

$\therefore$ $\boldsymbol{L_{1200\ °C} = 8.90\times10^{-7}\ \text{m}}$ or $\boldsymbol{0.89\ \mu\text{m}}$ (almost 10 times longer than at 1000 °C)

d Diffusion in polycrystalline Si would involve diffusion along grain boundaries, which is easier than diffusion in the bulk. The activation energy is smaller because it is easier for an atom to break bonds and jump to a neighboring site; there are vacancies or voids and strained bonds in a grain boundary.

1.12 Diffusion in SiO_2

The diffusion coefficient of P atoms in SiO_2 has an activation energy of 2.30 eV atom^{-1} and $D_o = 5.73 \times 10^{-9}$ m^2 s^{-1}. What is the rms distance diffused in one hour by P atoms in SiO_2 at 1200 °C?

Solution

Given:

Temperature: $T = 1200\ ^\circ\text{C} + 273 = 1473\ \text{K}$

Damping constant: $D_o = 5.73 \times 10^{-9}\ \text{m}^2/\text{s}$

Activation energy: $E_A = 2.30\ \text{eV/atom}$

Time: $t = (1\ \text{hr}) \times (3600\ \text{s/hr}) = 3600\ \text{s}$

Using the following equation, find the diffusion coefficient D:

$$D = D_o \exp\left(-\frac{E_A q}{kT}\right)$$

$$\therefore \quad D = \left(5.73\times10^{-9}\ \text{m}^2/\text{s}\right)\exp\left(-\frac{(2.30\ \text{eV})\left(1.602\times10^{-19}\ \text{J/eV}\right)}{\left(1.381\times10^{-23}\ \text{J K}^{-1}\right)(1473\ \text{K})}\right)$$

$$\therefore \quad D = 7.793 \times 10^{-17}\ \text{m}^2/\text{s}$$

The rms diffusion distance $(L_{1200\ ^\circ\text{C}})$ is given as:

$$L_{1200\ ^\circ\text{C}} = \sqrt{2(Dt)} = \sqrt{2\left(7.793\times10^{-17}\ \text{m}^2/\text{s}\right)(3600\ \text{s})}$$

$$\therefore \quad \boldsymbol{L_{1200\ ^\circ\text{C}} = 7.49 \times 10^{-7}\ \text{m}} \text{ or } \boldsymbol{0.75\ \mu\text{m}}$$

Comment: The P atoms at 1200 °C seems to be able to diffuse through an oxide thickness of about 0.75 μm.

1.13 BCC and FCC Crystals

a. Molybdenum has the BCC crystal structure, has a density of 10.22 g cm^{-3} and an atomic mass of 95.94 g mol^{-1}. What is the atomic concentration, lattice parameter *a*, and atomic radius of molybdenum?

b. Gold has the FCC crystal structure, a density of 19.3 g cm^{-3} and an atomic mass of 196.97 g mol^{-1}. What is the atomic concentration, lattice parameter *a*, and atomic radius of gold?

Solution

a. Since molybdenum has BCC crystal structure, there are 2 atoms in the unit cell. The density is

$$\rho = \frac{\text{Mass of atoms in unit cell}}{\text{Volume of unit cell}} = \frac{(\text{Number of atoms in unit cell}) \times (\text{Mass of one atom})}{\text{Volume of unit cell}}$$

that is,

$$\rho = \frac{2\left(\frac{M_{at}}{N_A}\right)}{a^3}.$$

Solving for the lattice parameter *a* we receive

$$a = \sqrt[3]{\frac{2M_{at}}{\rho N_A}} = \sqrt[3]{\frac{2\left(95.94\times10^{-3}\ \text{kg mol}^{-1}\right)}{\left(10.22\times10^{3}\ \text{kg m}^{-3}\right)\left(6.022\times10^{23}\ \text{mol}^{-1}\right)}} = \mathbf{3.147\ \times\ 10^{-10}\ m}$$

The Atomic concentration is 2 atoms in a cube of volume a^3, *i.e.*

$$n_{at} = \frac{2}{a^3} = \frac{2}{\left(3.147\times10^{-10}\ \text{m}\right)^3} = \mathbf{6.415\ \times\ 10^{22}\ cm^{-3} = 6.415\ \times\ 10^{28}\ m^{-3}}$$

For a BCC cell, the lattice parameter a and the radius of the atom R are in the following relation (listed in Table 1.3 in the textbook):

$$R = \frac{a\sqrt{3}}{4}$$

i.e.

$$R = \frac{\left(3.147\times10^{-10}\ \text{m}\right)\sqrt{3}}{4} = \mathbf{1.363\ \times\ 10^{-10}\ m}$$

b. Gold has the FCC crystal structure, hence, there are 4 atoms in the unit cell (as shown in Table 1.3 in the textbook).

The lattice parameter a is

$$a = \sqrt[3]{\frac{4M_{at}}{\rho N_A}} = \sqrt[3]{\frac{4\left(196.97\times10^{-3}\ \text{kg mol}^{-1}\right)}{\left(19.3\times10^{3}\ \text{kg m}^{-3}\right)\left(6.022\times10^{23}\ \text{mol}^{-1}\right)}} = \mathbf{4.077\ \times\ 10^{-10}\ m}$$

The atomic concentration is

$$n_{at} = \frac{4}{a^3} = \frac{4}{\left(4.077\times10^{-10}\ \text{m}\right)^3} = \mathbf{5.901\ \times\ 10^{22}\ cm^{-3} = 5.901\ \times\ 10^{28}\ m^{-3}}$$

For an FCC cell, the lattice parameter a and the radius of the atom R are in the following relation (shown in Table 1.3 in the textbook):

$$R = \frac{a\sqrt{2}}{4}$$

i.e.

$$R = \frac{\left(4.077\times10^{-10}\ m\right)\sqrt{2}}{4} = \mathbf{1.442\ \times\ 10^{-10}\ m}$$

*1.14 BCC and FCC crystals

a. Consider iron below 912 °C, where its structure is BCC. Given the density of iron as 7.86 g cm^{-3} and its atomic mass as 55.85 g/mol, calculate the lattice parameter of the unit cell and the radius of the Fe atom.

b. At 912 °C, iron changes from the BCC (α-Fe) to the FCC (γ-Fe) structure. The radius of the Fe atom correspondingly changes from 0.1258 nm to 0.1291 nm. Calculate the density of γ-Fe and explain whether there is a volume expansion or contraction during this phase change.

Solution

Given:

Density of iron at room temperature: $\rho = 7.86 \times 10^3$ kg/m^3

Atomic mass of iron: $M_{at} = 55.85$ g/mole

a For the BCC structure, the density is given by:

$$\rho = \frac{2\dfrac{M_{at} \times \left(10^{-3} \text{ kg/g}\right)}{N_A}}{a^3}$$

Thus the lattice parameter a is:

$$a = \left(\frac{1}{\left(500 \text{ g/kg}\right)} \frac{M_{at}}{N_A \rho} \right)^{\frac{1}{3}}$$

$$\therefore \quad a = \left(\frac{1}{\left(500 \text{ g/kg}\right)} \frac{\left(55.85 \text{ g/mol}\right)}{\left(6.022 \times 10^{23} \text{ mol}^{-1}\right)\left(7.86 \times 10^3 \text{ kg/m}^3\right)} \right)^{\frac{1}{3}}$$

$$\therefore \quad \boldsymbol{a = 2.87 \times 10^{-10} \text{ m}}$$

The radius of the Fe atom, R, and the lattice parameter, a, are related.

$$a\sqrt{3} = 4R$$

$$\therefore \quad R = \frac{1}{4}\sqrt{3}a = \frac{1}{4}\sqrt{3}\left(2.87 \times 10^{-10} \text{ m}\right)$$

$$\therefore \quad \boldsymbol{R = 1.24 \times 10^{-10} \text{ m}}$$

b Fe has a BCC structure just below 912 °C (α-Fe). An Fe atom in the α-Fe state has a radius of $R_{BCC} = 0.1258 \times 10^{-9}$ m. The density of α-Fe is therefore:

$$\rho_{BCC} = \frac{2\dfrac{M_{at} \times \left(10^{-3} \text{ kg/g}\right)}{N_A}}{\left(\dfrac{4R_{BCC}}{\sqrt{3}}\right)^3} = \frac{2\dfrac{\left(55.85 \text{ g/mol}\right)\left(10^{-3} \text{ kg/g}\right)}{\left(6.022 \times 10^{23} \text{ mol}^{-1}\right)}}{\left(\dfrac{4\left(0.1258 \times 10^{-9} \text{ m}\right)}{\sqrt{3}}\right)^3}$$

$$\therefore \quad \boldsymbol{\rho_{BCC} = 7564 \text{ kg/m}^3} \text{ (less than at room temperature)}$$

Fe has a FCC structure just above 912 °C (γ-Fe). An Fe atom in the γ-Fe state has a radius of $R_{FCC} = 0.1291 \times 10^{-9}$ m (Remember that for a FCC structure, $a\sqrt{2} = 4R_{FCC}$). The density of γ-Fe is therefore:

$$\rho_{FCC} = \frac{4\dfrac{M_{at} \times \left(10^{-3} \text{ kg/g}\right)}{N_A}}{\left(\dfrac{4R_{FCC}}{\sqrt{2}}\right)^3} = \frac{4\dfrac{\left(55.85 \text{ g/mol}\right)\left(10^{-3} \text{ kg/g}\right)}{\left(6.022 \times 10^{23} \text{ mol}^{-1}\right)}}{\left(\dfrac{4\left(0.1291 \times 10^{-9} \text{ m}\right)}{\sqrt{2}}\right)^3}$$

$\therefore$ ρ_{FCC} **= 7620 kg/m^3**

As the density increases, the volume must contract (the Fe retains the same mass).

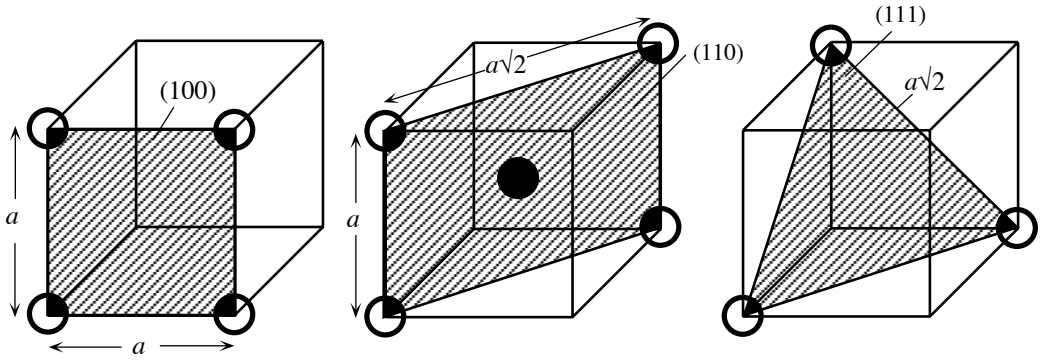
1.15 Planar and surface concentrations

Niobium (Nb) has the BCC crystal with a lattice parameter a = 0.3294 nm. Find the planar concentrations as the number of atoms per nm^2 of the (100), (110) and (111) planes. Which plane has the most concentration of atoms per unit area? Sometimes the number of atoms per unit area $n_{surface}$ on the surface of a crystal is estimated by using the relation $n_{surface} = n_{bulk}^{2/3}$ where n_{bulk} is the concentration of atoms in the bulk. Compare $n_{surface}$ values with the planar concentrations calculated above and comment on the difference. [Note: The BCC (111) plane does not cut through the center atom and the (111) has 1/6th of an atom at each corner]

Solution

Planar concentration (or density) is the number of atoms per unit area on a given plane in the crystal. It is the surface concentration of atoms on a given plane. To calculate the planar concentration $n_{(hkl)}$ on a given (*hkl*) plane, we consider a bound area *A*. Only atoms whose centers line on *A* are involved in $n_{(hkl)}$. For each atom, we then evaluate what portion of the atomic cross section cut by the plane (*hkl*) is contained within *A*.

For the BCC crystalline structure the planes (100), (110) and (111) are drawn in Figure 1Q15-1.



(100), (110), (111) planes in the BCC crystal

Figure 1Q15-1

Consider the (100) plane.

Number of atoms in the area a × a , which is the cube face

= (4 corners) × (1/4th atom at corner) = 1.

Planar concentration is

$$n_{(100)} = \frac{4\left(\frac{1}{4}\right)}{a^2} = \frac{1}{\left(3.294\times10^{-10}\text{ m}\right)^2} = \mathbf{9.216\times10^{18}\ atoms\ m^{-2}}$$

The most populated plane for BCC structure is (110).

Number of atoms in the area $a \times a\sqrt{2}$ defined by two face-diagonals and two cube-sides

= (4 corners) × (1/4th atom at corner) + 1 atom at face center = 2

Planar concentration is

$$n_{(110)} = \frac{4\left(\frac{1}{4}\right)+1}{a^2\sqrt{2}} = \frac{2}{\left(3.294\times10^{-10}\ \text{m}\right)^2\sqrt{2}} = \mathbf{1.303\ \times\ 10^{19}\ atoms\ \ m^{-2}}$$

The plane (111) for the BCC structure is the one with rarest population. The area of interest is an equilateral triangle defined by face diagonals of length $a\sqrt{2}$ (Figure 1Q15-1). The height of the triangle is $a\sqrt{\frac{3}{2}}$ so that the triangular area is $\frac{1}{2}\times a\sqrt{2}\times a\sqrt{\frac{3}{2}} = \frac{a^2\sqrt{3}}{2}$. An atom at a corner only contributes a fraction (60°/360°=1/6) to this area.

So the planar concentration is

$$n_{(111)} = \frac{\frac{1}{6}(3)}{\frac{a^2\sqrt{3}}{2}} = \frac{1}{a^2\sqrt{3}} = \frac{1}{\left(3.294\times10^{-10}\ \text{m}\right)^2\sqrt{3}} = \mathbf{5.321\ \times\ 10^{18}\ atoms\ \ m^{-2}}$$

For the BCC structure there are two atoms in unit cell and the bulk atomic concentration is

$$n_{bulk} = \frac{\text{number of atoms in unit cell}}{vo\text{volume of the cell}} = \frac{2}{a^3} = \frac{2}{\left(3.294\times10^{-10}\ \text{m}\right)^3} = \mathbf{5.596\ \times\ 10^{28}\ atoms\ \ m^{-3}}$$

and the surface concentration is

$$n_{surface} = \left(n_{bulk}\right)^{\frac{2}{3}} = \left(5.596\times10^{28}\ m^{-3}\right)^{\frac{2}{3}} = \mathbf{1.463\ \times\ 10^{19}\ atoms\ \ m^{-2}}$$

1.16 Diamond and zinc blende

Si has the diamond and GaAs has the zinc blende crystal structure. Given the lattice parameters of Si and GaAs, a = 0.543 nm and a = 0.565 nm, respectively, and the atomic masses of Si, Ga, and As as 28.08 g/mol, 69.73 g/mol, and 74.92 g/mol, respectively, calculate the density of Si and GaAs. What is the atomic concentration (atoms per unit volume) in each crystal?

Solution

Referring to the diamond crystal structure in Figure 1Q16-1, we can identify the following types of atoms:

8 corner atoms labeled C,

6 face center atoms (labeled FC) and

4 inside atoms labeled 1,2,3,4.

The effective number of atoms within the unit cell is:

(8 Corners) × ($^1/_8$ C-atom) + (6 Faces) × ($^1/_2$ FC-atom) + 4 atoms within the cell (1,2,3,4) = 8

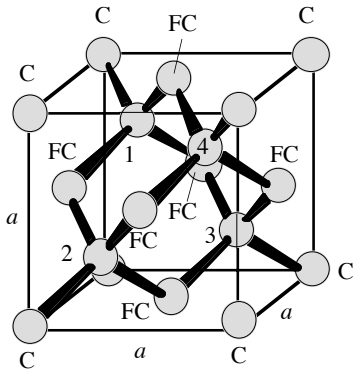


Figure 1Q16-1 The diamond crystal structure.

The lattice parameter (length of a cube side) of the unit cell is a. Thus the atomic concentration in the Si crystal (n_{Si}) is

$$\boldsymbol{n_{Si}} = \frac{8}{a^3} = \frac{8}{(0.543\times10^{-9}\,\text{m})^3} = \mathbf{5.0\times10^{28}\ atoms\ per\ m^{-3}}$$

If M_{at} is the atomic mass in the Periodic Table then the mass of the atom (m_{at}) in kg is

$$m_{at} = (10^{-3}\ \text{kg/g})M_{at}/N_A \qquad (1)$$

where N_A is Avogadro's number. For Si, $M_{at} = M_{Si} = 28.09$ g/mol, so then the density of Si is

$$\rho = (\text{number of atoms per unit volume}) \times (\text{mass per atom}) = n_{Si}\, m_{at}$$

or
$$\rho = \left(\frac{8}{a^3}\right)\left[\frac{(10^{-3}\ \text{kg/g})M_{Si}}{N_A}\right]$$

i.e.
$$\rho = \left[\frac{8}{(0.543\times10^{-9}\ \text{m})^3}\right]\left[\frac{(10^{-3}\ \text{kg/g})(28.09\ \text{g mol}^{-1})}{(6.022\times10^{23}\ \text{mol}^{-1})}\right]$$

calculating, $\rho = \mathbf{2.33\times10^3\ kg\ m^{-3}}$ or $\mathbf{2.33\ g\ cm^{-3}}$

In the case of GaAs, it is apparent that there are 4 Ga and 4 As atoms in the unit cell. The concentration of Ga (or As) atoms per unit volume (n_{Ga}) is

$$n_{Ga} = \frac{4}{a^3} = \frac{4}{(0.565\times10^{-9}\,\text{m})^3} = 2.22\times10^{28}\ \text{m}^{-3}$$

Total atomic concentration (counting both Ga and As atoms) is twice n_{Ga}.

$$\boldsymbol{n_{Total}} = 2n_{Ga} = \mathbf{4.44\times10^{28}\ m^{-3}}$$

There are 2.22×10^{28} Ga-As pairs per m^3. We can calculate the mass of the Ga and As atoms from their relative atomic masses in the Periodic Table using Equation (1) with $M_{at} = M_{Ga} = 69.72$ g/mol for Ga and $M_{at} = M_{As} = 74.92$ g/mol for As. Thus,

$$\rho = \left(\frac{4}{a^3}\right)\left[\frac{(10^{-3}\ \text{kg/g})(M_{Ga}+M_{As})}{N_A}\right]$$

or
$$\rho = \left[\frac{4}{(0.565\times10^{-9}\ \text{m})^3}\right]\left[\frac{(10^{-3}\ \text{kg/g})(69.72\ \text{g/mol}+74.92\ \text{g/mol})}{6.022\times10^{23}\ \text{mol}^{-1}}\right]$$

i.e. $\rho = \mathbf{5.33\times10^3\ kg\ m^{-3}}$ or $\mathbf{5.33\ g\ cm^{-3}}$

1.17 Zinc blende, NaCl and CsCl

a. InAs is a III-V semiconductor that has the zinc blende structure with a lattice parameter of 0.606 nm. Given the atomic masses of In (114.82 g mol^{-1}) and As (74.92 g mol^{-1}) find the density.

b. CdO has the NaCl crystal structure with a lattice parameter of 0.4695 nm. Given the atomic masses of Cd (112.41 g mol^{-1}) and O (16.00 g mol^{-1}) find the density.

c. KCl has the same crystal structure as NaCl. The lattice parameter *a* of KCl is 0.629 nm. The atomic masses of K and Cl are 39.10 g mol^{-1} and 35.45 g mol^{-1} respectively. Calculate the density of KCl.

Solution

a For zinc blende structure there are 8 atoms per unit cell (as shown in Table 1.3 in the textbook). In the case of InAs, it is apparent that there are 4 In and 4 As atoms in the unit cell. The density of InAs is then

$$d_{InAs} = \frac{4\left(\frac{M_{at\,In}}{N_A}\right) + 4\left(\frac{M_{at\,As}}{N_A}\right)}{a^3} = \frac{4\left(M_{at\,In} + M_{at\,As}\right)}{N_A a^3} =$$

$$= \frac{4(114.82 + 74.92) \times \left(10^{-3}\ \text{kg mol}^{-1}\right)}{\left(6.022 \times 10^{23}\ \text{mol}^{-1}\right)\left(0.606 \times 10^{-9}\ \text{m}\right)^3} = \mathbf{5.663 \times 10^3\ kg\ m^{-3} = 5.663\ g\ cm^{-3}}$$

b For NaCl crystal structure, there are 4 cations and 4 anions per unit cell. For the case of CdO we have 4 Cd atoms and 4 O atoms per unit cell and the density of CdO is

$$d_{CdO} = \frac{4\left(\frac{M_{at\,Cd}}{N_A}\right) + 4\left(\frac{M_{at\,O}}{N_A}\right)}{a^3} = \frac{4\left(M_{at\,Cd} + M_{at\,O}\right)}{N_A a^3} =$$

$$= \frac{4(112.41 + 16.00) \times \left(10^{-3}\ \text{kg mol}^{-1}\right)}{\left(6.022 \times 10^{23}\ \text{mol}^{-1}\right)\left(0.4695 \times 10^{-9}\ \text{m}\right)^3} = \mathbf{8.241 \times 10^3\ kg\ m^{-3} = 8.241\ g\ cm^{-3}}$$

c Analogously to **b**, for the density of KCl we receive

$$d_{KCl} = \frac{4\left(\frac{M_{at\,K}}{N_A}\right) + 4\left(\frac{M_{at\,Cl}}{N_A}\right)}{a^3} = \frac{4\left(M_{at\,K} + M_{at\,Cl}\right)}{N_A a^3} =$$

$$= \frac{4(39.1 + 35.45) \times \left(10^{-3}\ \text{kg mol}^{-1}\right)}{\left(6.022 \times 10^{23}\ \text{mol}^{-1}\right)\left(0.629 \times 10^{-9}\ \text{m}\right)^3} = \mathbf{1.99 \times 10^3\ kg\ m^{-3} = 1.99\ g\ cm^{-3}}$$

1.18 Crystallographic directions and planes

Consider the cubic crystal system.

a. Show that the line [*hkl*] is perpendicular to the (*hkl*) plane.

b. Show that the spacing between adjacent (*hkl*) planes is given by

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

Solution

This problem assumes that students are familiar with three dimensional geometry and vector products.

Figure 1Q18-1(a) shows a typical [*hkl*] line, labeled as **ON**, and a (*hkl*) plane in a cubic crystal. $\mathbf{u}_x$, $\mathbf{u}_y$ and $\mathbf{u}_z$ are the unit vectors along the *x*, *y*, *z* coordinates. This is a cubic lattice so we have Cartesian coordinates and $\mathbf{u}_x \cdot \mathbf{u}_x = 1$ and $\mathbf{u}_x \cdot \mathbf{u}_y = 0$ etc.

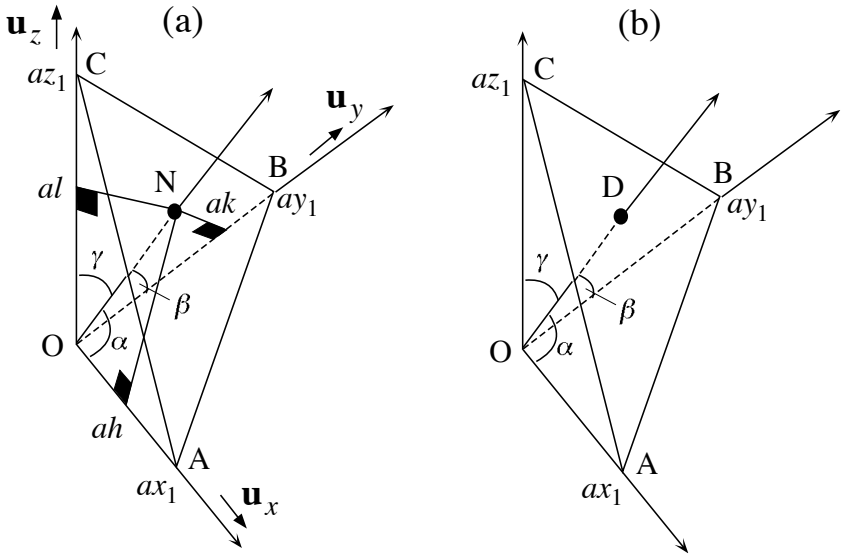


Figure 1Q18-1 Crystallographic directions and planes

a Given *a* = lattice parameter, then from the definition of Miller indices ($h = 1/x_1$, $k = 1/y_1$ and $l = 1/z_1$) , the plane has intercepts: $x_o = ax_1 = a/h$; $y_o = ay_1 = a/k$; $z_o = az_1 = a/l$.

The vector $\mathbf{ON} = ah\mathbf{u}_x + ak\mathbf{u}_y + al\mathbf{u}_z$

If **ON** is perpendicular to the (*hkl*) plane then the product of this vector with any vector in the (*hkl*) plane will be zero. We only have to choose 2 non-parallel vectors (such as **AB** and **BC**) in the plane and show that the dot product of these with **ON** is zero.

$$\mathbf{AB} = \mathbf{OB} - \mathbf{OA} = (a/k)\mathbf{u}_y - (a/h)\mathbf{u}_x$$

$$\mathbf{ON} \bullet \mathbf{AB} = (ah\mathbf{u}_x + ak\mathbf{u}_y + al\mathbf{u}_z) \bullet ((a/k)\mathbf{u}_y - (a/h)\mathbf{u}_x) = a^2 - a^2 = 0$$

Remember that: $\mathbf{u}_x\mathbf{u}_x = \mathbf{u}_y\mathbf{u}_y = 1$ and $\mathbf{u}_x\mathbf{u}_y = \mathbf{u}_x\mathbf{u}_z = \mathbf{u}_y\mathbf{u}_z = 0$

Similarly, $\mathbf{ON} \bullet \mathbf{BC} = (ah\mathbf{u}_x + ak\mathbf{u}_y + al\mathbf{u}_z) \bullet ((a/l)\mathbf{u}_z - (a/k)\mathbf{u}_y) = 0$

Therefore **ON** or [*hkl*] is normal to the (*hkl*) plane.

b Suppose that **OD** is the normal from the plane to the origin as shown in Figure 1Q18-1(b). Shifting a plane by multiples of lattice parameters does not change the miller indices. We can therefore assume the adjacent plane passes through O. The separation between the adjacent planes is then simply the distance **OD** in Figure 1Q18-1(b).

Let α, β and γ be the angles of **OD** with the *x*, *y* and *z* axes. Consider the direction cosines of the line **OD**: $\cos\alpha = d/(ax_1) = dh/a$; $\cos\beta = d/(ay_1) = dk/a$; $\cos\gamma = d/(az_1) = dl/a$

But in 3 dimensions, $(\cos\alpha)^2 + (\cos\beta)^2 + (\cos\gamma)^2 = 1$

Thus, $(d^2h^2/a^2) + (d^2k^2/a^2) + (d^2l^2/a^2) = 1$

Rearranging, $d^2 = a^2 / [h^2 + k^2 + l^2]$

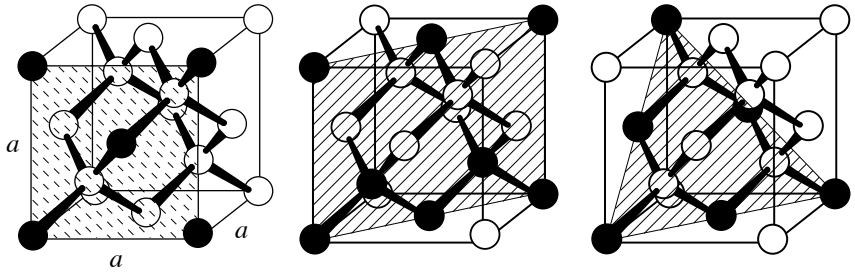
or, $\boldsymbol{d = a / [h^2 + k^2 + l^2]^{1/2}}$

1.19 Si and SiO_2

a. Given the Si lattice parameter $a = 0.543$ nm, calculate the number of Si atoms per unit volume, in nm^{-3}.

b. Calculate the number of atoms per m^2 and per nm^2 on the (100), (110) and (111) planes in the Si crystal as shown on Figure 1Q19-1. Which plane has the most number of atoms per unit area?

c. The density of SiO_2 is 2.27 g cm^{-3}. Given that its structure is amorphous, calculate the number of molecules per unit volume, in nm^{-3}. Compare your result with (a) and comment on what happens when the surface of an Si crystal oxidizes. The atomic masses of Si and O are 28.09 g/mol and 16 g/mol, respectively.



(100) plane (110) plane (111) plane

Figure 1Q19-1 Diamond cubic crystal structure and planes. Determine what portion of a black-colored atom belongs to the plane that is hatched.

Solution

a Si has the diamond crystal structure with 8 atoms in the unit cell, and we are given the lattice parameter $a = 0.543 \times 10^{-9}$ m and atomic mass $M_{at} = 28.09 \times 10^{-3}$ kg/mol. The concentration of atoms per unit volume (n) in nm^{-3} is therefore:

$$n = \frac{8}{a^3}\frac{1}{\left(10^9 \text{ nm/m}\right)^3} = \frac{8}{\left(0.543\times10^{-9} \text{ m}\right)^3}\frac{1}{\left(10^9 \text{ nm/m}\right)^3}$$

$\therefore$ $\boldsymbol{n = 50.0 \text{ atoms/nm}^3}$

If desired, the density ρ can be found as follows:

$$\rho = \frac{8\dfrac{M_{at}}{N_A}}{a^3} = \frac{8\dfrac{28.09\times10^{-3} \text{ kg/mol}}{6.022\times10^{23} \text{ mol}^{-1}}}{\left(0.543\times10^{-9} \text{ m}\right)^3}$$

$\therefore$ $\rho = 2331$ kg m^{-3} or 2.33 g cm^{-3}

b The (100) plane has 4 shared atoms at the corners and 1 unshared atom at the center. The corner atom is shared by 4 (100) type planes. Number of atoms per square nm of (100) plane area (n) is shown in Fig. 1Q19-2:

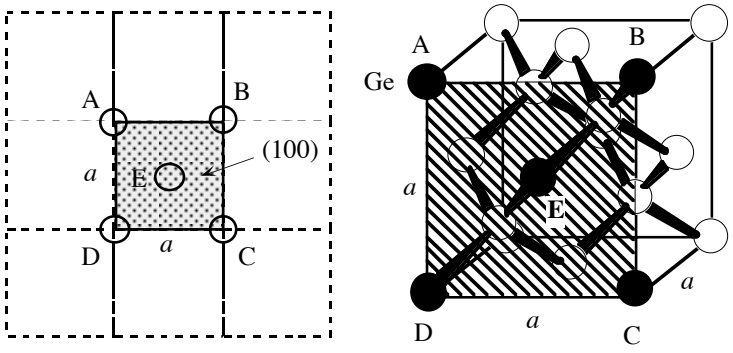


Figure 1Q19-2 The (100) plane of the diamond crystal structure.

The number of atoms per nm^2, n_{100}, is therefore:

$$n_{100} = \frac{4\left(\frac{1}{4}\right)+1}{a^2}\frac{1}{\left(10^9 \text{ nm/m}\right)^2} = \frac{4\left(\frac{1}{4}\right)+1}{\left(0.543\times10^{-9} \text{ m}\right)^2}\frac{1}{\left(10^9 \text{ nm/m}\right)^2}$$

$\therefore$ $\boldsymbol{n_{100} = 6.78 \text{ atoms/nm}^2}$ or $\boldsymbol{6.78 \times 10^{18} \text{ atoms/m}^2}$

The (110) plane is shown below in Fig. 1Q19-3. There are 4 atoms at the corners and shared with neighboring planes (hence each contributing a quarter), 2 atoms on upper and lower sides shared with upper and lower planes (hence each atom contributing 1/2) and 2 atoms wholly within the plane.

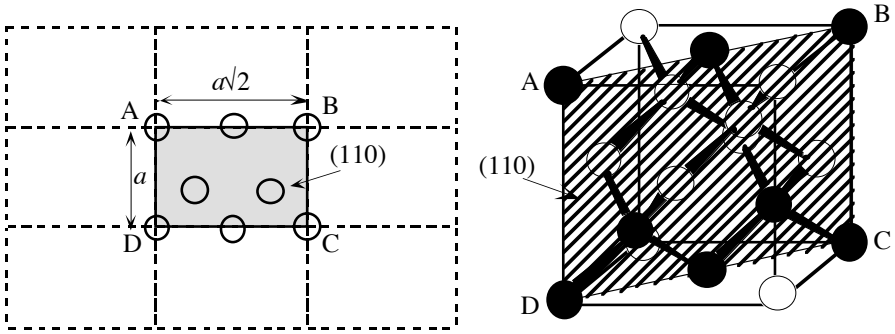


Figure 1Q19-3 The (110) plane of the diamond crystal structure.

The number of atoms per nm^2, n_{110}, is therefore:

$$n_{110} = \frac{4\left(\frac{1}{4}\right)+2\left(\frac{1}{2}\right)+2}{\left[a\left(a\sqrt{2}\right)\right]}\left(\frac{1}{\left(10^9 \text{ nm/m}\right)^2}\right)$$

$\therefore$
$$n_{110} = \frac{4\left(\frac{1}{4}\right)+2\left(\frac{1}{2}\right)+2}{\left[\left(0.543\times10^{-9} \text{ m}\right)\left(\left(0.543\times10^{-9} \text{ m}\right)\sqrt{2}\right)\right]}\left(\frac{1}{\left(10^9 \text{ nm/m}\right)^2}\right)$$

$\therefore$ $\boldsymbol{n_{110} = 9.59 \text{ atoms/nm}^2}$ or $\boldsymbol{9.59 \times 10^{18} \text{ atoms/m}^2}$

This is the most crowded plane with the most number of atoms per unit area.

The (111) plane is shown below in Fig. 1Q19-4:

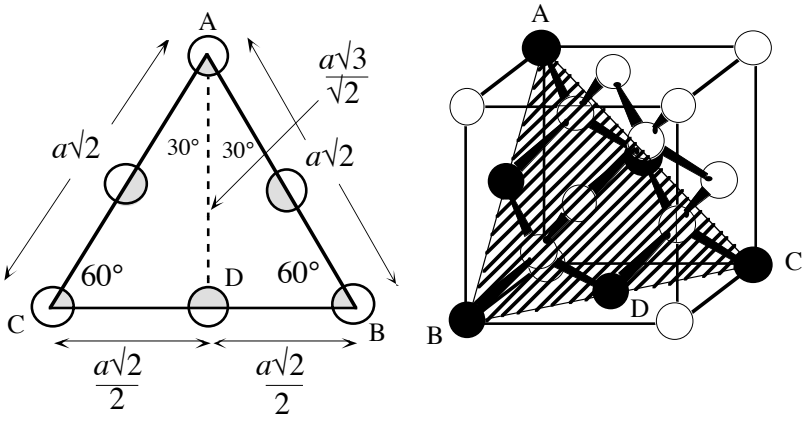


Figure 1Q19-4 The (111) plane of the diamond crystal structure

The number of atoms per nm^2, n_{111}, is therefore:

$$n_{111} = \frac{3\left(\frac{60}{360}\right)+3\left(\frac{1}{2}\right)}{\left[2\left(\frac{1}{2}\right)\left(a\frac{\sqrt{2}}{2}\right)\left(a\frac{\sqrt{3}}{\sqrt{2}}\right)\right]}\left(\frac{1}{\left(10^9 \text{ nm/m}\right)^2}\right)$$

$$\therefore \quad n_{111} = \frac{3\left(\frac{60}{360}\right)+3\left(\frac{1}{2}\right)}{\left[2\left(\frac{1}{2}\right)\left(\left(0.543\times10^{-9} \text{ m}\right)\frac{\sqrt{2}}{2}\right)\left(\left(0.543\times10^{-9} \text{ m}\right)\frac{\sqrt{3}}{\sqrt{2}}\right)\right]}\left(\frac{1}{\left(10^9 \text{ nm/m}\right)^2}\right)$$

$$\therefore \quad \boldsymbol{n_{111} = 7.83 \text{ atoms/nm}^2} \text{ or } \boldsymbol{7.83 \times 10^{18} \text{ atoms/m}^2}$$

c Given:

Molar mass of SiO_2: $M_{at} = 28.09 \times 10^{-3}$ kg/mol + $2 \times 16 \times 10^{-3}$ kg/mol = 60.09×10^{-3} kg/mol

Density of SiO_2: $\rho = 2.27 \times 10^3$ kg m^{-3}

Let n be the number of SiO_2 molecules per unit volume, then:

$$\rho = n\frac{M_{at}}{N_A}$$

$$\therefore \quad n = \frac{N_A\rho}{M_{at}} = \frac{\left(6.022\times10^{23} \text{ mol}^{-1}\right)\left(2.27\times10^3 \text{ kg m}^{-3}\right)}{\left(60.09\times10^{-3} \text{ kg/mol}\right)}$$

$$\therefore \quad n = 2.27 \times 10^{28} \text{ molecules per m}^3$$

Or, converting to molecules per nm^3:

$$\boldsymbol{n} = \frac{2.27\times10^{28} \text{ molecules/m}^3}{\left(10^9 \text{ nm/m}\right)^3} = \boldsymbol{22.7 \text{ molecules per nm}^3}$$

Oxide has less dense packing so it has a more open structure. For every 1 micron of oxide formed on the crystal surface, only about 0.5 micron of Si crystal is consumed.

1.20 Vacancies

Estimate the equilibrium vacancy concentration in the Si crystal at room temperature and at 1200 °C, given that the energy of vacancy formation is 3.6 eV.

Solution

To find the vacancy concentration in Si at $T = 27\ °C + 273 = 300$ K:

Si has a diamond crystal structure (as shown in Table 1.3 in the textbook) and therefore the concentration of atoms per unit volume (n) is given as:

$$n = \frac{8}{a^3} = \frac{8}{\left(0.543\times10^{-9}\ \text{m}\right)^3} = 4.997\times10^{28}\ \text{atoms/m}^3$$

where $a = 0.543$ nm is the lattice parameter of Si, given in Q1.19.

We are given that the energy of vacancy formation (E_V) in Si is 2.4 eV, or:

$$E_V = (2.4\ \text{eV/atom})(1.602\times10^{-19}\ \text{J/eV}) = 3.845\times10^{-19}\ \text{J/atom}$$

The concentration of vacancies (n_V) is then given by:

$$n_V = n\exp\left(-\frac{E_V}{kT}\right)$$

$$\therefore \quad n_V = \left(4.997\times10^{28}\ \text{m}^3\right)\exp\left[-\frac{3.845\times10^{-19}\ \text{J}}{\left(1.381\times10^{-23}\ \text{J K}^{-1}\right)(300\ \text{K})}\right]$$

$$\therefore \quad \boldsymbol{n_V = 2.47\times10^{-12}\ \text{vacancies / m}^3}\ \text{(at room temperature)}$$

At $T' = 1200\ °C + 273 = 1473$ K:

$$n_V' = n\exp\left(-\frac{E_V}{kT'}\right)$$

$$\therefore \quad n_V = \left(4.997\times10^{28}\ \text{m}^{-3}\right)\exp\left[-\frac{3.845\times10^{-19}\ \text{J}}{\left(1.381\times10^{-23}\ \text{J K}^{-1}\right)(1473\ \text{K})}\right]$$

$$\therefore \quad \boldsymbol{n_V = 3.09\times10^{20}\ \text{vacancies / m}^3}\ \text{(at 1200 °C)}$$

Note: Strictly we should use the concentration of Si (n) at 1473 K by taking into account the unit cell expansion and recalculating $n = 8/a^3$ to get a better value for n'. Given that $n_V = n\exp(-E_V/kT)$ has the entropic pre-exponential term missing (i.e. it is already an approximation) the calculation above is more than sufficient.

1.21 Pb-Sn solder

Consider the soldering of two copper components. When the solder melts, it wets both metal surfaces. If the surfaces are not clean or have an oxide layer, the molten solder cannot wet the surfaces and the soldering fails. Assume that soldering takes place at 250 °C, and consider the diffusion of Sn atoms into the copper (the Sn atom is smaller than the Pb atom and hence diffuses more easily).

a. The diffusion coefficient of Sn in Cu at two temperatures is $D = 1.69 \times 10^{-9}$ cm^2 hr^{-1} at 400 °C and $D = 2.48 \times 10^{-7}$ cm^2 hr^{-1} at 650 °C. Calculate the rms distance diffused by an Sn atom into the copper, assuming the cooling process takes 10 seconds.

b. What should be the composition of the solder if it is to begin freezing at 250 °C?

c. What are the components (phases) in this alloy at 200 °C? What are the compositions of the phases and their relative weights in the alloy?

d. What is the microstructure of this alloy at 25 °C? What are weight fractions of the α and β phases assuming near equilibrium cooling?

Solution

a Given information:

Temperatures: $T_1 = 400\ ^\circ\text{C} + 273 = 673\ \text{K}$ $T_2 = 650\ ^\circ\text{C} + 273 = 923\ \text{K}$

Diffusion coefficients: $D_1 = 1.69 \times 10^{-9}$ cm^2/hr = $(1.69 \times 10^{-9}\ \text{cm}^2/\text{hr})(0.01\ \text{m/cm})^2 / (1\ \text{hr}) \times (3600\ \text{sec/hr})$

$$D_1 = 4.694 \times 10^{-17}\ \text{m}^2/\text{s}$$

$D_2 = 2.48 \times 10^{-7}$ cm^2/hr = $(2.48 \times 10^{-7}\ \text{cm}^2/\text{hr})(0.01\ \text{m/cm})^2 / (1\ \text{hr}) \times (3600\ \text{sec/hr})$

$$D_2 = 6.889 \times 10^{-15}\ \text{m}^2/\text{s}$$

The diffusion coefficients at certain temperatures are given by:

$$D_1 = D_o \exp\left(-\frac{E_A q}{kT_1}\right) \qquad D_2 = D_o \exp\left(-\frac{E_A q}{kT_2}\right)$$

where E_A is the activation energy in eV/atom. We can take the ratio of the diffusion coefficients to express them in terms of the activation energy (E_A):

$$\therefore \quad \frac{D_1}{D_2} = \frac{D_o \exp\left(-\frac{E_A q}{kT_1}\right)}{D_o \exp\left(-\frac{E_A q}{kT_2}\right)} = \exp\left(\frac{E_A q}{kT_2} - \frac{E_A q}{kT_1}\right) = \exp\left(\frac{q[T_1 - T_2]E_A}{T_1 T_2 k}\right)$$

$$\therefore \quad E_A = \frac{T_1 T_2 k \ln\left(\frac{D_1}{D_2}\right)}{q(T_1 - T_2)}$$

$$\therefore \quad E_A = \frac{(673\ \text{K})(923\ \text{K})\left(1.381 \times 10^{-23}\ \text{J K}^{-1}\right)\ln\left(\frac{4.694 \times 10^{-17}\ \text{m}^2/\text{s}}{6.889 \times 10^{-15}\ \text{m}^2/\text{s}}\right)}{\left(1.602 \times 10^{-19}\ \text{J/eV}\right)(673\ \text{K} - 923\ \text{K})}$$

$$\therefore \quad E_A = 1.068\ \text{eV/atom}$$

Now the diffusion coefficient D_o can be found as follows:

$$D_1 = D_o \exp\left(-\frac{E_A q}{kT_1}\right)$$

$$\therefore \qquad D_o = \frac{D_1}{\exp\left(-\frac{E_A q}{kT_1}\right)} = \frac{4.694\times10^{-17}\ \text{m}^2/\text{s}}{\exp\left(-\frac{(1.068\ \text{eV})(1.602\times10^{-19}\ \text{J/eV})}{(1.381\times10^{-23}\ \text{J K}^{-1})(673\ \text{K})}\right)}$$

$$\therefore \qquad D_o = 4.638\times10^{-9}\ \text{m}^2/\text{s}$$

To check our value for D_o, we can substitute it back into the equation for D_2 and compare values:

$$D_2 = D_o \exp\left(-\frac{E_A q}{kT_2}\right) = (4.638\times10^{-9}\ \text{m}^2/\text{s})\exp\left(-\frac{(1.068\ \text{eV})(1.602\times10^{-19}\ \text{J/eV})}{(1.381\times10^{-23}\ \text{J K}^{-1})(923\ \text{K})}\right)$$

$$\therefore \qquad D_2 = 6.870\times10^{-15}\ \text{m}^2/\text{s}$$

This agrees with the given value of 6.889×10^{-15} m^2/s for D_2.

Now we must calculate the diffusion coefficient D_3 at T_3 = 250 °C + 273 = 523 K (temperature at which soldering is taking place).

$$D_3 = D_o \exp\left(-\frac{E_A q}{kT_3}\right) = (4.638\times10^{-9}\ \text{m}^2/\text{s})\exp\left(-\frac{(1.068\ \text{eV})(1.602\times10^{-19}\ \text{J/eV})}{(1.381\times10^{-23}\ \text{J K}^{-1})(523\ \text{K})}\right)$$

$$\therefore \qquad D_3 = 2.391\times10^{-19}\ \text{m}^2/\text{s}$$

The rms distance diffused by the Sn atom in time t = 10 s (L_{rms}) is given by:

$$L_{rms} = \sqrt{2D_3 t} = \sqrt{2(2.391\times10^{-19}\ \text{m}^2/\text{s})(10\ \text{s})}$$

$$\therefore \qquad \boldsymbol{L_{rms} = 2.19\times10^{-9}\ \text{m}} \text{ or } \boldsymbol{2\ \text{nm}}$$

b From Figure 1Q21-1, 250 °C cuts the liquidus line approximately at 33 wt.% Sn composition (C_o).

$$\therefore \qquad \boldsymbol{C_o = 0.33} \quad \text{(Sn)}$$

c For α-phase and liquid phase (L), the compositions as wt.% of Sn from Figure 1Q21-1 are:

$$C_\alpha = 0.18 \qquad\qquad C_L = 0.56$$

The weight fraction of α and L phases are:

$$\boldsymbol{W_\alpha} = \frac{C_L - C_o}{C_L - C_\alpha} = \frac{0.56-0.33}{0.56-0.18} = \boldsymbol{0.605} \text{ or } \boldsymbol{60\ \text{wt.\%}\ \alpha\text{-phase}}$$

$$\boldsymbol{W_L} = \frac{C_o - C_\alpha}{C_L - C_\alpha} = \frac{0.33-0.18}{0.56-0.18} = \boldsymbol{0.395} \text{ or } \boldsymbol{39.5\ \text{wt.\%}\ \text{liquid phase}}$$

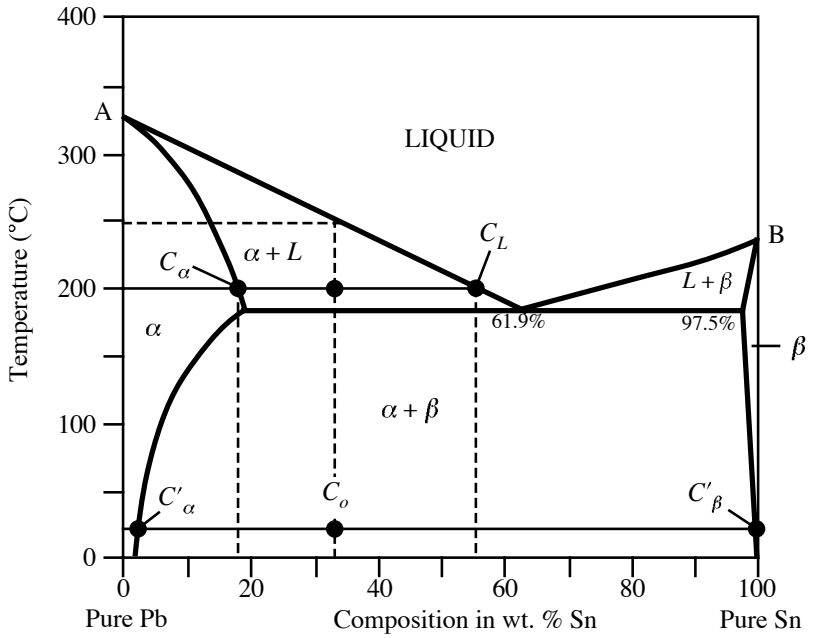


Figure 1Q21-1 The equilibrium phase diagram of the Pb-Sn alloy.

d The microstructure is a primary α-phase and a eutectic solid ($\alpha + \beta$) phase. There are two phases present, $\alpha + \beta$. See Figure 1Q21-2.



Figure 1Q21-2 Microstructure of Pb-Sn at temperatures less than 183°C.

Assuming equilibrium concentrations have been reached:

$$C'_\alpha = 0.02 \qquad C'_\beta = 1$$

The weight fraction of α in the whole alloy is then:

$$W'_\alpha = \frac{C'_\beta - C_o}{C'_\beta - C'_\alpha} = \frac{1 - 0.33}{1 - 0.02} = \mathbf{0.684} \text{ or } \mathbf{68.4\ wt.\%\ \alpha\text{-phase}}$$

The weight fraction of β in the whole alloy is:

$$W'_\beta = \frac{C'_o - C_\alpha}{C'_\beta - C'_\alpha} = \frac{0.33 - 0.02}{1 - 0.02} = \mathbf{0.316} \text{ or } \mathbf{31.6\ wt.\%\ \beta\text{-phase}}$$

1.22 Pb-Sn solder

Consider the 50% Pb-50% Sn solder.

a. Sketch the temperature-time profile and the microstructure of the alloy at various stages as it is cooled from the melt.

b. At what temperature does the solid melt?

c. What is the temperature range over which the alloy is a mixture of melt and solid? What is the structure of the solid?

d. Consider the solder at room temperature following cooling from 182 °C. Assume that the rate of cooling from 182 °C to room temperature is faster than the atomic diffusion rates needed to change the compositions of the α and β phases in the solid. Assuming the alloy is 1 kg, calculate the masses of the following components in the solid:

1. The primary α.
2. α in the whole alloy.
3. α in the eutectic solid.
4. β in the alloy (Where is the β-phase?).

e. Calculate the specific heat of the solder given the atomic masses of Pb (207.2) and Sn (118.71).

Solution

a

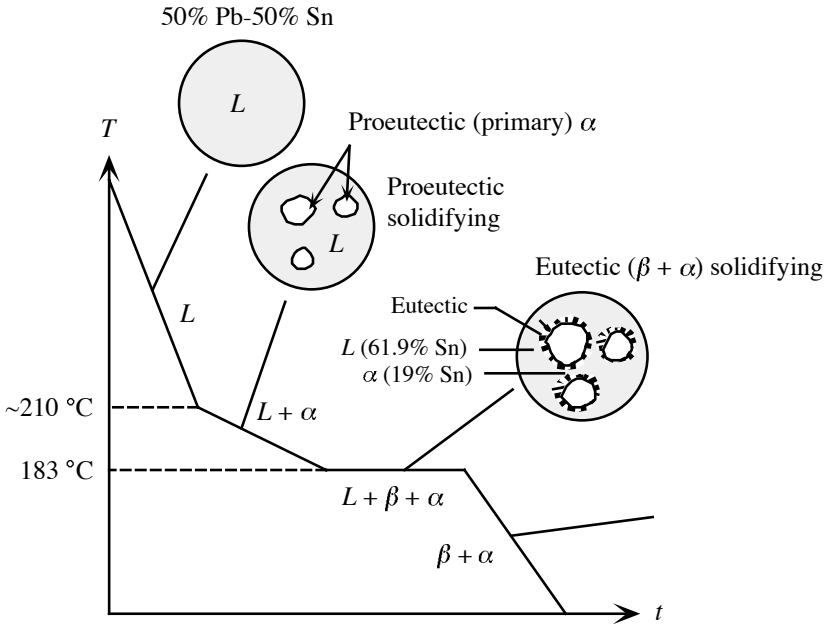


Figure 1Q22-1 Temperature - time profile and microstructure diagram of 50% Pb-50% Sn. All compositions are in weight %.

b When 50% Pb-50% Sn is cooled from the molten state down to room temperature, it begins to solidify at point A at about 210 °C. Therefore at 210 °C, the solid begins to melt.

c Between 210 °C and 183 °C, the liquid has the eutectic composition and undergoes the eutectic transformation to become the eutectic solid. Below 183 °C, all the liquid has solidified, and the structure is a combination of the solid α-phase and the eutectic structure (which is composed of α and β layers).

d At 182 °C, the composition of the proeutectic or primary α is given by the solubility limit of Sn in α: **19.2% Sn**.

The primary or proeutectic α (pro-α) exists just above and below 183 °C (eutectic temperature), i.e. it is stable just above and below 183 °C. Thus the mass of pro-α at 182 °C is the same as at 184 °C. Applying the lever rule (at 50% Sn):

$$W_{pro-\alpha} = \frac{C_L - C_o}{C_L - C_\alpha} = \frac{61.9 - 50}{61.9 - 19.2} = \mathbf{0.279}$$

Assume that the whole alloy is 1 kg. The mass of the primary or proeutectic α is thus **27.9% of the whole alloy**; or **0.279 kg**. The mass of the total eutectic solid ($\alpha + \beta$) is thus 1 - 0.279 = **0.721** or **72.1%** or **0.721 kg**.

If we apply the lever rule at 182 °C at 50% Sn, we obtain the weight percentage of α in the whole solid:

$$W_\alpha = \frac{C_\beta - C_o}{C_\beta - C_\alpha} = \frac{97.5 - 50}{97.5 - 19.2} = \mathbf{0.607}$$

Thus the mass of α in the whole solid is **0.607 kg**. Of this, 0.28 kg is in the primary (proeutectic) α phase. Thus 0.607 - 0.279 or **0.328 kg** of α is in the eutectic solid.

Since the total mass of α in the solid is 0.607 kg, the remainder of the mass must be the β-phase. Thus the mass of the β-phase is 1 kg - 0.607 kg or **0.393 kg**. The β-phase is present solely in the eutectic solid, with no primary β, because the solid is forming from a point where it consisted of the α and liquid phases only.

e Suppose that n_A and n_B are atomic fractions of A and B in the whole alloy,

$$n_A + n_B = 1$$

Suppose that we have 1 mole of the alloy. Then it has n_A moles of A and n_B moles of B (atomic fractions also represent molar fractions in the alloy). Suppose that we consider 1 gram of the alloy. Since w_A is the weight fraction of A, w_A is also the mass of A in grams in the alloy. The number of moles of A in the alloy is then w_A/M_A where M_A is the atomic mass of A. Thus,

Number of moles of $A = w_A/M_A$.

Number of moles of $B = w_B/M_B$.

Number of moles of the whole alloy = $w_A/M_A + w_B/M_B$.

Molar fraction of A is the same as n_A. Thus,

$$n_A = \frac{w_A / M_A}{\frac{w_A}{M_A} + \frac{w_B}{M_B}} \qquad \text{and} \qquad n_B = \frac{w_B / M_B}{\frac{w_A}{M_A} + \frac{w_B}{M_B}}$$

We are given the molar masses of Pb and Sn:

M_{Pb} = 207.2 g/mol $\qquad$ M_{Sn} = 118.71 g/mol

Let n = mole (atomic) fraction and W = weight fraction. Then W_{Pb} = weight fraction of Pb in the alloy. We know this to be 0.5 (50% Pb).

The mole fractions of Pb and Sn are n_{Pb} and n_{Sn} and are given by:

$$n_{Pb} = -\frac{M_{Sn} W_{Pb}}{(M_{Pb} - M_{Sn}) W_{Pb} - M_{Pb}}$$

$$\therefore \qquad n_{Pb} = -\frac{(118.71 \text{ g/mol})(0.5)}{(207.2 \text{ g/mol} - 118.71 \text{ g/mol})(0.5) - 207.2 \text{ g/mol}} = 0.3642$$

$$n_{Sn} = 1 - n_{Pb} = 1 - 0.3642 = 0.6358$$

The mole fraction of Pb is 0.3642 and that of Sn is 1 - 0.3642 = 0.6358.

The heat capacity of a metal is $3R$ per mole, where R is the gas constant. We want the specific heat capacity C_s (heat capacity per gram of alloy). 1 mole of the alloy has a mass M_{PbSn}:

$$M_{PbSn} = n_{Pb} M_{Pb} + n_{Sn} M_{Sn}$$

Thus the specific heat capacity C_s (i.e. heat capacity per gram) is:

$$C_s = \frac{3R}{M_{PbSn}} = \frac{3R}{n_{Pb} M_{Pb} + n_{Sn} M_{Sn}}$$

$$\therefore \qquad C_s = \frac{3(8.315 \text{ J K}^{-1} \text{ mol}^{-1})}{(0.3642)(207.2 \text{ g/mol}) + (0.6358)(118.71 \text{ g/mol})}$$

$$\therefore \qquad \boldsymbol{C_s = 0.165 \text{ J K}^{-1} \text{ g}^{-1}}$$

"If your experiment needs statistics, you ought to have done a better experiment."

Ernest Rutherford (1871-1937)

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Chapter 2

2.1 Electrical conduction

Na is a monovalent metal (BCC) with a density of 0.9712 g cm^{-3}. Its atomic mass is 22.99 g mol^{-1}. The drift mobility of electrons in Na is 53 $cm^2\ V^{-1}\ s^{-1}$.

a. Consider the collection of conduction electrons in the solid. If each Na atom donates one electron to the electron sea, *estimate* the mean separation between the electrons?

b. What is the approximate mean separation between an electron (e^-) and a metal ion (Na^+), assuming that most of the time the electron prefers to be between two neighboring Na^+ ions. What is the approximate Coulombic interaction energy (in eV) between an electron and an Na^+ ion?

c. How does this electron/metal-ion interaction energy compare with the average thermal energy per particle, according to the kinetic molecular theory of matter? Do you expect the kinetic molecular theory to be applicable to the conduction electrons in Na? If the mean electron/metal-ion interaction energy is of the same order of magnitude as the mean *KE* of the electrons, what is the mean speed of electrons in Na? Why should the mean kinetic energy be comparable to the mean electron/metal-ion interaction energy?

d. Calculate the electrical conductivity of Na and compare this with the experimental value of 2.1×10^7 $\Omega^{-1}\ m^{-1}$ and comment on the difference.

Solution

a If D is the density, M_{at} is the atomic mass and N_A is Avogadro's number, then the atomic concentration n_{at} is

$$n_{at} = \frac{DN_A}{M_{at}} = \frac{(971.2\ \text{kg m}^{-3})(6.022\times10^{23}\ \text{mol}^{-1})}{(22.99\times10^{-3}\ \text{kg mol}^{-1})}$$

i.e. $n_{at} = 2.544 \times 10^{28}\ m^{-3}$

which is also the electron concentration, given that each Na atom contributes 1 conduction electron.

If d is the mean separation between the electrons then d and n_{at} are related by (see Chapter 1 Solutions, Q1.5; this is only an estimate)

$$\boldsymbol{d} \approx \frac{1}{n_{at}^{1/3}} = \frac{1}{(2.544\times10^{28}\ \text{m}^{-3})^{1/3}} = \mathbf{3.40 \times 10^{-10}\ m}\ \text{or}\ \mathbf{0.34\ nm}$$

b Na is BCC with 2 atoms in the unit cell. So if a is the lattice constant (side of the cubic unit cell), the density is given by

$$D = \frac{(\text{atoms in unit cell})(\text{mass of 1 atom})}{\text{volume of unit cell}} = \frac{2\left(\frac{M_{at}}{N_A}\right)}{a^3}$$

isolate for a: $$a = \left[\frac{2M_{at}}{DN_A}\right]^{1/3} = \left[\frac{2(22.99\times10^{-3}\ \text{kg mol}^{-1})}{(0.9712\times10^{3}\ \text{kg m}^{-3})(6.022\times10^{23}\ \text{mol}^{-1})}\right]^{1/3}$$

so that $a = 4.284 \times 10^{-10}$ m or 0.4284 nm

For the BCC structure, the radius of the metal ion R and the lattice parameter a are related by $(4R)^2 = 3a^2$, so that,

$$R = (1/4)\sqrt{[3a^2]} = 1.855 \times 10^{-10} \text{ m or } 0.1855 \text{ nm}$$

If the electron is somewhere roughly between two metal ions, then the mean electron to metal ion separation $d_{\text{electron-ion}}$ is roughly R. If $d_{\text{electron-ion}} \approx R$, the electrostatic potential energy PE between a conduction electron and one metal ion is then

$$PE = \frac{(-e)(+e)}{4\pi\varepsilon_o d_{\text{electron-ion}}} = \frac{(-1.602\times10^{-19}\ \text{C})(+1.602\times10^{-19}\ \text{C})}{4\pi(8.854\times10^{-12}\ \text{F m}^{-1})(1.855\times10^{-10}\ \text{m})} \qquad (1)$$

$\therefore$ $\boldsymbol{PE = -1.24 \times 10^{-18}\ \text{J}}$ or $\boldsymbol{-7.76\ \text{eV}}$

c This electron-ion PE is much larger than the average thermal energy expected from the kinetic theory for a collection of "free" particles, that is $E_{\text{average}} = KE_{average} = 3(kT/2) \approx 0.039$ eV at 300 K. In the case of Na, the electron-ion interaction is very strong so we cannot assume that the electrons are moving around freely as if in the case of free gas particles in a cylinder. If we assume that the mean KE is roughly the same order of magnitude as the mean PE,

$$KE_{average} = \tfrac{1}{2} m_e u^2 \approx |PE|_{average} = -1.24\times10^{-18}\ \text{J} \qquad (2)$$

where u is the mean speed (strictly, u = root mean square velocity) and m_e is the electron mass.

Thus, $$u \approx \left[\frac{2PE_{average}}{m_e}\right]^{1/2} = \left[\frac{2(1.24\times10^{-18}\ \text{J})}{(9.109\times10^{-31}\ \text{kg})}\right]^{1/2} \qquad (3)$$

so that $\boldsymbol{u = 1.65 \times 10^{6}\ \text{m/s}}$

There is a theorem in classical physics called the *Virial theorem* which states that if the interactions between particles in a system obey the inverse square law (as in Coulombic interactions) then the magnitude of the mean KE is equal to the magnitude of the mean PE. The Virial Theorem states that:

$$KE_{average} = -\frac{1}{2} PE_{average}$$

Indeed, using this expression in Eqn. (2), we would find that $u = 1.05 \times 10^6$ m/s. If the conduction electrons were moving around freely and obeying the kinetic theory, then we would expect $(1/2)m_e u^2 = (3/2)kT$ and $u = 1.1 \times 10^5$ m/s, a much lower mean speed. Further, kinetic theory predicts that u increases as $T^{1/2}$ whereas according to Eqns. (1) and (2), u is insensitive to the temperature. The experimental linear dependence between the resistivity ρ and the absolute temperature T for most metals (non-magnetic) can only be explained by taking u = constant as implied by Eqns. (1) and (2).

d If μ is the drift mobility of the conduction electrons and n is their concentration, then the electrical conductivity of Na is $\sigma = en\mu$. Assuming that each Na atom donates one conduction electron ($n = n_{at}$), we have

$$\sigma = en\mu = (1.602\times10^{-19}\ \text{C})(2.544\times10^{28}\ \text{m}^{-3})(53\times10^{-4}\ \text{m}^2\ \text{V}^{-1}\ \text{s}^{-1})$$

i.e. $\boldsymbol{\sigma = 2.16 \times 10^{7}\ \Omega^{-1}\ \text{m}^{-1}}$

which is quite close to the experimental value.

Nota Bene: If one takes the Na^+-Na^+ separation $2R$ to be roughly the mean electron-electron separation then this is 0.37 nm and close to $d = 1/(n^{1/3}) = 0.34$ nm. In any event, all calculations are only approximate to highlight the main point. The interaction PE is substantial compared with the mean thermal energy and we cannot use $(3/2)kT$ for the mean KE!

2.2 Electrical conduction

The resistivity of aluminum at 25 °C has been measured to be 2.72×10^{-8} Ω m. The thermal coefficient of resistivity of aluminum at 0 °C is 4.29×10^{-3} K^{-1}. Aluminum has a valency of 3, a density of 2.70 g cm^{-3}, and an atomic mass of 27.

a. Calculate the resistivity of aluminum at –40 °C.

b. What is the thermal coefficient of resistivity at –40 °C?

c. Estimate the mean free time between collisions for the conduction electrons in aluminum at 25 °C, and hence estimate their drift mobility.

d. If the mean speed of the conduction electrons is ~1.5×10^{6} m s^{-1}, calculate the mean free path and compare this with the interatomic separation in Al (Al is FCC). What should be the thickness of an Al film that is deposited on an IC chip such that its resistivity is the same as that of bulk Al?

e. What is the percentage change in the power loss due to Joule heating of the aluminum wire when the temperature drops from 25 °C to –40 °C?

Solution

a Apply the equation for temperature dependence of resistivity, $\rho(T) = \rho_o[1 + \alpha_o(T-T_o)]$. We have the temperature coefficient of resistivity, α_o, at T_o where T_o is the reference temperature. The two given reference temperatures are 0 °C or 25 °C, depending on choice. Taking T_o = 0 °C + 273 = 273 K,

$$\rho(\text{-40 °C + 273 = 233 K}) = \rho_o[1 + \alpha_o(\text{233 K - 273 K})]$$

$$\rho(\text{25 °C + 273 = 298 K}) = \rho_o[1 + \alpha_o(\text{298 K - 273 K})]$$

Divide the above two equations to eliminate ρ_o,

$$\rho(\text{-40 °C})/\rho(\text{25 °C}) = [1 + \alpha_o(\text{-40 K})] \,/\, [1 + \alpha_o(\text{25 K})]$$

Next, substitute the given values ρ(25 °C) = 2.72×10^{-8} Ω m and $\alpha_o = 4.29 \times 10^{-3}$ K^{-1} to obtain

$$\rho(\text{-40 °C}) = (2.72 \times 10^{-8}\ \Omega\ \text{m}) \frac{[1 + (4.29 \times 10^{-3}\ \text{K}^{-1})(\text{-40 K})]}{[1 + (4.29 \times 10^{-3}\ \text{K}^{-1})(\text{25 K})]}$$

i.e. $\boldsymbol{\rho(\text{-40 °C}) = 2.03 \times 10^{-8}\ \Omega\ \text{m}}$

b In $\rho(T) = \rho_o[1 + \alpha_o(T - T_o)]$ we have α_o at T_o where T_o is the reference temperature, for example, 0° C or 25 °C depending on choice. We will choose T_o to be first at 0 °C = 273 K and then at -40 °C = 233 K so that

$$\rho(\text{-40 °C}) = \rho(\text{0 °C})[1 + \alpha_o(\text{233 K - 273 K})]$$

and $$\rho(\text{0 °C}) = \rho(\text{-40 °C})[1 + \alpha_{-40}(\text{273 K - 233 K})]$$

Multiply and simplify the two equations above to obtain

$$[1 + \alpha_o(\text{233 K - 273 K})][1 + \alpha_{-40}(\text{273 K - 233 K})] = 1$$

or $$[1 - 40\alpha_o][1 + 40\alpha_{-40}] = 1$$

Rearranging,

$$\alpha_{-40} = (1 / [1 - 40\alpha_o] - 1)(1 / 40)$$

$\therefore$ $$\alpha_{-40} = \alpha_o / [1 - 40\alpha_o]$$

i.e. $$\alpha_{\mathbf{-40}} = (4.29 \times 10^{-3}\ \text{K}^{-1}) / [1 - (40\ \text{K})(4.29 \times 10^{-3}\ \text{K}^{-1})] = \mathbf{5.18 \times 10^{-3}\ K^{-1}}$$

Alternatively, consider, $\rho(25\ ^\circ\text{C}) = \rho(-40\ ^\circ\text{C})[1 + \alpha_{-40}(298\ \text{K} - 233\ \text{K})]$ so that

$$\alpha_{-40} = [\rho(25\ ^\circ\text{C}) - \rho(-40\ ^\circ\text{C})] / [\rho(-40\ ^\circ\text{C})(65\ \text{K})]$$

$\therefore$ $$\alpha_{-40} = [2.72 \times 10^{-8}\ \Omega\ \text{m} - 2.03 \times 10^{-8}\ \Omega\ \text{m}] / [(2.03 \times 10^{-8}\ \Omega\ \text{m})(65\ \text{K})]$$

$\therefore$ $$\alpha_{\mathbf{-40}} = \mathbf{5.23 \times 10^{-3}\ K^{-1}}$$

c We know that $1/\rho = \sigma = en\mu$ where σ is the electrical conductivity, e is the electron charge, and μ is the electron drift mobility. We also know that $\mu = e\tau/m_e$, where τ is the mean free time between electron collisions and m_e is the electron mass. Therefore,

$$1/\rho = e^2 n\tau/m_e$$

$\therefore$ $$\tau = m_e/\rho e^2 n \qquad (1)$$

Here n is the number of conduction electrons per unit volume. But, from the density d and atomic mass M_{at}, atomic concentration of Al is

$$n_{\text{Al}} = \frac{N_A d}{M_{at}} = \frac{\left(6.022 \times 10^{23}\ \text{mol}^{-1}\right)\left(2700\ \text{kg/m}^3\right)}{\left(0.027\ \text{kg/mol}\right)} = 6.022 \times 10^{28}\ \text{m}^{-3}$$

so that $$n = 3n_{\text{Al}} = 1.807 \times 10^{29}\ \text{m}^{-3}$$

assuming that each Al atom contributes 3 "free" conduction electrons to the metal. Substitute into Eqn. (1):

$$\tau = \frac{m_e}{\rho e^2 n} = \frac{(9.109 \times 10^{-31}\ \text{kg})}{(2.72 \times 10^{-8}\ \Omega\ \text{m})(1.602 \times 10^{-19}\ \text{C})^2 (1.807 \times 10^{29}\ \text{m}^{-3})}$$

$\therefore$ $$\tau = \mathbf{7.22 \times 10^{-15}\ s}$$

(Note: If you do not convert to meters and instead use centimeters you will not get the correct answer because seconds is an SI unit.)

The relation between the drift mobility μ_d and the mean scattering time is given by Equation 2.5 (in textbook), so that

$$\mu_d = \frac{e\tau}{m_e} = \frac{\left(1.602 \times 10^{-19}\ C\right)\left(7.22 \times 10^{-15}\ s\right)}{\left(9.109 \times 10^{-31}\ kg\right)}$$

$\therefore$ $$\mu_d = \mathbf{1.27 \times 10^{-3}\ m^2\ V^{-1}s^{-1}} = \mathbf{12.7\ cm^2\ V^{-1}s^{-1}}$$

d The mean free path is $l = u\tau$ where u is the mean speed. With $u \approx 1.5 \times 10^6$ m s^{-1} we find the mean free path:

$$\boldsymbol{l} = u\tau = (1.5 \times 10^6\ \text{m s}^{-1})(7.22 \times 10^{-15}\ \text{s}) \approx \mathbf{1.08 \times 10^{-8}\ m} \approx \mathbf{10.8\ nm}$$

A thin film of Al must have a much greater thickness than l to show bulk behavior. Otherwise, scattering from the surfaces will increase the resistivity by virtue of Matthiessen's rule.

e Power $P = I^2R$ and is proportional to resistivity ρ, assuming the rms current level stays relatively constant. Then we have

$$[P(-40\ °C) - P(25\ °C)] / P(25\ °C) = P(-40\ °C) / P(25\ °C) - 1$$

$$\therefore \quad = \rho(-40\ °C) / \rho(25\ °C) - 1$$

$$\therefore \quad = (2.03 \times 10^{-8}\ \Omega\ m / 2.72 \times 10^{-8}\ \Omega\ m) - 1$$

$$\therefore \quad = \mathbf{-0.254}, \text{ or } \mathbf{-25.4\%}$$

(Negative sign means a reduction in the power loss).

2.3 TCR and Matthiessen's rule

Determine the temperature coefficient of resistivity of pure iron and of electrotechnical steel (Fe with 4% C), which are used in various electrical machinery, at two temperatures: 0 °C and 500 °C. Comment on the similarities and differences in the resistivity versus temperature behavior shown in Figure 2Q3-1 for the two materials.

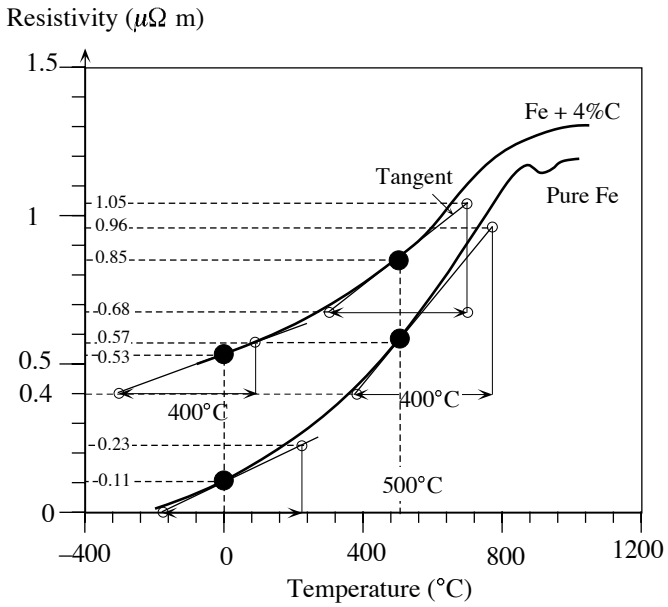


Figure 2Q3-1 Resistivity versus temperature for pure iron and 4% C steel.

Solution

The temperature coefficient of resistivity α_o (TCR) is defined as follows:

$$\alpha_o = \frac{1}{\rho_o}\left(\frac{d\rho}{dT}\right)_{T_o} = \frac{\text{Slope}_o}{\rho_o}$$

where the slope is $d\rho/dT$ at $T = T_o$ and ρ_o is the resistivity at $T = T_o$.

To find the slope, we draw a tangent to the curve at $T = T_o$ ($T_o = 0$ °C and then $T_o = 500$ °C) and obtain $\Delta\rho/\Delta T \approx d\rho/dT$. One convenient way is to define $\Delta T = 400$ °C and find $\Delta\rho$ on the tangent line and then calculate $\Delta\rho/\Delta T$.

Iron at 0 °C,

$\text{Slope}_o \approx (0.23 \times 10^{-6}\ \Omega\ m - 0\ \Omega\ m) / (400\ °C) = 5.75 \times 10^{-10}\ \Omega\ m\ °C^{-1}$

Since $\rho_o \approx 0.11 \times 10^{-6}$ Ω m, $\alpha_o = \text{Slope}_o/\rho_o \approx$ **0.00523 °C $^{-1}$**

Fe + 4% C at 0 °C,

$\text{Slope}_o \approx (0.57 \times 10^{-6}\ \Omega\text{ m} - 0.4 \times 10^{-6}\ \Omega\text{ m}) / (400\ ^\circ\text{C}) = 4.25 \times 10^{-10}\ \Omega\text{ m }^\circ\text{C}^{-1}$

Since $\rho_o \approx 0.53 \times 10^{-6}$ Ω m, $\alpha_o = \text{Slope}_o/\rho_o \approx$ **0.00802 °C $^{-1}$**

Iron at 500 °C,

$\text{Slope}_o \approx (0.96 \times 10^{-6}\ \Omega\text{ m} - 0.4 \times 10^{-6}\ \Omega\text{ m}) / (400\ ^\circ\text{C}) = 1.40 \times 10^{-9}\ \Omega\text{ m }^\circ\text{C}^{-1}$

Since $\rho_o \approx 0.57 \times 10^{-6}$ Ω m, $\alpha_o = \text{Slope}_o/\rho_o \approx$ **0.00245 °C $^{-1}$**

Fe + 4% C at 500 °C,

$\text{Slope}_o \approx (1.05 \times 10^{-6}\ \Omega\text{ m} - 0.68 \times 10^{-6}\ \Omega\text{ m}) / (400\ ^\circ\text{C}) = 9.25 \times 10^{-10}\ \Omega\text{ m }^\circ\text{C}^{-1}$

Since $\rho_o \approx 0.85 \times 10^{-6}$ Ω m, $\alpha_o = \text{Slope}_o/\rho_o \approx$ **0.00109 °C $^{-1}$**

*2.4 TCR of isomorphous alloys

a. Show that for an isomorphous alloy *A*%-*B*% (*B*% solute in *A*% solvent), the temperature coefficient of resistivity α_{AB} is given by

$$\alpha_{AB} \approx \frac{\alpha_A \rho_A}{\rho_{AB}}$$

where ρ_{AB} is the resistivity of the alloy (*AB*) and ρ_A and α_A are the resistivity and TCR of pure *A*. What are the assumptions behind this equation?

b. Estimate the composition of the Cu-Ni alloy that will have a TCR of 4×10^{-4} K^{-1}, that is, a TCR that is an order of magnitude less than that of Cu.

Solution

a By the Nordheim rule, the resistivity of the alloy is $\rho_{\text{alloy}} = \rho_o + CX(1–X)$. We can find the TCR of the alloy from its definition

$$\alpha_{\text{alloy}} = \frac{1}{\rho_{\text{alloy}}}\frac{d\rho_{\text{alloy}}}{dT} = \frac{1}{\rho_{\text{alloy}}}\frac{d}{dT}[\rho_o + CX(1 \pm X)]$$

To obtain the desired equation, **we must assume that *C* is temperature independent** (i.e. the increase in the resistivity depends on the lattice distortion induced by the impurity) so that $d[CX(1 - X)]/dT = 0$, enabling us to substitute for $d\rho_o/dT$ using the definition of the TCR: $\alpha_o = (d\rho_o/dT)/\rho_o$. Substituting into the above equation:

$$\alpha_{\text{alloy}} = \frac{1}{\rho_{\text{alloy}}}\frac{d\rho_o}{dT} = \frac{1}{\rho_{\text{alloy}}}\alpha_o\rho_o$$

i.e. $\alpha_{\text{alloy}}\rho_{\text{alloy}} = \alpha_o\rho_o$ or $\alpha_{AB}\rho_{AB} = \alpha_A\rho_A$

Remember that all values for the alloy and pure substance must all be taken at the same temperature, or the equation is invalid.

b Assume room temperature $T = 293$ K. Using values for copper from Table 2.1 in Equation 2.17 (both in the textbook), $\rho_{\text{Cu}} = 17.1$ nΩ m and $\alpha_{\text{Cu}} = 4.0 \times 10^{-3}$ K^{-1}, and from Table 2.3 (in the textbook)

the Nordheim coefficient of Ni dissolved in Cu is C = 1570 nΩ m. We want to find the composition of the alloy such that $\alpha_{CuNi} = 4 \times 10^{-4}$ K^{-1}. Then,

$$\rho_{alloy} = \frac{\alpha_{Cu}\rho_{Cu}}{\alpha_{alloy}} = \frac{(0.0040\ \text{K}^{-1})(17.1\ \text{n}\Omega\ \text{m})}{(0.0004\ \text{K}^{-1})} = 171.0\ \text{n}\Omega\ \text{m}$$

Using Nordheim's rule:

$$\rho_{alloy} = \rho_{Cu} + CX(1 - X)$$

i.e. $$171.0\ \text{n}\Omega\ \text{m} = 17.1\ \text{n}\Omega\ \text{m} + (1570\ \text{n}\Omega\ \text{m})X(1 - X)$$

$\therefore$ $$X^2 - X + 0.0879 = 0$$

solving the quadratic, we find $\boldsymbol{X = 0.11}$.

Thus the composition is **89% Cu-11% Ni**. However, this value is in *atomic percent* as the Nordheim coefficient is in atomic percent. Note that as Cu and Ni are very close in the Periodic Table this would also be the weight percentage. Note: the quadratic will produce another value, namely X = 0.86. However, using this number to obtain a composition of 11% Cu-89% Ni is incorrect because the values we used in calculations corresponded to a solution of Ni dissolved in Cu, not vice-versa (i.e. Ni was taken to be the impurity).

2.5 Constantan

Constantan has the composition 45% Ni-55% Cu. Cu-Ni alloys show complete solid solubility. As an alloy, constantan is widely used in resistor applications (up to 500 °C), in strain gauges, and as one of the thermocouple metal pairs. Given that the resistivity and TCR of copper at 20 °C are 17 nΩ m and 0.004 K^{-1}, respectively, and the Nordheim coefficient of Ni dissolved in Cu is 1570 nΩ m, calculate the resistivity ρ, TCR (α), and thermal conductivity κ of constantan and compare the values with the experimental measurements: ρ(20 °C) = 5×10^{-7} Ω m, α (20 °C) = 2×10^{-5} K^{-1}, κ = 21 W m^{-1} K^{-1}. What are the reasons for the differences between the calculated and experimental values?

Solution

Take the given composition to be atomic percentage (approximately true for Cu and Ni in constantan). Otherwise convert 45 wt. % to 47 at. % in calculations.

Given ρ_o = 17 nΩ m, C = 1570 nΩ m and X = 0.45, the alloy resistivity is

$$\rho_{alloy} = \rho_o + CX(1–X) = 17\ \text{n}\Omega\ \text{m} + (1570\ \text{n}\Omega\ \text{m})(0.45)(1–0.45)$$

$\therefore$ $$\boldsymbol{\rho_{alloy} = 405.6\ \text{n}\Omega\ \text{m}}$$

(X = 0.47 gives $\boldsymbol{\rho_{alloy} = 408.1\ \text{n}\Omega\ \text{m}}$). The experimental value is 500 nΩ m. The listed value of C in the tables is therefore not very good in predicting the experimental value. In addition, C values are typically determined for very dilute alloys.

$$\alpha_{alloy} = \frac{\alpha_{Cu}\rho_{Cu}}{\rho_{alloy}} = \frac{(0.004\ \text{K}^{-1})(17\ \text{n}\Omega\ \text{m})}{(405.6\ \text{n}\Omega\ \text{m})} = \boldsymbol{1.68 \times 10^{-4}\ \text{K}^{-1}}$$

This is about an order of magnitude greater than the experimental value. Using ρ_{alloy} = 500 nΩ m, we obtain $\alpha_{alloy} = 1.36 \times 10^{-4}$ K^{-1}, only about 20% lower than the theoretical value. Obviously the expression $\rho_{alloy}\alpha_{alloy} = \alpha_A\rho_A$ does not work well in this case.

The conductivity is $\sigma_{alloy} = 1/\rho_{alloy}$. The thermal conductivity of the alloy can be found from the Wiedemann-Franz-Lorenz law,

$$\kappa_{alloy} = \sigma_{alloy} TC_{WFL} = (405.6 \times 10^{-9}\ \Omega\ m)^{-1}(20 + 273\ K)(2.44 \times 10^{-8}\ W\ \Omega\ K^{-2})$$

i.e. $$\kappa_{alloy} = \mathbf{17.6\ W\ m^{-1}\ K^{-1}}$$

The experimental value is about 21 W K^{-1} m^{-1}.

Addendum:

We know the resistivity to be $\rho_{alloy} = \rho_o + CX(1–X)$. If we were to plot $\rho_{alloy} - \rho_o$ versus $X(1 - X)$, we would obtain a straight line with slope equal to C, the Nordheim coefficient. Table 2Q5-1 lists values taken from Figure 2Q5-1 for resistivity of Ni-Cu alloy in varying compositions, along with corresponding values of $X(1 - X)$:

Table 2Q5-1

% of Ni in Ni-Cu	at. % of Ni in Ni-Cu	Resistivity of alloy (nΩ m)	Change in resistivity ($\rho_{alloy} - \rho_o$)	X(1-X)
0	0	17	0	0
2	2.162	50	33	0.0211
6	6.464	100	83	0.0605
10	10.74	191	174	0.0958
11	11.80	150	133	0.104
20	21.30	266	249	0.168
22	23.39	300	283	0.179
30	31.69	375	358	0.216
43	44.96	500	483	0.247
45	46.97	500	483	0.249

Using these values, we obtain the following plot:

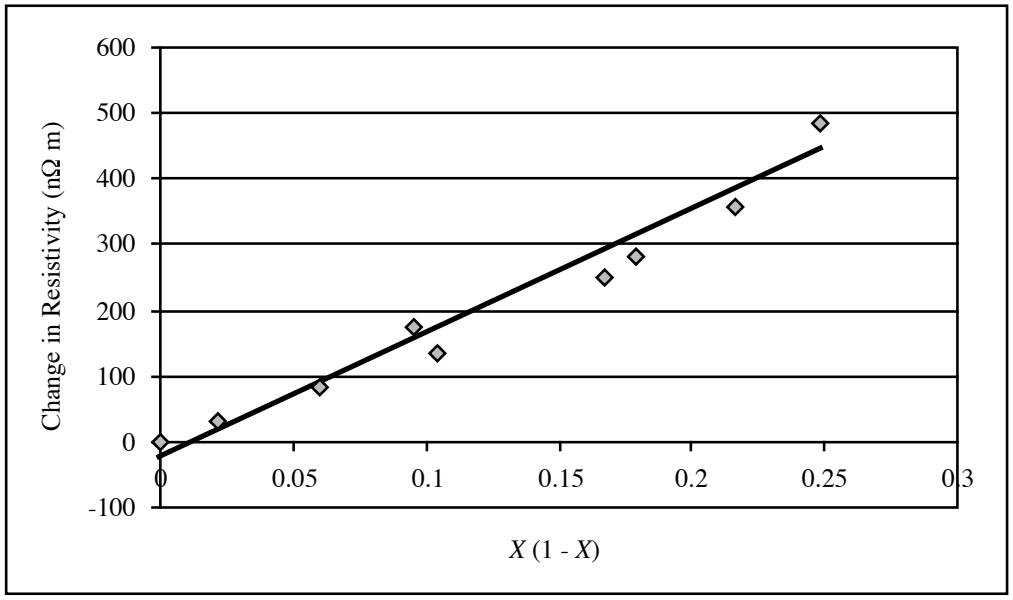


Figure 2Q5-1 Plot of Change in resistivity versus $X(1 - X)$

The plot reveals a straight line with slope equal to 1882.3 nΩ m. This is close to the value of C used before. Using this new value of C, the calculated values would come out closer to the experimental values given.

2.6 Experimental Nordheim coefficient for Pd in Ag

Silver and palladium form a complete solid solution and, as alloys, they are widely used in various electrical switches. Pd improves the wear resistance of Ag and the alloy has good fabricability. Table 2Q6-1 shows the resistivity of Ag-Pd alloys for various compositions. Using a suitable plot, obtain the Nordheim coefficient for Pd in Ag. [Note: The atomic masses of Pd and Ag are very close, 106.42 g/mol and 107.87 g/mol, respectively. You can therefore assume that the atomic % = weight %.]

Table 2Q6-1

wt.% Pd in Ag	0	1	3	10	30
Resistivity, nΩ m	16.1	21.8	38.3	63.9	149.3

Solution

Using the data given in Table 2Q6-1, plot $\Delta\rho$ versus $[X(1 - X)]$, where $\Delta\rho$ is the increase in resistivity due to alloying, namely $\Delta\rho = \rho - \rho_o$, and X is the atomic fraction of Pd ≈ weight fraction of Pd (ρ_o = 16.1 nΩ m from Table 2Q6-1). Since the increase in resistivity is $\Delta\rho = C[X(1 - X)]$, plotting $\Delta\rho$ versus $[X(1 - X)]$ should yield a straight line graph with slope equal to the Nordheim coefficient C.

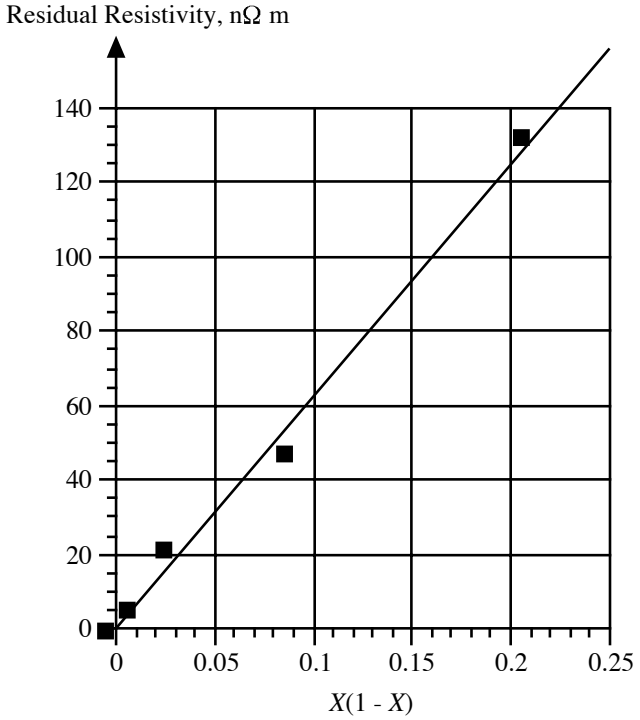


Figure 2Q6-1 Plot of change in resistivity versus $[X(1 - X)]$. The slope of the line is the Nordheim coefficient.

The equation of the best fit line is

$$\Delta\rho = 623.76[X(1 - X)] - 0.5102$$

Therefore the Nordheim coefficient is $\boldsymbol{C}$ **= 623 nΩ m** or **62.3** μ**Ω cm**.

2.7 Electrical and thermal conductivity of In

Electron drift mobility in indium has been measured to be 6 $cm^2\ V^{-1}\ s^{-1}$. The room temperature (27 °C) resistivity of In is $8.37 \times 10^{-8}\ \Omega$ m, and its atomic mass and density are 114.82 amu or g mol^{-1} and 7.31 g cm^{-3}, respectively.

a. Based on the resistivity value, determine how many free electrons are donated by each In atom in the crystal. How does this compare with the position of In in the Periodic Table (Group IIIB)?

b. If the mean speed of conduction electrons in In is 1.74×10^8 cm s^{-1}, what is the mean free path?

c. Calculate the thermal conductivity of In. How does this compare with the experimental value of 81.6 W $m^{-1}\ K^{-1}$?

Solution

a From $\sigma = en\mu_d$ (σ is the conductivity of the metal, e is the electron charge, and μ_d is the electron drift mobility) we can calculate the concentration of conduction electrons (n):

$$n = \frac{\sigma}{e\mu_d} = \frac{(8.37\times10^{-8}\ \ \Omega\ \ \text{m})^{-1}}{(1.602\times10^{-19}\ \ \text{C})(6\times10^{-4}\ \ \text{m}^2\text{V}^{-1}\text{s}^{-1})}$$

i.e. $n = 1.243 \times 10^{29}\ m^{-3}$

Atomic concentration n_{at} is

$$n_{at} = \frac{dN_A}{M_{at}} = \frac{(7.31\times10^{3}\ \ \text{kg m}^{-3})(6.022\times10^{23}\ \ \text{mol}^{-1})}{(114.82\times10^{-3}\ \ \text{kg mol}^{-1})}$$

i.e. $n_{at} = 3.834 \times 10^{28}\ m^{-3}$

Effective number of conduction electrons donated per In atom (n_{eff}) is:

$$\boldsymbol{n_{eff}} = n / n_{at} = (1.243 \times 10^{29}\ m^{-3}) / (3.834 \times 10^{28}\ m^{-3}) = \mathbf{3.24}$$

Conclusion: There are therefore about three electrons per atom donated to the conduction-electron sea in the metal. This is in good agreement with the position of the In element in the Periodic Table (III) and its valency of 3.

b If τ is the mean scattering time of the conduction electrons, then from $\mu_d = e\tau/m_e$ (m_e = electron mass) we have:

$$\tau = \frac{\mu_d m_e}{e} = \frac{(6\times10^{-4}\ \ \text{m}^2\ \text{V}^{-1}\ \text{s}^{-1})(9.109\times10^{-31}\ \ \text{kg})}{(1.602\times10^{-19}\ \ \text{C})} = 3.412 \times 10^{-15}\ \text{s}$$

Taking the mean speed $u \approx 1.74 \times 10^6$ m s^{-1}, the mean free path (l) is given by

$$\boldsymbol{l} = u\tau = (1.74 \times 10^6\ \text{m s}^{-1})(3.412 \times 10^{-15}\ \text{s}) = \mathbf{5.94 \times 10^{-9}\ m}\ \text{or}\ \mathbf{5.94\ nm}$$

c From the Wiedemann-Franz-Lorenz law, thermal conductivity is given as:

$$\kappa = \sigma T C_{\text{WFL}} = (8.37 \times 10^{-8}\ \Omega\ \text{m})^{-1}(20\ ^\circ\text{C} + 273\ \text{K})(2.44 \times 10^{-8}\ \text{W}\ \Omega\ \text{K}^{-2})$$

i.e. $\boldsymbol{\kappa = 85.4\ W\ m^{-1}\ K^{-1}}$

This value reasonably agrees with the experimental value.

2.8 Electrical and thermal conductivity of Ag

The electron drift mobility in silver has been measured to be 56 $cm^2\ V^{-1}\ s^{-1}$ at 27 °C. The atomic mass and density of Ag are given as 107.87 amu or g mol^{-1} and 10.50 g cm^{-3}, respectively.

a. Assuming that each Ag atom contributes one conduction electron, calculate the resistivity of Ag at 27 °C. Compare this value with the measured value of 1.6×10^{-8} Ω m at the same temperature and suggest reasons for the difference.

b. Calculate the thermal conductivity of silver at 27 °C and at 0 °C.

Solution

a Atomic concentration n_{at} is

$$n_{at} = \frac{dN_A}{M_{at}} = \frac{(10.50 \times 10^3 \text{ kg m}^{-3})(6.022 \times 10^{23} \text{ mol}^{-1})}{(107.87 \times 10^{-3} \text{ kg mol}^{-1})}$$

i.e. $n_{at} = 5.862 \times 10^{28}\ m^{-3}$

If we assume there is one conduction electron per Ag atom, the concentration of conduction electrons (n) is $5.862 \times 10^{28}\ m^{-3}$, and the conductivity is therefore:

$$\sigma = en\mu_d = (1.602 \times 10^{-19}\ C)(5.862 \times 10^{28}\ m^{-3})(56 \times 10^{-4}\ m^2\ V^{-1}s^{-1})$$

$\therefore$ $\sigma = 5.259 \times 10^7\ \Omega^{-1}\ m^{-1}$

and the resistivity $\rho = 1/\sigma = \mathbf{19.0\ n\Omega\ m}$

The experimental value of ρ is 16 nΩ m. We assumed that exactly 1 "free" electron per Ag atom contributes to conduction - this is not necessarily true. We need to use energy bands to describe conduction more accurately and this is addressed in Chapter 4 (in the textbook).

b From the Wiedemann-Franz-Lorenz law at 27 °C,

$$\kappa = \sigma T C_{WFL} = (5.259 \times 10^7\ \Omega^{-1}\ m^{-1})(27 + 273\ K)(2.44 \times 10^{-8}\ W\ \Omega\ K^{-2})$$

i.e. $\kappa = \mathbf{385\ W\ m^{-1}\ K^{-1}}$

For pure metals such as Ag this is nearly independent of temperature (same at 0 °C).

2.9 Mixture rules

A 70% Cu - 30% Zn brass electrical component has been made of powdered metal and contains 15 vol. % porosity. Assume that the pores are dispersed randomly. Estimate the effective electrical resistivity of the brass component.

Solution:

Assume room temperature $T = 293$ K. From Table 2.1 and Equation 2.17 in the textbook, ρ_o for Cu is 17 nΩ m, and from Table 2.3 (in the textbook) $C = 300$ nΩ m. The atomic fraction of the impurity Zn is given as $X = 0.3$ (we can take the weight fraction given as the atomic fraction because Cu and Zn have very close atomic weights). The brass metal resistivity is then:

$$\rho = \rho_o + CX(1-X) = 17\ n\Omega\ m + (300\ n\Omega\ m)(0.3)(1 - 0.3)$$

$\therefore \qquad \rho = 80.0\ \text{n}\Omega\ \text{m}$

The component has 15% air pores. Apply the empirical mixture rule in Equation 2.24 (in the textbook). The fraction of volume with air pores is $\chi = 0.15$. Then,

$$\rho_{eff} = \rho\frac{(1+\frac{1}{2}\chi)}{(1-\chi)} = (80.0\ \text{n}\Omega\ \text{m})\frac{(1+\frac{1}{2}0.15)}{(1-0.15)} = \mathbf{101\ n\Omega\ m}$$

2.10 Thermal conduction

a. An 80 at % Cu–20 at % Zn brass disk of 40 mm diameter and 5 mm thickness is used to conduct heat from a heat source to a heat sink.

1. Calculate the thermal resistance of the brass disk.
2. If the disk is conducting heat at a rate of 100 W, calculate the temperature drop along the disk.

b. What should be the composition of brass if the temperature drop across the disk is to be halved?

Solution

a

(1) Assume $T = 20\ ^\circ\text{C} = 293$ K. Apply Equation 2.20 (in the textbook) to find the resistivity of the brass in the disk with $\rho_{Cu} = 17.1\ \text{n}\Omega\ \text{m}$ (Table 2.1 and Equation 2.17, in the textbook) and $X_{Zn} = 0.20$:

$$\rho_{brass} = \rho_{Cu} + C_{Zn\ in\ Cu}X_{Zn}(1 - X_{Zn})$$

i.e. $$\rho_{brass} = 17.1\ \text{n}\Omega\ \text{m} + (300\ \text{n}\Omega\ \text{m})(0.20)(1 - 0.20)$$

$\therefore \qquad \rho_{brass} = 65.1\ \text{n}\Omega\ \text{m}$

We know that the thermal conductivity is given by $\kappa/\sigma_{brass} = C_{FWL}T$ where σ_{brass} is the conductivity of the disk, C_{FWL} is the Lorenz number and T is the temperature. This equation can also be written as $\kappa\rho_{brass} = C_{FWL}T$ so that $\kappa = C_{FWL}T/\rho$. Applying this equation,

$$\kappa(20\ ^\circ\text{C}) = (2.44 \times 10^{-8}\ \text{W}\ \Omega\ \text{K}^{-2})(293\ \text{K}) / (6.51 \times 10^{-8}\ \Omega\ \text{m})$$

$\therefore \qquad \kappa(20\ ^\circ\text{C}) = 109.8\ \text{W K}^{-1}\ \text{m}^{-1}$

The thermal resistance is $\theta = L/(\kappa A)$, where L is the thickness of the disk and A is the cross-sectional area of the disk.

$$\theta = L/(\kappa A) = (5 \times 10^{-3}\ \text{m})/[(109.8\ \text{W K}^{-1}\ \text{m}^{-1})(\pi)(2 \times 10^{-2}\ \text{m})^2] = \mathbf{0.0362\ K\ W^{-1}}$$

(2) From $dQ/dt = A\kappa\Delta T/\Delta x = \Delta T/\theta$ (Δx can be taken to be the same as L), and $dQ/dt = P$ (power conducted), we can substitute to obtain:

$$\Delta T = P\theta = (100\ \text{W})(3.62 \times 10^{-2}\ \text{K W}^{-1}) = \mathbf{3.62\ K}\ \text{or}\ \mathbf{3.62\ ^\circ C}$$

Note: Change in temperature is the same in either Kelvins or degrees Celsius - i.e. $\Delta T = T_1 - T_2 = (T_1 + 273) - (T_2 + 273)$.

b Since $\Delta T = P\theta$, to get half ΔT, we need half θ or double κ or double σ or half ρ. We thus need $^1/_2\rho_{brass}$ or $^1/_2(65.1\ \text{n}\Omega\ \text{m})$ which can be attained if the brass composition is X_{new} so that

$$\rho_{new} = \rho_{Cu} + C_{Zn\ in\ Cu}X_{new}(1 - X_{new})$$

i.e. $^1/_2(65.1\ \text{n}\Omega\ \text{m}) = 17\ \text{n}\Omega\ \text{m} + (300\ \text{n}\Omega\ \text{m})X_{new}(1 - X_{new})$

Solving this quadratic equation we get $X_{new} = 0.0545$, or 5.5% Zn. Thus we need **94.5% Cu-5.5% Zn** brass.

2.11 Thermal resistance

Consider a thin insulating disc made of mica to electrically insulate a semiconductor device from a conducting heat sink. Mica has $\kappa = 0.75\ \text{W m}^{-1}\text{K}^{-1}$. The disk thickness is 0.1 mm, and the diameter is 10 mm. What is the thermal resistance of the disk. What is the temperature drop across the disk if the heat current through it is 25W.

Solution

The thermal resistance of the mica disk can be calculated directly from Equation 2.37 (in the textbook)

$$\theta = \frac{L}{\kappa A} = \frac{4L}{\kappa \pi d^2} = \frac{4\left(1\times10^{-3}\ \text{m}\right)}{\pi\left(0.75\ \text{W m}^{-1}\ \text{K}^{-1}\right)\left(1\times10^{-2}\right)^2} = \mathbf{1.698\ K\ W^{-1}}$$

The temperature drop across the disk according to Equation 2.36 (in the textbook) is

$$\Delta T = Q'\theta = \mathbf{42.4\ °C}$$

*2.12 Thermal resistance

Consider a coaxial cable operating under steady state conditions when the current flow through the inner conductor generates Joule heat at a rate $P = I^2R$. The heat generated per second by the core conductor flows through the dielectric; $Q' = I^2R$. The inner conductor reaches a temperature T_i whereas the outer conductor is at T_o. Show that the thermal resistance θ of the hollow cylindrical insulation is

$$\theta = \frac{\left(T_i - T_o\right)}{Q©} = \frac{\ln\left(\frac{b}{a}\right)}{2\pi\kappa L}$$ *Thermal resistance of hollow cylinder*

where a is the inside (core conductor) radius, b is the outside radius (outer conductor), κ is the thermal conductivity of the insulation, and L is the cable length. Consider a coaxial cable that has a copper core conductor and polyethylene (PE) dielectric with following properties: Core conductor resistivity ρ = 19 nΩ m, core radius, a = 4 mm, dielectric thickness, b -a = 3.5 mm, dielectric thermal conductivity κ = 0.3 W m^{-1} K^{-1}. The outside temperature T_o is 25 °C. The cable is carrying a current of 500 A. What is the temperature of the inner conductor?

Solution

Consider a thin cylindrical shell of thickness dr as shown in Figure 2Q12-1. The temperature difference across dr is dT. The surface area of this shell is $2\pi rL$. Thus, from Fourier's law,

$$Q' = -(2\pi rL)\kappa\frac{dT}{dr}$$

which we can integrate with respect to r from $r = a$ where $T = T_i$ to $r = b$ where $T = T_o$,

$$Q'\int_a^b \frac{dr}{r} = -2\pi L\kappa \int_{T_i}^{T_o} dT$$

i.e.

$$Q' = (T_i - T_o)\frac{2\pi\kappa L}{\ln\left(\frac{b}{a}\right)}$$

Thus the thermal resistance of the hollow cylindrical insulation is

$$\theta = \frac{(T_i - T_o)}{Q'} = \frac{\ln\left(\frac{b}{a}\right)}{2\pi\kappa L}$$

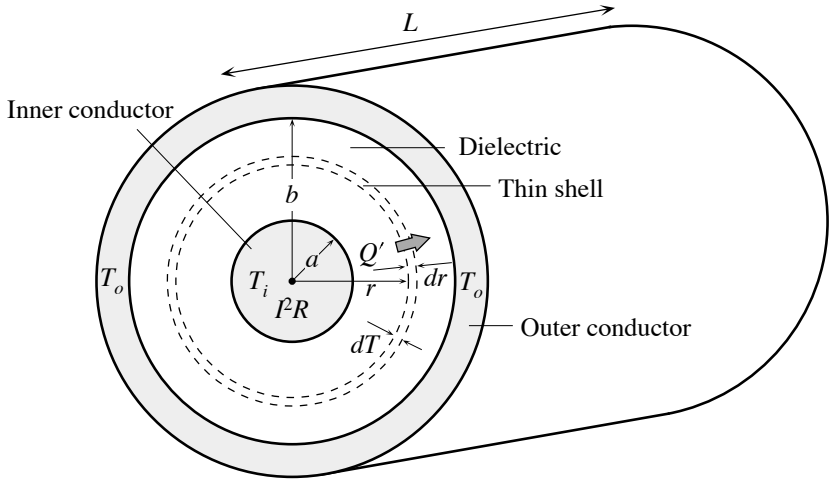


Figure 2Q12-1 Thermal resistance of a hollow cylindrical shell. Consider an infinitesimally thin cylindrical shell of radius *r* and thickness *dr* in the dielectric and concentrically around the inner conductor. The surface area is $2\pi rL$.

The actual length of the conductor does not affect the calculations as long as the length is sufficiently long that there is no heat transfer along the length; heat flows radially from the inner to the outer conductor. We consider a portion of length L of a very long cable and we set $L = 1$ m so that calculations are per unit length. The joule heating per unit second (power) generated by the current I through the core conductor is

$$Q' = I^2\frac{\rho L}{\pi a^2} = (500^2)\frac{(19\times10^{-9})(1)}{\pi(4\times10^{-3})^2} = \mathbf{94.5\ \ W}$$

The thermal resistance of the insulation is,

$$\theta = \frac{\ln\left(\frac{b}{a}\right)}{2\pi\kappa L} = \frac{\ln\left(\frac{(4+3.5)\times10^{-3}}{4\times10^{-3}}\right)}{2\pi(0.3)(1)} = \mathbf{0.33\ °C/W}$$

Thus, the temperature difference ΔT due to Q' flowing through θ is,

$$\Delta T = Q'\theta = (94.5\text{ W})(0.33\text{ °C/W}) = \mathbf{31.2\ °C}.$$

The inner temperature is therefore,

$$T_i = T_o + \Delta T = 25 + 31.2 = \mathbf{56.2\ °C}.$$

Note that for simplicity we assumed that the inner conductor resistivity ρ and thermal conductivity κ are constant (do not change with temperature).

2.13 The Hall effect

Consider a rectangular sample, a metal or an *n*-type semiconductor, with a length L, width W, and thickness D. A current I is passed along L, perpendicular to the cross-sectional area WD. The face $W \times L$ is exposed to a magnetic field density B. A voltmeter is connected across the width, as shown in Figure 2Q13-1, to read the Hall voltage V_H.

a. Show that the Hall voltage recorded by the voltmeter is

$$V_H = \frac{IB}{Den} \qquad \textit{Hall voltage}$$

b. Consider a 1-micron-thick strip of gold layer on an insulating substrate that is a candidate for a Hall probe sensor. If the current through the film is maintained at constant 100 mA, what is the magnetic field that can be recorded per μV of Hall voltage?

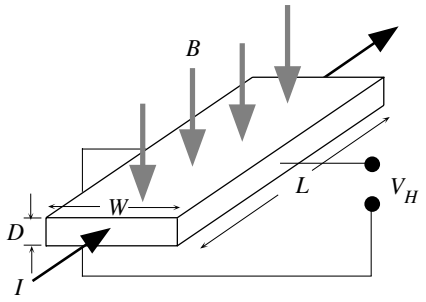


Figure 2Q13-1 Hall effect in a rectangular material with length L, width W, and thickness D. The voltmeter is across the width W.

Solution

a The Hall coefficient, R_H, is related to the electron concentration, n, by $R_H = -1 / (en)$, and is defined by $R_H = \mathcal{E}_y / (JB)$, where $\mathcal{E}_y$ is the electric field in the y-direction, J is the current density and B is the magnetic field. Equating these two equations:

$$-\frac{1}{en} = \frac{\mathcal{E}_y}{JB}$$

$$\therefore \qquad \mathcal{E}_y = -\frac{JB}{en}$$

This electric field is in the opposite direction of the Hall field ($\mathcal{E}_H$) and therefore:

$$\mathcal{E}_H = -\mathcal{E}_y = \frac{JB}{en} \qquad (1)$$

The current density perpendicular (going through) the plane $W \times D$ (width by depth) is:

$$J = \frac{I}{WD}$$

Isolating for W: $$W = \frac{I}{JD} \qquad (2)$$

The Hall voltage (V_H) across W is:

$$V_H = W\mathcal{E}_H$$

If we substitute expressions (1) and (2) into this equation, the following will be obtained:

$$V_H = \frac{IB}{Den}$$

Note: this expression only depends on the thickness and not on the length of the sample.

In general, the Hall voltage will depend on the specimen shape. In the elementary treatment here, the current flow lines were assumed to be nearly parallel from one end to the other end of the sample. In an irregularly shaped sample, one has to consider the current flow lines. However, if the specimen thickness is uniform, it is then possible to carry out meaningful Hall effect measurements using the van der Pauw technique as discussed in advanced textbooks.

b We are given the depth of the film D = 1 micron = 1 μm and the current through the film I = 100 mA = 0.1 A. The Hall voltage can be taken to be V_H = 1 μV, since we are looking for the magnetic field B per μV of Hall voltage. To be able to use the equation for Hall voltage in part (a), we must find the electron concentration of gold. Appendix B in the textbook contains values for gold's atomic mass (M_{at} =196.97 g/mol) and density (d = 19.3 g/cm^3 = 19300 kg/m^3). And since gold has a valency of 1 electron, the concentration of free electrons is equal to the concentration of Au atoms. Therefore:

$$n = \frac{dN_A}{M_{at}} = \frac{\left(19300 \text{ kg m}^{-3}\right)\left(6.022\times10^{23} \text{ mol}^{-1}\right)}{\left(196.97\times10^{-3} \text{ kg mol}^{-1}\right)} = 5.901\times10^{28} \text{ m}^{-3}$$

Now the magnetic field B can be found using the equation for Hall voltage:

$$V_H = \frac{IB}{Den}$$

$$\therefore \quad B = \frac{V_H Den}{I} = \frac{\left(1\times10^{-6} \text{ V}\right)\left(1\times10^{-6} \text{ m}\right)\left(1.602\times10^{-19} \text{ C}\right)\left(5.901\times10^{28} \text{ m}^{-3}\right)}{(0.1 \text{ A})}$$

$$\therefore \quad \boldsymbol{B = 0.0945 \text{ T}}$$

As a side note, the power (P) dissipated in the film could be found very easily. Using the value for resistivity of Au at T = 273 K, ρ = 22.8 nΩ m, the resistance of the film is:

$$R = \frac{\rho L}{A} = \frac{\rho L}{WD} = \frac{\left(22.8\times10^{-9} \ \Omega \text{ m}\right)(0.001 \text{ m})}{(0.0001 \text{ m})\left(1\times10^{-6} \text{ m}\right)} = 0.228 \ \Omega$$

The power dissipated is then:

$$P = I^2R = (0.1 \text{ A})^2(0.228 \ \Omega) = 0.00228 \text{ W}$$

2.14 The strain gauge

A **strain gauge** is a transducer attached to a body to measure its fractional elongation $\Delta L/L$ under an applied load (force) F. The gauge is a grid of many folded runs of a thin, resistive wire glued to a flexible backing, as depicted in Figure 2Q14-1. The gauge is attached to the body under test such that the resistive wire length is parallel to the strain.

a. Assume that the elongation does not change the resistivity and show that the change in the resistance ΔR is related to the strain, $\varepsilon = \Delta L/L$ by

$$\Delta R \approx R(1+2\upsilon)\varepsilon \qquad \textit{Strain gauge equation}$$

where υ is the Poisson ratio, which is defined by

$$v = -\frac{\text{Transverse strain}}{\text{Longitudinal strain}} = -\frac{\varepsilon_t}{\varepsilon_l} \qquad \textit{Position ratio}$$

where ε_l is the strain along the applied load, that is, $\varepsilon_l = \Delta L/L = \varepsilon$, and ε_t is the strain in the transverse direction, that is, $\varepsilon_t = \Delta D/D$, where D is the diameter (thickness) of the wire.

b. Explain why a nichrome wire would be a better choice than copper for the strain gauge (consider the TCR).

c. How do temperature changes affect the response of the gauge? Consider the effect of temperature on ρ. Also consider the differential expansion of the specimen with respect to the gauge wire such that even if there is no applied load, there is still strain, which is determined by the differential expansion coefficient, $\lambda_{\text{specimen}} - \lambda_{\text{gauge}}$, where λ is the thermal coefficient of linear expansion: $L = L_o[1 + \lambda(T - T_o)]$, where T_o is the reference temperature.

d. The gauge factor for a transducer is defined as the fractional change in the measured property $\Delta R/R$ per unit input signal (ε). What is the gauge factor for a metal-wire strain gauge, given that for most metals, $\upsilon \approx 1/3$?

e. Consider a strain gauge that consists of a nichrome wire of resistivity 1 $\mu\Omega$ m, a total length of 1 m, and a diameter of 25 μm. What is ΔR for a strain of 10^{-3}? Assume that $\upsilon \approx 1/3$.

f. What will ΔR be if constantan wire is used (see Question 2.5)?

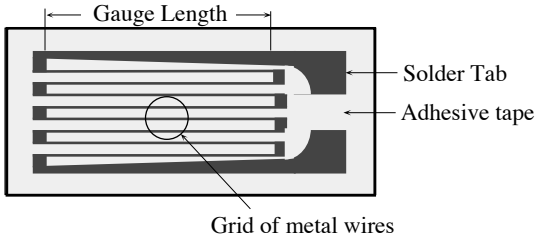


Figure 2Q14-1 The strain gauge consists of a long, thin wire folded several times along its length to form a grid as shown and embedded in a self-adhesive tape. The ends of the wire are attached to terminals (solder pads) for external connections. The tape is stuck on the component for which the strain is to be measured.

Solution

a Consider the resistance R of the gauge wire,

$$R = \frac{\rho L}{\pi\left(\frac{D}{2}\right)^2} \qquad (1)$$

where L and D are the length and diameter of the wire, respectively. Suppose that the applied load changes L and D by δL and δD which change R by δR. The total derivative of a function R of two variables L and D can be found by taking partial differentials (like those used for error calculations in physics labs). Assuming that ρ is constant,

$$\delta R = \left(\frac{\partial R}{\partial L}\right)\delta L + \left(\frac{\partial R}{\partial D}\right)\delta D = \left(\frac{\rho}{\frac{\pi}{4}D^2}\right)\delta L - 2\left(\frac{\rho L}{\frac{\pi}{4}D^3}\right)\delta D$$

so that the fractional change is

$$\frac{\delta R}{R} = \frac{\left(\frac{\rho}{\frac{\pi}{4}D^2}\right)}{\left(\frac{\rho L}{\frac{\pi}{4}D^2}\right)}\delta L - 2\frac{\left(\frac{\rho L}{\frac{\pi}{4}D^3}\right)}{\left(\frac{\rho L}{\frac{\pi}{4}D^2}\right)}\delta D$$

i.e. $$\frac{\delta R}{R} = \frac{\delta L}{L} - 2\frac{\delta D}{D} \qquad (2)$$

We can now use the definitions of longitudinal and transverse strain,

$$\frac{\delta L}{L} = \varepsilon_l \qquad \text{and} \qquad \frac{\delta D}{D} = \varepsilon_t = -\upsilon\varepsilon_l$$

in the expression for $\delta R/R$ in Eqn. (2) to obtain,

$$\frac{\delta R}{R} = \varepsilon_l - 2(-\upsilon\varepsilon_l) = (1 + 2\upsilon)\varepsilon \qquad (3)$$

where $\varepsilon = \varepsilon_l$.

b The change in R was attributed to changes in L and D due to their extension by the applied load. There are two reasons why a nichrome wire will be a better choice. First is that nichrome has a higher resistivity ρ which means that its resistance R will be higher than that of a similar size copper wire and hence nichrome wire will exhibit a greater change in R (δR is easier to measure). Secondly nichrome has a very small TCR which means that ρ and hence R do not change significantly with temperature. If we were to use a Cu wire, any small change in the temperature will most likely overwhelm any change due to strain.

c Consider a change δR in R due to a change δT in the temperature. We can differentiate R with respect to T by considering that ρ, L and D depend on T. It is not difficult to show (See Question 2.15) that if α is the temperature coefficient of resistivity and λ is the linear expansion coefficient , then

$$\left(\frac{1}{R}\right)\frac{dR}{dT} = \alpha - \lambda \qquad (4)$$

Typically $\lambda \approx 2 \times 10^{-5}$ K^{-1}, and for pure metals $\alpha \approx 1/273$ K^{-1} or 3.6×10^{-3} K^{-1}. A 1 °C fluctuation in the temperature will result in $\delta R/R = 3.6 \times 10^{-3}$ which is about the same as $\delta R/R$ from a strain of $\varepsilon = 2 \times 10^{-3}$ at a constant temperature. It is clear that temperature fluctuations would not allow sensible strain measurements if we were to use a pure metal wire. To reduce the temperature fluctuation effects, we need $\alpha \approx \lambda$ which means we have to use a metal alloy such as a nichrome wire or a constantan wire.

Even if we make α - $\lambda = 0$, the temperature change still produces a change in the resistance because the metal wire and specimen expand by different amounts and this creates a strain and hence a

change in the resistance. Suppose that λ_{gauge} and $\lambda_{specimen}$ are the linear expansion coefficients of the gauge wire and the specimen, then the differential expansion will be $\lambda_{specimen}$ - $\lambda_{specimen}$ and this can only be zero if $\lambda_{gauge} = \lambda_{specimen}$.

d The gauge factor is defined as

$$\text{GF} = \frac{\text{Fractional change in gauge property}}{\text{Input signal}} = \frac{\delta R / R}{\varepsilon} = \frac{(1+2\upsilon)\varepsilon}{\varepsilon}$$

i.e. $\mathbf{GF} = 1 + 2\upsilon = 1 + 2(1/3) = \mathbf{1.67}$

which applies to metals inasmuch as $\upsilon \approx 1/3$.

e For nichrome wire, given that $\rho = 10^{-6}\ \Omega$ m, $L = 1$ m, and $D = 25 \times 10^{-6}$ m,

we have, $$R = \frac{\rho L}{\pi\left(\frac{D}{2}\right)^2} = \frac{(10^{-6}\ \ \Omega\ \ \text{m})(1\ \text{m})}{\pi\left(\frac{25\times10^{-6}\ \ \text{m}}{2}\right)^2} = 2037\ \Omega$$

For a strain of $\varepsilon = 10^{-3}$,

$$\delta\boldsymbol{R} = R(1 + 2\upsilon)\varepsilon = (2037\ \Omega)[1 + 2(1/3)](10^{-3}) = \mathbf{3.40}\ \Omega$$

This can be easily measured with an ohm-meter, though instrumentation engineers would normally use the gauge in an electrical bridge circuit. Note that $\delta R/R = (3.4\ \Omega)/(2037\ \Omega) = 0.0017$, or 0.17 %.

f For a constantan wire, given that $\rho = 5 \times 10^{-7}\ \Omega$ m, $L = 1$ m, and $D = 25 \times 10^{-6}$ m,

we have, $$R = \frac{(5\times10^{-7}\ \ \Omega\ \ \text{m})(1\ \text{m})}{\pi\left(\frac{25\times10^{-6}\ \ \text{m}}{2}\right)^2} = 1019\ \Omega$$

so that $\delta\boldsymbol{R} = R(1 + 2\upsilon)\varepsilon = (1019\ \Omega)[1 + 2(1/3)](10^{-3}) = \mathbf{1.70}\ \Omega$

2.15 Thermal coefficients of expansion and resistivity

a. Consider a thin metal wire of length L and diameter D. Its resistance is $R = \rho L/A$, where $A = \pi D^2/4$. By considering the temperature dependence of L, A, and ρ individually, show that

$$\frac{1}{R}\frac{dR}{dT} = \alpha_o - \lambda_o$$

where α_o is the temperature coefficient of resistivity (TCR), and λ_o is the temperature coefficient of linear expansion (thermal expansion coefficient or expansivity), that is,

$$\lambda_o = L_o^{-1}\left(\frac{dL}{dT}\right)_{T=T_o} \qquad \text{or} \qquad \lambda_o = D_o^{-1}\left(\frac{dD}{dT}\right)_{T=T_o}$$

Note: Consider differentiating $R = \rho L/[(\pi D^2)/4]$ with respect to T with each parameter, ρ, L, and D, having a temperature dependence.

Given that typically, for most pure metals, $\alpha_o \approx 1/273\ \text{K}^{-1}$ and $\lambda_o \approx 2 \times 10^{-5}\ \text{K}^{-1}$, confirm that the temperature dependence of ρ controls R, rather than the temperature dependence of the geometry. Is it necessary to modify the given equation for a wire with a noncircular cross section?

b. Is it possible to design a resistor from a suitable alloy such that its temperature dependence is almost nil? Consider the TCR of an alloy of two metals *A* and *B*, for which $\alpha_{AB} \approx \alpha_A\rho_A/\rho_{AB}$.

Solution

a Consider the resistance *R* of the wire,

$$R = \frac{\rho L}{\pi\left(\frac{D}{2}\right)^2} \tag{1}$$

Consider a change δR in *R* due to a change δT in the temperature. We can differentiate *R* with respect to *T* by considering that ρ, *L*, and *D* depend on *T*,

$$\frac{dR}{dT} = \frac{d}{dT}\left(\frac{\rho L}{\frac{\pi}{4}D^2}\right) = \left(\frac{L}{\frac{\pi}{4}D^2}\right)\frac{d\rho}{dT} + \left(\frac{\rho}{\frac{\pi}{4}D^2}\right)\frac{dL}{dT} - 2\left(\frac{\rho L}{\frac{\pi}{4}D^3}\right)\frac{dD}{dT}$$

divide by *R*:

$$\left(\frac{1}{R}\right)\frac{dR}{dT} = \left(\frac{1}{\rho}\right)\frac{d\rho}{dT} + \left(\frac{1}{L}\right)\frac{dL}{dT} - 2\left(\frac{1}{D}\right)\frac{dD}{dT} \tag{2}$$

At $T = T_o$ we have,

$$\left(\frac{1}{R}\right)\frac{dR}{dT} = \left(\frac{1}{\rho_o}\right)\frac{d\rho}{dT} + \left(\frac{1}{L_o}\right)\frac{dL}{dT} - 2\left(\frac{1}{D_o}\right)\frac{dD}{dT} \tag{3}$$

where the derivatives are at $T = T_o$. Recall that the temperature coefficient of resistivity, TCR (α_o), and the linear expansion coefficient λ_o are defined as follows,

$$\alpha_0 = \left(\frac{1}{\rho_0}\right)\frac{d\rho}{dT} \qquad \text{and} \qquad \lambda_0 = \left(\frac{1}{L_0}\right)\frac{dL}{dT} = \left(\frac{1}{D_0}\right)\frac{dD}{dT} \tag{4}$$

which reduces Eqn. (3) to

$$\left(\frac{1}{R}\right)\frac{dR}{dT} = \alpha_0 - \lambda_0 \tag{5}$$

Typically $\lambda_o \approx 2 \times 10^{-5}$ K^{-1}, and for pure metals $\alpha_o \approx 1/273$ K^{-1} or 3.6×10^{-3} K^{-1}. Thus,

$$\left(\frac{1}{R}\right)\frac{dR}{dT} = 3.6\times10^{-3}\ \text{K}^{-1} - 2\times10^{-5}\ \text{K}^{-1} \approx 3.6\times10^{-3}\ \text{K}^{-1} \approx \alpha_0$$

Since α_o is much larger than λ_o, it dominates the change in *R*.

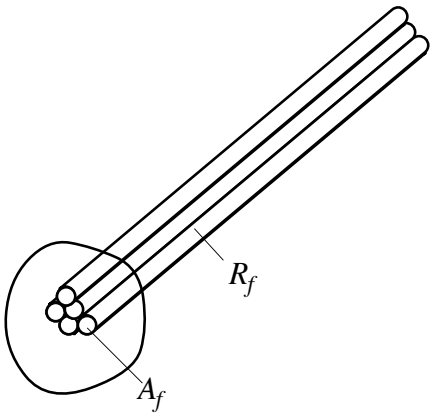


Figure 2Q15-1 Wire with non-circular cross section.

There is no need to modify Eqn. (5) for a non-circular cross sectional area. You can derive the same expression for a rectangular or an elliptic cross section, or, indeed, any arbitrary cross section. One can consider the wire to be made up of N thin fibers each of *circular* cross section A_f. Imagine holding a bunch of these in your hand and then sliding them into any cross section you like as in the Figure 2Q15-1. In all cases $A = \Sigma A_f$. However, since the fibers are in parallel, the total resistance is given by $R^{-1} = \Sigma R_f^{-1} = NR_f^{-1}$. Thus $R = R_f / N$. For each fiber, $\delta R_f = R_f(\alpha_o - \lambda_o)\delta T$ as we have derived above. Then,

$$\delta R = (\delta R_f)/N = [R_f(\alpha_o - \lambda_o)\delta T]/N = R(\alpha_o - \lambda_o)\delta T$$

[There are a few assumptions such as the resistivity is homogeneous and the cross section does not change along the wire!]

b For pure metals, $\alpha_o >> \lambda_o$. Alloying metal A with metal B reduces the TCR (temperature coefficient of resistivity) α_A of metal A to $\alpha_{AB} = \alpha_A(\rho_A/\rho_{AB})$. The $\alpha_o - \lambda_o$ can be brought to zero by using a suitable composition alloy for which $\alpha_o = \lambda_o$. Since strictly α_o depends (however slightly) on T, the condition $\alpha_o - \lambda_o$ can only be exactly true at one temperature or approximately true in a small region around that temperature. In practice this temperature range may be sufficient to cover a typical application range in which, for most practical purposes, $\alpha_o - \lambda_o$ is negligible.

2.16 Einstein relation and ionic conductivity

In the case of ionic conduction, ions have to jump from one interstice to the neighboring one. This process involves overcoming a potential energy barrier just like atomic diffusion, and drift and diffusion are related. The drift mobility μ of ions is proportional to the diffusion coefficient D because drift is limited by the atomic diffusion process. The **Einstein relation** relates the two by

$$\frac{D}{\mu} = \frac{kT}{e} \qquad \textit{Einstein relation}$$

The diffusion coefficient of the Na^+ ion in sodium silicate (Na_2O-SiO_2) glasses at 400 °C is 3.4×10^{-9} $cm^2\ s^{-1}$. The density of such glasses is 2.4 g cm^{-3}. Calculate the ionic conductivity and resistivity of (17.5 mol% Na_2O)(82.5 mol% SiO_2) sodium silicate glass at 400 °C and compare your result with the experimental value of about 10^4 Ω cm for the resistivity.

Solution

We can apply the conductivity expression $\sigma = q_i n_i \mu_i$, where q_i is the charge of the ion, Na^+, so that it is +e, n_i is the concentration of the Na^+ ions in the glass and μ_i is their mobility.

We can think of the glass as built from $[(Na_2O)_{0.175}(SiO_2)_{0.825}]$ units. The atomic masses of Na, O and Si are 23, 16 and 28.1 g mole^{-1} respectively. The atomic mass of one unit is

$$M_{at} = 0.175[2\times(23)+16] + 0.825[28.1+2\times(16)] = \mathbf{60.43\ g\ mol^{-1}} \text{ of } [(Na_2O)_{0.175}(SiO_2)_{0.825}]$$

The number of $[(Na_2O)_{0.175}(SiO_2)_{0.825}]$ units per unit volume can be found from the density d by

$$n = \frac{dN_A}{M_{at}} = \frac{\left(2.4\,\text{g cm}^{-3}\right)\left(6.02\times10^{23}\ \text{mol}^{-1}\right)}{\left(60.43\ \text{g mol}^{-1}\right)} = \mathbf{2.39\times10^{22}\ units\ cm^{-3}}$$

The concentration of Na^+ ions is equal to the concentration of Na atoms and then

$$n_i = n_{Na} = (\text{atomic fraction of Na in the unit})\times n =$$

$$= \left[\frac{0.175\times(2)}{0.175(2+1)+0.825(1+2)}\right]\left(2.39\times10^{22}\ \text{cm}^{-3}\right) = \mathbf{2.79\times10^{21}\ cm^{-3}}$$

The mobility of sodium ions can be calculated from Einstein relation

$$\mu_i = \frac{eD}{kT} = \frac{\left(1.602\times10^{-19}\ \text{C}\right)\left(3.4\times10^{-9}\ \text{cm}^2\ \text{s}^{-1}\right)}{\left(1.38\times10^{-23}\ \text{J K}^{-1}\right)\left((400+273)\ \text{K}\right)} = \mathbf{6.04\times10^{-8}\ cm^2\ V^{-1}\ s^{-1}}$$

Thus the conductivity of the glass is

$$\sigma = en_i\mu_i = \left(1.602\times10^{-19}\ \text{C}\right)\left(2.79\times10^{21}\ \text{cm}^{-3}\right)\left(6.04\times10^{-8}\ \text{cm}^2\ \text{V}^{-1}\ \text{s}^{-1}\right) = \mathbf{2.7\times10^{-5}\ \Omega^{-1}\ cm^{-1}}$$

and its resistivity is

$$\rho = \frac{1}{\sigma} = \frac{1}{\left(2.7\times10^{-5}\ \Omega^{-1}\ \text{cm}^{-1}\right)} = \mathbf{3.8\times10^{4}\ \Omega\ cm}$$

2.17 Skin effect

a. What is the skin depth for a copper wire carrying a current at 60 Hz? The resistivity of copper at 27 °C is 17 nΩ m. Its relative permeability is $\mu_r \approx 1$. Is there any sense in using a conductor for power transmission with a diameter of more than 2 cm?

b. What is the skin depth for an iron wire carrying a current at 60 Hz? The resistivity of iron at 27 °C is 97 nΩ m. Assume that its relative permeability is $\mu_r \approx 700$. How does this compare with the copper wire? Discuss why copper is preferred over iron for power transmission even though the iron is nearly 100 times cheaper than copper.

Solution

a The conductivity is $1/\rho$. The relative permeability (μ_r) for copper is 1, thus $\mu_{Cu} = \mu_o$. The angular frequency is $\omega = 2\pi f = 2\pi(60\text{ Hz})$. Using these values in the equation for skin depth (δ):

$$\delta = \frac{1}{\sqrt{\frac{1}{2}\omega_o \frac{1}{\rho}\mu_{\text{Cu}}}} = \frac{1}{\sqrt{\frac{1}{2}\left(2\pi 60\ \text{s}^{-1}\right)\frac{\left(4\pi\times10^{-7}\ \text{H m}^{-1}\right)}{\left(17\times10^{-9}\ \ \Omega\ \text{m}\right)}}}$$

$\therefore$ δ **= 0.00847 m** or **8.47 mm**

This is the depth of current flow. If the radius of wire is 10 mm or more, no current flows through the core region and it is wasted. There is no point in using wire much thicker than a radius of 10 mm (diameter of 20 mm).

b The conductivity is $1/\rho$. The relative permeability (μ_r) for Iron is 700, thus $\mu_{\text{Fe}} = 700\mu_o$. The angular frequency is $\omega = 2\pi f = 2\pi(60\text{ Hz})$. Using these values in the equation for skin depth (δ):

$$\delta_{\text{Fe}} = \frac{1}{\sqrt{\frac{1}{2}\omega_o \frac{1}{\rho}\mu_{\text{Fe}}}} = \frac{1}{\sqrt{\frac{1}{2}\left(2\pi 60\ \text{s}^{-1}\right)\frac{(700)\left(4\pi\times10^{-7}\ \text{H m}^{-1}\right)}{\left(97\times10^{-9}\ \ \Omega\ \text{m}\right)}}}$$

$\therefore$ δ **= 0.000765 m** or **0.765 mm**

Thus the skin depth is 0.765 mm, about 11 times less than that for copper.

To calculate the resistance we need the cross sectional area for conduction. The material cross sectional area is πr^2 where r is the radius of the wire. But the current flow is within depth δ. We deduct the area of the core, $\pi (r - \delta)^2$, from the overall area, πr^2, to obtain the cross sectional area for conduction.

Comparison of Cu and Fe based on <u>solid core wires</u>

The resistance per unit length of the solid core Fe wire (R_{Fe}) is:

$$R_{\text{Fe}} = \frac{\rho_{\text{Fe}}}{A_{Fe}} = \frac{\rho_{\text{Fe}}}{\pi r_{\text{Fe}}{}^2 - \pi\left(r_{\text{Fe}} - \delta_{\text{Fe}}\right)^2} = \frac{\rho_{\text{Fe}}}{2\pi r_{\text{Fe}}\delta_{\text{Fe}} - \pi\delta_{\text{Fe}}{}^2}$$

The resistance per unit length of solid core Cu wire is:

$$R_{\text{Cu}} = \frac{\rho_{\text{Cu}}}{A_{Cu}} = \frac{\rho_{\text{Cu}}}{\pi r_{\text{Cu}}{}^2 - \pi\left(r_{\text{Cu}} - \delta_{\text{Cu}}\right)^2} = \frac{\rho_{\text{Cu}}}{2\pi r_{\text{Cu}}\delta_{\text{Cu}} - \pi\delta_{\text{Cu}}{}^2}$$

If we equate these two resistances, we can make a comparison between Fe and Cu:

$$R_{\text{Fe}} = R_{\text{Cu}}$$

$\therefore$ $$\frac{\rho_{\text{Fe}}}{2\pi r_{\text{Fe}}\delta_{\text{Fe}} - \pi\delta_{\text{Fe}}{}^2} = \frac{\rho_{\text{Cu}}}{2\pi r_{\text{Cu}}\delta_{\text{Cu}} - \pi\delta_{\text{Cu}}{}^2}$$

$\therefore$ $2\pi\rho_{\text{Cu}} r_{\text{Fe}}\delta_{\text{Fe}} = \rho_{\text{Fe}}\left(2\pi r_{\text{Cu}}\delta_{\text{Cu}} - \pi\delta_{\text{Cu}}{}^2\right)$ (Neglect the $\delta_{\text{Fe}}{}^2$ term which is small)

$\therefore$ $$r_{\text{Fe}} = \frac{\rho_{\text{Fe}}\left(2\pi r_{\text{Cu}}\delta_{\text{Cu}} - \pi\delta_{\text{Cu}}{}^2\right)}{2\pi\rho_{\text{Cu}}\delta_{\text{Fe}}}$$

We can assume a value for r_{Cu} for calculation purposes, r_{Cu} = 10 mm. The resistivity ρ_{Cu} is given as 17 nΩ m and the skin depth of Cu is known to be δ_{Cu} = 8.47 mm. The resistivity of Fe is given as ρ_{Fe} = 97 nΩ m and its skin depth was just calculated to be δ_{Fe} = 0.765 mm. We can substitute these values into the above equation to determine the radius of Fe wire that would be equivalent to 10 mm diameter Cu wire.

$$r_{\text{Fe}} = \frac{\left(97\times10^{-9}\ \ \Omega\ \text{m}\right)\left[2\pi(0.010\ \text{m})(0.00847\ \text{m}) - \pi(0.00847\ \text{m})^2\right]}{2\pi\left(17\times10^{-9}\ \ \Omega\ \text{m}\right)(0.000765\ \text{m})}$$

$\therefore$ $\boldsymbol{r_{\text{Fe}} = 0.364\ \text{m}}$

Now compare the volume (V) of Fe per unit length to the volume of Cu per unit length:

$$\frac{V_{\text{Fe}}}{V_{\text{Cu}}} = \frac{\pi r_{\text{Fe}}{}^2(1\ \text{m})}{\pi r_{\text{Cu}}{}^2(1\ \text{m})} = \frac{(0.364\ \text{m})^2}{(0.010\ \text{m})^2} = \mathbf{1325}$$

Even though Fe costs 100 times less than Cu, we need about 1300 times the volume of Cu if Fe is used. The cost disadvantage is 13 times in addition to weight disadvantage.

ADDENDUM JANUARY 2001

Submitted by George Belev

Comparison of Cu and Fe wires: any shape and any number

To determine if it is worthwhile to use iron rather than copper, we must compare the amount of iron needed to perform the equivalent task of some amount of copper (i.e. have the same resistance).

Let us first assume that by choosing a proper shape for the conductors we can *eliminate* the influence of the skin effect on conduction.

The resistance per unit length of the Fe wire (R_{Fe}) is:

$$R_{\text{Fe}} = \frac{\rho_{\text{Fe}}}{A_{Fe}}$$

The resistance per unit length of Cu wire is

$$R_{\text{Cu}} = \frac{\rho_{\text{Cu}}}{A_{Cu}}$$

If we equate these two resistances, we can make a direct comparison between Fe and Cu:

$$R_{\text{Fe}} = R_{\text{Cu}}$$

$\therefore$ $$A_{Fe} = \frac{\rho_{Fe}}{\rho_{Cu}} A_{Cu}$$

and the volumes of iron and copper per unit length will be in the same ratio

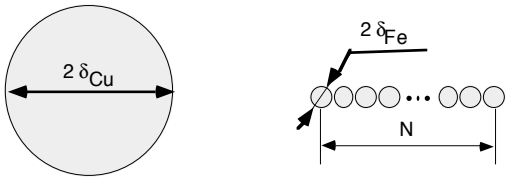
$$V_{Fe} = \frac{\rho_{Fe}}{\rho_{Cu}} V_{Cu}$$

Let us compare the masses of Fe and Cu needed per unit length

$$\frac{M_{Fe}}{M_{Cu}} = \frac{V_{Fe}D_{Fe}}{V_{Cu}D_{Cu}} = \frac{\rho_{Fe}}{\rho_{Cu}}\frac{D_{Fe}}{D_{Cu}} = \frac{97\ \text{n}\Omega\ \text{m}}{17\ \text{n}\Omega\ \text{m}} \cdot \frac{7.86\ \text{g cm}^{-3}}{8.96\ \text{g cm}^{-3}} \approx 5$$

Since Fe costs 100 times less than Cu, if we use iron conductors, we will reduce the cost for wire by 100/5 = 20 times. So it seems that the use of Fe will have great economic advantage if we can find a reasonable way to eliminate the influence of the skin effect on conduction.

There is no sense in making the conductor with a radius bigger than the skin depth, so let us consider a single copper conductor with radius δ_{Cu}, and using N iron conductors each with radius δ_{Cu} as shown bellow:



As we have calculated above both conductors will have equal resistance per unit length if

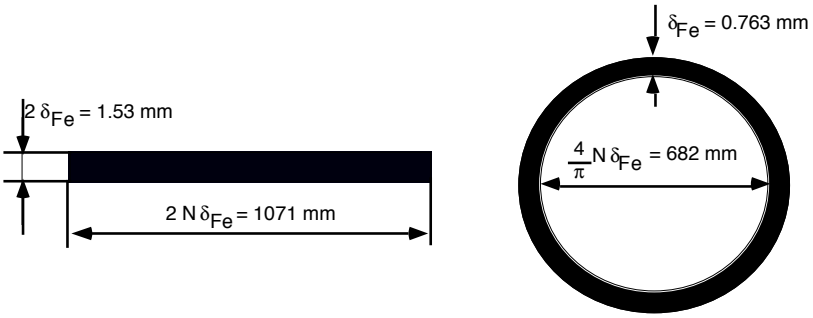
$$\frac{A_{Fe}}{A_{Cu}} = \frac{N\,\pi\delta_{Fe}^2}{\pi\delta_{Cu}^2} = \frac{\rho_{Fe}}{\rho_{Cu}}$$

and we can calculate the number N of Fe wires which should run in parallel

$$N = \frac{\rho_{Fe}}{\rho_{Cu}}\frac{\delta_{Cu}^2}{\delta_{Fe}^2} \approx 700$$

Thus, copper as a single conductor has 700 times better performance than a single iron conductor.

It is not necessary to manufacture 700 Fe wires and run them in parallel. The iron conductor can be produced more conveniently in the shapes shown bellow and it will be cheaper than the Cu conductor.



But it will be of impractical size, it will have poor mechanical properties and will be 5 times heavier compared with the single Cu wire. A power grid based on Cu conductors will be much cheaper and much smaller than the one based on Fe conductors.

2.18 Temperature of a light bulb

a. Consider a 100 W, 120 V incandescent bulb (lamp). The tungsten filament has a length of 0.579 m and a diameter of 63.5 μm. Its resistivity at room temperature is 56 nΩ m. Given that the resistivity of the filament can be represented as

$$\rho = \rho_0\left[\frac{T}{T_0}\right]^n \qquad \textit{Resistivity of W}$$

where T is the temperature in K, ρ_o is the resistance of the filament at T_o K, and n = 1.2, estimate the temperature of the bulb when it is operated at the rated voltage, that is, directly from the mains outlet. Note that the bulb dissipates 100 W at 120 V.

b. Suppose that the electrical power dissipated in the tungsten wire is totally radiated from the surface of the filament. The radiated power at the absolute temperature T can be described by Stefan's Law

$$P_{\text{radiated}} = \epsilon\,\sigma_s A(T^4 - T_o^4) \qquad \textit{Radiated power}$$

where σ_s is Stefan's constant (5.67×10^{-8} W m^{-2} K^{-4}), ϵ is the emissivity of the surface (0.35 for tungsten), A is the surface area of the tungsten filament, and T_o is the room temperature (293 K). Obviously, for $T > T_o$, $P_{\text{radiated}} = \epsilon\,\sigma_s A T^4$.

Assuming that all of the electrical power is radiated from the surface, estimate the temperature of the filament and compare it with your answer in part (a).

c. If the melting temperature of W is 3407 °C, what is the voltage that guarantees that the light bulb will blow?

Solution

a First, find the current through the bulb at 100 W and 120 V.

$$P = VI$$

$\therefore$ $$I = P/V = (100\text{ W})/(120\text{ V}) = 0.8333\text{ A}$$

From Ohm's law the resistance of the bulb can be found:

$$R = V/I = (120\text{ V})/(0.8333\text{ A}) = 144.0\ \Omega$$

The values for length of the filament ($L = 0.579$ m) and diameter of the filament ($D = 63.5\ \mu$m) at operating temperature are given. Using these values we can find the resistivity of the filament when the bulb is on (ρ_1).

$$R = \frac{\rho_1 L}{\frac{\pi}{4}D^2}$$

$\therefore$ $$\rho_1 = \frac{R\frac{\pi}{4}D^2}{L} = \frac{(144.0\ \Omega)\frac{\pi}{4}\left(63.5\times10^{-6}\ \text{m}\right)^2}{(0.579\ \text{m})} = 7.876\times10^{-7}\ \Omega\ \text{m}$$

Now the bulb's operating temperature (T_1) can be found using our obtained values in the equation for resistivity of *W* (assuming room temperature $T_o = 293$ K and given $n = 1.2$):

$$\rho = \rho_0\left[\frac{T}{T_0}\right]^n$$

isolate: $$\boldsymbol{T_1} = T_0\left[\frac{\rho_1}{\rho_0}\right]^{\frac{1}{n}} = (293\ \text{K})\left[\frac{7.876\times10^{-7}\ \Omega\ \text{m}}{56\times10^{-9}\ \Omega\ \text{m}}\right]^{\frac{1}{1.2}} = \mathbf{2652\ K}$$

b First we need the surface area *A* of the Tungsten filament. Since it is cylindrical in shape:

$$A = L(\pi D) = (0.579\text{ m})(\pi)(63.5\times10^{-6}\text{ m}) = 0.0001155\text{ m}^2$$

Now the temperature of the filament T_1 can be found by isolating it in Stefan's equation and substituting in the given values for emissivity ($\epsilon = 0.35$), Stefan's constant ($\sigma_s = 5.67\times10^{-8}$ W m^{-2} K^{-4}) and room temperature ($T_o = 293$ K).

$$P = \epsilon\,\sigma_s A\left(T_1^4 - T_0^4\right)$$

$\therefore$ $$T_1 = \left(\frac{P}{\epsilon\,\sigma_s A} + T_0^4\right)^{\frac{1}{4}}$$

$\therefore$ $$T_1 = \left[\frac{100\ \text{W}}{(0.35)\left(5.67\times10^{-8}\ \text{W m}^{-2}\ \text{K}^{-1}\right)\left(0.0001155\ \text{m}^2\right)} + (293\ \text{K})^4\right]^{\frac{1}{4}}$$

$\therefore$ $\boldsymbol{T_1 = 2570\ K}$

Note: We can even ignore T_o to get the same temperature since $T_o << T_1$:

$$P = \in \sigma_s A T_1^4$$

$$\therefore \quad T_1 = \left(\frac{P}{\in \sigma_s A} \right)^{\frac{1}{4}} = \left[\frac{100\ \text{W}}{(0.35)\left(5.67 \times 10^{-8}\ \text{W m}^{-2}\ \text{K}^{-1}\right)\left(0.0001155\ \text{m}^2\right)} \right]^{\frac{1}{4}}$$

$\therefore$ $\boldsymbol{T_1 = 2570\ K}$

These values are fairly close to the answer obtained in part (a).

c Let V be the voltage and R be the resistance when the filament is at temperature T_m. We are given the temperature T_m = 3407 °C + 273 = 3680 K. Since we know the following:

$$R = \frac{L}{\frac{\pi}{4} D^2} \rho \qquad \rho = \rho_0 \left[\frac{T_m}{T_0} \right]^n$$

We can make a substitution for ρ and use the values given for the light bulb filament to find the resistance of the filament at temperature T_m.

$$R = \frac{L}{\frac{\pi}{4} D^2} \rho_0 \left[\frac{T_m}{T_0} \right]^n = \frac{(0.579\ \text{m})}{\frac{\pi}{4}\left(63.5 \times 10^{-6}\ \text{m}\right)^2} \left(56 \times 10^{-9}\ \Omega\ \text{m}\right) \left[\frac{3680\ \text{K}}{293\ \text{K}} \right]^{1.2}$$

$\therefore$ $R = 213.3\ \Omega$

Assuming that all electrical power is radiated from the surface of the bulb, we can use Stefan's law and substitute in V^2/R for power P:

$$\frac{V^2}{R} = \in \sigma_s A \left(T_1^4 - T_0^4 \right)$$

$$\therefore \quad V^2 = R \in \sigma_s A \left(T_m^{\ 4} - T_0^{\ 4} \right)$$

$$\therefore \quad V^2 = (213.3\ \Omega)(0.35)\left(5.67 \times 10^{-8}\ \text{W m}^{-2}\ \text{K}^{-4}\right)(0.0001155\ \text{m}) \left[(3680\ \text{K})^4 - (293\ \text{K})^4 \right]$$

$\therefore$ $\boldsymbol{V = 299\ V}$

"I think there is a world market for maybe five computers"
Thomas Watson (Chairman of IBM, 1943)

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Chapter 3

3.1 Photons and photon flux

a. Consider a 1 kW AM radio transmitter at 700 kHz. Calculate the number of photons emitted from the antenna per second.

b. The average intensity of sunlight on Earth's surface is about 1 kW m^{-2}. The maximum intensity is at a wavelength around 800 nm. Assuming that all the photons have an 800 nm wavelength, calculate the number of photons arriving on Earth's surface per unit time per unit area. What is the magnitude of the electric field in the sunlight?

c. Suppose that a solar cell device can convert each sunlight photon into an electron, which can then give rise to an external current. What is the maximum current that can be supplied per unit area (m^2) of this solar cell device?

Solution

a Given: power $P = 1000$ W and frequency $\upsilon = 700 \times 10^3\ s^{-1}$.

Then the photon energy is

$$E_{ph} = h\upsilon = (6.626 \times 10^{-34}\ \text{J s})(700 \times 10^3\ s^{-1})$$

$\therefore$ $$E_{ph} = 4.638 \times 10^{-28}\ \text{J/photon or } 2.895 \times 10^{-9}\ \text{eV/photon}$$

The number of photons emitted from the antenna per unit time (N_{ph}) is therefore:

$$N_{ph} = \frac{P}{E_{ph}} = \frac{1000\ \text{W}}{4.638 \times 10^{-28}\ \text{J}} = \mathbf{2.16 \times 10^{30}\ photons\ per\ second}$$

b Average intensity $\mathcal{I}_{\text{average}} = 1000\ \text{W/m}^2$ and maximum wavelength $\lambda_{\text{max}} = 800$ nm. The photon energy at λ_{max} is,

$$E_{ph} = hc/\lambda_{\text{max}} = (6.626 \times 10^{-34}\ \text{J s})(3.0 \times 10^8\ \text{m s}^{-1})/(800 \times 10^{-9}\ \text{m})$$

$\therefore$ $$E_{ph} = 2.485 \times 10^{-19}\ \text{J or } 1.551\ \text{eV}$$

The photon flux Γ_{ph} is the number of photons arriving per unit time per unit area,

$$\Gamma_{ph} = \frac{\mathcal{I}_{\text{average}}}{E_{ph}} = \frac{1000\ \text{W m}^{-2}}{2.485 \times 10^{-19}\ \text{J}} = \mathbf{4.02 \times 10^{21}\ photons\ m^{-2}\ s^{-1}}$$

For the electric field we use classical physics. $\mathcal{I}_{\text{average}} = (1/2)c\varepsilon_o\mathcal{E}^2$, so that

$$\mathcal{E} = \sqrt{\frac{2\mathcal{I}_{\text{average}}}{c\varepsilon_o}} = \sqrt{\frac{2(1000\ \text{W m}^{-2})}{(3.0 \times 10^8\ \text{m s}^{-1})(8.854 \times 10^{-12}\ \text{F m}^{-1})}} = \mathbf{868\ V\ m^{-1}}$$

c If each photon gives rise to one electron then the current density J is:

$$J = e\Gamma_{ph} = (1.602 \times 10^{-19}\ \text{C})(4.02 \times 10^{21}\ \text{m}^{-2}\ \text{s}^{-1})$$

$\therefore$ $$\mathbf{J = 644\ A\ m^{-2}}$$

*3.2 Human eye

Photons passing through the pupil are focused by the lens onto the retina and are detected by two types of photosensitive cells, called rods and cones. Rods are highly sensitive photoreceptors with a peak response at the wavelength 500 nm. They do not register color, but they are responsible for our vision under dimmed light conditions, which is termed **scotopic vision.** Cones are color sensitive and are responsible for our daytime vision, called **photopic vision.** There are three types of cone photoreceptors, which are sensitive to the blue, green, and red wavelengths 430 nm, 535 nm and 575 nm, respectively. All three cones have an overall peak response of 555 nm.

a. Calculate the photon energy (in eV) for the peak responsivity of each photoreceptor in the eye.

b. The fovea is a retina region on the visual axis; images are focused onto this region. The density of the cones in the fovea is on the order of 150,000/mm^2. Below a light intensity of the order of 100 μW m^{-2}, cones are not functional and rods take over the vision. (Due to marked variation from person to person, these values are very approximate.)

1. Estimate the minimum photon flux for color vision assuming that photopic vision occurs roughly at 555 nm.
2. If a visual sensation persists for a time (1/5)th of a second, how many photons does the eye need per cone for a visual color sensation?
3. If the eye is 10 percent efficient overall, due to photon reflections, etc., how many photons are actually absorbed per cone to generate a colored visual sensation?

Solution

Minimum light intensity $\mathcal{I}_{\min}$ is given to be 100×10^{-6} W m^{-2}.

a The photon energy for peak responsivity can be found as follows (firstly for blue light):

$$E_{ph} = hc/\lambda = (6.626 \times 10^{-34} \text{ J s})(3.0 \times 10^{8} \text{ m/s})/(430 \times 10^{-9} \text{ m})$$

$\therefore$ $$\boldsymbol{E_{ph} = 4.623 \times 10^{-19} \text{ J}} \text{ or } \mathbf{2.886 \text{ eV}}$$

b(1) The photon flux can then be found (blue light):

$$\Gamma = \mathcal{I}_{\min}/E_{ph} = (100 \times 10^{-6} \text{ W m}^{-2})/(4.623 \times 10^{-19} \text{ J}) = 2.163 \times 10^{14} \text{ m}^{-2} \text{ s}^{-1}$$

The values for the other wavelengths are listed in Table 3Q2-1:

Table 3Q2-1 Summarized values of photon energy and photon flux for different wavelengths of light.

	Blue	Green	Red	Peak response
Wavelength, λ (nm)	430	535	575	555
a) Photon energy ($\times 10^{-19}$) $E_{ph} = hc/\lambda$ (J)	4.623	3.716	3.457	3.582
Photon energy $E_{ph} \times 1/q$ (eV)	**2.886**	**2.319**	**2.158**	**2.236**
b) (1) **Photon flux needed** ($\times 10^{14}$) $\Gamma = I_{\min}/E_{ph}$ (m^{-2} s^{-1})	**2.163**	**2.691**	**2.893**	**2.792**

The minimum photon flux $\Gamma_{\mathbf{min}}$ needed for visual color sensation **is** $\mathbf{2.792 \times 10^{14}\ m^{-2}\ s^{-1}}$.

(2) The number of photons needed for a visual color sensation is (where D_{cone} is the density of the cones in the fovea):

$$N_{\mathbf{ph-cone}} = \frac{\Gamma_{\min} t}{D_{\text{cone}}} = \frac{\left(2.163 \times 10^{14}\ \text{m}^{-2}\ \text{s}^{-1}\right)\left(\frac{1}{5}\ \text{s}\right)}{1.5 \times 10^{11}\ \text{m}^{-2}} = \mathbf{288\ \ photons}$$

(3) As the eye is only 10% efficient because most of the photons are reflected, the actual number of ***absorbed photons*** that give rise to a visual signal **is 29**.

3.3 Photoelectric effect

A photoelectric experiment indicates that violet light of wavelength 420 nm is the longest wavelength radiation that can cause photoemission of electrons from a particular multialkali photocathode surface.

a. What is the work function in eV?

b. If a UV radiation of wavelength 300 nm is incident upon the photocathode, what will be the maximum kinetic energy of the photoemitted electrons, in eV?

c. Given that the UV light of wavelength 300 nm has an intensity of 20 mW/cm^2, if the emitted electrons are collected by applying a positive bias to the opposite electrode, what will be the photoelectric current density in mA cm^{-2} ?

Solution

a We are given $\lambda_{\max} = 420$ nm. The work function is then:

$$\Phi = h\upsilon_o = hc/\lambda_{\max} = (6.626 \times 10^{-34}\ \text{J s})(3.0 \times 10^{8}\ \text{m s}^{-1})/(420 \times 10^{-9}\ \text{m})$$

$\therefore$ Φ **= 4.733 × 10^{-19} J** or **2.96 eV**

b Given $\lambda = 300$ nm, the photon energy is then:

$$E_{ph} = h\upsilon = hc/\lambda = (6.626 \times 10^{-34}\ \text{J s})(3.0 \times 10^{8}\ \text{m s}^{-1})/(300 \times 10^{-9}\ \text{m})$$

$\therefore$ $E_{ph} = 6.626 \times 10^{-19}\ \text{J} = 4.14\ \text{eV}$

The kinetic energy *KE* of the emitted electron can then be found:

$$\boldsymbol{KE} = E_{ph} - \Phi = 4.14\ \text{eV} - 2.96\ \text{eV} = \mathbf{1.18\ eV}$$

c The photon flux Γ_{ph} is the number of photons arriving per unit time per unit area. If I_{light} is the light intensity (light energy flowing through unit area per unit time) then,

$$\Gamma_{ph} = \frac{I_{\text{light}}}{E_{ph}}$$

Suppose that each photon creates a single electron, then

J = Charge flowing per unit area per unit time = Charge × Photon Flux

$\therefore$ $$\boldsymbol{J} = e\Gamma_{ph} = \frac{eI_{\text{light}}}{E_{ph}} = \frac{(1.602 \times 10^{-19}\ \text{C})(200\ \text{W m}^{-2})}{(6.626 \times 10^{-19}\ \text{J})} = \mathbf{48.36\ A\ m^{-2} = 4.836\ mA\ cm^{-2}}$$

3.4 Photoelectric effect and quantum efficiency.

Cesium metal is to be used as the photocathode material in a photoemissive electron tube because electrons are relatively easily removed from a cesium surface. The work function of a clean cesium surface is 1.9 eV.

a. What is the longest wavelength of radiation which can result in photoemission?

b. If blue radiation of wavelength 450 nm is incident onto the Cs photocathode, what will be the kinetic energy of the photoemitted electrons in eV? What should be the voltage required on the opposite electrode to extinguish the external photocurrent?

c. **Quantum efficiency** (*QE*) of a photocathode is defined by,

$$\text{Quantum efficiency} = \frac{\text{Number of photoemitted electrons}}{\text{Number of incident photons}}$$

QE is 100% if each incident photon ejects one electron. Suppose that blue light of wavelength 450 nm with an intensity of 30 mW cm^{-2} is incident on a Cs photocathode that is a circular disk of diameter 6 mm. If the emitted electrons are collected by applying a positive bias voltage to the anode, and the photocathode has a *QE* of 25%, what will be the photocurrent?

Solution

a The longest wavelength will be $\Phi = h\upsilon_o = hc/\lambda_{\max}$

$$\lambda_{\max} = \frac{hc}{\Phi} = \frac{\left(6.626\times10^{-34}\ \text{J s}\right)\left(3\times10^{8}\ \text{m s}^{-1}\right)}{\left(1.9\times1.602\times10^{-19}\ \text{J}\right)} = \mathbf{653.9\times10^{-7}\ m = 653.9\ nm}$$

b The energy of the photon at 450 nm is

$$E_{ph} = \frac{hc}{\lambda} = \frac{\left(6.626\times10^{-34}\ \text{J s}\right)\left(3\times10^{8}\ \text{m s}^{-1}\right)}{\left(450\times10^{-9}\ \text{m}\right)} = \mathbf{4.417\times10^{-19}\ J = 2.76\ eV}$$

The excess energy over Φ goes to kinetic energy of the photoelectron. Thus the electron kinetic energy is

$$KE = E_{ph} - \Phi = (2.76\ \text{eV}) - (1.9\ \text{eV}) = \mathbf{0.86\ eV}$$

and the stopping voltage for this wavelength will be **-0.86 V**.

c . The number of photons arriving per unit area per unit time at the photocathode is

$$\Gamma_{ph} = \frac{I}{E_{ph}} = \frac{\left(30\times10^{-3}\times10^{4}\ \text{J s}^{-1}\ \text{m}^{2}\right)}{\left(4.417\times10^{-19}\ \text{J}\right)} = 6.792\times10^{20}\ \text{s}^{-1}\ \text{m}^{-2}$$

The current density is then simply

$$J = e\Gamma_{ph}QE = \left(1.602\times10^{-19}\ \text{C}\right)\left(6.792\times10^{20}\ \text{s}^{-1}\ \text{m}^{-2}\right)(0.25) = 27.2\ \text{A m}^{-2}$$

The photoelectric current is then

$$I = AJ = \frac{\pi d^2}{4}J = \frac{\pi\left(6\times10^{-3}\ \text{m}\right)^2}{4}\left(27.2\ \text{A m}^{-2}\right) = \mathbf{7.691\times10^{-4}\ A = 0.769\ mA}$$

3.5 Diffraction by X-rays and an electron beam

Diffraction studies on a polycrystalline Al sample using X-rays gives the smallest diffraction angle (2θ) of 29.5° corresponding to diffraction from the (111) planes. The lattice parameter a of Al (FCC), is 0.405 nm. If we wish to obtain the same diffraction pattern (same angle) using an electron beam, what should be the voltage needed to accelerate the electron beam? Note that the interplanar separation d for planes (h,k,l) and the lattice parameter a for cubic crystals are related by

$$d = a / [h^2+k^2+l^2]^{1/2}.$$

Solution

The plane indices are given as $h = 1$, $k = 1$ and $l = 1$, and the lattice parameter is given as $a =$ 0.405 nm. Therefore,

$$d = \frac{a}{\sqrt{h^2+k^2+l^2}} = \frac{0.405 \text{ nm}}{\sqrt{1^2+1^2+1^2}} = 0.2338 \text{ nm}$$

From the Bragg condition with $\theta = 29.5° / 2 = 14.75°$ we have,

$$\lambda = 2d\sin(\theta) = 2(0.2338 \text{ nm})\sin(14.75°) = 0.1191 \text{ nm}$$

The voltage V required to accelerate the electron is (see Example 3.4 in text),

$$\boldsymbol{V} = \left(\frac{1.226 \text{ nm}}{\lambda}\right)^2 = \left(\frac{1.226 \text{ nm}}{0.1191 \text{ nm}}\right)^2 = \mathbf{106\ V}$$

3.6 Heisenberg's uncertainty principle

Show that if the uncertainty in the position of a particle is on the order of its de Broglie wavelength, then the uncertainty in its momentum is about the same as the momentum value itself.

Solution

The de Broglie wavelength is

$$\lambda = \frac{h}{p}$$

where p is the momentum. From Heisenberg's uncertainty principle we have,

$$\Delta x \Delta p \approx \hbar$$

If we take the uncertainty in the position to be of the order of the wavelength, $\Delta x \sim \lambda$, then

$$\Delta p \approx \frac{\hbar}{\Delta x} \approx \left(\frac{1}{2\pi}\right)\frac{h}{\lambda} = \left(\frac{1}{2\pi}\right)p \sim p$$

so that the uncertainty in the momentum will be of the same order as the momentum itself.

3.7 Heisenberg's uncertainty principle

An excited electron in an Na atom emits radiation at a wavelength 589 nm and returns to the ground state. If the mean time for the transition is about 20 ns, calculate the inherent width in the emission line. What is the length of the photon emitted?

Solution

The emitted wavelength is λ= 589 nm and the transition time is Δt = 20 ns. This time corresponds to an uncertainty in the energy that is ΔE. Using the uncertainty principle we can calculate ΔE,

$$\Delta t \Delta E \approx \hbar$$

$$\therefore \quad \Delta E \approx \frac{\hbar}{\Delta t} = \frac{\frac{1}{2\pi}\left(6.626\times10^{-34}\ \text{J s}\right)}{20\times10^{-9}\ \text{s}} = 5.273\times10^{-27}\ \text{J or } 3.29\times10^{-8}\ \text{eV}$$

The corresponding uncertainty in the emitted frequency $\Delta\upsilon$ is

$$\Delta\upsilon = \frac{\Delta E}{h} = \frac{(5.273\times10^{-27}\ \text{J})}{\left(6.626\times10^{-34}\ \text{J s}\right)} = 7.958\times10^{6}\ \text{s}^{-1}$$

To find the corresponding spread in wavelength and hence the line width $\Delta\lambda$, we can differentiate $\lambda = c/\upsilon$.

$$\frac{d\lambda}{d\upsilon} = -\frac{c}{\upsilon^2} = -\frac{\lambda^2}{c}$$

Thus, $$\Delta\lambda = \frac{\lambda^2}{c}\Delta\upsilon = \frac{(589\times10^{-9}\ \text{m})^2}{3.0\times10^{8}\ \text{m/s}}(7.958\times10^{6}\ \text{s}^{-1}) = \mathbf{9.20\times10^{-15}\ m}\ \text{or}\ \mathbf{9.20\ fm}$$

The length of the photon l is

$$\boldsymbol{l} = (\text{light velocity})\times(\text{emission duration}) = (3.0\times10^{8}\ \text{m/s})(20\times10^{-9}\ \text{s}) = \mathbf{6.0\ m}$$

3.8 Tunneling

a. Consider the phenomenon of tunneling through a potential energy barrier of height V_o and width a, as shown in Figure 3Q8-1. What is the probability that the electron will be reflected? Given the transmission coefficient T, can you find the reflection coefficient R? What happens to R as a or V_o or both become very large?

b. For a wide barrier ($\alpha a >> 1$), show that T_o can at most be 4 and that $T_o = 4$ when $E = (1/2)V_o$.

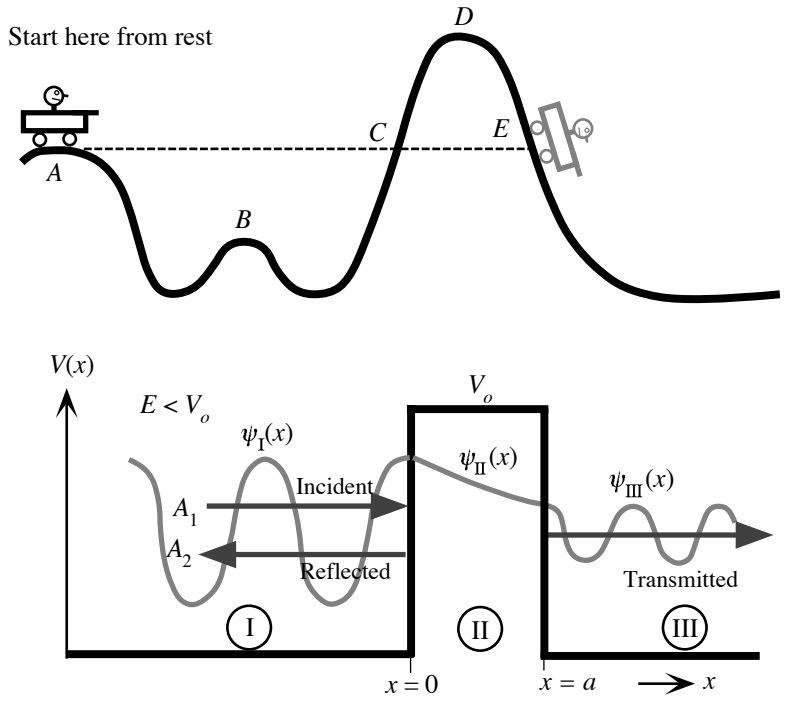


Figure 3Q8-1

(a) The roller coaster released from *A* can at most make it to *C*, but not to *E*. Its *PE* at *A* is less than the *PE* at *D*. When the car is at the bottom, its energy is totally *KE*. *CD* is the energy barrier that prevents the car from making it to *E*. In quantum theory, on the other hand, there is a chance that the car could tunnel (leak) through the potential energy barrier between *C* and *E* and emerge on the other side of the hill at *E*.

(b) The wavefunction for the electron incident on a potential energy barrier (V_o). The incident and reflected waves interfere to give $\psi_I(x)$. There is no reflected wave in region III. In region II, the wavefunction decays with *x* because $E < V_o$.

Solution

a The relative reflection probability or reflection coefficient *R* is given as the ratio of the square of the amplitude of the reflected wave to that of the incident wave, that is:

$$R = \frac{A_2^2}{A_1^2}$$

Also, *R* can be found from the transmission coefficient *T* by the equation $R = 1 - T$, as stated in Equation 3.28 (in the textbook). From Equation 3.24 (in the textbook), *T* is given as:

$$T = \frac{1}{1 + D\left(\sinh[a\alpha]\right)^2}$$

where *a* is the width of the potential energy barrier, α is the rate of decay, and *D* is given by:

$$D = \frac{V_o^2}{4E\left(V_o - E\right)}$$

To determine the behavior of *R* as *a* or V_o or both become very large, we can use the equation $R = 1 - T$ to express *R* in terms of *a* and *D* (remember *D* is a function of V_o).

$$R = 1 - T = 1 - \frac{1}{1 + D\left(\sinh[a\alpha]\right)^2}$$

$$\therefore \quad R = \frac{D\left(\sinh[a\alpha]\right)^2}{1 + D\left(\sinh[a\alpha]\right)^2}$$

We know that sinh(∞) = ∞, and also that 1 / ∞ = 0. Therefore, as V_o becomes large, so does D, which leads to $T = 0$ and $R = 1$, meaning total reflection occurs. If a becomes large then sinh(∞) = ∞ and $T = 0$, making $R = 1$ for total reflection.

b We need to find the maximum value of T_o. Since T_o depends on the energy E, we can differentiate it with respect to E, set the result to 0 and isolate E.

$$T_o = \frac{16E\left(V_o - E\right)}{V_o^2} \qquad \text{(Eqn. 3.27 in the textbook)}$$

$$\therefore \quad \frac{\delta T_o}{\delta E} = 16\left(\frac{-2E + V_o}{V_o^2}\right) = 0$$

$$\therefore \quad E = \frac{1}{2}V_o$$

Thus T_o is maximum when $E = V_o / 2$. If this expression for energy is substituted back into the equation for T_o to find its maximum value (T_o'):

$$T_o' = \frac{16E\left(V_o - E\right)}{V_o^2} = \frac{16\left(\frac{1}{2}V_o\right)\left(V_o - \frac{1}{2}V_o\right)}{V_o^2}$$

$$\therefore \quad \boldsymbol{T_o'} = \frac{16V_o^2}{4V_o^2} = \mathbf{4}$$

3.9 Electron impact excitation

a. A projectile electron of kinetic energy 12.2 eV collides with a hydrogen atom in a gas discharge tube. Find the *n*-th energy level to which the electron in the hydrogen atom gets excited.

b. Calculate the possible wavelengths of radiation (in nm) that will be emitted from the excited H atom in part (a) as the electron returns to its ground state. Which one of these wavelengths will be in the visible spectrum?

c. In neon street lighting tubes, gaseous discharge in the Ne tube involves electrons accelerated by the electric field impacting Ne atoms and exciting some of them to the $2p^5 3p^1$ states, as shown in Figure 3Q9-1. What is the wavelength of emission? Can the Ne atom fall from the $2p^5 3p^1$ state to the ground state by spontaneous emission?

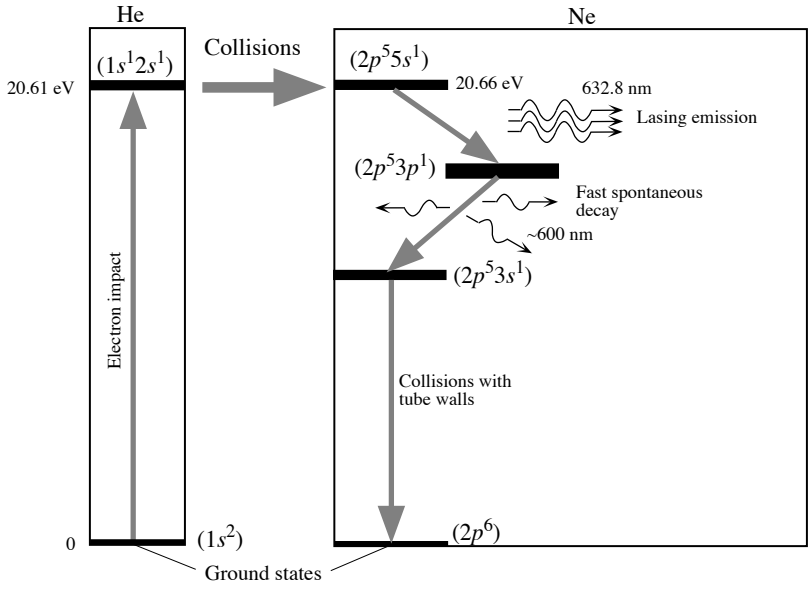


Figure 3Q9-1 The principle of operation of the HeNe laser. Important HeNe laser energy levels (for 632.8 nm emission).

Solution

a The energy of the electron at the *n*th level is given by the equation:

$$E_n = \frac{-13.6\ \text{eV}}{n^2} Z^2$$

where *Z* is the atomic number, in this case 1 for hydrogen. Also, the kinetic energy present in the electron will be equal to the original energy state of the electron (in this case the ground energy), subtracted from the energy of the *n*th level at which the electron resides. That is:

$$KE = E_n - E_{\text{ground}}$$

The ground energy of the Hydrogen electron is known to be -13.6 eV. We can then substitute this value and the equation for E_n given above into the expression for *KE* and isolate for *n*.

$$KE = \frac{-13.6\ \text{eV}}{n^2} + 13.6\ \text{eV}$$

$$\therefore \quad n = \sqrt{\frac{13.6\ \text{eV}}{13.6\ \text{eV} - KE}}$$

Substituting in the given value for *KE* of 12.2 eV:

$$\boldsymbol{n} = \sqrt{\frac{13.6\ \text{eV}}{13.6\ \text{eV} - 12.2\ \text{eV}}} = \mathbf{3.12}$$

Therefore the electron is at the 3rd energy level in the hydrogen atom (since *n* must be an integer). The amount of energy absorbed by the electron in the excitation is:

$$\Delta E = (\text{-}13.6\ \text{eV}) / 3^2 - E_{\text{ground}} = 12.09\ \text{eV}$$

and the projectile electron that hit it originally leaves the collision site with a new *KE* that is:

$$KE' = KE - \Delta E = 12.2\ \text{eV} - 12.09\ \text{eV} = 0.11\ \text{eV}$$

b The electron can return down to the ground level either from $n = 3$ to $n = 1$, or from $n = 3$ to $n = 2$ then to $n = 1$. For the first transition ($n = 3$ to $n = 1$):

$$\Delta E_{31} = \frac{-13.6\ \text{eV}}{3^2} - \left(\frac{-13.6\ \text{eV}}{1^2}\right) = 12.09\ \text{eV}$$

This value can be put into the following equation to find the emitted wavelength:

$$\Delta E_{31} = h\upsilon = h\frac{c}{\lambda_{31}}$$

$$\therefore \quad \lambda_{31} = h\frac{c}{\Delta E_{31}} = \left(6.626\times10^{-34}\ \text{J s}\right)\frac{3.0\times10^{8}\ \text{m/s}}{(12.09\ \text{eV})\left(1.602\times10^{-19}\ \text{J/eV}\right)}$$

$$\therefore \quad \boldsymbol{\lambda_{31} = 1.03\times10^{-7}\ \text{m}}\ \text{or}\ \boldsymbol{103\ \text{nm}}$$

This wavelength is not in the visible spectrum. For the $n = 3$ to $n = 2$ to $n = 1$ transition, two different wavelengths of light will be released, one in the first transition from 3 to 2 and the other in the second.

$$\Delta E_{32} = \frac{-13.6\ \text{eV}}{3^2} - \left(\frac{-13.6\ \text{eV}}{2^2}\right) = 1.889\ \text{eV}$$

$$\therefore \quad \lambda_{32} = h\frac{c}{\Delta E_{32}} = \left(6.626\times10^{-34}\ \text{J s}\right)\frac{3.0\times10^{8}\ \text{m/s}}{(1.889\ \text{eV})\left(1.602\times10^{-19}\ \text{J/eV}\right)}$$

$$\therefore \quad \boldsymbol{\lambda_{32} = 6.57\times10^{-7}\ \text{m}}\ \text{or}\ \boldsymbol{657\ \text{nm}}$$

This wavelength is visible (red). For the 2 to 1 transition:

$$\Delta E_{21} = \frac{-13.6\ \text{eV}}{2^2} - \left(\frac{-13.6\ \text{eV}}{1^2}\right) = 10.2\ \text{eV}$$

$$\therefore \quad \lambda_{21} = h\frac{c}{\Delta E_{21}} = \left(6.626\times10^{-34}\ \text{J s}\right)\frac{3.0\times10^{8}\ \text{m/s}}{(10.2\ \text{eV})\left(1.602\times10^{-19}\ \text{J/eV}\right)}$$

$$\therefore \quad \boldsymbol{\lambda_{21} = 1.22\times10^{-7}\ \text{m}}\ \text{or}\ \boldsymbol{122\ \text{nm}}$$

This wavelength is not visible.

c **The wavelength is** given in Figure 3Q9-1, as 600 nm approximately which is in the red. The Ne atom must change its orbital quantum number ℓ in a transition involving a photon (emission or absorption). The transition ($2p^5 3p^1$) to ($2p^6$) does not change ℓ and hence it is NOT allowed in a photon-emission transition.

3.10 Line spectra of hydrogenic atoms

Spectra of hydrogen-like atoms are classified in terms of electron transitions to a common lower energy level.

a. All transitions from energy levels n = 2, 3, ... to n = 1 (the K shell) are labeled K lines and constitute the **Lyman series.** The spectral line corresponding to the smallest energy difference (n = 2 to n = 1) is labeled the K_α line, next is labeled K_β, and so on. The transition from $n = \infty$ to n = 1 has the largest energy difference and defines the greatest photon energy (shortest wavelength) in the K series; hence it is called the absorption edge $K_{\alpha e}$. What is the range of wavelengths for the K lines? What is $K_{\alpha e}$? Where are these lines with respect to the visible spectrum?

b. All transitions from energy levels $n = 3, 4, \ldots$ to $n = 2$ (L shell) are labeled L lines and constitute the **Balmer series.** What is the range of wavelengths for the L lines (i.e., L_α and $L_{\alpha e}$)? Are these in the visible range?

c. All transitions from energy levels $n = 4, 5, \ldots$ to $n = 3$ (M shell) are labeled M lines and constitute the **Paschen series.** What is the range of wavelengths for the M lines? Are these in the visible range?

d. How would you expect the spectral lines to depend on the atomic number Z?

Solution

Consider the expression for the energy change for a transition from $n = n_2$ to $n = n_1$,

$$\Delta E = -Z^2 E_I \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right) \tag{1}$$

where $E_I = 13.6$ eV is the ionization energy from the ground state ($n = 1$) and Z is the atomic number; for hydrogen $Z = 1$.

The absorption wavelength is given by $\lambda = hc / \Delta E$.

a For the Lyman series (K-shell), $n_1 = 1$. For K_α, $n_2 = 2$, which gives $\Delta E = 10.2$ eV from Eqn. (1) and

$$\lambda(\boldsymbol{K}_\alpha) = \frac{hc}{\Delta E} = \frac{\left(6.626 \times 10^{-34} \text{ J s}\right)\left(3.0 \times 10^{8} \text{ m s}^{-1}\right)}{(10.2 \text{ eV})\left(1.602 \times 10^{-19} \text{ J/eV}\right)} = \mathbf{122 \quad nm}$$

For $K_{\alpha e}$, $n_2 = \infty$ and $\Delta E = 13.6$ eV and

$$\lambda(\boldsymbol{K}_{\alpha e}) = \frac{hc}{\Delta E} = \frac{\left(6.626 \times 10^{-34} \text{ J s}\right)\left(3.0 \times 10^{8} \text{ m s}^{-1}\right)}{(13.6 \text{ eV})\left(1.602 \times 10^{-19} \text{ J/eV}\right)} \mathbf{= 91.2 \quad nm}$$

The wavelengths range from **91.2 nm to 122 nm; much shorter than the visible spectrum.**

b For the Balmer series (L-shell), $n_1 = 2$. For L_a, $n_2 = 3$, which gives $\Delta E = 1.889$ eV and

$$\lambda(\boldsymbol{L}_\alpha) = \frac{hc}{\Delta E} = \frac{\left(6.626 \times 10^{-34} \text{ J s}\right)\left(3.0 \times 10^{8} \text{ m s}^{-1}\right)}{(1.889 \text{ eV})\left(1.602 \times 10^{-19} \text{ J/eV}\right)} = \mathbf{657 \text{ nm}}$$

For $L_{\alpha e}$, $n_2 = \infty$ and $\Delta E = 3.4$ eV and

$$\lambda(\boldsymbol{L}_{\alpha e}) = \frac{hc}{\Delta E} = \frac{\left(6.626 \times 10^{-34} \text{ J s}\right)\left(3.0 \times 10^{8} \text{ m s}^{-1}\right)}{(3.4 \text{ eV})\left(1.602 \times 10^{-19} \text{ J/eV}\right)} = \mathbf{365 \text{ nm}}$$

The wavelengths range from **365 nm to 657 nm. Part of this is in the visible spectrum (blue and green).**

c For the Paschen series (M-shell), $n_1 = 3$. For M_a, $n_2 = 4$, which gives $\Delta E = 0.6611$ eV and

$$\lambda(\boldsymbol{M}_\alpha) = \frac{hc}{\Delta E} = \frac{\left(6.626 \times 10^{-34} \text{ J s}\right)\left(3.0 \times 10^{8} \text{ m s}^{-1}\right)}{(0.6611 \text{ eV})\left(1.602 \times 10^{-19} \text{ J/eV}\right)} = \mathbf{1877 \text{ nm}}$$

For $M_{\alpha e}$, $n_2 = \infty$ and $\Delta E = 1.511$ eV and

$$\lambda(M_{\alpha e}) = \frac{hc}{\Delta E} = \frac{(6.626\times10^{-34}\ \text{J s})(3.0\times10^{8}\ \text{m s}^{-1})}{(1.511\ \text{eV})(1.602\times10^{-19}\ \text{J/eV})} = \mathbf{821\ nm}$$

The wavelengths range from **821 nm to 1877 nm which is in the infrared region.**

d The transition energy depends on Z^2, and therefore the emitted photon wavelength of the spectral lines depends inversely on Z^2.

*3.11 X-rays and the Moseley relation

X-rays are photons with wavelengths in the range 0.01 nm-10 nm, with typical energies in the range 100 eV to 100 keV. When an electron transition occurs in an atom from the L to the K shell, the emitted radiation is generally in the X-ray spectrum. For all atoms with atomic number $Z > 2$, the K shell is full. Suppose that one of the electrons in the K shell has been knocked out by an energetic projectile electron impacting the atom (the projectile electron would have been accelerated by a large voltage difference). The resulting vacancy in the K shell can then be filled by an electron in the L shell transiting down and emitting a photon. The emission resulting from the L to K shell transition is labeled the K_α line. The table shows the K_α line data obtained for various materials.

Table 3Q11-1 Values for atomic number Z and wavelength of K_α emissions.

Material	Mg	Al	S	Ca	Cr	Fe	Cu	Rb	W
Z	12	13	16	20	24	26	29	37	74
K_α line (nm)	0.987	0.834	0.537	0.335	0.229	0.194	0.154	0.093	0.021

a. If υ is the frequency of emission, plot $\upsilon^{1/2}$ against the atomic number Z of the element.

b. H.G. Moseley, while still a graduate student of E. Rutherford in 1913, found the empirical relationship

$$\upsilon^{1/2} = B\,(Z - C) \qquad \textit{Moseley relation}$$

where B and C are constants. What are B and C from the plot? Can you give a simple explanation as to why K_α absorption should follow this relationship?

Solution

a If $\lambda(K_\alpha)$ is the wavelength of the K_α emission, then the corresponding frequency is

$$\upsilon = c\ /\ \lambda(K_\alpha)$$

For example, for Mg, $\lambda(K_\alpha) = 0.987$ nm and

$$\upsilon = c\ /\ \lambda(K_\alpha) = (3.0\times10^{8}\ \text{m s}^{-1})/(0.987\times10^{-9}\ \text{m}) = 3.04\times10^{17}\ \text{s}^{-1}$$

We can then calculate υ for each metal and plot $\sqrt{(\upsilon)}$ vs. Z (atomic number) as in the figure below. The result is a straight line indicating that

$$(\upsilon)^{1/2} = (5.210\times10^{7}\ \text{s}^{-1/2})Z - 9.626\times10^{7}\ \text{s}^{-1/2}$$

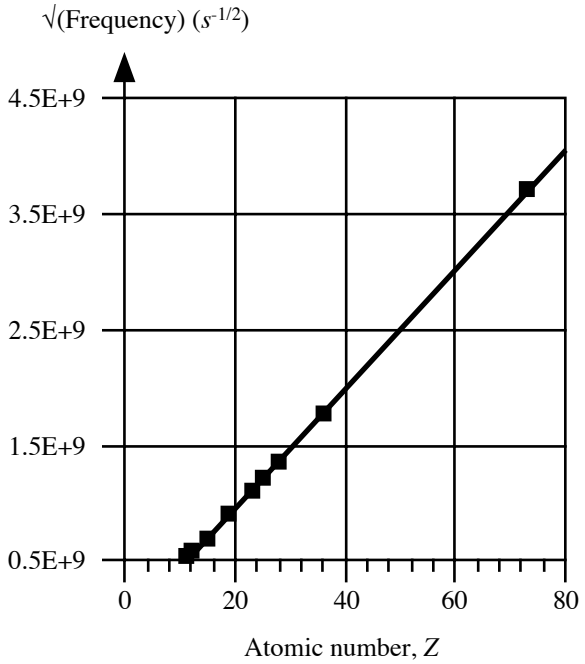


Figure 3Q11-1 Plot of square root of frequency versus atomic number.

Table 3Q11-2 Summarized values for frequency of K_α emissions.

Metal	Mg	Al	S	Ca	Cr	Fe	Cu	Rb	W
Z	12	13	16	20	24	26	29	37	74
$\upsilon\,(K_\alpha) \times 10^{17}$ s^{-1}	3.04	3.60	5.59	8.96	13.1	15.5	19.5	32.3	143

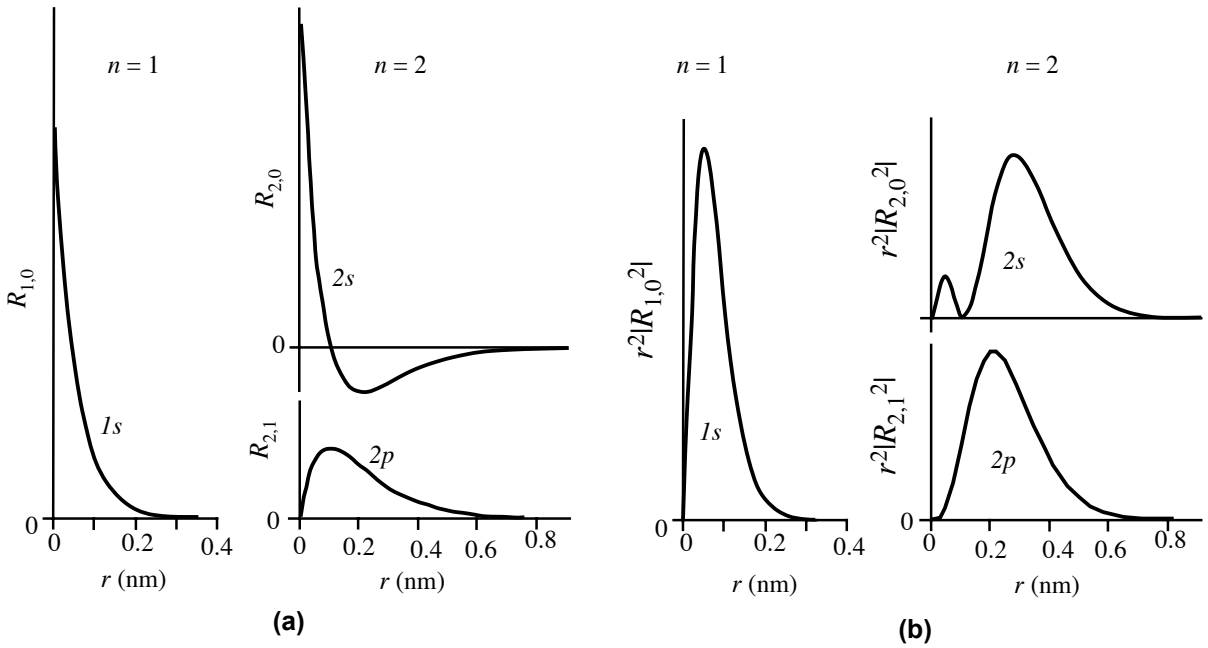


Figure 3Q11-2

(a) Radial wavefunctions of the electron in a hydrogenic atom for various n and ℓ values.

(b) $r^2\left|R^2_{n,\ell}\right|$ gives the radial probability density. Vertical axis scales are linear in arbitrary units.

b The first energy level, given by Equation 3.37a (in the textbook), depends on Z^2. This is for a hydrogenic atom i.e., only 1 electron in the atom but a nuclear charge of $+Ze$. In the present case, the transiting electron in the L shell sees an effective charge of $+Ze - 1e$ because there is 1 electron in the K-shell shielding the nucleus. Therefore the photon energy should be proportional to $(Z-1)^2$. However, some of the L-shell electrons also spend time near the nucleus (see Figure 3Q11-2) and therefore provide some additional shielding. This means the photon energy should be proportional to $(Z-C)^2$ where $C > 1$ due to the additional shielding effect. It seems that the additional shielding is not negligible.

The best fit line is $(\upsilon)^{1/2} = (5.210 \times 10^7\ s^{-1/2})Z - 9.626 \times 10^7\ s^{-1/2}$

$\therefore$ $\boldsymbol{B = 5.210 \times 10^7\ s^{-1/2}}$

$\therefore$ $\boldsymbol{C} = (9.626 \times 10^7\ s^{-1/2})/B = \mathbf{1.85}$

3.12 The He atom

Suppose that for the He atom, zero energy is taken to be the two electrons stationary at infinity (and infinitely apart) from the nucleus (He^{++}). Estimate the energy (in eV) of the electrons in the He atom by neglecting the electron-electron repulsion, that is, neglecting the potential energy due to the mutual Coulombic repulsion between the electrons. How does this compare with the experimental value of -79 eV? How strong is the electron-electron repulsion energy?

Solution

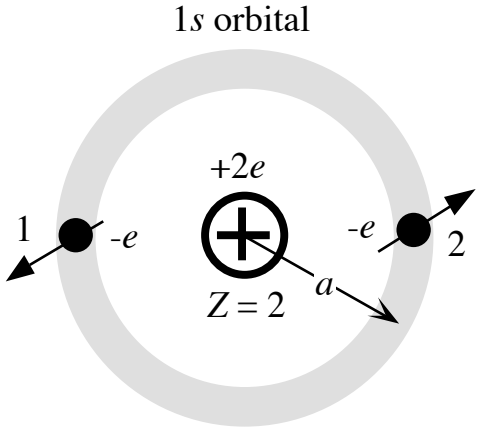


Figure 3Q12-1 Basic diagram of the He atom.

In He there are 2 electrons and the nucleus has a positive charge of $+2e$. Both electrons occupy the same orbital and hence have the same spatial distribution (same n, l, m_l). The two electrons, however, have opposite spins, i.e. different spin quantum numbers (Pauli exclusion principle). Due to their Coulombic repulsion, we envisage the two electrons as doing their best to avoid each other as shown. Let a be the distance to the maximum of the probability distribution. Then, $a = a_o / Z$ where a_o is the Bohr radius. In the hydrogen atom ($Z = 1$), a is simply the Bohr radius (a_o).

Energy in eV of an electron in some energy state of a hydrogenic atom is given by:

$$E_n = -\frac{13.6\ \text{eV}}{n^2} Z^2$$

where $n = 1$ corresponds to the ground energy state. This energy is with respect to the electron and nucleus infinitely separated. It also assumes that there is also no other electron so that there is no electron-electron repulsion energy.

Electron 1 sees a net charge of +2*e* in the nucleus so that $Z = 2$. Obviously the electron is in the ground state (1*s*), thus $n = 1$.

$$E_1 = -\frac{13.6 \text{ eV}}{1^2}(2)^2 = -54.4 \text{ eV}$$

Similarly in the absence of electron 1, electron 2 will have the energy:

$$E_2 = -\frac{13.6 \text{ eV}}{1^2}(2)^2 = -54.4 \text{ eV}$$

Thus neglecting electron-electron interactions, the total energy of the electrons in the He atom is:

$$E_{\text{total}} = E_1 + E_2 = \text{-108.8 eV}$$

As we would expect, this value is negative, corresponding to an attractive force (between a positive and negative charge). However, when this value is compared to the experimentally obtained value of -79 eV, it is apparent that ignoring the electron-electron repulsion gives a substantially different number. The experimental value is different because it includes the interaction energy between the two electrons. Using the value for E_{total} and the experimental, the interaction energy can be found:

$$\boldsymbol{E_{\text{interaction}}} = E_{\text{experiment}} - E_{\text{total}} = \mathbf{29.8\ eV}$$

As expected this is a positive quantity corresponding to repulsion (two negative charges).

As a footnote to demonstrate the power of intuition, let us estimate this repulsion energy differently ("back of an envelope" type calculation).

Both electrons occupy the same orbital and thus have the time averaged same spatial distribution except that due to electron-electron repulsion they will be avoiding each other. They will correlate their motions to avoid each other as follows: At one instant when one is at the extreme right the other will be at the extreme left and vice versa later on. The maximum probability radius for the 1*s* state for $Z = 2$ is given as:

$$a = a_o / Z = (5.29 \times 10^{-11} \text{ m}) / (2) = 2.645 \times 10^{-11} \text{ m}$$

And the potential energy is then:

$$PE = \frac{e^2}{4\pi\varepsilon_o(2a)}\left(\frac{1}{q}\right)$$

$$\therefore \quad PE = \frac{\left(1.602\times10^{-19}\ \text{C}\right)^2}{4\pi\left(8.854\times10^{-12}\ \text{F/m}\right)\left(2\times2.645\times10^{-11}\ \text{m}\right)}\left(\frac{1}{1.602\times10^{-19}\ \text{J/eV}}\right)$$

$$\therefore \quad \boldsymbol{PE = 27.2\ eV}$$

This is very close to what we found above (29.8 eV).

3.13 Hund's rule

For each of the following atoms and ions, sketch the electronic structure, using a box for an orbital wavefunction and an arrow (up or down) for an electron:

a. Manganese, [Ar]$3d^5 4s^2$

b. Cobalt, [Ar]$3d^7 4s^2$

c. Iron ion, Fe^{+2}, given that Fe is [Ar]$3d^6 4s^2$

d. Neodymium ion, Nd^{+3}, given that Nd is [Xe]$4f^4 6s^2$.

Solution

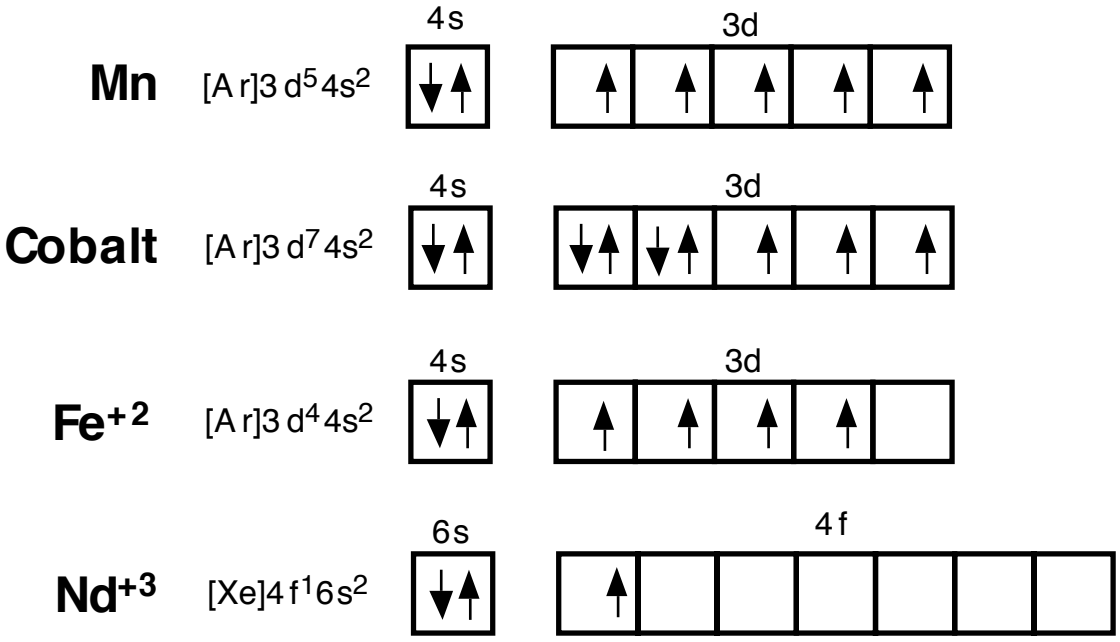


Figure 3Q13-1 Electronic structures of various atoms.

3.14 The HeNe Laser

A particular HeNe laser operating at 632.8 nm has a tube that is 40 cm long. The operating gas temperature is about 130 °C.

a. Calculate the Doppler-broadened linewidth $\Delta\lambda$ in the output spectrum.

b. What are the n values that satisfy the resonant cavity condition? How many modes are therefore allowed?

c. Calculate the frequency separation and the wavelength separation of the laser modes. How do these change as the tube warms up during operation? Taking the linear expansion coefficient to be 10^{-6} K^{-1}, estimate the change in the mode frequency separation.

Solution

Operating wavelength = $\lambda_o = 632.8 \times 10^{-9}$ m

Length of the tube = $L = 0.4$ m

Operating temperature = $T = 403$ K

Room temperature = 293 K (20 °C)

a $M_{at} = 20.2 \times 10^{-3}$ kg mol^{-1}. The mass of the Ne atom is M_{at}/N_A. The effective velocity v_x of the Ne atoms along the x-direction can be found from $(1/2)Mv_x^2 = (1/2)kT$,

$$v_x = \sqrt{\frac{kT}{M}} = \sqrt{\frac{kT}{M_{at}/N_A}} = \sqrt{\frac{(1.381\times10^{-23}\ \text{J K}^{-1})(403\ \text{K})}{(20.18\times10^{-3}\ \text{kg mol}^{-1})/(6.022\times10^{23}\ \text{mol}^{-1})}}$$

$\therefore$ $v_x = 407.5 \text{ m s}^{-1}$

The wavelength $\lambda_o = 632.8$ nm corresponds to a frequency υ_o:

$$\upsilon_o = \frac{c}{\lambda_o} = \frac{(3.0\times10^{8} \text{ m s}^{-1})}{(632.8\times10^{-9} \text{ m})} = 4.741 \times 10^{14} \text{ s}^{-1}$$

The width in the frequency spectrum, $\Delta\upsilon$, is given by (see Example 3.21, in the textbook):

$$\Delta\upsilon = \frac{2\upsilon_o v_x}{c} = \frac{2(4.741\times10^{14} \text{ s}^{-1})(407.5 \text{ m s}^{-1})}{(3.0\times10^{8} \text{ m s}^{-1})} = 1.288 \times 10^{9} \text{ s}^{-1}$$

From Example 3.21 (in the textbook),

$$\Delta\lambda = \frac{\lambda_o}{\upsilon_o}\Delta\upsilon = \frac{(632.8\times10^{-9} \text{ m})}{(4.741\times10^{14} \text{ s}^{-1})}(1.288\times10^{9} \text{ s}^{-1})$$

$\therefore$ $\Delta\lambda$ **= 1.72 × 10^{-12} m** or **0.00172 nm**

The velocity v_x we used is the root mean square (rms) velocity along one dimension (the laser tube axis) as given by the kinetic theory, *i.e.* $(1/2)Mv_x^2 = (1/2)kT$. Intuitively we would therefore expect the linewidth $\Delta\upsilon$ or $\Delta\lambda$ to correspond to the width between the rms points in the Gaussian output spectrum.

b Let n be the mode number corresponding to λ_o, then

$$n = \frac{L}{\frac{1}{2}\lambda_o} = \frac{(40\times10^{-2} \text{ m})}{\frac{1}{2}(632.8\times10^{-9} \text{ m})} = 1264222.5; \text{ a very large number.}$$

However n must be an integer so 1264222 and 126223 would be two possible modes that are within the spectrum.

The n_1 and n_2 values corresponding to the shortest λ_1 and longest λ_2 wavelengths are

$$n_1 = \frac{L}{\frac{1}{2}(\lambda_o - \frac{1}{2}\Delta\lambda)} = \frac{(40\times10^{-2} \text{ m})}{\frac{1}{2}(632.8\times10^{-9} \text{ m} \pm \frac{1}{2}1.7195\times10^{-12} \text{ m})} = 1264224.2$$

$$n_2 = \frac{L}{\frac{1}{2}(\lambda_o + \frac{1}{2}\Delta\lambda)} = \frac{(40\times10^{-2} \text{ m})}{\frac{1}{2}(632.8\times10^{-9} \text{ m} + \frac{1}{2}1.7195\times10^{-12} \text{ m})} = 1264220.8$$

Thus *n* values are: 1264221, 1264222, 1264223, 1264224. There are about four modes.

Alternatively we can find the number of modes as follows. Changes in n and λ are related by the above equation; $n = 2L/\lambda$. We can relate the change in n to the change in λ by differentiating $n = 2L/\lambda$. Then over the width $\Delta\lambda$ of the spectrum, n changes by Δn :

$$\frac{dn}{d\lambda} = -2L\frac{1}{\lambda^2}$$

substitute: $$|\Delta n| = 2\left(\frac{n\lambda}{2}\right)\frac{1}{\lambda^2}|\Delta\lambda|$$

$\therefore$ $$|\Delta n| = \frac{n}{\lambda}|\Delta\lambda| = \frac{(1.264\times10^{6})}{(632.8\times10^{-9} \text{ m})}(1.72\times10^{-12} \text{ m}) = 3.44$$

Since Δn must be an integer Δn =3.

The number of modes, however, is Δn + 1 or 4 (there are 4 numbers between 13 and 16 inclusively although 16 - 13 = 3).

Note: These 4 modes are within the central region, *i.e*. between the rms points, of the Gaussian output spectrum, that is within $\Delta\lambda$. If we wished to find *all* the modes, we can estimate this by considering the full "extent" of the spectrum. Suppose that the full extent of the spectrum is roughly $2\Delta\lambda$. Then there would be ~8 modes. The situation is more complicated by the fact that the exact number of modes depends on how the emission spectrum and the cavity modes coincide. Further, we need to find the net optical gain, that is the optical gain minus cavity losses not just the optical gain curve. All these fine points are far beyond the scope of this text (see S. O. Kasap, *Applied Optoelectronics: Principles and Practices*, Prentice Hall, 1999).

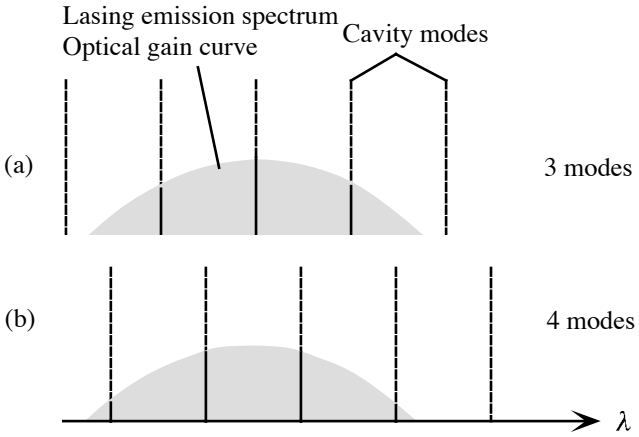


Figure 3Q14-1 Number of laser modes depends on how the cavity modes intersect the optical gain curve (lasing emission spectrum). In (a) there are 3 modes and in (b) there are 4 modes. The width of the emission spectrum here is roughly between rms (root mean square points of the intensity vs. wavelength behavior.

c The separation $\Delta\upsilon$ of two neighboring modes can be found by finding the change in υ corresponding to a change of 1 in n, that is $\Delta n = 1$. Substitute $\lambda = c/\upsilon$ in $n(\lambda/2) = L$ to find $\upsilon = cn/(2L)$. Differentiate υ with respect to n to find

$$\Delta\upsilon = \frac{c}{2L}\Delta n = \frac{(3.0\times10^{8}\ \text{m s}^{-1})}{2(0.40\ \text{m})}(1) = \mathbf{3.75\ \times\ 10^{8}\ s^{-1}}\ \text{or}\ \mathbf{375\ \ MHz}$$

When the tube warms up, L expands and $\Delta\upsilon$ decreases. The new length L' of the tube at 130 °C is given by

$$L' = L[1 + \alpha(T - T_o)] = (0.4\ \text{m})[1 + (10^{-6}\ ^\circ\text{C}^{-1})(130 - 20)\ ^\circ\text{C}] = 0.400044\ \text{m}$$

where α is the linear expansion coefficient, T_o is room temperature assumed to be 20 °C. The new frequency separation is

$$\Delta\upsilon' = \frac{c}{2L'}\Delta n = \frac{(3.0\times10^{8}\ \text{m s}^{-1})}{2(0.400044\ \text{m})}(1) = \mathbf{3.74959\ \times\ 10^{8}\ s^{-1}}$$

The change in the frequency separation is

$$\Delta\upsilon' - \Delta\upsilon = \mathbf{-4.10\ \times\ 10^{4}\ s^{-1}}\ \text{or}\ \mathbf{41.0\ \ kHz\ \ (decrease)}$$

"It was not unusual in theoretical physics to spend a lot of time in speculative notion that turns out to be wrong. I do it all the time. Having a lot of crazy ideas is the secret of my success. Some of them turned out to be right!"

Sheldon L. Glashow (Higgins Professor of Physics at Harvard University; Nobel Laureate, 1979)

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Chapter 4

4.1 Phase of an atomic orbital

a. What is the functional form of a 1*s* wavefunction, $\psi(r)$? Sketch schematically the atomic wavefunction $\psi_{1s}(r)$ as a function of distance from the nucleus.

b. What is the total wavefunction $\Psi_{1s}(r,t)$?

c. What is meant by two wavefunctions $\Psi_{1s}(A)$ and $\Psi_{1s}(B)$ that are out of phase?

d. Sketch schematically the two wavefunctions $\Psi_{1s}(A)$ and $\Psi_{1s}(B)$ at one instant.

Solution

a $\psi_{1s}(r)$ decays exponentially as $\exp(-r/a_o)$, where a_o is the Bohr radius (Table 3.2 in the textbook).

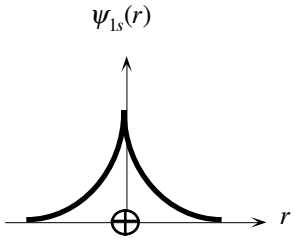


Figure 4Q1-1 Atomic wavefunction as a function of distance from the nucleus.

b $\Psi_{1s}(r,t) = \psi_{1s}(r) \times \exp(-jEt/\hbar) = \psi_{1s}(r) \times \exp(-j\omega t)$ where $\omega = E/\hbar$ is an angular frequency (Section 3.2.2 of the textbook). The total wavefunction is harmonic in time. Recall that $\exp(j\theta) = \cos\theta + j\sin\theta$ so that $\exp(j\omega t) = \cos\omega t + j\sin\omega t$.

c Two sine waves of the same frequency will have a certain phase difference which represents the time delay between the time oscillations of the two waves. Two waves will be in phase if their maxima coincide and out of phase if the maximum of one coincides with the minimum of the other. Two 1*s* wavefunctions, $\Psi_{1s}(A)$ and $\Psi_{1s}(B)$ will have the same frequency. Like two sine waves of the same frequency, the waves can be in phase or out of phase. When they are out of phase, if $\Psi_{1s}(A) = 1$ then $\Psi_{1s}(B) = -1$ and vice versa.

d

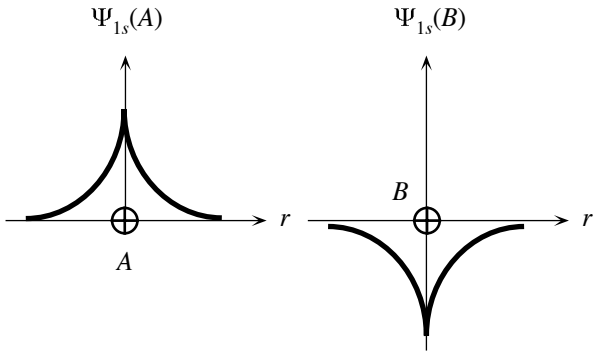


Figure 4Q1-2 Two *1s* wavefunctions which are out of phase (sketches at the same instant)

4.2 Molecular orbitals and atomic orbitals

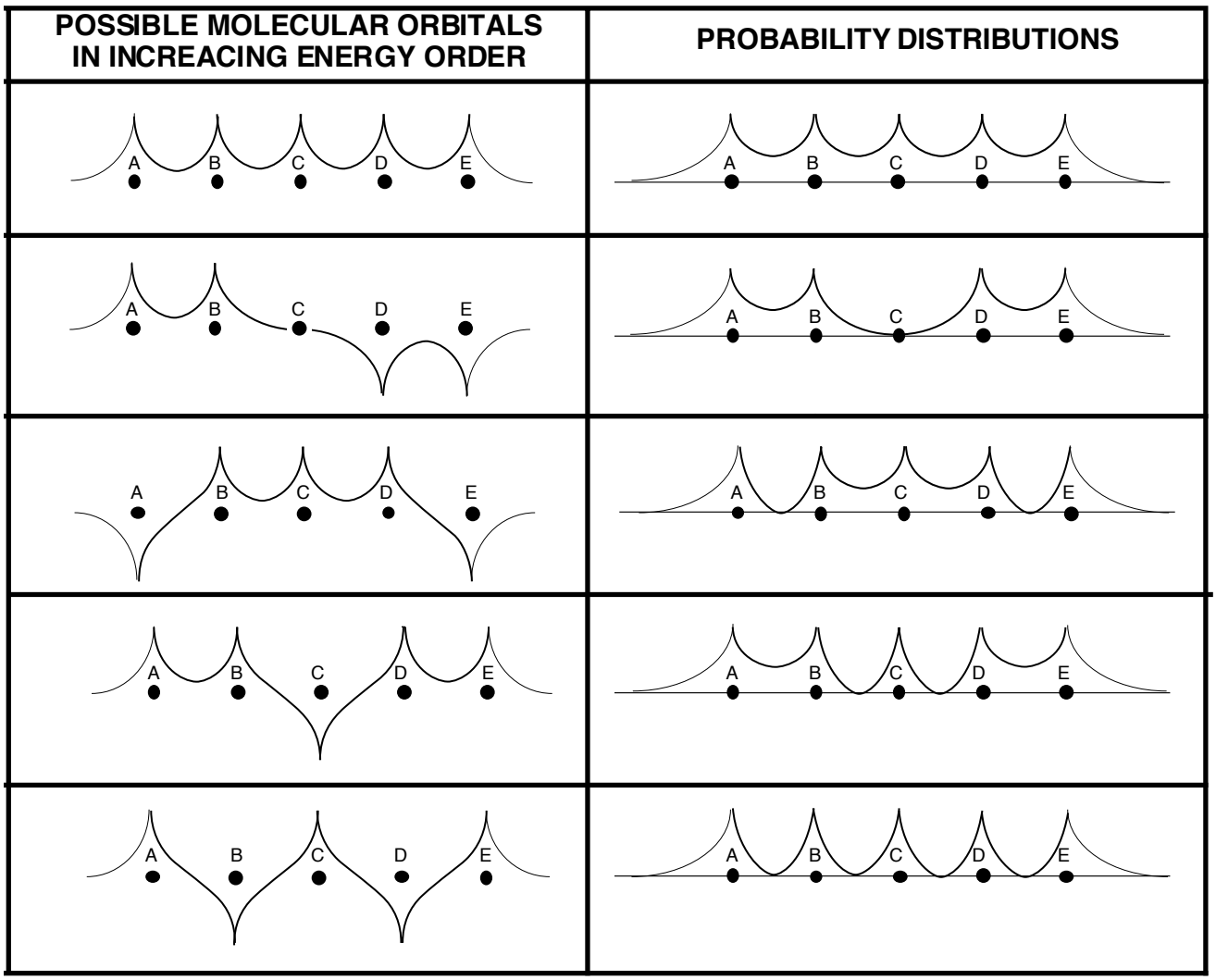
Consider a linear chain of five identical atoms representing a hypothetical molecule. Suppose that each atomic wavefunction is 1s wavefunction. This system of identical atoms has a center of symmetry *C* with respect to the center of molecule (third atom from the end) and all molecular wavefunctions must be either symmetric or antisymmetric about C.

a. Using LCAO principle, sketch the possible molecular orbitals.

b. Sketch the probability distribution $|\psi|^2$

c. If more nodes in the wavefunction lead to greater energies, order the energies of the molecular orbitals.

Solution



4.3 Diamond and Tin

Germanium, silicon, and diamond have the same crystal structure, that of diamond. Bonding in each case involves sp^3 hybridization. The bonding energy decreases as we go from C to Si to Ge, as noted in Table 4Q3-1.

Table 4Q3-1

Property	Diamond	Silicon	Germanium	Tin
Melting temperature, ^{0}C	3800	1417	937	232
Covalent radius, nm	0.077	0.117	0.122	0.146
Bond energy, eV	3.60	1.84	1.7	1.2
First ionization energy, eV	11.26	8.15	7.88	7.33
Bandgap, eV	?	1.12	0.67	?

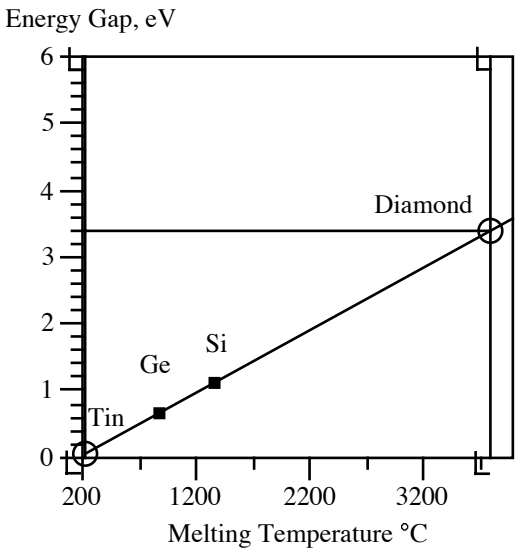
a. What would you expect for the band gap of diamond? How does it compare with the experimental value of 5.5 eV?

b. Tin has a tetragonal crystal structure, which makes it different than its group members, diamond, silicon, and germanium.

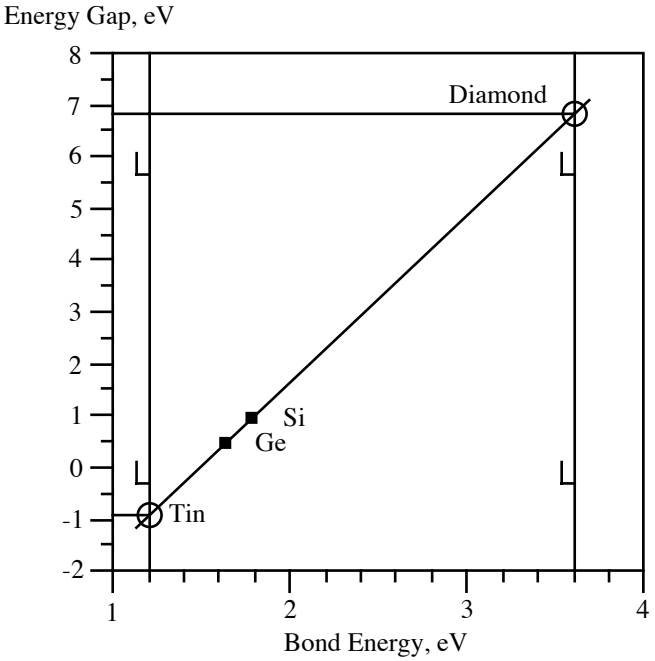
1. Is it a metal or a semiconductor?
2. What experiments do you think would expose its semiconductor properties?

Solution

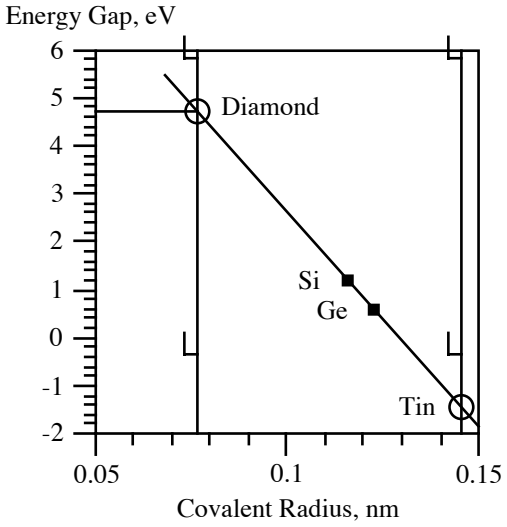
Given the properties in Table 4Q3-1, we have the following plots:



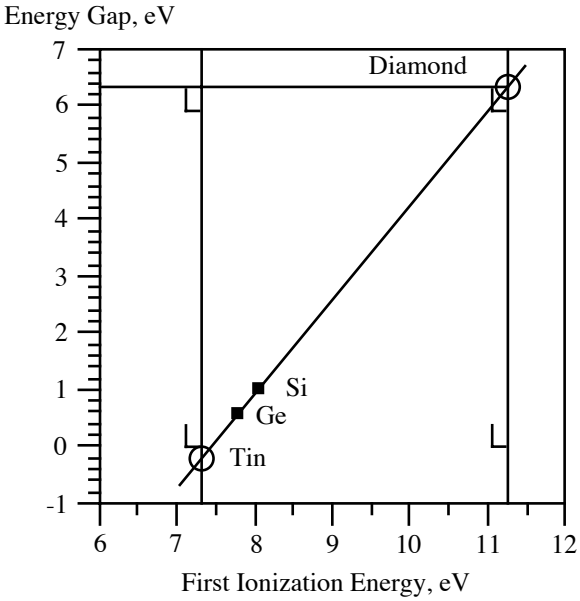
(a) Energy gap versus melting temperature. From the plot, it seems that diamond has an E_g = 3.4 eV and Sn has $E_g \approx 0$ or is a metal



(b) Energy gap versus bond energy. From the plot, it seems that diamond has an E_g = 6.8 eV and Sn has $E_g \approx -0.9$ or is a metal



(c) Energy gap versus covalent radius. From the plot, it seems that diamond has an E_g = 4.7 eV and Sn has $E_g \approx -1.5$ eV or is a metal.



(d) Energy gap versus first ionization energy. From the plot, it seems that diamond has an E_g = 6.3 eV and Sn has $E_g \approx -0.3$ eV or is a metal.

Figure 4Q3-1

Each is a plot of the band gap E_g (or energy gap) vs. some property. The straight line in each is drawn to pass through Si and Ge. Diamond and tin points are then located on this straight line at the intersections with the vertical lines representing the corresponding properties of diamond and tin.

a Diamond has an E_g greater than Si and Ge. Averaging the four E_g for diamond we find 5.3 eV which is close to the experimental value of 5.5 eV.

b (1) All the four properties indicate that tin has $E_g \leq 0$ or is a metal.

(2) Tin's semiconductivity can be tested by examining its electrical conductivity and optical absorption (see Chapter 5 in the textbook). For example, for metals the conductivity should NOT be thermally activated over a wide temperature range, whereas for semiconductors there will be an Arrhenius temperature dependence over at least some temperature range. Further, semiconductors have an absorption edge that corresponds to $h\upsilon > E_g$ (Chapter 5 in the textbook).

4.4 Compound III-V semiconductors

Indium as an element is a metal. It has a valency of III. Sb as an element is a metal and has a valency of V. InSb is a semiconductor, with each atom bonding to four neighbors, just like in silicon. Explain how this is possible and why InSb is a semiconductor and not a metal alloy. (Consider the electronic structure and sp^3 hybridization for each atom.)

Solution

The one *s* and three *p* orbitals hybridize to form 4 ψ_{hyb} orbitals. In Sb there are 5 valence electrons. One ψ_{hyb} has two paired electrons and 3 ψ_{hyb} have 1 electron each as shown in Figure 4Q4-1. In In there are 3 electrons so one ψ_{hyb} is empty. This empty ψ_{hyb} of In can overlap the full ψ_{hyb} of Sb. The overlapped orbital, the bonding orbital, then has two paired electrons. This is a bond between In and Sb even though the electrons come from Sb (this type of bonding is called *dative bonding*). It is a bond because the electrons in the overlapped orbital are shared by both Sb and In. The other 3 ψ_{hyb} of Sb can overlap 3 ψ_{hyb} of neighboring In to form "normal bonds". Repeating this in three dimensions generates the InSb crystal where each atom bonds to four neighboring atoms as shown. As all the bonding orbitals are full, the valence band formed from these orbitals is also full. The crystal structure is reminiscent of that of Si, as all the valence electrons are in bonds. Since it is similar to Si, InSb is a semiconductor.

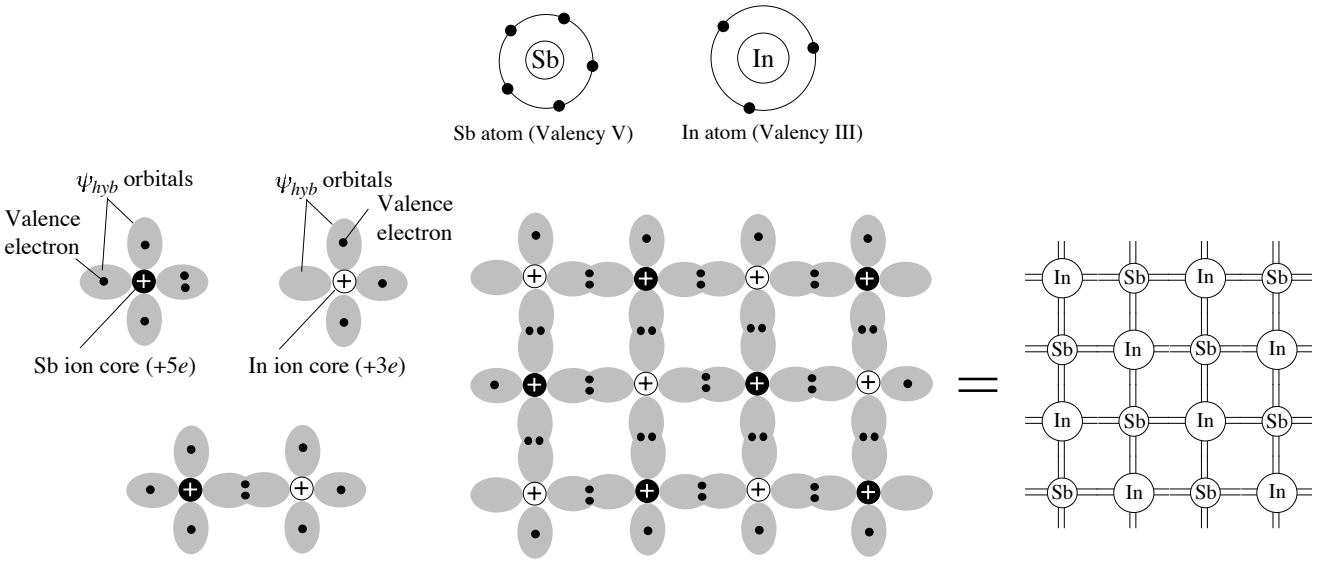


Figure 4Q4-1 Bonding structure of InSb.

4.5 Compound II-VI semiconductors

CdTe is a semiconductor, with each atom bonding to four neighbors, just like in silicon. In terms of covalent bonding and the positions of Cd and Te in the Periodic Table, explain how this is possible. Would you expect the bonding in CdTe to have more ionic character than that in III-V semiconductors?

Solution

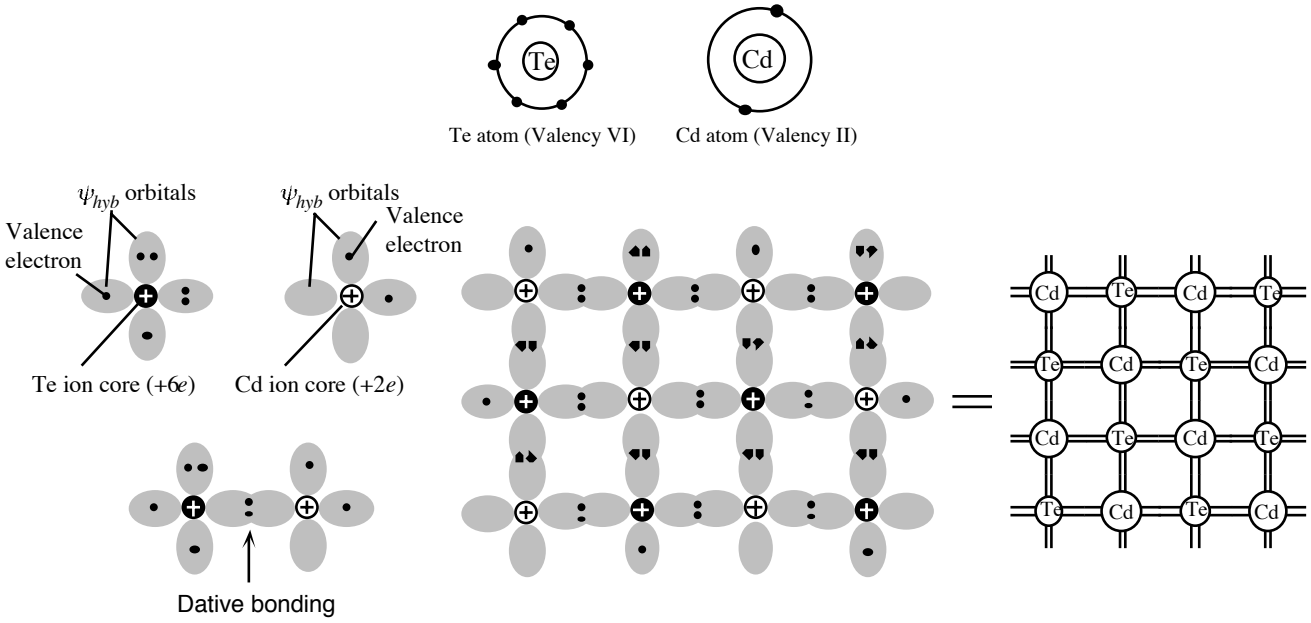


Figure 4Q5-1 Bonding structure of CdTe.

In CdTe one would expect a mixture of covalent and ionic bonding. Transferring 2 Cd electrons to Te would generate Te^{2-} and Cd^{2+} which then bond ionically. In covalent bonding we expect hybridization of *s* and *p* orbitals. The one *s* and three *p* orbitals hybridize to form 4 ψ_{hyb} orbitals. In Te there are 6 valence electrons. Two ψ_{hyb} have two paired electrons each and two ψ_{hyb} have 1 electron each as shown. In Cd there are 2 electrons so two ψ_{hyb} are empty. An empty ψ_{hyb} of Cd can overlap a full ψ_{hyb} of Te. The overlapped orbital, the bonding orbital, then has two paired electrons. This is a bond between Cd and Te even though the electrons come from Te (this type of bonding is called *dative bonding*). It is a bond because the electrons in the overlapped orbital are shared by both Te and Cd. The other full ψ_{hyb} of Te can similarly overlap another empty ψ_{hyb} of a different neighboring Cd to form another dative bond. The half occupied orbitals (two on Te and two on Cd) overlap and form "normal bonds". Repeating two dative bonds and two normal bonds in three dimensions generates the CdTe crystal where each atom bonds to four neighboring atoms as shown. Since all the bonding orbitals are full, the valence band formed from these orbitals is also full. Strictly the bonding is neither fully covalent nor fully ionic but a mixture.

*4.6 Density of states for a two-dimensional electron gas

Consider a two-dimensional electron gas in which the electrons are restricted to move freely within a square area a^2 in the *xy* plane. Following the procedure in Section 4.5 (in the textbook), show that the density of states $g(E)$ is constant (independent of energy).

Solution

For a two dimensional electron gas confined within a square region of sides a we have:

$$E = \frac{h^2}{8m_e a^2}\left(n_1^{\,2} + n_2^{\,2}\right)$$

Only positive n_1 and n_2 are allowed. Each n_1 and n_2 combination is an orbital state. Define a new variable n as:

$$n^2 = n_1^{\,2} + n_2^{\,2}$$

substitute:
$$E = \frac{h^2}{8m_e a^2} n^2$$

Let us consider how many states there are with energies less than E'. E' corresponds to $n \le n'$.

$$E' = \frac{h^2}{8m_e a^2} n'^2$$

$\therefore$
$$n' = \sqrt{\frac{8a^2 m_e E'}{h^2}}$$

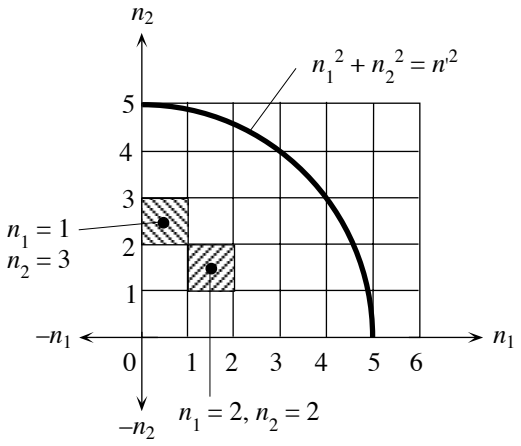


Figure 4Q6-1 Each state, or electron wavefunction in the crystal, can be represented by a box at n_1, n_2.

Consider Figure 4Q6-1. All states within the quarter arc defined by n' have $E < E'$. The area of this quarter arc is the total number of orbital states. The total number of states, S, including spin is twice as many,

$$S = 2\left(\frac{1}{4}\pi n'^2\right) = 2\left(\frac{1}{4}\pi\left[\sqrt{\frac{8a^2 m_e E'}{h^2}}\right]^2\right)$$

$\therefore$
$$S = \frac{4\pi a^2 m_e E'}{h^2}$$

The density of states g is defined as the number of states per unit area per unit energy. Therefore,

$$g = \frac{1}{a^2}\frac{dS}{dE'} = \frac{1}{a^2}\frac{4\pi a^2 m_e}{h^2} = \frac{4\pi m_e}{h^2}$$

Thus, for a two dimensional gas, the density of states is constant.

4.7 Fermi energy of Cu

The Fermi energy of electrons in copper at room temperature is 7.0 eV. The electron drift mobility in copper, from Hall effect measurements, is 33 $cm^2\ V^{-1}\ s^{-1}$.

a. What is the speed v_F of conduction electrons with energies around E_F in copper? By how many times is this larger than the average thermal speed $v_{thermal}$ of electrons, if they behaved like an ideal gas (Maxwell-Boltzmann statistics)? Why is v_F much larger than $v_{thermal}$?

b. What is the De Broglie wavelength of these electrons? Will the electrons get diffracted by the lattice planes in copper, given that interplanar separation in Cu = 2.09 Å? (Solution guide: Diffraction of waves occurs when $2d\sin\theta = \lambda$, which is the Bragg condition. Find the relationship between λ and d that results in $\sin\theta > 1$ and hence no diffraction.)

c. Calculate the mean free path of electrons at E_F and comment.

Solution

a The Fermi speed v_F is given by:

$$E_F = \frac{1}{2}m_e v_F^{\ 2}$$

$$\therefore \quad v_F = \sqrt{2\frac{E_F}{m_e}} = \sqrt{2\frac{(7\ \text{eV})\left(1.602\times10^{-19}\ \text{J/eV}\right)}{\left(9.109\times10^{-31}\ \text{kg}\right)}}$$

$$\therefore \quad \boldsymbol{v_F = 1.57 \times 10^6\ \text{m/s}}$$

Maxwell - Boltzmann statistics predicts an effective velocity (rms velocity) which is called "thermal velocity" given by (assume room temperature T = 20 °C = 293 K):

$$\frac{1}{2}m_e v_{\text{thermal}}^{\ 2} = \frac{3}{2}kT$$

$$\therefore \quad v_{\text{thermal}} = \sqrt{\frac{3Tk}{m_e}} = \sqrt{\frac{3(293\ \text{K})\left(1.381\times10^{-23}\ \text{J/K}\right)}{\left(9.109\times10^{-31}\ \text{kg}\right)}}$$

$$\therefore \quad \boldsymbol{v_{\text{thermal}} = 1.15 \times 10^5\ \text{m/s}}$$

Comparing the two values:

$$\textbf{Ratio} = v_F/v_{\text{thermal}} = \textbf{13.7}$$

v_F is about 14 times greater than $v_{thermal}$. This is because $v_{thermal}$ assumes that electrons do not interact and obey Maxwell-Boltzmann statistics ($E_{av} = 3/2kT$). However, in a metal there are many conduction electrons. They interact with the metal ions and obey the Pauli exclusion principle, i.e. Fermi-Dirac statistics. They extend to higher energies to avoid each other and thereby fulfill the Pauli exclusion principle.

b The De Broglie wavelength is $\lambda = h/p$ where $p = m_e v_F$ is the momentum of the electrons.

$$\lambda = \frac{h}{m_e v_F} = \frac{6.626\times10^{-34}\ \text{J s}}{(9.109\times10^{-31}\ \text{kg})(1.57\times10^{6}\ \text{m/s})}$$

$\therefore$ λ **= 4.63 × 10^{-10} m** or **4.63 Å**

The interplanar separation, d, is given as 2.09 Å. The diffraction condition is:

$$\lambda = 2d\sin\theta$$

$\therefore$ $$\sin\theta = \frac{1}{2}\frac{\lambda}{d} = \frac{1}{2}\frac{(4.63\ \text{Å})}{(2.09\ \text{Å})} = 1.11$$

Since this is greater than 1, and $\sin\theta$ cannot be greater than 1, the electrons will not be diffracted.

c The drift mobility is related to the mean scattering time τ by:

$$\mu = \frac{e\tau}{m_e}$$

$\therefore$ $$\tau = \frac{\mu m_e}{e} = \frac{(33\times10^{-4}\ \text{m}^2\ \text{V}^{-1}\ \text{s}^{-1})(9.109\times10^{-31}\ \text{kg})}{1.602\times10^{-19}\ \text{C}}$$

$\therefore$ $\tau = 1.876 \times 10^{-14}$ s

The mean free path, l_F, of electrons with speed, v_F is:

$$l_F = v_F\tau = (1.57 \times 10^{6}\ \text{m/s})(1.876 \times 10^{-14}\ \text{s}) = \mathbf{2.95 \times 10^{-8}\ m}\ \text{or}\ \mathbf{295\ Å}$$

The mean free path of those electrons with effective speeds v_e (close to mean speed) can be found as follows (E_F has little change with temperature, therefore $E_F \approx E_{FO}$):

$$\frac{1}{2}m_e v_e^{\,2} = \frac{3}{5}E_{FO} = \frac{3}{5}E_F$$

$\therefore$ $$v_e = \sqrt{\frac{6}{5}\frac{E_F}{m_e}} = \sqrt{\frac{6}{5}\frac{(7.0\ \text{eV})(1.602\times10^{-19}\ \text{J/eV})}{(9.109\times10^{-19}\ \text{kg})}} = 1.215\times10^{6}\ \text{m/s}$$

$\therefore$ $l_e = v_e\tau = (1.215 \times 10^{6}\ \text{m/s})(1.876 \times 10^{-14}\ \text{s}) = \mathbf{2.28 \times 10^{-8}\ m}\ \text{or}\ \mathbf{228\ Å}$

4.8 Fermi energy and electron concentration

Consider the metals in Table 4Q8-1 from groups I, II and III in the Periodic Table. Calculate the Fermi energies at absolute zero, and compare the values with the experimental values. What is your conclusion?

Table Q4.8-1

Metal	Group	M_{at} (g/mol)	Density (g cm^{-3})	E_F (eV) [Calculated]	E_F (eV) [Experiment]
Cu	I	63.55	8.96	-	6.5
Zn	II	65.38	7.14	-	11.0
Al	III	27	2.70	-	11.8

Solution

Since Cu is in group I, its valency (G) is also 1. The electron concentration n is then the atomic concentration multiplied by the group number, or:

$$n = G\frac{N_A D}{M_{at}} = (1)\frac{\left(6.022\times10^{23}\ \text{mol}^{-1}\right)\left(8.96\times10^{3}\ \text{kg/m}^3\right)}{63.55\times10^{-3}\ \text{kg/mol}} = 8.490\times10^{28}\ \text{m}^{-3}$$

Using Equation 4.22 in the text:

$$E_{FO} = \frac{h^2}{8m_e}\left(\frac{3n}{\pi}\right)^{\frac{2}{3}}\left(\frac{1}{q}\right)$$

$$\therefore \quad E_{FO} = \frac{\left(6.626\times10^{-34}\ \text{J s}\right)^2}{8\left(9.109\times10^{-31}\ \text{kg}\right)}\left(\frac{3\left(8.490\times10^{28}\ \text{m}^{-3}\right)}{\pi}\right)^{\frac{2}{3}}\left(\frac{1}{1.602\times10^{-19}\ \text{J/eV}}\right)$$

$$\therefore \quad \boldsymbol{E_{FO} = 7.04\ \text{eV}}$$

Comparing with the experimental value:

$$\left|\textbf{\% difference}\right| = \left|\frac{7.04\ \text{eV} - 6.5\ \text{eV}}{6.5\ \text{eV}}\right|\times100\% = \textbf{8.31\%}$$

E_{FO} can be calculated for Zn and Al in the same way (remember to take into account the different valencies). The values are summarized in the following table and it can be seen that calculated values are close to experimental values:

Table Q4.8-2 Summarized values for Fermi energy at absolute zero temperature.

Metal	n (m^{-3}) ($\times 10^{28}$)	E_{FO} (eV) (calculated)	E_{FO} (eV) (experimental)	% Difference
Cu	8.490	**7.04**	6.5	**8.31**
Zn	13.15	**9.43**	11.0	**14.3**
Al	18.07	**11.7**	11.8	**0.847**

4.9 Temperature dependence of the Fermi energy

a . Given that the Fermi energy for Cu is 7.0 eV at absolute zero, calculate the E_F at 300 K. What is the percentage change in E_F and what is your conclusion?

b. Given the Fermi energy for Cu at absolute zero, calculate the average energy and mean speed per conduction electron at absolute zero and 300 K, and comment.

Solution

a The Fermi energy in eV at 0 K is given as 7.0 eV. The temperature dependence of E_F is given by Equation 4.23 in the textbook. Remember that E_{FO} is given in eV.

$$E_F = E_{FO}\left(1 - \frac{\pi^2}{12}\left[\frac{kT}{E_{FO}}\right]^2\right)$$

$$\therefore \quad \boldsymbol{E_F} = (7.0\ \text{eV})\left(1 - \frac{\pi^2}{12}\left[\frac{(1.381\times10^{-23}\ \text{J/K})(300\ \text{K})}{(7.0\ \text{eV})(1.602\times10^{-19}\ \text{J/eV})}\right]^2\right) = \mathbf{6.999921\ \ eV}$$

$$\therefore \quad |\textbf{\% difference}| = \left|\frac{6.999921\ \text{eV} - 7.0\ \text{eV}}{7.0\ \text{eV}}\right| \times 100\% = \mathbf{0.00129\%}$$

This is a very small change. The Fermi energy appears to be almost unaffected by temperature.

b The average energy per electron at 0 K is:

$$\boldsymbol{E_{av}}\textbf{(0 K)} = 3/5\ (E_{FO}) = \mathbf{4.2\ eV}$$

The average energy at 300 K can be calculated from Equation 4.26 (in the textbook):

$$E_{av}(T) = \frac{3}{5}E_{FO}\left(1 + \frac{5\pi^2}{12}\left[\frac{kT}{E_{FO}q}\right]^2\right)$$

$$\therefore \quad E_{av}(300\ \text{K}) = \frac{3}{5}(7.0\ \text{eV})\left(1 + \frac{5\pi^2}{12}\left[\frac{(1.381\times10^{-23}\ \text{J/K})(300\ \text{K})}{(7.0\ \text{eV})(1.602\times10^{-19}\ \text{J/eV})}\right]^2\right)$$

$$\therefore \quad \boldsymbol{E_{av}}\textbf{(300 K) = 4.200236 eV}$$

This is a very small change.

Assume that the mean speed will be close to the effective speed v_e. Effective speed at absolute zero is denoted as v_{eo}, and is given by:

$$E_{av}(0\ \text{K}) \times q = \frac{1}{2}m_e {v_{eo}}^2$$

$$\therefore \quad \boldsymbol{v_{eo}} = \sqrt{2\frac{qE_{av}(0\ \text{K})}{m_e}} = \sqrt{2\frac{(1.602\times10^{-19}\ \text{J/eV})(4.2\ \text{eV})}{(9.109\times10^{-31}\ \text{kg})}} = \mathbf{1215446\ \ m/s}$$

At 300 K, the effective speed is v_e:

$$\boldsymbol{v_e} = \sqrt{2\frac{qE_{av}(300\ \text{K})}{m_e}} = \sqrt{2\frac{(1.602\times10^{-19}\ \text{J/eV})(4.200236\ \text{eV})}{(9.109\times10^{-31}\ \text{kg})}} = \mathbf{1215480\ \ m/s}$$

Comparing the values:

$$|\textbf{\% difference}| = \left|\frac{1215480\ \text{m/s} - 1215446\ \text{m/s}}{1215446\ \text{m/s}}\right| \times 100\% = \mathbf{0.002797\%}$$

The mean speed has increased by a negligible amount (0.003%) from 0 K to 300 K.

Note: For thermal conduction this tiny increase in the velocity is sufficient to transport energy from hot regions to cold regions. This very small increase in the velocity also allows the electrons to diffuse from hot to cold regions giving rise to the Seebeck effect.

4.10 X-ray emission spectrum from sodium

Structure of Na atom is [Ne]$3s^1$. Figure 4Q10-1a shows formation of the 3*s* and 3*p* energy bands in Na as a function of internuclear separation. Figure 4Q10-1b shows the x-ray emission spectrum (called the *L*-band) from crystalline sodium in the soft x-ray range as explained in Example 4.6 (in the textbook).

a. From Figure 4Q10-1, estimate the nearest neighbor equilibrium separation between Na atoms in the crystal if some electrons in the 3*s* band spill over into the states in the 3*p* band.

b. Explain the origin of X-ray emission band in Figure 4Q10-1b and the reason for calling it the *L*-band.

c. What is the Fermi energy of the electrons in Na from Figure 4Q10-1b ?

d. Taking the valency of Na to be I, what is the expected Fermi energy and how does it compare with that in (c)?

Solution

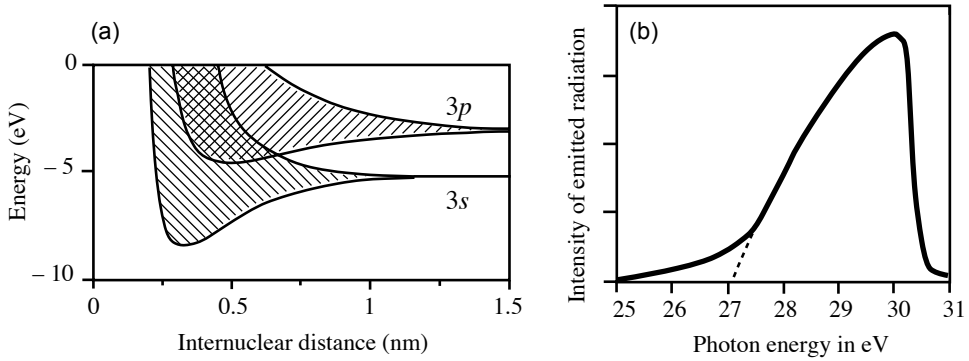


Figure 4Q10-1

(a) Energy band formation in sodium.

(b) *L*-emission band of X-rays from sodium.

SOURCE: (b) Data extracted from W. M. Cadt and D. H. Tomboulian, *Phys. Rev.*, **59,** 1941, p.381).

a As represented in Figure 4Q10-1a, the estimated nearest interatomic separation is about

0.36 ÷ 0.37 nm.

b When an electron, for some reason, is leaving the closed inner *L*-shell of an atom, an empty state is created there. An electron from the energy band of the metal drops into the *L*-shell to fill the vacancy and emits a soft X-ray photon in this process. The spectrum of this X-ray emission from metal involves a range of energies, corresponding to transitions from the bottom of the band and from the Fermi level to the *L*-shell. So all the X-ray photons emitted from the electrons during their transitions to the *L*-shell will have energies laying in that range (band) and because all of them are emitted due to a transition to the *L*-shell, this X-ray band is called *L*-band.

c As explained in **b**, the width of the X-ray band corresponds to the distance from the bottom of the energy band to the Fermi level. As shown in Figure 4Q10-1b, the position of the Fermi level with respect to the bottom of the energy band is approximately **3.2 eV**.

d Theoretically, the position of the Fermi level is given by Equation 4.23 (in the textbook). Since the temperature dependence of the Fermi level is really very weak, we can neglect it. Then $E_F(T) = E_{F0}$ and using Equation 4.22 (in the textbook), we receive

$$E_F = \frac{h^2}{8m_e}\left(\frac{3n}{\pi}\right)^{\frac{2}{3}}$$

The electron concentration in Na can be calculated, assuming that each Na atom donates exactly one electron to the crystal. Taking from Appendix B (in the textbook) the density of Na d (0.97 g cm^{-3}) and its atomic mass M_{at} (22.99 g mol^{-1}), we receive

$$n = \frac{d\,N_A}{M_{at}} = \frac{\left(0.97\ \text{g cm}^{-3}\right)\left(6.022\times10^{23}\ \text{mol}^{-1}\right)}{\left(22.99\ \text{g mol}^{-1}\right)} = \mathbf{2.54 \times 10^{22}\ cm^{-3} = 2.54 \times 10^{28}\ m^{-3}}$$

Thus for the Fermi level is

$$E_F = \frac{h^2}{8m_e}\left(\frac{3n}{\pi}\right)^{\frac{2}{3}} = \frac{\left(6.626\times10^{-34}\ \text{J s}\right)^2}{8\left(9.1\times10^{-31}\ \text{kg}\right)}\left[\frac{3\left(2.54\times10^{28}\ \text{m}^3\right)}{\pi}\right]^{\frac{2}{3}} = \mathbf{5.05 \times 10^{-19}\ J}$$

Or $\boldsymbol{E_F = 3.15\ eV}$,

which is very close to the value obtained from the X-ray spectrum.

4.11 Thermoelectric effects and E_F

Consider a thermocouple pair that consists of gold and aluminum. One junction is at 100 °C and the other is at 0 °C. A voltmeter (with a very large input resistance) is inserted into the aluminum wire. Use the properties of Au and Al in Table 4Q11-1 below to estimate the emf registered by the voltmeter and identify the positive end.

Table 4Q11-1

	Au	**Al**
Atomic Mass, M_{at}, g/mol	197	27.0
Density, g cm^{-3}	19.3	2.7
Conduction electrons per atom	1	3
x (Mott-Jones index)	-1.48	2.78

Solution

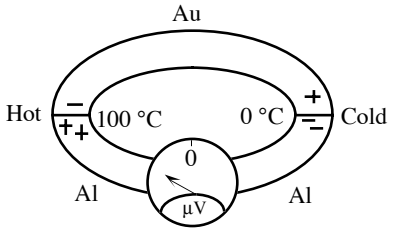


Figure 4Q11-1 The Al-Au thermocouple. The cold end is maintained at 0 °C which is the reference temperature. The other junction is used to sense the temperature. In this example it is heated to 100 °C.

We essentially have the arrangement shown above. For each metal there will be a voltage across it given by integrating the Seebeck coefficient. From the Mott-Jones equation:

$$\Delta V = \int_{T_0}^{T} S dT = \int_{T_0}^{T} -\frac{x\pi^2 k^2 T}{3eE_{FO}}\,dT = -\frac{x\pi^2 k^2}{6eE_{FO}}\left(T^2 - T_0^2\right)$$

The emf (V_{AB}) available is the difference in ΔV for the two metals labeled A (= Al) and B (= Au) so that

$$V_{AB} = \Delta V_A - \Delta V_B$$

where in this example, $T = 373$ K and $T_0 = 273$ K. We can calculate E_{FAO} and E_{FBO} for each metal as in Example 4.9 (in the textbook) by using

$$E_{FO} = \frac{h^2}{8m_e}\left(\frac{3n}{\pi}\right)^{2/3}$$

where n is the electron concentration:

$n =$ atomic concentration (n_{at}) $\times$ number of conduction electrons per atom

From the density d and atomic mass M_{at}, the atomic concentration of Al is:

$$n_{\text{Al}} = \frac{N_A d}{M_{at}} = \frac{\left(6.022\times10^{23}\ \text{mol}^{-1}\right)\left(2700\ \text{kg/m}^3\right)}{\left(0.027\ \text{kg/mol}\right)} = 6.022\times10^{28}\ \text{m}^{-3}$$

so that $n = 3n_{\text{Al}} = 1.807 \times 10^{29}\ \text{m}^{-3}$

which leads to: $$E_{FAO} = \frac{h^2}{8m_e}\left(\frac{3n}{\pi}\right)^{2/3} = \frac{\left(6.626\times10^{-34}\right)^2}{8\left(9.109\times10^{-31}\right)}\left(\frac{3\left(1.807\times10^{29}\right)}{\pi}\right)^{2/3}$$

$E_{FAO} = 1.867 \times 10^{-18}$ J or 11.66 eV

Similarly for Au, we find $E_{FBO} = 5.527$ eV.

Substituting x and E_F values for A (Al) and B (Au) we find,

$$\Delta V_A = -\frac{\pi^2 k^2 x_A}{6eE_{FAO}}\left(T^2 - T_0^2\right) = \text{-188.4 } \mu\text{V}$$

and $$\Delta V_B = -\frac{\pi^2 k^2 x_B}{6eE_{FBO}}\left(T^2 - T_0^2\right) = 211.3\ \mu\text{V}$$

so that the magnitude of the voltage difference is

$$|V_{AB}| = |\text{-188.3 } \mu\text{V} - 211.3\ \mu\text{V}| = \mathbf{399.7\ \mu V}$$

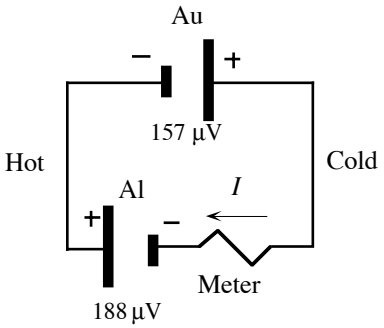


Figure 4Q11-2

To find which end is positive, we put in the resistance of the voltmeter and replace each metal by its emf and determine the direction of current flow as in the figure. For the particular circuit shown, the cold connected side of the voltmeter is positive.

4.12 The thermocouple equation

Although inputting the measured emf for V in the thermocouple equation $V = a\Delta T + b(\Delta T)^2$ leads to a quadratic equation, which in principle can be solved for ΔT, in general ΔT is related to the measured emf via

$$\Delta T = a_1 V + a_2 V^2 + a_3 V^3 + ...$$

with the coefficients a_1, a_2 etc., determined for each pair of TCs. By carrying out a Taylor's expansion of TC equation, find the first two coefficients a_1 and a_2. Using an emf table for the K-type thermocouple or Figure 4Q12-1, evaluate a_1 and a_2.

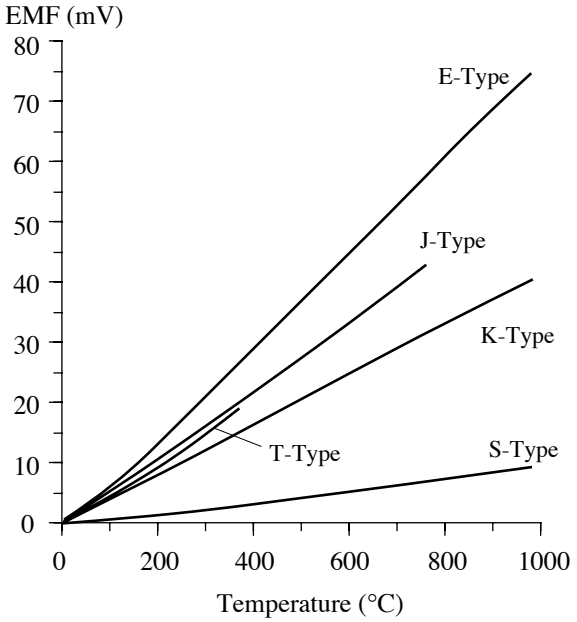


Figure 4Q12-1 Output emf versus temperature (°C) for various thermocouples between 0 to 1000 °C.

Solution

From Example 4.11 (in the textbook), the emf voltage (V) can be expressed:

$$V = \frac{\pi^2 k^2}{4e}\left(\frac{1}{E_{FAO}} - \frac{1}{E_{FBO}}\right)\left(T^2 - T_o^2\right)$$

Since we know that $\Delta T = T - T_o$, and therefore $T = \Delta T + T_o$, we can make the following substitution:

$$V = \frac{\pi^2 k^2}{4e}\left(\frac{1}{E_{FAO}} - \frac{1}{E_{FBO}}\right)\left(\left[\Delta T + T_o\right]^2 - T_o^2\right)$$

expanding,

$$V = \frac{\pi^2 k^2 T_o \Delta T}{2eE_{FAO}} - \frac{\pi^2 k^2 T_o \Delta T}{2eE_{FBO}} + \frac{\pi^2 k^2 (\Delta T)^2}{4eE_{FAO}} - \frac{\pi^2 k^2 (\Delta T)^2}{4eE_{FBO}}$$

factoring,

$$V = \frac{\pi^2 k^2 T_o}{2e}\left(\frac{1}{E_{FAO}} - \frac{1}{E_{FBO}}\right)\Delta T + \frac{\pi^2 k^2}{4e}\left(\frac{1}{E_{FAO}} - \frac{1}{E_{FBO}}\right)(\Delta T)^2$$

Upon inspection, it can be seen that this equation is in the form of the thermocouple equation, $V = a\Delta T + b\Delta T^2$, and therefore we know that the coefficients a and b are equal to:

$$a = \frac{\pi^2 k^2 T_o}{2e}\left(\frac{1}{E_{FAO}} - \frac{1}{E_{FBO}}\right) \qquad b = \frac{\pi^2 k^2}{4e}\left(\frac{1}{E_{FAO}} - \frac{1}{E_{FBO}}\right)$$

Continuing with the thermocouple equation, we can rearrange it as follows to obtain a quadratic equation:

$$-b(\Delta T)^2 - a\Delta T + V = 0$$

This is a quadratic equation in ΔT. The solution is

$$\Delta T = \frac{a - \sqrt{a^2 + 4bV}}{-2b} = -\frac{a}{2b} + \frac{a\sqrt{1 + 4bV/a^2}}{2b}$$

$\therefore$ $$\Delta T = -\frac{a}{2b} + \frac{a}{2b}\left(1 + 4bV/a^2\right)^{\frac{1}{2}}$$

$\therefore$ $$\Delta T = -\frac{a}{2b} + \frac{a}{2b}\sum_{n=0}^{\infty}\binom{-1/2}{n}\left(1 + \frac{4bV}{a^2}\right)^n$$

Taylor expansion: $$\Delta T = -\frac{a}{2b} + \frac{a}{2b}\left[1 + \frac{1}{2}\left(\frac{4bV}{a^2}\right) + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)}{2!}\left(\frac{4bV}{a^2}\right)^2 + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!}\left(\frac{4bV}{a^2}\right)^3 + \ldots\right]$$

$\therefore$ $$\Delta T = \frac{V}{a} - \frac{bV^2}{a^3} + 2\frac{b^2V^3}{a^5} + \ldots$$

The positive root is not used because $V = 0$ must give $\Delta T = 0$, and with the positive root the $a/2b$ terms will not cancel out. This equation is of the form given in the question,

$$\Delta T = a_1 V + a_2 V^2 + a_3 V^3 + \ldots$$

such that $$a_1 = \frac{1}{a} \qquad \text{and} \qquad a_2 = \frac{-b}{a^3}$$

$\therefore$ $$a_1 = \frac{1}{a} = \frac{1}{\frac{\pi^2 k^2 T_o}{2e}\left(\frac{1}{E_{FAO}} - \frac{1}{E_{FBO}}\right)} = \frac{2eE_{FAO}E_{FBO}}{\pi^2 k^2 T_o\left(E_{FBO} - E_{FAO}\right)}$$

and $$a_2 = -\frac{b}{a^3} = -\frac{\frac{\pi^2 k^2}{4e}\left(\frac{1}{E_{FAO}} - \frac{1}{E_{FBO}}\right)}{\left[\frac{\pi^2 k^2 T_o}{2e}\left(\frac{1}{E_{FAO}} - \frac{1}{E_{FBO}}\right)\right]^3} = -\frac{2e^2 {E_{FAO}}^2 {E_{FBO}}^2}{\pi^4 k^4 {T_o}^3\left(E_{FBO} - E_{FAO}\right)^2}$$

From Figure 4Q12-1, at $\Delta T = 200$ °C, $V = 8$ mV, and at $\Delta T = 800$ °C, $V = 33$ mV. Using these values and neglecting the effect of a_3, we have two simultaneous equations we can solve:

$$\Delta T = a_1 V + a_2 V^2$$

$\Delta T = 200$ °C: $$200\ ^\circ\text{C} = a_1(8\text{ mV}) + a_2(8\text{ mV})^2$$

$\therefore$ $$a_1 = (-(64\text{ mV}^2)a_2 + 200\ ^\circ\text{C}) / (8\text{ mV})$$

ΔT = 800 ˚C: $800\ ^\circ\text{C} = a_1(33\ \text{mV}) + a_2(33\ \text{mV})^2$

substitute for a_1: $800\ ^\circ\text{C} = [(-(64\ \text{mV}^2)a_2 + 200\ ^\circ\text{C}) / (8\ \text{mV})](33\ \text{mV}) + a_2(33\ \text{mV})^2$

isolate a_2: $\boldsymbol{a_2 = -0.03030\ ^\circ\text{C/mV}^2}$

$\therefore$ $200\ ^\circ\text{C} = a_1(8\ \text{mV}) + (-0.03030\ ^\circ\text{C/mV}^2)(8\ \text{mV})^2$

$\therefore$ $\boldsymbol{a_1 = 25.24\ ^\circ\text{C/mV}}$

We can check our calculation by calculating ΔT when V = 20 mV and comparing the value we obtain from examining Figure 4Q12-1.

$$\Delta T = (25.24\ ^\circ\text{C/mV})(20\ \text{mV}) + (-0.03030\ ^\circ\text{C/mV}^2)(20\ \text{mV})^2 = 492.7\ ^\circ\text{C}$$

This is close to that shown in Figure 4Q12-1. Obviously to get a more accurate determination, we need a_3 as well but then we must solve a cubic equation to find a_1, a_2 and a_3 or use numerical techniques.

4.13 Thermionic emission

A vacuum tube is required to have a cathode operating at 800 °C and providing an emission (saturation) current of 10 A. What should be the surface area of the cathode for the two materials in Table 4Q13-1? What should be the operating temperature for the Th on W cathode, if it is to have the same surface area as the oxide-coated cathode?

Table 4Q13-1

	B_e (A m^{-2} K^{-2})	Φ (eV)
Th on W	3×10^4	2.6
Oxide coating	100	1

Solution

Operating temperature T is given as 800 ˚C = 1073 K and emission current I is given as 10 A. The temperature and current of the tube are related to its area by Equation 4.44 (in the textbook):

$$J = \frac{I}{A} = B_e T^2 \exp\left[\frac{-\left(\Phi - \beta_s\sqrt{\mathcal{E}}\right)}{kT}\right]$$

$\therefore$

$$A = \frac{I}{B_e T^2 \exp\left[\dfrac{-\left(\Phi - \beta_s\sqrt{\mathcal{E}}\right)}{kT}\right]}$$

Assuming there is no assisting field emission, the area needed for Th on W is:

$$A = \frac{10\ \text{A}}{\left(3\times10^4\ \text{A m}^{-2}\ \text{K}^{-2}\right)(1073\ \text{K})^2 \exp\left[\dfrac{-(2.6\ \text{eV})\left(1.602\times10^{-19}\ \text{J/eV}\right)}{\left(1.381\times10^{-23}\ \text{J K}^{-1}\right)(1073\ \text{K})}\right]}$$

$\therefore$ $\boldsymbol{A = 467\ \text{m}^2}$ (large tube)

For the oxide coating:

$$A = \frac{10\ \text{A}}{(100\ \text{A m}^{-2}\ \text{K}^{-2})(1073\ \text{K})^2 \exp\left[\dfrac{-(1\ \text{eV})(1.602\times10^{-19}\ \text{J/eV})}{(1.381\times10^{-23}\ \text{J K}^{-1})(1073\ \text{K})}\right]}$$

$\therefore$ $\boldsymbol{A = 0.00431\ \text{m}^2}$ (small and practical tube)

To find the temperature that the Th on W cathode would have to work at to have the same surface area as the oxide coated cathode, the area of the oxide cathode can be used in the current equation and the temperature can be solved for. It is a more difficult equation, but can be solved through graphical methods.

$$I = AB_e T^2 \exp\left[\frac{-\Phi}{kT}\right]$$

The plot of thermionic emission current I versus temperature T is shown below, with A = 0.00431 m^2, $B_e = 3 \times 10^{-4}$ A m^{-2} K^{-2}, and Φ = (2.6 eV)(1.602 × 10^{-19} J/eV).

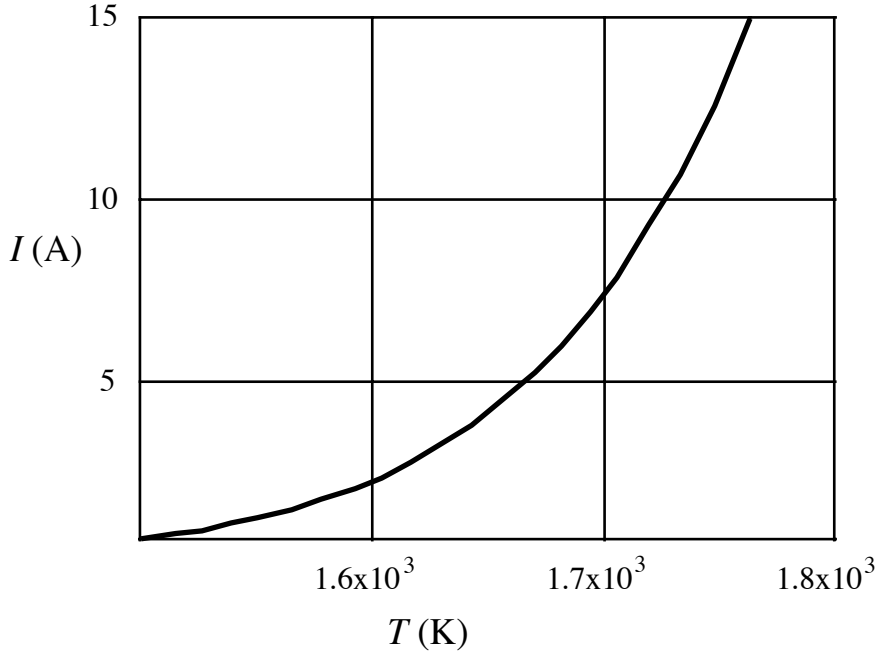


Figure 4Q13-1 Behavior of current versus temperature for the Th on W cathode.

From the graph, it appears that at 10 A of current the cathode will be operating at a temperature of $\boldsymbol{T = 1725\ \text{K}}$ or **1452 ˚C**.

4.14 Field-assisted emission in MOS device

Metal-oxide-semiconductor (MOS) transistors in microelectronics have metal gate on SiO_2 insulating layer on the surface of doped Si crystal. Consider this as a parallel plate capacitor. Suppose the gate is an Al electrode of area 50 μm × 50 μm and has a voltage of 10 V with respect of the Si crystal. Consider two thicknesses for the SiO_2, (a) 100 Å and (b) 40 Å, where (1 Å = 10^{-10} m). The workfunction of Al is 4.2 eV, but this refers to electron emission in vacuum, whereas in this case, the electron is emitted into the oxide. Given that the potential energy barrier Φ_B between Al and SiO_2 is about 3.1 eV, and the field emission current density in Equation 4.47 (in the textbook) in full is

$$J_{field-emission} = \left(\frac{e^3}{8\pi h\Phi_B}\right)E^2 \exp\left(-\frac{E_c}{E}\right); \qquad E_c = \frac{8\pi\left(2m_e\Phi_B^3\right)^{1/2}}{3eh}$$

calculate the field emission current for the two cases. For simplicity take m_e to be the electron mass in free space. What is your conclusion?

Solution

We can begin the calculation of field emission current finding the values of the field independent constants E_c, $B = \frac{e^3}{8\pi h\Phi_B}$ and the area A of the Al electrode.

Thus

$$E_c = \frac{8\pi\left(2m_e\Phi_B^3\right)^{1/2}}{3eh} = \frac{8\pi\left[2\left(9.1\times10^{-31}\ \text{kg}\right)\left(3.1\times1.602\times10^{-19}\ \text{J}\right)^3\right]^{\frac{1}{2}}}{3\left(1.602\times10^{-19}\ \text{C}\right)\left(6.626\times10^{-34}\ \text{J s}\right)} = \mathbf{3.726\times10^{10}\ V\ m^{-1}}$$

$$B = \frac{e^3}{8\pi h\Phi_B} = \frac{\left(1.602\times10^{-19}\ \text{C}\right)^3}{8\pi\left(6.626\times10^{-34}\ \text{J s}\right)\left(3.1\times1.602\times10^{-19}\ \text{J}\right)} = \mathbf{4.971\times10^{-7}\ A\ V^{-2}}$$

$$A = \left(50\times10^{-6}\ \text{m}\right)\times\left(50\times10^{-6}\ \text{m}\right) = \mathbf{2.5\times10^{-9}\ m^2}$$

When the thickness of SiO_2 layer d is 100 Å, the field in the MOS device is

$$E = \frac{10\ \text{V}}{100\times10^{-10}\ \text{m}} = \mathbf{1\times10^{9}\ V\ m^{-1}}$$

and the field emission current is

$$I = AJ_{field-emissiom} = ABE^2\exp\left(-\frac{E_c}{E}\right) =$$

$$= \left(2.5\times10^{-9}\ \text{m}^2\right)\left(4.971\times10^{-7}\ \text{A V}^{-2}\right)\left(1\times10^{9}\ \text{V m}^{-1}\right)^2\exp\left[-\frac{\left(3.726\times10^{10}\ \text{V m}^{-1}\right)}{\left(1\times10^{9}\ \text{V m}^{-1}\right)}\right]$$

$$= \mathbf{8.18\times10^{-14}\ A.}$$

In the second case the SiO_2 layer is 2.5 times thinner (40 Å) and the field in the device is 2.5 times stronger.

$$E = \frac{10\ \text{V}}{40\times10^{-10}\ \text{m}} = \mathbf{2.5\times10^{9}\ V\ m^{-1}}$$

The current in this case is

$$I = AJ_{field-emissiom} = ABE^2\exp\left(-\frac{E_c}{E}\right) =$$

$$= \left(2.5\times10^{-9}\ \text{m}^2\right)\left(4.971\times10^{-7}\ \text{A V}^{-2}\right)\left(2.5\times10^{9}\ \text{V m}^{-1}\right)^2\exp\left[-\frac{\left(3.726\times10^{10}\ \text{V m}^{-1}\right)}{\left(2.5\times10^{9}\ \text{V m}^{-1}\right)}\right]$$

$$= \mathbf{2.62\times10^{-3}\ A.}$$

So as predicted by equation 4.47 (in the textbook), the field-assisted emission current is a very strong function of the electric field.

4.15 Lattice waves and heat capacity

a. Consider an aluminum sample. The nearest separation $2R$ ($2 \times$ atomic radius) between the Al-Al atoms in the crystal is 0.286 nm. Taking a to be $2R$, and given the sound velocity in Al as 5100 m s^{-1}, calculate the force constant β in Equation 4.66 (in the textbook). Use the group velocity v_g from the actual dispersion relation, Equation 4.55 (in the textbook), to calculate the "sound velocity" at wavelengths of Λ = 1 mm, 1 μm and 1 nm. What is your conclusion?

b. Aluminum has a Debye temperature of 394 K. Calculate its specific heat capacity in the summer at 25 °C and in winter at -10 °C.

c. Calculate the specific heat capacity of a Ge crystal at 25 °C and compare it with the experimental value.

Solution

a The group velocity of lattice waves is given by Equation 4.55 (in the textbook). For sufficiently small K, or long wavelengths, such that $1/2Ka \rightarrow 0$, the expression for the group velocity can be simplified like in Equation 4.66 (in the textbook) to

$$v_g = a\sqrt{\frac{\beta}{M}}$$

From here we cam calculate the force constant β

$$\beta = M\left(\frac{v_g}{a}\right)^2$$

The mass of one Al atom is

$$M = \frac{M_{at}}{N_A}$$

and finally for the force constant we receive

$$\beta = \frac{M_{at}}{N_A}\left(\frac{v_g}{a}\right)^2 = \frac{\left(27\times10^{-3}\ \text{kg mol}^{-1}\right)}{\left(6.022\times10^{23}\ \text{mol}^{-1}\right)}\left(\frac{5100\ \text{m s}^{-1}}{0.286\times10^{-9}\ \text{m}}\right)^2 = \mathbf{14.26\ kg\ s^{-2}}$$

Now considering the dispersion relation $K = \frac{2\pi}{\Lambda}$ and Equation 4.55 (in the textbook) we receive

$$v_g(\Lambda) = a\left(\frac{\beta N_A}{M_{at}}\right)^{\frac{1}{2}}\cos\left(\frac{\pi a}{\Lambda}\right)$$

Performing the calculations for the given wavelengths, we receive the following results:

$$v_g\left(10^{-3}\ m\right) = \mathbf{5100\ m\ s^{-1}}$$

$$v_g\left(10^{-6}\ m\right) = \mathbf{5099.998\ m\ s^{-1}}$$

$$v_g\left(10^{-9}\ m\right) = \mathbf{3176.22\ m\ s^{-1}}$$

It is evident that for the first two wavelengths, $1/2Ka \rightarrow 0$ and we can use the approximation in Equation 4.66 (in the textbook). For the third wavelength, this is not true and we have to use the exact dispersion relation when calculating the group velocity.

b In summer, the temperature is given to be T = 25 °C = 298 K and T/T_D is 298/394 = 0.756. From Figure 4Q15-1, the molar heat capacity of Al during the summer is

$$C_m = 0.92 \times (3R) = \mathbf{22.95\ J\ K^{-1}\ mol^{-1}}$$

The corresponding specific heat capacity is

$$c_s = \frac{C_m}{M_{at}} = \frac{\left(22.95\ \text{J K}^{-1}\ \text{mol}^{-1}\right)}{\left(27\ \text{g mol}^{-1}\right)} = \mathbf{0.85\ J\ K^{-1}\ g^{-1}}$$

In similar way during the winter we have

$C_m = 0.898 \times (3R) = \mathbf{22.40\ J\ K^{-1}\ mol^{-1}}$ and $c_s = \frac{C_m}{M_{at}} = \frac{\left(22.40\ \text{J K}^{-1}\ \text{mol}^{-1}\right)}{\left(27\ \text{g mol}^{-1}\right)} = \mathbf{0.83\ J\ K^{-1}\ g^{-1}}$

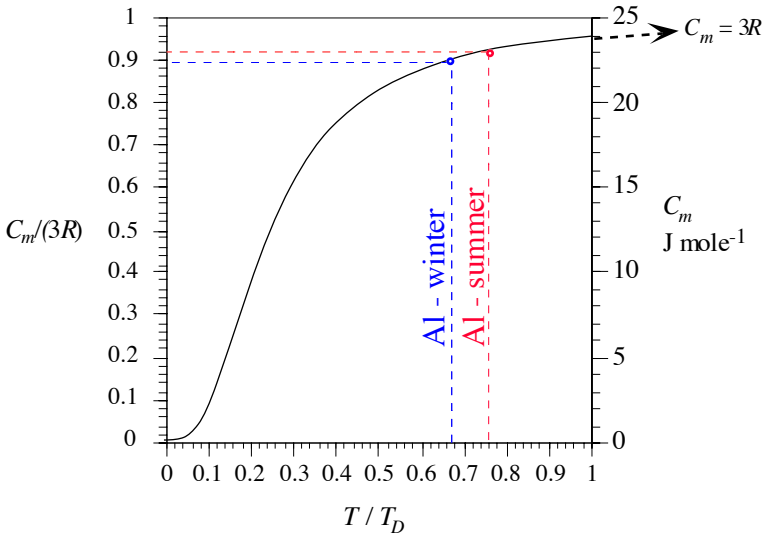


Figure 4Q15-1 Heat capacity of Al at -10 °C and at 25 °C.

c We can find the heat capacity of Ge in the way described in **b**. Alternatively, we can find C_m performing the integration in Equation 4.64 (in the textbook) numerically

$$C_m = 9R\left(\frac{T}{T_D}\right)^3 \int_0^{\frac{T_D}{T}} \frac{x^4 e^x}{\left(e^x - 1\right)^2} dx = 3R\left[3\left(\frac{298}{360}\right)^3 \int_0^{\frac{360}{298}} \frac{x^4 e^x}{\left(e^x - 1\right)^2} dx\right] = 3R(0.931) = \mathbf{23.22\ J\ K^{-1}\ mol^{-1}}$$

Thus the specific heat capacity is:

$$c_s = \frac{C_m}{M_{at}} = \frac{\left(23.22\ \text{J K}^{-1}\ \text{mol}^{-1}\right)}{\left(72.59 \times 10^{-3}\ \text{kg mol}^{-1}\right)} = \mathbf{319.9\ J\ K^{-1}\ kg^{-1}}$$

The difference between the calculated value and the experimental one is less than 1%.

4.16 Thermal conductivity

a. Given that silicon has Young's modulus of about 110 GPa density of 2.3 g cm^{-3}, calculate the mean free path of phonons in Si at room temperature.

b. Diamond has the same crystal structure as Si but has a very large thermal conductivity, about 1000 W m^{-1} K^{-1} at room temperature. Given that diamond has a specific heat capacity c_s of 0.50 J K^{-1} g^{-1}, Young's modulus of 830 GPa, and density ρ of 0.35 g cm^{-3}, calculate the mean free path of phonons in diamond.

c. GaAs has a thermal conductivity of 200 W m^{-1} K^{-1} at 100 K and 80 W m^{-1} K^{-1} at 200 K. Calculate its thermal conductivity at 25 °C and compare with the experimental value of 44 W m^{-1} K^{-1}. (Hint: Consider Figure 4Q16-1.)

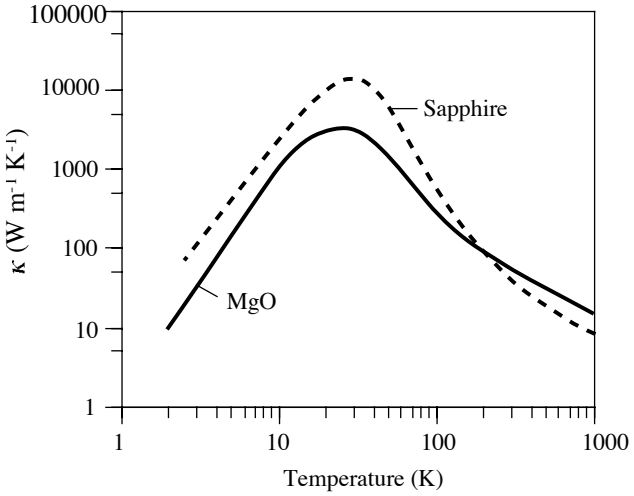


Figure 4Q16-1 Thermal conductivity of sapphire and MgO as a function of temperature.

Solution

a Assume room temperature of 25 °C (298 K). For this temperature from Table 4.5 (in the textbook), we can find the thermal conductivity κ (κ = 148 W m^{-1} K^{-1}) for silicon and its specific heat capacity C_s (C_s = 0.703 J K^{-1} g^{-1}). We can calculate the phonon mean free path at this temperature from Equation 4.68 (in the textbook):

$$\ell_{ph} = \frac{3\kappa}{C_V \upsilon_{ph}}$$

where C_V is the heat capacity per unit volume. C_V can be found from the specific heat capacity $C_V = \rho C_s$ and the phonon velocity can be obtained from Equation 4.67 (in the textbook) - $\upsilon_{ph} \approx \sqrt{\frac{Y}{\rho}}$.

Thus the phonon mean free path in Si at 25 °C is

$$\ell_{ph} = \frac{3\kappa}{\rho C_s}\sqrt{\frac{\rho}{Y}} = \frac{3\kappa}{C_s\sqrt{\rho Y}} =$$

$$= \frac{3\left(148\ \text{W m}^{-1}\ \text{K}^{-1}\right)}{\left(0.703\times10^{3}\ \text{J K}^{-1}\ \text{kg}^{-1}\right)\sqrt{\left(2.3\times10^{3}\ \text{kg m}^{-3}\right)\left(110\times10^{9}\ \text{Pa}\right)}} = \mathbf{3.971 \times 10^{-8}\ m}$$

b The mean free path of phonons in diamond is

$$\ell_{ph} = \frac{3\kappa}{C_s\sqrt{\rho Y}} = \frac{3\left(1000\ \text{W m}^{-1}\ \text{K}^{-1}\right)}{\left(0.5\times10^{3}\ \text{J K}^{-1}\ \text{kg}^{-1}\right)\sqrt{\left(3.5\times10^{3}\ \text{kg m}^{-3}\right)\left(830\times10^{9}\ \text{Pa}\right)}} = \mathbf{1.113 \times 10^{-7}\ m}$$

c The temperatures at which the thermal conductivity κ is given can be considered as relatively high. For this temperature range, we can assume that C_V is almost constant and since the phonon velocity is approximately independent from temperature according to Equation 4.68 (in the textbook) the thermal conductivity is proportional to the mean free path of phonons ℓ_{ph}. Since the phonon concentration increases with temperature, $n_{ph} \propto T$, the mean free path decreases as $\ell_{ph} \propto \frac{1}{T}$. Thus, κ decreases in the same manner with temperature as in Figure 4Q16-1.

We can assume that the temperature dependence of the thermal conductivity is given by:

$$\kappa(T) = \frac{A}{T} + B$$

Then we have two equations and two unknowns

$$200 = \frac{A}{100} + B$$

$$80 = \frac{B}{200} + B$$

and for the coefficients A and B we receive: A = $\mathbf{2.4 \times 10^{4}\ W\ m^{-1}}$ and B = $\mathbf{-40\ W\ m^{-1}\ K^{-1}}$

The thermal conductivity at 25 °C (298 K) is

$$\kappa = \frac{2.4\times10^{4}\ \text{W m}^{-1}}{298\ \text{K}} - 40\ \text{W m}^{-1}\ \text{K}^{-1} = \mathbf{40.5\ W\ m^{-1}\ K^{-1}}$$

which is close to the experimental value**.**

*4.17 Overlapping bands

Consider Cu and Ni with their density of states as schematically sketched in Figure 4Q17-1. Both have overlapping 3*d* and 4*s* bands, but the 3*d* band is very narrow compared to the 4*s* band. In the case of Cu the band is full, whereas in Ni, it is only partially filled.

a. In Cu, do the electrons in the 3*d* band contribute to electrical conduction? Explain.

b. In Ni, do electrons in both bands contribute to conduction? Explain.

c. Do electrons have the same effective mass in the two bands? Explain.

d. Can an electron in the 4*s* band with energy around E_F become scattered into the 3*d* band as a result of a scattering process? Consider both metals.

e. Scattering of electrons from the 4*s* band to the 3*d* band and vice versa can be viewed as an additional scattering process. How would you expect the resistivity of Ni to compare with that of Cu, even though Ni has 2 valence electrons and nearly the same density as Cu? In which case would you expect a stronger temperature dependence for the resistivity?

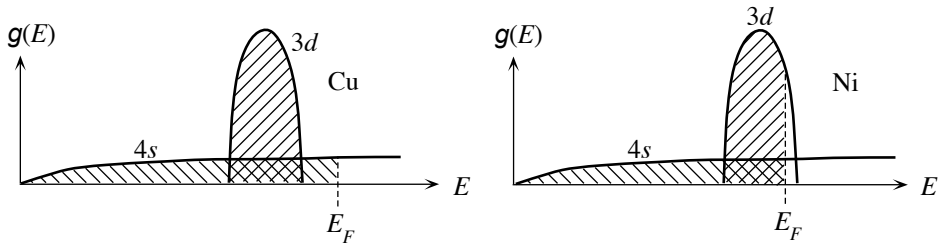


Figure 4Q17-1 Density of states and electron filling in Cu and Ni.

Solution

a In Cu the 3*d* band is full so the electrons in this band do not contribute to conduction.

b In Ni both the 3*d* and 4*s* bands are partially filled so electrons in both bands can gain energy from the field and move to higher energy levels. Thus both contribute to electrical conductivity.

c No, because the effective mass depends on how easily the electron can gain energy from the field and accelerate or move to higher energy levels. The energy distributions in the two bands are different. In the 4*s* band, the concentration of states is increasing with energy whereas in the 3*d* band, it is decreasing with energy. One would therefore expect different inertial resistances to acceleration, different effective mass and hence different drift mobility for electrons in these bands.

d Not in copper because the 3*d* band is full and cannot take electrons. In Ni the electrons can indeed be scattered from one band to the other, e.g. an electron in the 4*s* band can be scattered into the 3*d* band. Its mobility will then change. Electrons in the 3*d* band are very sluggish (low drift mobility) and contribute less to the conductivity.

e Ni should be more resistive because of the additional scattering mechanism from the 4*s* to the 3*d* band (Matthiessen's rule). This scattering is called *s-d* scattering. One may at first think that this *s-d* scattering de-emphasizes the importance of scattering from lattice vibrations and hence, overall, the resistivity should be less temperature dependent. In reality, electrons in Ni also get scattered by magnetic interactions with Ni ion magnetic moments (Nickel is ferromagnetic; Ch. 8 in the textbook) which has a stronger temperature dependence than $\rho \sim T$.

"After a year's research, one realises that it could have been done in a week."
Sir William Henry Bragg

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Chapter 5

5.1 Bandgap and photodetection

a. Determine the maximum value of the energy gap that a semiconductor, used as a photoconductor, can have if it is to be sensitive to yellow light (600 nm).

b. A photodetector whose area is 5×10^{-2} cm^2 is irradiated with yellow light whose intensity is 2 mW cm^{-2}. Assuming that each photon generates one electron-hole pair, calculate the number of pairs generated per second.

c. From the known energy gap of the semiconductor GaAs (E_g = 1.42 eV), calculate the primary wavelength of photons emitted from this crystal as a result of electron-hole recombination.

d. Is the above wavelength visible?

e. Will a silicon photodetector be sensitive to the radiation from a GaAs laser? Why?

Solution

a We are given the wavelength $\lambda = 600$ nm, therefore we need $E_{ph} = h\upsilon = E_g$ so that,

$$E_g = hc/\lambda = (6.626 \times 10^{-34}\ \text{J s})(3.0 \times 10^{8}\ \text{m s}^{-1}) / (600 \times 10^{-9}\ \text{m})$$

$\therefore$ $$\boldsymbol{E_g = 3.31 \times 10^{-19}\ \text{J}}\ \text{or}\ \boldsymbol{2.07\ \text{eV}}$$

b Area $A = 5 \times 10^{-2}$ cm^2 and light intensity $\mathcal{I}_{\text{light}} = 2 \times 10^{-3}$ W/cm^2. The received power is:

$$P = A\mathcal{I}_{\text{light}} = (5 \times 10^{-2}\ \text{cm}^2)(2 \times 10^{-3}\ \text{W/cm}^2) = 1.0 \times 10^{-4}\ \text{W}$$

N_{ph} = number of photons arriving per second = P/E_{ph}

$\therefore$ $$N_{ph} = (1.0 \times 10^{-4}\ \text{W}) / (3.31 \times 10^{-19}\ \text{J})$$

$\therefore$ $$N_{ph} = 3.02 \times 10^{14}\ \text{Photons s}^{-1}$$

Since the each photon contributes one electron-hole pair (EHP), the number of EHPs is then:

$$\boldsymbol{N_{EHP} = 3.02 \times 10^{14}\ \text{EHP s}^{-1}}$$

c For GaAs, E_g = 1.42 eV and the corresponding wavelength is

$$\lambda = hc/E_g = (6.626 \times 10^{-34}\ \text{J s})(3.0 \times 10^{8}\ \text{m s}^{-1}) / (1.42\ \text{eV} \times 1.602 \times 10^{-19}\ \text{J/eV})$$

$\therefore$ $$\boldsymbol{\lambda = 8.74 \times 10^{-7}\ \text{m}}\ \text{or}\ \boldsymbol{874\ \text{nm}}$$

The wavelength of emitted radiation due to electron-hole pair (EHP) recombination is therefore **874 nm.**

d It is not in the visible region (it is in the infrared).

e From Table 5.1 (in the textbook), for Si, E_g = 1.10 eV and the corresponding cut-off wavelength is,

$$\lambda_g = hc/E_g = (6.626 \times 10^{-34}\ \text{J s})(3.0 \times 10^{8}\ \text{m s}^{-1}) / (1.1\ \text{eV} \times 1.602 \times 10^{-19}\ \text{J/eV})$$

$\therefore$ $\lambda_g = \mathbf{1.13 \times 10^{-6}\ m}$ or **1130 nm**

Since the 874 nm wavelength of the GaAs laser is shorter than the cut-off wavelength of 1130 nm, the Si photodetector can detect the 874 nm radiation (Put differently, the photon energy corresponding to 874 nm, 1.42 eV, is larger than the E_g, 1.10 eV, of Si which means that the Si photodetector can indeed detect the 874 nm radiation).

5.2 Minimum conductivity

a. Consider the conductivity of a semiconductor, $\sigma = en\mu_e + ep\mu_h$. Will doping always increase the conductivity?

b. Show that the minimum conductivity for Si is obtained when it is *p*-type doped such that the hole concentration is

$$p_m = n_i\sqrt{\frac{\mu_e}{\mu_h}}$$

and the corresponding minimum conductivity (maximum resistivity) is

$$\sigma_{\min} = 2en_i\sqrt{\mu_e\mu_h}$$

c. Calculate p_m and $\sigma_{\min}$ for Si and compare with intrinsic values.

Solution

a Doping does not always increase the conductivity. Suppose that we have an intrinsic sample with $n = p$ but the hole drift mobility is smaller. If we dope the material very slightly with *p*-type then $p > n$. However, this would decrease the conductivity because it would create more holes with lower mobility at the expense of electrons with higher mobility. Obviously with further doping *p* increases sufficiently to result in the conductivity increasing with the extent of doping.

b To find the minimum conductivity, first consider the mass action law:

$$np = n_i^2$$

isolate *n*: $n = n_i^2/p$

Now substitute for *n* in the equation for conductivity:

$$\sigma = en\mu_e + ep\mu_h$$

$\therefore$ $$\sigma = \frac{en_i^2\mu_e}{p} + \mu_h ep$$

To find the value of *p* that gives minimum conductivity (p_m), differentiate the above equation with respect to *p* and set it equal to zero:

$$\frac{d\sigma}{dp} = -\frac{en_i^2\mu_e}{p^2} + \mu_h e$$

$\therefore$ $$-\frac{en_i^2\mu_e}{p_m^2} + \mu_h e = 0$$

Isolate p_m and simplify: $p_m = n_i\sqrt{\frac{\mu_e}{\mu_h}}$

Substituting this expression back into the equation for conductivity will give the minimum conductivity:

$$\sigma_{\min} = \frac{en_i^2\mu_e}{p_m} + \mu_h e p_m = \frac{en_i^2\mu_e}{n_i\sqrt{\mu_e/\mu_h}} + \mu_h e n_i \sqrt{\frac{\mu_e}{\mu_h}}$$

$$\therefore \quad \sigma_{\min} = en_i\mu_e\sqrt{\frac{\mu_h}{\mu_e}} + en_i\sqrt{\mu_e\mu_h} = en_i\sqrt{\mu_e\mu_h} + en_i\sqrt{\mu_e\mu_h}$$

$$\therefore \quad \sigma_{\min} = 2en_i\sqrt{\mu_e\mu_h}$$

c From Table 5.1, for Si: μ_e = 1350 cm^2 V^{-1} s^{-1}, μ_h = 450 cm^2 V^{-1} s^{-1} and $n_i = 1.45 \times 10^{10}$ cm^{-3}. Substituting into the equations for p_m and $\sigma_{\min}$:

$$p_m = n_i\sqrt{\frac{\mu_e}{\mu_h}} = \left(1.45\times10^{10}\ \text{cm}^{-3}\right)\sqrt{\frac{1350\ \text{cm}^2\ \text{V}^{-1}\ \text{s}^{-1}}{450\ \text{cm}^2\ \text{V}^{-1}\ \text{s}^{-1}}}$$

$$\therefore \quad \boldsymbol{p_m = 2.51 \times 10^{10}\ \text{cm}^{-3}}$$

$$\sigma_{\min} = 2en_i\sqrt{\mu_e\mu_h}$$

$$\therefore \quad \sigma_{\min} = 2\left(1.602\times10^{-19}\ \text{C}\right)\left(1.45\times10^{10}\ \text{cm}^{-3}\right)\sqrt{\left(1350\ \text{cm}^2\ \text{V}^{-1}\ \text{s}^{-1}\right)\left(450\ \text{cm}^2\ \text{V}^{-1}\ \text{s}^{-1}\right)}$$

$$\therefore \quad \boldsymbol{\sigma_{min} = 3.62 \times 10^{-6}\ \Omega^{-1}\ \text{cm}^{-1}}$$

The corresponding maximum resistivity is:

$$\rho_{\max} = 1 / \sigma_{\min} = 2.76 \times 10^5\ \Omega\ \text{cm}$$

The intrinsic value corresponding to p_m is simply n_i (= 1.45×10^{10} cm^{-3}). Comparing it to p_m:

$$\frac{p_m}{n_i} = \frac{2.51\times10^{10}\ \text{cm}^{-3}}{1.45\times10^{10}\ \text{cm}^{-3}} = 1.73$$

The intrinsic conductivity is:

$$\sigma_{\text{int}} = en_i(\mu_e + \mu_h)$$

$$\therefore \quad \sigma_{\text{int}} = (1.602 \times 10^{-19}\ \text{C})(1.45 \times 10^{10}\ \text{cm}^{-3})(1350\ \text{cm}^2\ \text{V}^{-1}\ \text{s}^{-1} + 450\ \text{cm}^2\ \text{V}^{-1}\ \text{s}^{-1})$$

$$\therefore \quad \sigma_{\text{int}} = 4.18 \times 10^{-6}\ \Omega^{-1}\ \text{cm}^{-1}$$

Comparing this value to the minimum conductivity:

$$\frac{\sigma_{\text{int}}}{\sigma_{\min}} = \frac{3.62\times10^{-6}\ \text{W}^{-1}\ \text{cm}^{-1}}{4.18\times10^{-6}\ \text{W}^{-1}\ \text{cm}^{-1}} = 0.866$$

Sufficient *p*-type doping that increases the hole concentration by 73% decreases the conductivity by 15% to its minimum value.

5.3 Compensation doping in Si

a. A Si wafer has been doped *n*-type with 10^{17} As atoms cm^{-3}.

1. Calculate the conductivity of the sample at 27 °C.

2. Where is the Fermi level in this sample at 27 °C with respect to the Fermi level (E_{Fi}) in intrinsic Si?

3. Calculate the conductivity of the sample at 127 °C.

b. The above *n*-type Si sample is further doped with 9×10^{16} boron atoms (*p*-type dopant) per centimeter cubed.

1. Calculate the conductivity of the sample at 27 °C.

2. Where is the Fermi level in this sample with respect to the Fermi level in the sample in (a) at 27 °C? Is this an *n*-type or *p*-type Si?

Solution

a Given temperature $T = 27$ °C = 300 K, concentration of donors $N_d = 10^{17}$ cm^{-3}, and drift mobility $\mu_e \approx 800$ cm^2 V^{-1} s^{-1} (from Figure 5Q3-1). At room temperature the electron concentration $n = N_d >> p$ (hole concentration).

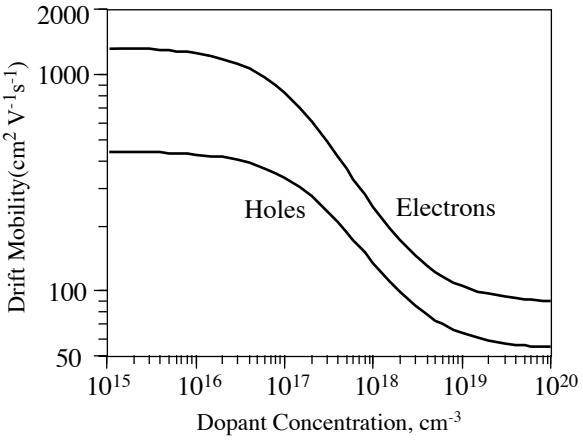


Figure 5Q3-1 The variation of the drift mobility with dopant concentration in Si for electrons and holes at 300 K.

(1) The conductivity of the sample is:

$$\sigma = eN_d\mu_e \approx (1.602 \times 10^{-19}\text{ C})(10^{17}\text{ cm}^{-3})(800\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}) = \mathbf{12.8\ \Omega^{-1}\ cm^{-1}}$$

(2) In intrinsic Si, $E_F = E_{Fi}$,

$$n_i = N_c\exp[-(E_c - E_{Fi})/kT] \quad (1)$$

In doped Si, $n = N_d$, $E_F = E_{Fn}$,

$$n = N_d = N_c\exp[-(E_c - E_{Fn})/kT] \quad (2)$$

Eqn. (2) divided by Eqn. (1) gives,

$$\frac{N_d}{n_i} = \exp\left(\frac{E_{Fn} - E_{Fi}}{kT}\right) \quad (3)$$

$\therefore$

$$\ln\left(\frac{N_d}{n_i}\right) = \frac{E_{Fn} - E_{Fi}}{kT}$$

$\therefore$ $$\Delta E_F = E_{Fn} - E_{Fi} = kT \ln(N_d/n_i) \quad (4)$$

Substituting we find ($n_i = 1.45 \times 10^{10}$ cm^{-3} from Table 5.1 in the textbook),

$$\Delta E_F = (8.617 \times 10^{-5} \text{ eV/K})(300 \text{ K})\ln[(10^{17} \text{ cm}^{-3})/ (1.45 \times 10^{10} \text{ cm}^{-3})]$$

$\therefore$ $$\boldsymbol{\Delta E_F = 0.407 \text{ eV above } E_{fi}}$$

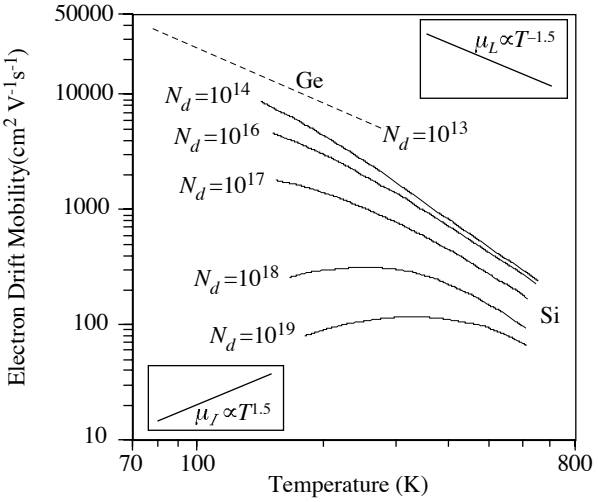


Figure 5Q3-2 Log-log plot for drift mobility versus temperature for *n*-type Ge and *n*-type Si samples. Various donor concentrations for Si are shown, N_d are in cm^{-3}. The upper right insert is the simple theory for lattice limited mobility whereas the lower left inset is the simple theory for impurity scattering limited mobility.

(3) At $T_i = 127$ °C = 400 K, $\mu_e \approx 450$ cm^2 V^{-1} s^{-1} (from Figure 5Q3-2). The semiconductor is still *n*-type (check that $N_d >> n_i$ at 400 K), then

$$\sigma = eN_d\mu_e \approx (1.602 \times 10^{-19} \text{ C})(10^{17} \text{ cm}^{-3})(450 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}) = \mathbf{7.21\ \Omega^{-1}\ cm^{-1}}$$

b The sample is further doped with $N_a = 9 \times 10^{16}$ cm^{-3} $= 0.9 \times 10^{17}$ cm^{-3} acceptors. Due to compensation, the net effect is still an *n*-type semiconductor but with an electron concentration given by,

$$n = N_d - N_a = 10^{17} \text{ cm}^{-3} - 0.9 \times 10^{17} \text{ cm}^{-3} = 1 \times 10^{16} \text{ cm}^{-3} \ (>> n_i)$$

We note that the electron scattering now occurs from $N_a + N_d$ (1.9×10^{17} cm^{-3}) number of ionized centers so that $\mu_e \approx 700$ cm^2 V^{-1} s^{-1} (Figure 5Q3-1).

(1) $$\sigma = eN_d\mu_e \approx (1.602 \times 10^{-19} \text{ C})(10^{16} \text{ cm}^{-3})(700 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}) = \mathbf{1.12\ \Omega^{-1}\ cm^{-1}}$$

(2) Using Eqn. (3) with $n = N_d - N_a$ we have

$$\frac{N_d - N_a}{n_i} = \exp\left(\frac{E'_{Fn} - E_{Fi}}{kT}\right)$$

so that $$\Delta E'_\Phi = E_{Fn}' - E_{Fi} = (0.02586 \text{ eV})\ln[(10^{16} \text{ cm}^{-3}) / (1.45 \times 10^{10} \text{ cm}^{-3})]$$

$\therefore$ $$\boldsymbol{\Delta E'_\Phi = 0.348 \text{ eV above } E_{Fi}}$$

The Fermi level from (a) and (b) has shifted "down" by an amount 0.059 eV. Since the energy is still above the Fermi level, this an *n*-type Si.

5.4 Temperature dependence of conductivity

An *n*-type Si sample has been doped with 10^{15} phosphorus atoms cm^{-3}. The donor energy level for P in Si is 0.045 eV below the conduction band edge energy.

a. Calculate the room temperature conductivity of the sample.

b. Estimate the temperature above which the sample behaves as if intrinsic.

c. Estimate to within 20% the lowest temperature above which all the donors are ionized.

d. Sketch schematically the dependence of the electron concentration in the conduction band on the temperature as log(*n*) versus 1/*T*, and mark the various important regions and critical temperatures. For each region draw an energy band diagram that clearly shows from where the electrons are excited into the conduction band.

e. Sketch schematically the dependence of the conductivity on the temperature as log(σ) versus 1/*T* and mark the various critical temperatures and other relevant information.

Solution

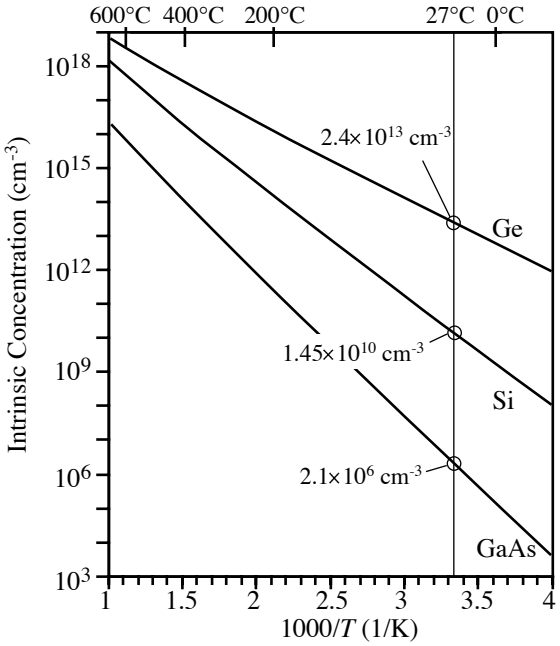


Figure 5Q4-1 The temperature dependence of the intrinsic concentration.

a The conductivity at room temperature $T = 300$ K is ($\mu_e = 1350 \times 10^{-4}$ m^2 V^{-1} s^{-1} can be found in Table 5.1 in the textbook):

$$\sigma = eN_d\mu_e$$

$$\therefore \quad \sigma = (1.602 \times 10^{-19}\ C)(1 \times 10^{21}\ m^{-3})(1350 \times 10^{-4}\ m^2\ V^{-1}\ s^{-1}) = \mathbf{21.6\ \Omega^{-1}\ m^{-1}}$$

b At $T = T_i$, the intrinsic concentration $n_i = N_d = 1 \times 10^{15}$ cm^{-3}. From Figure 5Q4-1, the graph of $n_i(T)$ vs. 1/*T*, we have:

$$1000 / T_i = 1.9\ K^{-1}$$

$\therefore$ $\quad T_i = 1000 / (1.9\ \text{K}^{-1}) =$ **526 K** or **253 ˚C**

c The ionization region ends at $T = T_s$ when all donors have been ionized, i.e. when $n = N_d$. From Example 5.7, at $T = T_s$:

$$n = N_d = \left(\frac{1}{2} N_c N_d\right)^{\frac{1}{2}} \exp\left(\frac{-\Delta E}{2kT_s}\right)$$

$\therefore$
$$T_s = \frac{-\Delta E}{2k \ln\left(\frac{N_d}{\sqrt{\frac{1}{2} N_c N_d}}\right)} = \frac{-\Delta E}{2k \ln\left(\sqrt{\frac{2N_d}{N_c}}\right)}$$

$\therefore$
$$T_s = \frac{\Delta E}{k \ln\left(\frac{N_c}{2N_d}\right)}$$

Take $N_c = 2.8 \times 10^{19}$ cm^{-3} at 300 K from Table 5.1 (in the textbook), and the difference between the donor energy level and the conduction band energy is $\Delta E = 0.045$ eV. Therefore our first approximation to T_s is:

$$T_s = \frac{\Delta E}{k \ln\left(\frac{N_c}{2N_d}\right)} = \frac{(0.045\ \text{eV})(1.602 \times 10^{-19}\ \text{J/eV})}{(1.381 \times 10^{-23}\ \text{J/K}) \ln\left(\frac{(2.8 \times 10^{19}\ \text{cm}^{-3})}{2(10^{15}\ \text{cm}^{-3})}\right)} = 54.68\ \text{K}$$

Find the new N_c at this temperature, N_c':

$$N_c' = N_c \left(\frac{T_s}{300}\right)^{\frac{3}{2}} = (2.8 \times 10^{19}\ \text{cm}^{-3}) \left(\frac{54.68\ \text{K}}{300\ \text{K}}\right)^{\frac{3}{2}} = 2.179 \times 10^{18}\ \text{cm}^{-3}$$

Find a better approximation for T_s by using this new N_c':

$$T_s' = \frac{\Delta E}{k \ln\left(\frac{N_c'}{2N_d}\right)} = \frac{(0.045\ \text{eV})(1.602 \times 10^{-19}\ \text{J/eV})}{(1.381 \times 10^{-23}\ \text{J/K}) \ln\left(\frac{(2.179 \times 10^{18}\ \text{cm}^{-3})}{2(10^{15}\ \text{cm}^{-3})}\right)} = 74.64\ \text{K}$$

$\therefore$
$$N_c'' = N_c \left(\frac{T_s'}{300}\right)^{\frac{3}{2}} = (2.8 \times 10^{19}\ \text{cm}^{-3}) \left(\frac{74.64\ \text{K}}{300\ \text{K}}\right)^{\frac{3}{2}} = 3.475 \times 10^{18}\ \text{cm}^{-3}$$

A better approximation to T_s is:

$$T_s'' = \frac{\Delta E}{k \ln\left(\frac{N_c''}{2N_d}\right)} = \frac{(0.045\ \text{eV})(1.602 \times 10^{-19}\ \text{J/eV})}{(1.381 \times 10^{-23}\ \text{J/K}) \ln\left(\frac{(3.475 \times 10^{18}\ \text{cm}^{-3})}{2(10^{15}\ \text{cm}^{-3})}\right)} = 69.97\ \text{K}$$

$\therefore$
$$N_c''' = N_c \left(\frac{T_s''}{300}\right)^{\frac{3}{2}} = (2.8 \times 10^{19}\ \text{cm}^{-3}) \left(\frac{69.97\ \text{K}}{300\ \text{K}}\right)^{\frac{3}{2}} = 3.154 \times 10^{18}\ \text{cm}^{-3}$$

$$\therefore \qquad T_s''' = \frac{\Delta E}{k \ln\left(\dfrac{N_c'''}{2N_d}\right)} = \frac{(0.045 \text{ eV})(1.602\times10^{-19} \text{ J/eV})}{(1.381\times10^{-23} \text{ J/K})\ln\left(\dfrac{(3.154\times10^{18} \text{ cm}^{-3})}{2(10^{15} \text{ cm}^{-3})}\right)} = 70.89 \text{ K}$$

We can see that the change in T_s is very small, and for all practical purposes we can consider the calculation as converged. Therefore $\boldsymbol{T_s}$ **= 70.9 K = -202.1 ˚C**.

d and **e** See Figures 5Q4-2 and 5Q4-3.

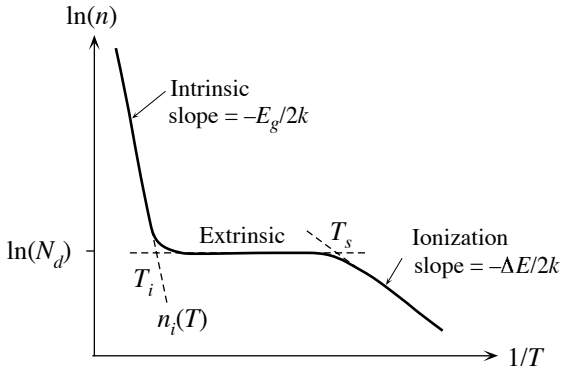


Figure 5Q4-2 The temperature dependence of the electron concentration in an *n*-type semiconductor.

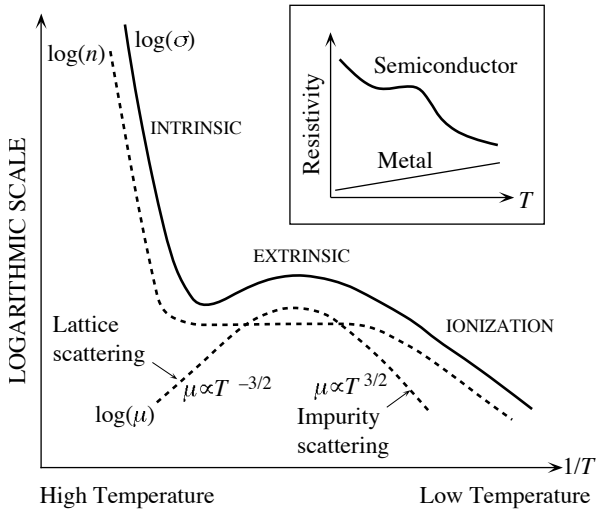


Figure 5Q4-3 Schematic illustration of the temperature dependence of electrical conductivity for a doped (*n*-type) semiconductor.

5.5 GaAs

Ga has a valency of III and As has V. When Ga and As atoms are brought together to form the GaAs crystal, as depicted in Figure 5Q5-1, the 3 valence electrons in each Ga and the 5 valence electrons in each As are all shared to form four covalent bonds per atom. In the GaAs crystal with some 10^{23} or so equal numbers of Ga and As atoms, we have an average of four valence electrons per atom, whether Ga or As, so we would expect the bonding to be similar to that in the Si crystal: four bonds per atom. The crystal structure, however, is not that of diamond but rather that of zinc blende (Chapter 1 of the textbook).

a. What is the average number of valence electrons per atom for a pair of Ga and As atoms and in the GaAs crystal?

b. What will happen if Se or Te, from Group VI, are substituted for an As atom in the GaAs crystal?

c. What will happen if Zn or Cd, from Group II, are substituted for a Ga atom in the GaAs crystal?

d. What will happen if Si, from Group IV, is substituted for an As atom in the GaAs crystal?

e. What will happen if Si, from Group IV, is substituted for a Ga atom in the GaAs crystal? What do you think **amphoteric dopant** means?

f. Based on the above discussion ,what do you think the crystal structures of the III-V compound semiconductors AlAs, GaP, InAs, InP, and InSb will be?

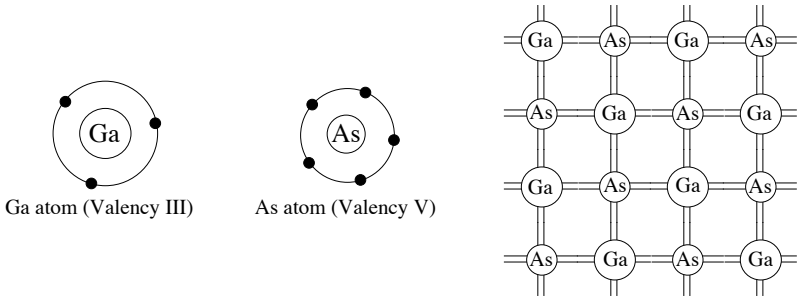
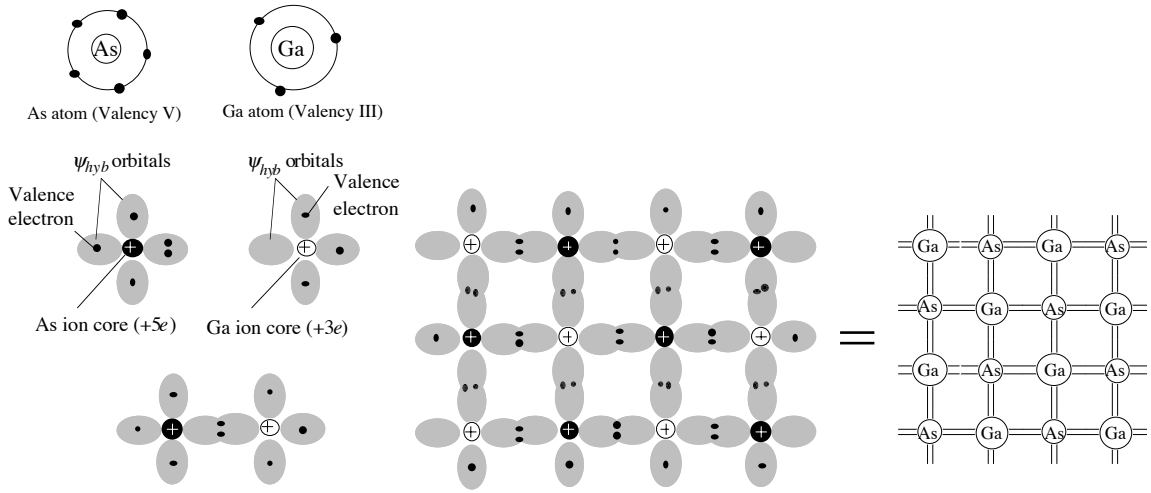


Figure 5Q5-1 The GaAs crystal structure in two dimensions. Average number of valence electrons per atom is four. Each Ga atom covalently bonds with four neighboring As atoms and vice versa.

Solution



Explanation of bonding in GaAs: The one *s* and three *p* orbitals hybridize to form 4 ψ_{hyb} orbitals. In As there are 5 valence electrons. One ψ_{hyb} has two paired electrons and 3 ψ_{hyb} have 1 electron each as shown. In Ga there are 3 electrons so one ψ_{hyb} is empty. This empty ψ_{hyb} of Ga can overlap the full ψ_{hyb} of As. The overlapped orbital, the bonding orbital, then has two paired electrons. This is a bond between Ga and As even though the electrons come from As (this type of bonding is called *dative bonding*). It is a bond because the electrons in the overlapped orbital are shared by both As and Ga. The other 3 ψ_{hyb} of As can overlap 3 ψ_{hyb} of neighboring Ga to form "normal bonds". Repeating this in three dimensions generates the GaAs crystal where each atom bonds to four neighboring atoms as shown. Because all the bonding orbitals are full, the valence band formed from these orbitals is also full. The crystal structure is reminiscent of that of Si. GaAs is a semiconductor.

a The average number of valence electrons is 4 electrons per atom.

b Se or Te replacing As will have one additional electron that cannot be involved in any of the four bonds. Hence Se and Te will act as a donor.

c Zn or Cd replacing Ga will have one less electron than the substituted Ga atom. This creates a hole in a bond. Zn and Cd will act as acceptors.

d The Si atom has 1 less electron than the As atom and when it substitutes for an As atom in GaAs there is a "hole" in one of the four bonds. This creates a hole, or the Si atom acts as an acceptor.

e The Si atom has 1 more electron than the Ga atom and when it substitutes for a Ga atom in GaAs there is an additional electron that cannot enter any of the four bonds and is therefore donated into the CB (given sufficiently large temperature). Si substituting for Ga therefore acts as a donor.

f All these compounds (AlAs, GaP, InAs, InP, InSb) are compounds of III elements and V elements so they will follow the example of GaAs.

5.6 Doped GaAs

Consider the GaAs crystal at 300 K.

a. Calculate the intrinsic conductivity and resistivity.

b. In a sample containing only 10^{15} cm^{-3} ionized donors, where is the Fermi level? What is the conductivity of the sample?

c. In a sample containing 10^{15} cm^{-3} ionized donors and 9×10^{14} cm^{-3} ionized acceptors, what is the free hole concentration?

Solution

a Given temperature, $T = 300$ K, and intrinsic GaAs.

From Table 5.1 (in the textbook), $n_i = 1.8 \times 10^6$ cm^{-3}, $\mu_e \approx 8500$ cm^2 V^{-1} s^{-1} and $\mu_h \approx 400$ cm^2 V^{-1} s^{-1}. Thus, $\sigma = en_i(\mu_e + \mu_h)$

$\therefore$ $\sigma = (1.602 \times 10^{-19}\text{ C})(1.8 \times 10^6\text{ cm}^{-3})(8500\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1} + 400\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1})$

$\therefore$ $\boldsymbol{\sigma = 2.57 \times 10^{-9}\ \Omega^{-1}\ \text{cm}^{-1}}$

$\therefore$ $\boldsymbol{\rho = 1/\sigma = 3.89 \times 10^8\ \Omega\ \text{cm}}$

b Donors are now introduced. At room temperature, $n = N_d = 10^{15}$ cm^{-3} >> n_i >> p.

$$\sigma_n = eN_d\mu_e \approx (1.602 \times 10^{-19}\text{ C})(10^{15}\text{ cm}^{-3})(8500\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}) = \mathbf{1.36\ \Omega^{-1}\ cm^{-1}}$$

$\therefore$ $\boldsymbol{\rho_n = 1/\sigma_n = 0.735\ \Omega\ \text{cm}}$

In the intrinsic sample, $E_F = E_{Fi}$,

$$n_i = N_c\exp[-(E_c - E_{Fi})/kT] \quad (1)$$

In the doped sample, $n = N_d$, $E_F = E_{Fn}$,

$$n = N_d = N_c\exp[-(E_c - E_{Fn})/kT] \quad (2)$$

Eqn. (2) divided by Eqn. (1) gives,

$$\frac{N_d}{n_i} = \exp\left(\frac{E_{Fn} - E_{Fi}}{kT}\right) \quad (3)$$

$\therefore$ $\Delta E_F = E_{Fn} - E_{Fi} = kT\ln(N_d/n_i)$ (4)

Substituting we find,

$$\Delta E_F = (8.617 \times 10^{-5} \text{ eV/K})(300 \text{ K})\ln[(10^{15} \text{ cm}^{-3})/(1.8 \times 10^6 \text{ cm}^{-3})]$$

$\therefore$ ΔE_F **= 0.521 eV above** E_{Fi} **(intrinsic Fermi level)**

c The sample is further doped with $N_a = 9 \times 10^{14}$ cm^{-3} $= 0.9 \times 10^{15}$ cm^{-3} acceptors. Due to compensation, the net effect is still an *n*-type semiconductor but with an electron concentration given by,

$$n = N_d - N_a = 10^{15} \text{ cm}^{-3} - 0.9 \times 10^{15} \text{ cm}^{-3} = 1 \times 10^{14} \text{ cm}^{-3} \quad (>> n_i)$$

The sample is still *n*-type though there are less electrons than before due to the compensation effect. From the mass action law, the hole concentration is:

$$\boldsymbol{p} = n_i^2 / n = (1.8 \times 10^6 \text{ cm}^{-3})^2 / (1 \times 10^{14} \text{ cm}^{-3}) = \mathbf{0.0324 \text{ cm}^{-3}}$$

On average there are virtually no holes in 1 cm^3 of sample.

We can also calculate the new conductivity. We note that electron scattering now occurs from $N_a + N_d$ number of ionized centers though we will assume that $\mu_e \approx 8500$ cm^2 V^{-1} s^{-1}.

$$\sigma = en\mu_e \approx (1.602 \times 10^{-19} \text{ C})(10^{14} \text{ cm}^{-3})(8500 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}) = 0.136 \ \Omega^{-1} \text{ cm}^{-1}$$

5.7 Degenerate semiconductor

Consider the general exponential expression for the concentration of electrons in the CB,

$$n = N_c \exp\left[-\frac{(E_c - E_F)}{kT}\right]$$

and the mass action law, $np = n_i^2$. What happens when the doping level is such that n approaches N_c and exceeds it? Can you still use the above expressions for n and p?

Consider an *n*-type Si that has been heavily doped and the electron concentration in the CB is 10^{20} cm^{-3}. Where is the Fermi level? Can you use $np = n_i^2$ to find the hole concentration? What is its resistivity? How does this compare with a typical metal? What use is such a semiconductor?

Solution

Consider $n = N_c \exp[-(E_c - E_F)/kT]$ (1)

and $np = n_i^2$ (2)

These expressions have been derived using the Boltzmann tail ($E > E_F$ + a few kT) to the Fermi - Dirac (FD) function $f(E)$ as in Section 5.1.4 (in the textbook). Therefore the expressions are NOT valid when the Fermi level is within a few kT of E_c. In these cases, we need to consider the behavior of the FD function $f(E)$ rather than its tail and the expressions for n and p are complicated.

It is helpful to put the 10^{20} cm^{-3} doping level into perspective by considering the number of atoms per unit volume (atomic concentration, n_{Si}) in the Si crystal:

$$n_{at} = \frac{(\text{Density})N_A}{M_{at}} = \frac{(2.33 \times 10^3 \text{ kg m}^{-3})(6.022 \times 10^{23} \text{ mol}^{-1})}{(28.09 \times 10^{-3} \text{ kg mol}^{-1})}$$

i.e. $n_{at} = 4.995 \times 10^{28}$ m^{-3} or 4.995×10^{22} cm^{-3}

Given that the electron concentration $n = 10^{20}$ cm^{-3} (not necessarily the donor concentration!), we see that

$$n/n_{at} = (10^{20}\ \text{cm}^{-3}) / \ (4.995 \times 10^{22}\ \text{cm}^{-3}) = 0.00200$$

which means that if all donors could be ionized we would need 1 in 500 doping or 0.2% donor doping in the semiconductor (n is not exactly N_d for degenerate semiconductors). We cannot use Equation (1) to find the position of E_F. The Fermi level will be in the conduction band. The semiconductor is degenerate (see Figure 5Q7-1).

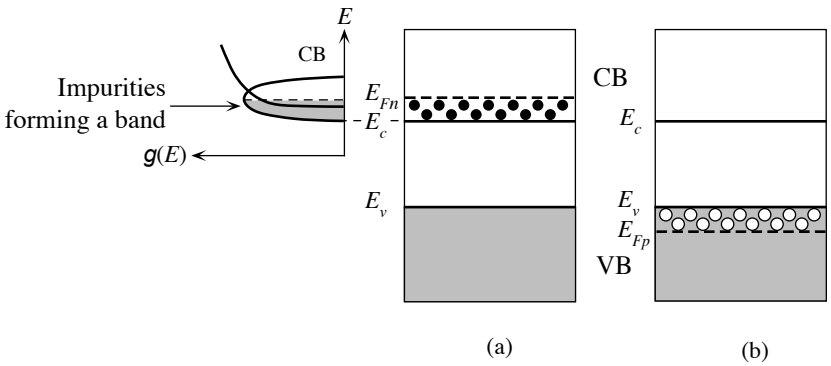


Figure 5Q7-1

(a) Degenerate n-type semiconductor. Large number of donors form a band that overlaps the CB.

(b) Degenerate p-type semiconductor.

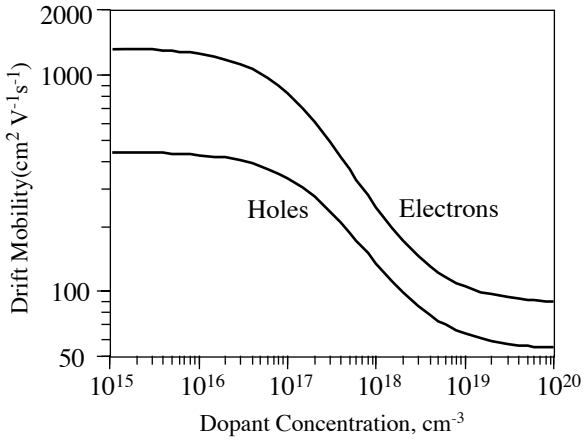


Figure 5Q7-2 The variation of the drift mobility with dopant concentration in Si for electrons and holes at 300 K.

Take T = 300 K, and $\mu_e \approx 900\ \text{cm}^2\ \text{V}^{-1}\ \text{s}^{-1}$ from Figure 5Q7-2. The resistivity is

$$\rho = 1/(en\mu_e) = 1/[(1.602 \times 10^{-19}\ \text{C})(10^{20}\ \text{cm}^{-3})(900\ \text{cm}^2\ \text{V}^{-1}\ \text{s}^{-1})]$$

$\therefore$ $$\rho = \mathbf{6.94 \times 10^{-5}\ \Omega\ cm} \text{ or } \mathbf{694 \times 10^{-7}\ \Omega\ m}$$

Compare this with a metal alloy such as nichrome which has ρ = 1000 nΩ m = $10 \times 10^{-7}\ \Omega$ m. The difference is only about a factor of 70.

This degenerate semiconductor behaves almost like a "metal". Heavily doped degenerate semiconductors are used in various MOS (metal- oxide- semiconductor) devices where they serve as the gate electrode (substituting for a metal) or interconnect lines.

5.8 Photoconductivity and speed

Consider two *p*-type Si samples both doped with 10^{15} B atoms cm^{-3}. Both have identical dimensions of length *L* (1 mm), width *W* (1 mm), and depth (thickness) *D* (0.1 mm). One sample, labeled *A*, has an electron lifetime of 1 μs whereas the other, labeled *B*, has an electron lifetime of 5 μs.

a. At time $t = 0$, a laser light of wavelength 750 nm is switched on to illuminate the surface ($L \times W$) of both the samples. The incident laser light intensity on both samples is 10 mW cm^{-2}. At time $t = 50$ μs, the laser is switched off. Sketch the time evolution of the minority carrier concentration for both samples on the same axes.

b. What is the photocurrent (current due to illumination alone) if each sample is connected to a 1 V battery?

Solution

a

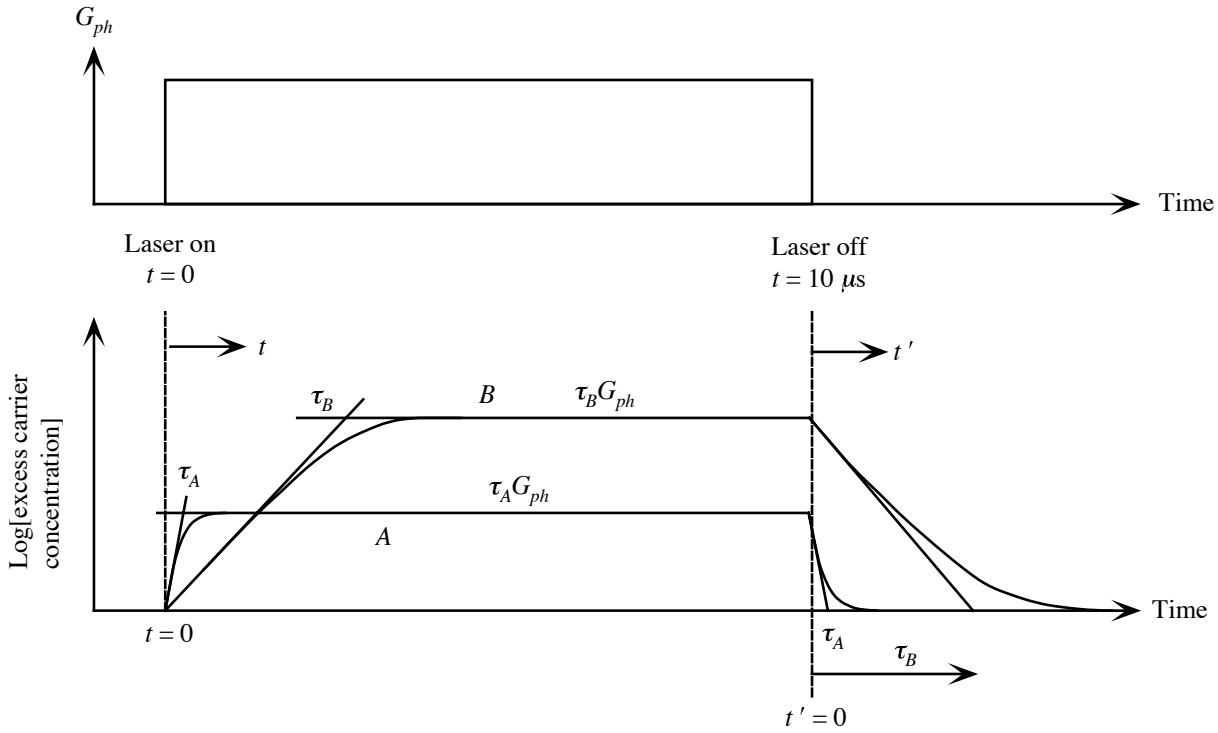


Figure 5Q8-1 Schematic sketch of the excess carrier concentration in the samples *A* and *B* as a function of time from the laser switch-on to beyond laser switch-off.

b From the mobility vs. dopant graph (Figure 5Q8-2), $\mu_h = 450 \times 10^{-4}$ m^2 V^{-1} s^{-1} and $\mu_e = 1300 \times 10^{-4}$ m^2 V^{-1} s^{-1}. Given are the wavelength of illumination $\lambda = 750 \times 10^{-9}$ m, light intensity $I = 100$ W/m^2, length $L = 1$ mm, width $W = 1$ mm, and depth (thickness) $D = 0.1$ mm.

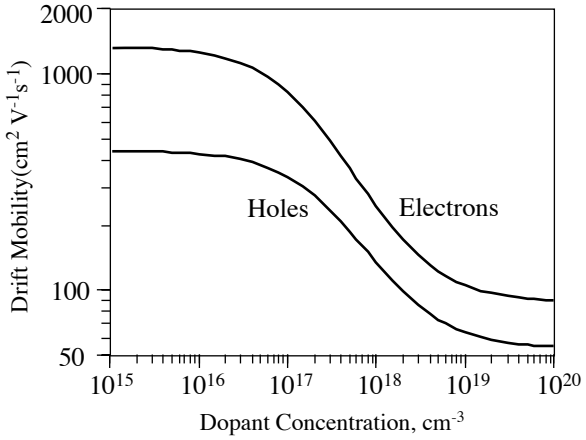


Figure 5Q8-2 The variation of the drift mobility with dopant concentration in Si for electrons and holes at 300 K.

The photoconductivity is given by (see Example 5.11 in the textbook):

$$\Delta\sigma = \frac{e\eta I\lambda\tau(\mu_e + \mu_h)}{hcD}$$

where $\eta = 1$ is the quantum efficiency. Assume all light intensity is absorbed (correct assumption as the absorption coefficient at this wavelength is large). The photocurrent density and hence the photo current is given by:

$$\Delta J = \Delta I/A = \mathcal{E}\Delta\sigma$$

Substitute:
$$\Delta I = (W \times D)\left(\frac{V}{L}\right)\frac{e\eta I\lambda\tau(\mu_e + \mu_h)}{hcD}$$

$\therefore$
$$\Delta I = \frac{WVe\eta I\lambda\tau(\mu_e + \mu_h)}{Lhc}$$

The photocurrent will travel perpendicular to the $W \times D$ direction, while the electric field $\mathcal{E}$ will be directed along L. Substituting the given values for sample A (electron lifetime $\tau_A = 10^{-6}$ s, voltage $V =$ 1 V):

$$\Delta I_A = \frac{(0.001\text{ m})(1\text{ V})(1.602\times10^{-19}\text{ C})(1)(100\text{ W/m}^2)(750\times10^{-9}\text{ m})(10^{-6}\text{ s})\left(0.13\ \frac{\text{m}^2}{\text{V s}} + 0.045\ \frac{\text{m}^2}{\text{V s}}\right)}{(0.001\text{ m})(6.626\times10^{-34}\text{ J s})(3.0\times10^{8}\text{ m/s})}$$

$\therefore$ $\mathbf{\Delta I_A = 1.06 \times 10^{-5}\ A}$

The photocurrent in sample B can be calculated with the same equation, using the given value of $\tau_B = 5 \times 10^{-6}$ s. After calculation:

$\mathbf{\Delta I_B = 5.29 \times 10^{-5}\ A}$

*5.9 Hall Effect in Semiconductors

The Hall effect in a semiconductor sample involves not only the electron and hole concentrations n and p, respectively, but also the electron and hole drift mobilities, μ_e and μ_h. The hall coefficient of a semiconductor is (see Chapter 2 of the textbook),

$$R_H = \frac{p - nb^2}{e(p + nb)^2}$$ *Hall coefficient of a semiconductor*

where $b = \frac{\mu_e}{\mu_h}$.

a. Given the mass action law, $pn = n_i^2$, find n for maximum $|R_H|$ (negative and positive R_H). Assume that the drift mobilities remain relatively unaffected as n changes (due to doping). Given the electron and hole drift mobilities, $\mu_e = 1350$ cm^2 V^{-1} s^{-1}, $\mu_h = 450$ cm^2 V^{-1} s^{-1} for silicon determine n for maximum $|R_H|$ in terms of n_i.

b. Taking $b = 3$, plot R_H as a function of electron concentration n/n_i from 0.1 to 10.

Solution

a Substituting the mass action law ($p = n_i^2/n$) into the given equation, we get

$$R_H = \frac{p - nb^2}{e(p + nb)^2} = \frac{\frac{n_i^2}{n} - nb^2}{e\left(\frac{n_i^2}{n} + nb\right)^2} = \frac{u}{v}$$

where u and v represent the numerator and denominator as a function of n.

Then $$\frac{dR_H}{dn} = \frac{u'v - uv'}{v^2} = 0$$

where primes are derivatives with respect to n. This means that $u'v - u\,v' = 0$, so that,

$$u'v - uv' = \left[-\frac{n_i^2}{n^2} - b^2\right]\left[e\left(\frac{n_i^2}{n} + nb\right)^2\right] - \left[\frac{n_i^2}{n} - nb^2\right]\left[2e\left(\frac{n_i^2}{n} + nb\right)\left(-\frac{n_i^2}{n^2} + b\right)\right] = 0$$

We can multiply through by n^3 and then combine terms and factor to obtain,

$$b^3n^4 - [3n_i^2b(1+b)]n^2 + n_i^4 = 0$$

or, $$b^3x^2 + [-3b(1 + b)]x + 1 = 0$$

where $x = (n/n_i)^2$. This is a quadratic equation in x. Its solution is,

$$x = \frac{n^2}{n_i^2} = \frac{3b(1+b) \pm \sqrt{9b^2(1+b)^2 - 4b^3}}{2b^3}$$

For Si, $b = 3$, and we have two solutions corresponding to,

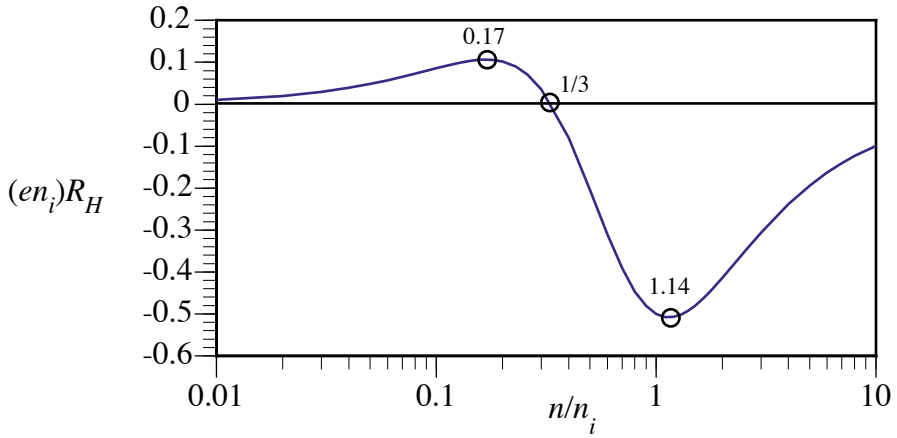
$$n/n_i = 1.14 \quad \text{and} \quad n/n_i = 0.169, \text{ or } p/n_i = 1/0.169 = 5.92$$

b Substituting the mass action law $p = n_i^2/n$ into the given equation and using a normalized electron concentration $x = n/n_i$, we get,

$$R_H = \frac{p - nb^2}{e(p + nb)^2} = \frac{\dfrac{n_i^2}{n} - nb^2}{e\left(\dfrac{n_i^2}{n} + nb\right)^2} = \frac{\dfrac{1}{x} - xb^2}{en_i\left(\dfrac{1}{x} + xb\right)^2}$$

or $$y = \frac{R_H}{(1/en_i)} = \frac{\dfrac{1}{x} - xb^2}{\left(\dfrac{1}{x} + xb\right)^2}$$

Normalized Hall coefficient $(en_i)R_H$ vs. normalized concentration n/ni obviously follows y vs. x which is shown in Figure 5Q9-1 for $b = 3$.



Normalized Hall coefficient vs. normalized electron concentration. Values 0.17, 1.14 and 0.33 shown are n/n_i values when the magnitude of R_H reaches maxima and zero respectively.

Figure 5Q9-1

*5.10 Compound semiconductor devices

Silicon and germanium crystalline semiconductors are what are called elemental group IV semiconductors. It is possible to have compound semiconductors from atoms in groups III and V. For example, GaAs is a compound semiconductor that has Ga from group III and As from group V, so that in the crystalline structure we have an "effective" or "mean" valency of IV per atom and the solid behaves like a semiconductor. Similarly GaSb (gallium antimonide) would be a III-V type semiconductor. Provided we have a stochiometric compound, the semiconductor will be ideally intrinsic. If, however, there is an excess of Sb atoms in the solid GaSb, then we will have nonstochiometry and the semiconductor will be extrinsic. In this case, excess Sb atoms will act as donors in the GaSb structure. There are many useful compound semiconductors, the most important of which is GaAs. Some can be doped both *n*- and *p*-type, but many are one type only. For example, ZnO is a II-VI compound semiconductor with a direct bandgap of 3.2 eV, but unfortunately, due to the presence of excess Zn, it is naturally *n*-type and cannot be doped to *p*-type.

a. GaSb (gallium antimonide) is an interesting direct bandgap semiconductor with an energy bandgap, $E_g = 0.67$ eV, almost equal to that of germanium. It can be used as an LED (light-emitting diode) or laser diode material. What would be the wavelength of emission from a GaSb LED? Will this be visible?

b. Calculate the intrinsic conductivity of GaSb at 300 K taking $N_c = 2.3 \times 10^{19}$ cm^{-3}, $N_v = 6.1 \times 10^{19}$ cm^{-3}, $\mu_e = 5000$ cm^2 V^{-1} s^{-1}, and $\mu_h = 1000$ cm^2 V^{-1} s^{-1}. Compare with the intrinsic conductivity of Ge.

c. Excess Sb atoms will make gallium antimonide nonstoichiometric, that is, $GaSb_{1+\delta}$, which will result in an extrinsic semiconductor. Given that the density of GaSb is 5.4 g cm^{-3}, calculate δ (excess Sb) that will result in GaSb having a conductivity of 100 Ω^{-1} cm^{-1}. Will this be an *n*- or *p*-type semiconductor? You may assume that the drift mobilities are relatively unaffected by the doping.

Solution

a Given $E_g = 0.67$ eV, the corresponding wavelength is:

$$\lambda = hc / E_g = (6.626 \times 10^{-34} \text{ J s})(3.0 \times 10^8 \text{ m s}^{-1}) / (0.67 \text{ eV} \times 1.602 \times 10^{-19} \text{ J/eV})$$

$\therefore$ λ **= 1.85 × 10^{-6} m** or **1850 nm**, not in the visible.

b Assume temperature, $T = 300$ K.

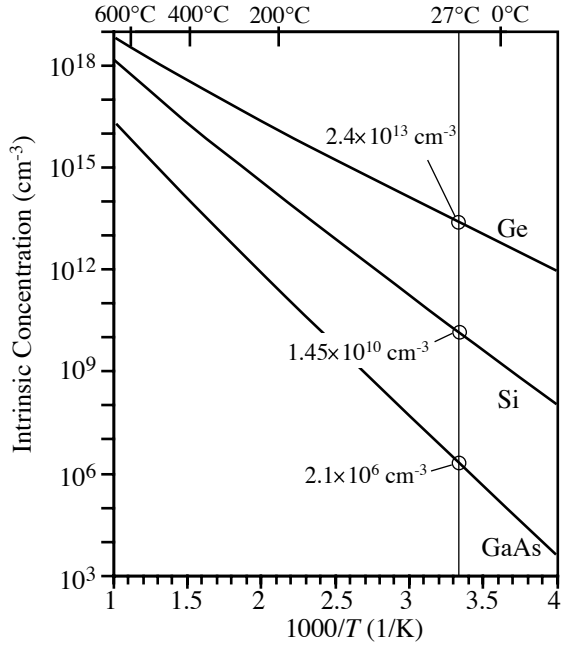


Figure 5Q10-1 The temperature dependence of the intrinsic concentration.

For GaSb, we are given $N_c = 2.3 \times 10^{25}$ m^{-3}; $N_v = 6.1 \times 10^{25}$ m^{-3}; $\mu_e = 5000 \times 10^{-4}$ m^2 V^{-1} s^{-1}; $\mu_h = 1000 \times 10^{-4}$ m^2 V^{-1} s^{-1}, therefore the intrinsic concentration is:

$$n_i = (N_c N_v)^{1/2} \exp[-E_g/2kT]$$

$\therefore$ $$n_i = [(2.3 \times 10^{25} \text{ m}^{-3})(6.1 \times 10^{25} \text{ m}^{-3})]^{1/2} \exp[-(0.67 \text{ eV})/(2 \times 8.617 \times 10^{-5} \text{ eV/K} \times 300 \text{ K})]$$

$\therefore$ $$n_i = 8.822 \times 10^{19} \text{ m}^{-3}$$

The intrinsic conductivity is therefore:

$$\sigma = e n_i (\mu_e + \mu_h)$$

$\therefore$ $\sigma = (1.602 \times 10^{-19}\ \text{C})(8.822 \times 10^{19}\ \text{m}^{-3})(0.5\ \text{m}^2\ \text{V}^{-1}\ \text{s}^{-1} + 0.1\ \text{m}^2\ \text{V}^{-1}\ \text{s}^{-1}) = \mathbf{8.48\ \Omega^{-1}\ m^{-1}}$

For Ge, from Figure 5Q10-1 $n_i = 2.4 \times 10^{19}\ \text{m}^{-3}$ and from Table 5.1 (in the textbook) we have $\mu_e = 0.39\ \text{m}^2\ \text{V}^{-1}\ \text{s}^{-1}$ and $\mu_h = 0.19\ \text{m}^2\ \text{V}^{-1}\ \text{s}^{-1}$. The intrinsic conductivity is then:

$$\sigma = en_i(\mu_e + \mu_h)$$

$\therefore$ $\sigma = (1.602 \times 10^{-19}\ \text{C})(2.4 \times 10^{19}\ \text{m}^{-3})(0.39\ \text{m}^2\ \text{V}^{-1}\ \text{s}^{-1} + 0.19\ \text{m}^2\ \text{V}^{-1}\ \text{s}^{-1}) = \mathbf{2.23\ \Omega^{-1}\ m^{-1}}$

Intrinsic Ge and GaSb have comparable conductivities (same order of magnitude).

(Typically, intrinsic conductivities of semiconductors are largely determined by the bandgap).

c Excess antimony will make the sample ***n*-type**. We will assume that the electron mobility is roughly the same. In the extrinsic sample, $n = N_d >> p$.

Given that $\sigma_n = 100\ \Omega^{-1}\ \text{m}^{-1}$, and $\sigma_n \approx eN_d\mu_e$:

$$N_d = \sigma_n / e\mu_e = (100\ \Omega^{-1}\ \text{m}^{-1})/[(1.602 \times 10^{-19}\ \text{C})(0.5\ \text{m}^2\ \text{V}^{-1}\ \text{s}^{-1})]$$

$\therefore$ $N_d = 1.248 \times 10^{21}\ \text{m}^{-3}\ (>> n_i)$

Now consider the atomic concentration.

GaSb has the GaAs crystal structure so that the unit cell (cubic) has 4 Ga atoms and 4 Sb atoms. The cube side is a. The density ρ is therefore given by,

$$\rho = \left(\frac{1}{a^3}\right)\left[\frac{(4M_{\text{Ga}} + 4M_{\text{Sb}})}{N_A}\right]$$

which means that the Sb atom concentration is

$$n_{\text{Sb}} = \frac{4}{a^3} = \frac{\rho N_A}{(M_{\text{Ga}} + M_{\text{Sb}})}$$

or $$n_{\text{Sb}} = \frac{(5.4 \times 10^3\ \text{kg m}^{-3})(6.022 \times 10^{23}\ \text{mol}^{-1})}{(10^{-3})(69.72\ \text{g mol}^{-1} + 121.75\ \text{g mol}^{-1})}$$

$\therefore$ $n_{\text{Sb}} = 1.698 \times 10^{28}\ \text{m}^{-3}$

Thus, $\delta = N_d/n_{\text{Sb}} = (1.248 \times 10^{21}\ \text{m}^{-3}) / (1.698 \times 10^{28}\ \text{m}^{-3}) = \mathbf{7.35 \times 10^{-8}}$

5.11 Semiconductor Strain Gauge

Suppose that an extrinsic semiconductor of conductivity $\sigma = eN_d\mu_e$ is strained in a certain direction by the application of uniaxial stress. Then there will be a change in the conductivity due to a change in the mobility as a result of the applied stress and the resulting strain ε, the effective mass m_e^* in the direction of strain changes which affects the mobility (why?) in that direction. The effect is strongest in the [100] direction but almost negligible in the [111] direction. The **Gauge factor** G of a strain gauge is defined as the fractional change in the resistance per unit strain. Show that for an extrinsic n-type semiconductor, G is given by

$$G = \frac{\left(\frac{\Delta R}{R}\right)}{\left(\frac{\Delta L}{L}\right)} = -\frac{1}{\mu_e}\left(\frac{\partial \mu_e}{\partial \varepsilon}\right)_T$$ *Semiconductor strain gauge*

where $\left(\frac{\partial \mu_e}{\partial \varepsilon}\right)_T$ is the mobility-strain coefficient that describes the effect of strain on the drift mobility at a constant temperature, due to a change in the effective mass with strain. Calculate the approximate gauge factor for a particular n-type Si that has $\left(\frac{\partial \mu_e}{\partial \varepsilon}\right)_T = 10^5$ cm^2 V^{-1} s^{-1} and compare this with a typical metal strain gauge factor of about 10 (see Question 2.14 in the textbook). Which would you recommend?

Solution

Differentiate $R = \ell/\sigma A$, to get $\Delta R/R = -\Delta\sigma/\sigma$. But $\sigma = eN_d\mu_e$ where μ_e is strain dependent, i.e., $\mu_e = \mu_e(e)$.

Thus, $$\partial\sigma/\partial\varepsilon = eN_d(\partial\mu_e/\partial\varepsilon) = (\sigma/\mu_e)(\partial\mu_e/\partial\varepsilon).$$

Clearly an incremental strain $\delta\varepsilon$ will result in $\delta\sigma = (\partial\sigma/\partial\varepsilon)\delta\varepsilon = (\sigma/\mu_e)(\partial\mu_e/\partial\varepsilon)\delta\varepsilon$, so that

$$G = (dR/R)/(dL/L) = -(\delta\sigma/\sigma)/(\delta\varepsilon) = -(\partial\mu_e/\partial\varepsilon)_T/\mu_e.$$

From above, $G = -(\partial\mu_e/\partial\varepsilon)_T/\mu_e = (10^5 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1})/(1350 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}) \approx$ **74**,

which is much higher than those values for metals.

5.12 Excess minority carrier concentration

Consider an *n*-type semiconductor and weak injection conditions. Assume that the minority carrier recombination time, τ_h, is constant (independent of injection - hence the weak injection assumption). The rate of change of the instantaneous hole concentration, $\partial p_n/\partial t$, due to recombination is given by

$$\frac{\partial p_n}{\partial t} = -\frac{p_n}{\tau_h}$$ *Recombination rate* **[1]**

The net rate of increase (change) in p_n is the sum of the total generation rate G and the rate of change due to recombination, that is,

$$\frac{dp_n}{dt} = G - \frac{p_n}{\tau_h}$$ **[2]**

By separating the generation term G into thermal generation G_o and photogeneration G_{ph} and considering the dark condition as one possible solution, show that

$$\frac{d\Delta p_n}{dt} = G_{ph} - \frac{\Delta p_n}{\tau_h}$$ *Excess carriers under uniform photogeneration and recombination* **[3]**

How does your derivation compare with Equation 5.27 (in the textbook)? What are the assumptions inherent in Equation 3?

Solution

We are given G as total generation rate or thermal generation (G_o) + photogeneration (G_{ph}), i.e. $G = G_o + G_{ph}$. Also, we know that $\Delta p_n = p_n - p_{no}$ and therefore $p_n = \Delta p_n + p_{no}$. From Equation 2:

$$\frac{dp_n}{dt} = G - \frac{p_n}{\tau}$$

Substitute for G: $$\frac{dp_n}{dt} = G_o + G_{ph} - \frac{p_n}{\tau} \qquad (4)$$

In the dark and in equilibrium we have no photogeneration ($G_{ph} = 0$), an equilibrium concentration of holes ($p_n = p_{no}$) and no net increase in the hole concentration ($dp_n/dt = 0$). Therefore:

$$0 = G_o - \frac{p_{no}}{\tau}$$

$\therefore$ $$G_o = \frac{p_{no}}{\tau}$$

Substitute this expression for G_o into Eqn. (4).

$$\frac{dp_n}{dt} = \frac{p_{no}}{\tau} + G_{ph} - \frac{p_n}{\tau}$$

Substitute for p_n in terms of excess hole concentration Δp_n.

$$\frac{d(\Delta p_n + p_{no})}{dt} = \frac{p_{no}}{\tau} + G_{ph} - \frac{\Delta p_n + p_{no}}{\tau}$$

$\therefore$ $$\frac{d\Delta p_n}{dt} = G_{ph} - \frac{\Delta p_n}{\tau}$$

This is Eqn. 3 which is the same as Eqn. 5.27 (in the textbook). The derivation above is more rigorous whereas that in Section 5.4.2 (in the textbook) is intuitive.

5.13 Schottky junction

a. Consider a Schottky junction diode between Au and *n*-Si, doped with 10^{16} donors cm^{-3}. The cross-sectional area is 1 mm^2. Given the work function of Au as 5.1 eV, what is the theoretical barrier height, Φ_B, from the metal to the semiconductor?

b. Given that the experimental barrier height Φ_B is about 0.8 eV, what is the reverse saturation current and the current when there is a forward bias of 0.3 V across the diode? (Use Equation 4.37 in the textbook.)

Solution

a The number of donors ($N_d = 10^{22}$ m^{-3}) is related to the energy difference from the Fermi level ($\Delta E = E_c - E_F$) by:

$$N_d = N_c \exp\left(-\frac{\Delta E}{kT}\right)$$

$$\therefore \qquad \Delta E = -kT\ln\left(\frac{N_d}{N_c}\right)$$

From Table 5.1 (in the textbook), $N_c = 2.8 \times 10^{25}$ m^{-3}, and we are given and $N_d = 10^{22}$ m^{-3}. Assuming temperature $T = 300$ K:

$$\Delta E = -\left(1.381\times10^{-23}\ \text{J/K}\right)(300\ \text{K})\ln\left(\frac{10^{22}\ \text{m}^{-3}}{2.8\times10^{25}\ \text{m}^{-3}}\right)\left(\frac{1}{1.602\times10^{-19}\ \text{J/eV}}\right)$$

$$\therefore \qquad \Delta E = 0.2053 \text{ eV}$$

The work function of *n*-Si ($\Phi_{n\text{Si}}$) must be less than that of Au ($\Phi_{\text{Au}} = 5.1$ eV) in order to have a Schottky junction. $\Phi_{n\text{Si}}$ is given as $\Phi_{n\text{Si}} = \Delta E + \chi$, where $\chi = 4.01$ eV (see Example 5.18 in the textbook) is the electron affinity of Si. Therefore:

$$\Phi_{n\text{Si}} = 0.2053 \text{ eV} + 4.01 \text{ eV} = 4.215 \text{ eV}$$

This is indeed less than Φ_{Au} and therefore we have a Schottky junction. The effective barrier height Φ_B is then:

$$\Phi_B = \Phi_{\text{Au}} - \chi = 5.1 \text{ eV} - 4.01 \text{ eV} = \mathbf{1.09\ eV}$$

b The experimental potential barrier is given as $\Phi_B = 0.8$ eV. Assume temperature $T = 300$ K. The reverse saturation current (I_o) is given by (see Example 5.18 in the textbook) (where $B_e = 1.2 \times 10^6$ A K^{-2} m^{-2} is the effective Richardson-Dushman constant):

$$I_o = AB_eT^2\exp\left(-\frac{\Phi_B}{kT}\right)$$

$$\therefore \qquad I_o = \left(10^{-6}\ \text{m}^2\right)\left(1.2\times10^6\ \text{A K}^{-2}\ \text{m}^{-2}\right)(300\ \text{K})^2\exp\left(-\frac{(0.8\ \text{eV})\left(1.602\times10^{-19}\ \text{J/eV}\right)}{\left(1.381\times10^{-23}\ \text{J/K}\right)(300\ \text{K})}\right)$$

$$\therefore \qquad \boldsymbol{I_o = 3.97 \times 10^{-9}\ A}$$

The forward bias voltage V_f is given as 0.3 V. The forward current I_f is then:

$$I_f = I_o\left[\exp\left(\frac{eV_f}{kT}\right)-1\right] = \left(3.97\times10^{-9}\ \text{A}\right)\left[\exp\left(\frac{\left(1.602\times10^{-19}\ \text{C}\right)(0.3\ \text{V})}{\left(1.381\times10^{-23}\ \text{J/K}\right)(300\ \text{K})}\right)-1\right]$$

$$\therefore \qquad \boldsymbol{I_f = 0.000433\ A}$$

5.14 Schottky and ohmic contacts

Consider an *n*-type Si sample doped with 10^{16} donors cm^{-3}. The length L is 100 μm; the cross-sectional area A is 10 μm × 10 μm. The two ends of the sample are labeled as B and C. The electron affinity (χ) of Si is 4.01 eV and the work functions, Φ, of four potential metals for contacts at B and C are listed in Table 5Q14-1.

a. Ideally, which metals will result in a Schottky contact?

b. Ideally, which metals will result in an Ohmic contact?

c. Sketch the *I-V* characteristics when both *B* and *C* are ohmic contacts. What is the relationship between *I* and *V*?

d. Sketch the *I-V* characteristics when *B* is ohmic and *C* is a Schottky junction. What is the relationship between *I* and *V*?

e. Sketch the *I-V* characteristics when both *B* and *C* are Schottky contacts. What is the relationship between *I* and *V*?

Table 5Q14-1 Work functions in eV

Cs	Li	Al	Au
1.8	2.5	4.25	5.1

Solution

We are given the concentration of donors, $N_d = 10^{22}$ m^{-3}. From Table 5.1 (in the textbook), the electron affinity of Si is 4.01 eV and the effective density of states at the conduction edges is 2.8×10^{25} m^{-3}. Assuming temperature $T = 300$ K,

$$N_d = N_c \exp\left(-\frac{\Delta E}{kT}\right)$$

$$\therefore \quad \Delta E = -kT \ln\left(\frac{N_d}{N_c}\right)$$

$$\therefore \quad \Delta E = -\left(1.381\times10^{-23}\ \text{J/K}\right)(300\ \text{K})\ln\left(\frac{10^{22}\ \text{m}^{-3}}{2.8\times10^{25}\ \text{m}^{-3}}\right)\left(\frac{1}{1.602\times10^{-19}\ \text{J/eV}}\right)$$

$$\therefore \quad \Delta E = 0.2053 \text{ eV}$$

The energy required to move an electron from the Si semiconductor is then:

$$\Phi_{n\text{Si}} = \Delta E + \chi = 0.2053 \text{ eV} + 4.01 \text{ eV} = 4.215 \text{ eV}$$

a For a Schottky contact you need $\Phi_m > \Phi_{n\text{Si}}$ so Au will result in a Schottky junction. Note however that for Al, $\Phi_m - \Phi_{n\text{Si}}$ is 0.04 eV, of the order of the thermal energy so Al / Si "should not" be a Schottky junction. This is not, however, necessarily the case as the junction depends very much on the surface conditions (surface states) as well as the crystal plane on to which the contact is made (Φ depends on the crystal surface).

b For an Ohmic contact you need $\Phi_m < \Phi_{n\text{Si}}$ so Cs and Li will result in Ohmic contacts.

c This is a straight line with slope equal to the conductance, or the inverse of the resistance ($\Delta I / \Delta V = 1/R$). The resistance can be found using the drift mobility of the electrons $\mu_e = 1350 \times 10^{-4}$ m^2 V^{-1} s^{-1} (from Table 5.1 in the textbook) and the equation for conductivity, $\sigma = eN_d\mu_e$:

$$R = \frac{L}{\sigma A} = \frac{L}{eN_d\mu_e A} = \frac{100\times10^{-6}\ \text{m}}{\left(1.602\times10^{-19}\ \text{C}\right)\left(10^{22}\ \text{m}^{-3}\right)\left(0.135\ \text{m}^2\ \text{V}^{-1}\ \text{s}^{-1}\right)\left(10^{-5}\ \text{m}\right)^2}$$

$$\therefore \quad \boldsymbol{R = 4620\ \Omega}$$

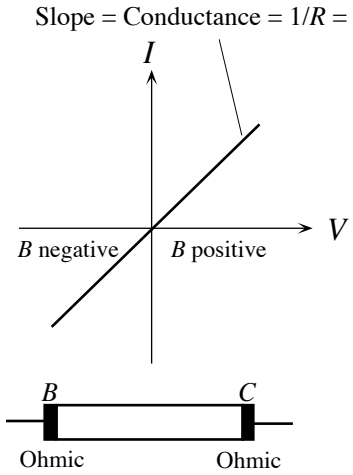


Figure 5Q14-1 *I-V* characteristic with *B* and *C* as Ohmic contacts.

d When *C* is reverse biased (*B* is positive), current is limited by thermionic emission over Φ_B between the metal and semiconductor at *C*. When *C* is forward biased (*B* negative), current is primarily limited by the resistance of the semiconductor, unless this is very small indeed in which case the *I-V* is the forward biased Schottky junction.

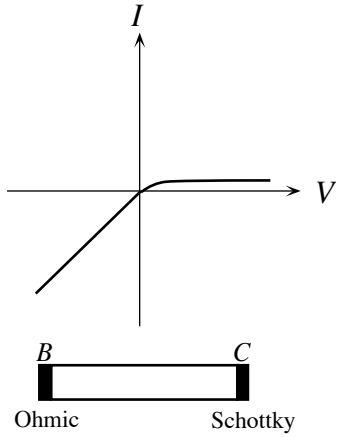


Figure 5Q14-2 *I-V* characteristic with *B* as Ohmic and *C* as Schottky.

We can then find the maximum saturation current (I_o) by first finding the barrier height, $\Phi_B = \Phi_{Au} - \chi = 5.1$ eV - 4.01 eV = 1.09 eV, and then assuming the maximum effective Richardson-Dushman constant, $B_e = 1.2 \times 10^6$ A m^{-2} K^{-2}. Using Equation 4.38 (in the textbook):

$$I_o = AB_eT^2 \exp\left(-\frac{\Phi_B}{kT}\right)$$

$$\therefore \quad I_o = \left(10\times10^{-6}\ \text{m}\right)^2\left(1.2\times10^{6}\ \text{A K}^{-2}\ \text{m}^{-2}\right)(300\ \text{K})^2 \exp\left(-\frac{(1.09\ \text{eV})\left(1.602\times10^{-19}\ \text{J/eV}\right)}{\left(1.381\times10^{-23}\ \text{J/K}\right)(300\ \text{K})}\right)$$

$$\therefore \quad \boldsymbol{I_o = 5.36 \times 10^{-18}\ \text{A}}$$

This is the maximum saturation current as we assumed maximum B_e.

e There is reverse saturation current in both directions. The current is saturated for even small voltages (above a few *kT*/*e*) and is the thermionic emission current over Φ_B. The calculation for reverse saturated current can be found as in part **d**.

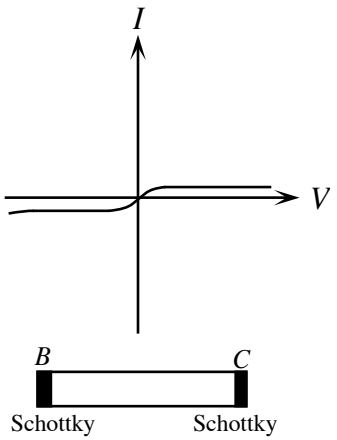


Figure 5Q14-3 *I-V* characteristic with *B* and *C* as Schottky contacts.

5.15 Peltier effect and electrical contacts

Consider the Schottky junction and the ohmic contact shown in Figures 5Q15-1 and 5Q15-2 between a metal and *n*-type semiconductor.

a. Is the Peltier effect similar in both contacts?

b. Is the sign in $Q' = \pm \Pi I$ the same for both contacts?

c. Which junction would you choose for a thermoelectric cooler? Give reasons.

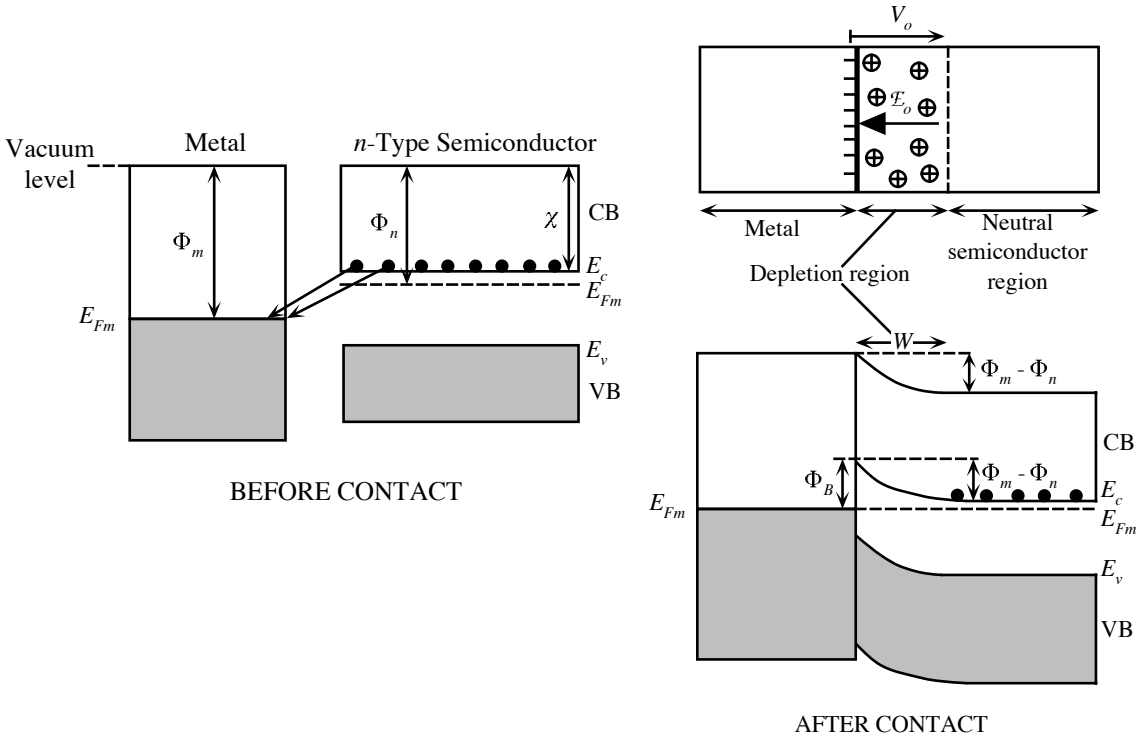


Figure 5Q15-1 Formation of a Schottky junction between a metal and an *n*-type semiconductor when $\Phi_m > \Phi_n$.

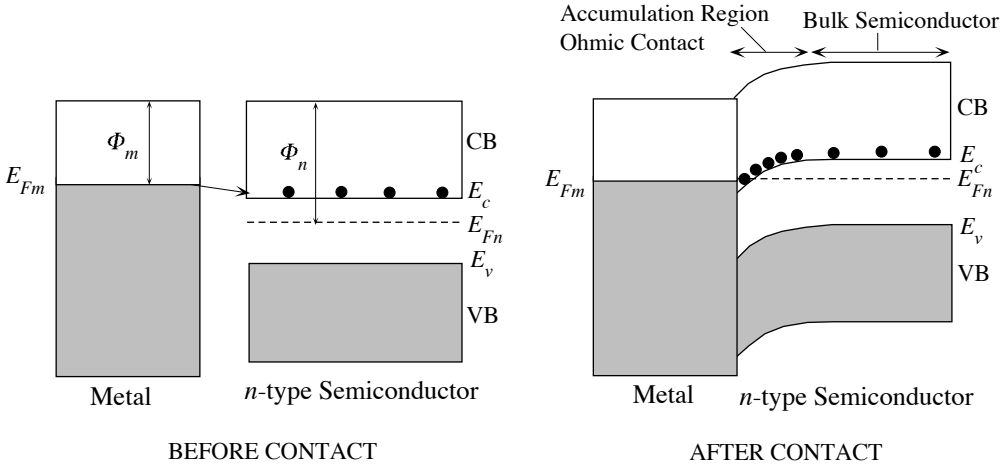
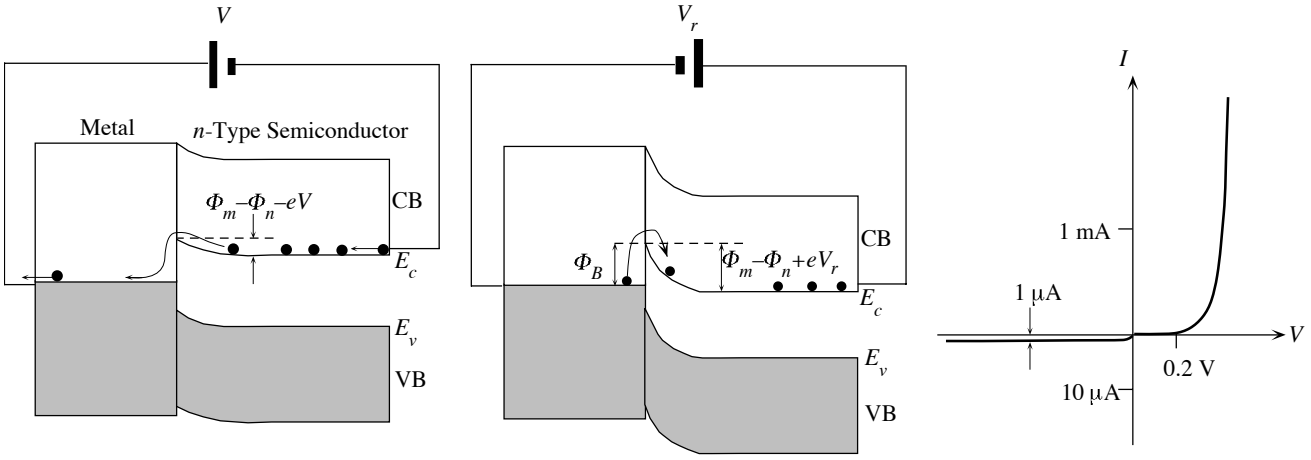


Figure 5Q15-2 When a metal with a smaller work function than an *n*-type semiconductor is put into contact with the n-type semiconductor, the resulting junction is an ohmic contact in the sense that it does not limit the current flow.

Solution

a, **b** Peltier effect is based on change in the energy of an electron in passing through a junction of two dissimilar materials. It will appear whether the junction is ohmic or Schottky if there is a change in the electron energy in going through the junction.



(a) Forward biased Schottky junction. Electrons in the CB of the semiconductor can eadily overcome the small *PE* barrier to enter the metal.

(b) Reverse biased Schottky junction. Electrons in the metal can not easily overcome the *PE* barrier Φ_B to enter the semiconductor.

(c) *I–V* Characteristics of a Schottky junction exhibits rectifying properties (negative current axis is in microamps)

Figure 5Q15-3 The Schottky junction.

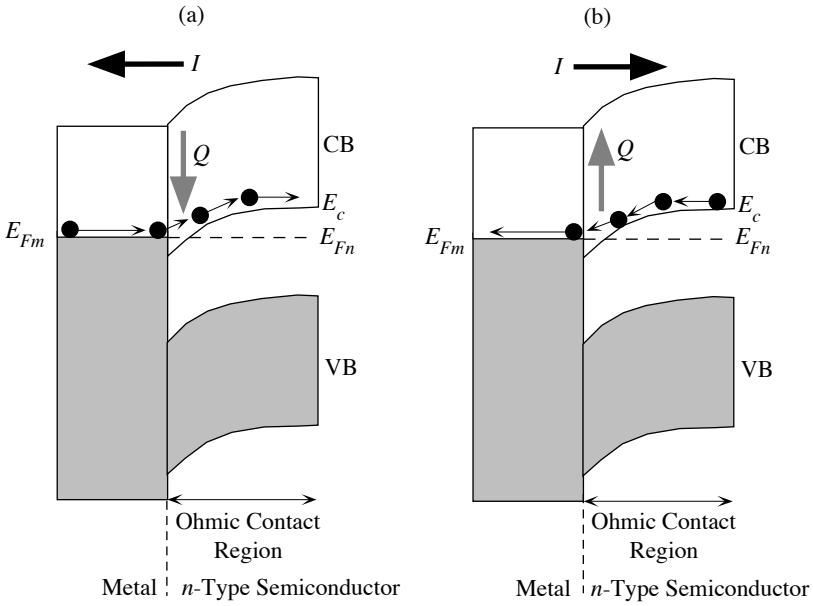


Figure 5Q15-4

(a) Current from an *n*-type semiconductor to the metal results in heat absorption at the junction.

(b) Current from the metal to an *n*-type semiconductor results in heat release at the junction.

Figure 5Q15-4 (a) is an *ohmic* contact between a metal and an *n*-type SC (semiconductor). Current from SC to metal in Figure 5Q15-4 (a) *absorbs* heat at the junction because the electron at the metal E_F has to absorb energy (heat) when entering the semiconductor E_c which is higher. In the Schottky junction in Figure 5Q15-3 (b) the current is from SC to metal and the electron from the metal to the semiconductor *absorbs* heat as in the ohmic case.

The current from metal to SC in Figure 5Q15-4 (b) *releases* heat because the electron at the semiconductor E_c has to lose energy when entering the metal E_F. In Figure 4Q15-3 (a) the current is from metal to SC and the electron from the SC *releases* heat at the junction because E_c has to lose energy in entering the metal E_F.

The sign in $Q_P' = \pm \prod I$ is therefore the same.

c One can only pass a small current though a semiconductor with Schottky junctions at both ends (M/SC/M where M = metal, SC = semiconductor) which is the reverse saturation current. This is because one Schottky junction is always reverse biased. If ohmic contacts are used then the contacts do not limit the current (which is determined by the bulk semiconductor) and Q' can be substantial and can be used practically as in a thermoelectric cooler. Note that it is possible to have a device with one junction ohmic and the other Schottky if the current is not limited by the Schottky contact but rather limited by the bulk resistance.

*5.16 Peltier coolers and figure of merit (FOM)

Consider the thermoelectric effect shown in Figure 5Q16-1 in which a semiconductor has two contacts at its ends and is conducting an electric current *I*. We assume that the cold junction is at a temperature T_c and the hot junction is at T_h and that there is a temperature difference of $\Delta T = T_h - T_c$ between the two ends of the semiconductor. The current *I* flowing through the cold junction absorbs Peltier heat at a rate Q'_P, given by

$$Q_P' = \Pi I \qquad [1]$$

where Π is the Peltier coefficient for the junction between the metal and semiconductor. The current *I* flowing through the semiconductor generates heat due to the Joule heating of the semiconductor. The rate of Joule heat generated through the bulk of the semiconductor is

$$Q'_J = \left(\frac{L}{\sigma A}\right) I^2 \qquad [2]$$

We assume that half of this heat flows to the cold junction.

In addition there is heat flow from the hot to the cold junction through the semiconductor, given by the thermal conduction equation

$$Q'_{TC} = \left(\frac{A\kappa}{L}\right)\Delta T \qquad [3]$$

The net rate of heat absorption (cooling rate) at the cold junction is then

$$Q'_{\text{net cool}} = Q'_P - {}^1/_2 Q'_J - Q'_{TC} \qquad [4]$$

By substituting from Equations 1 to 3 into Equation 4, obtain the net cooling rate in terms of the current *I*. Then by differentiating $Q'_{\text{net cool}}$ with respect to current, show that maximum cooling is obtained when the current is

$$I_m = \left(\frac{A}{L}\right)\Pi\sigma \qquad [5]$$

and the maximum cooling is

$$Q'_{\text{max cool}} = \frac{A}{L}\left[\frac{1}{2}\Pi^2\sigma - \kappa\Delta T\right] \qquad [6]$$

Under steady state operating conditions, the temperature difference, ΔT, reaches a steady-state value and the net cooling rate at the junction is then zero (ΔT is constant). From Equation 6 show that the maximum temperature difference achievable is

$$\Delta T_{\max} = \frac{1}{2}\frac{\Pi^2\sigma}{\kappa} \qquad \textit{Maximum temperature difference} \qquad [7]$$

The quantity $\Pi^2\sigma/k$ is defined as the **figure of merit** (FOM) for the semiconductor as it determines the maximum ΔT achievable. The same expression also applies to metals, though we will not derive it here.

Use Table 5Q16-1 to determine the FOM for various materials listed therein and discuss the significance of your calculations. Would you recommend a thermoelectric cooler based on a metal-to-metal junction?

Table 5Q16-1

Material	Π **V**	ρ **Ω m**	κ **W m^{-1} K^{-1}**	**FOM**
n-Bi_2Te_3	6.0×10^{-2}	10^{-5}	1.70	
p-Bi_2Te_3	7.0×10^{-2}	10^{-5}	1.45	
Cu	5.5×10^{-4}	1.7×10^{-8}	390	
W	3.3×10^{-4}	5.5×10^{-8}	167	

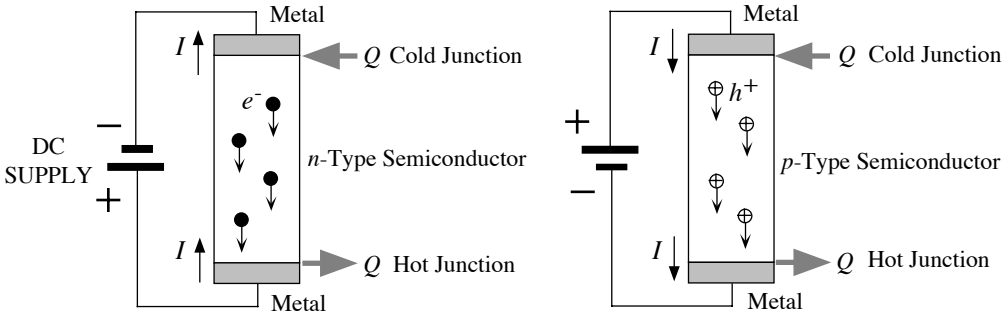


Figure 5Q16-1 When a dc current is passed through a semiconductor to which metal contacts have been made, one junction absorbs heat and cools (the cold junction) and the other releases heat and warms (the hot junction).

Solution

The net rate of heat absorption or cooling, Q', is given by Equation 4:

$$Q'_{\text{net cool}} = Q'_P - {}^1/_2 Q'_J - Q'_{TC}$$

Substituting for Q_p', Q_J', and Q_{TC}' as given by equations 1 to 3:

$$Q'_{\text{net cool}} = \Pi I - \frac{LI^2}{2\sigma A} - \frac{A\kappa\Delta T}{L}$$

Differentiating with respect to current:

$$\frac{dQ'_{\text{net cool}}}{dI} = \Pi - \frac{LI}{\sigma A}$$

This rate is maximum at the maximum current (I_m), when $dQ'_{\text{net cool}}/dI = 0$.

$$0 = \Pi - \frac{LI_m}{\sigma A}$$

$$\therefore \quad I_m = \left(\frac{A}{L}\right)\Pi\sigma$$

giving Equation 5. This expression can be then substituted back into the equation for $Q'_{\text{net cool}}$ to find the maximum cooling:

$$Q'_{\text{max cool}} = \Pi I_m - \frac{LI_m{}^2}{2\sigma A} - \frac{A\kappa\Delta T}{L}$$

$$\therefore \quad Q'_{\text{max cool}} = \frac{A\sigma\Pi^2}{L} - \frac{A\sigma\Pi^2}{2L} - \frac{A\kappa\Delta T}{L} = \frac{A\sigma\Pi^2}{2L} - \frac{A\kappa\Delta T}{L}$$

$$\therefore \quad Q'_{\text{max cool}} = \frac{A}{L}\left(\frac{\sigma\Pi^2}{2} - \kappa\Delta T\right)$$

which is the desired equation. In the steady state, the net cooling rate will be zero and the maximum temperature difference will be ΔT_{max}:

$$\frac{A}{L}\left(\frac{\sigma\Pi^2}{2} - \kappa\Delta T_{\text{max}}\right) = 0$$

$$\therefore \qquad \Delta T_{\max} = \frac{\sigma \Pi^2}{2\kappa}$$

The figure of merit (FOM) of the materials listed can now be found. From Table 5Q16-1, for *n*-Bi_2Te_3, $\Pi = 0.06$ V, $\sigma = 1/\rho = 1/(10^{-5}\ \Omega\ \text{m}) = 10^5\ \Omega^{-1}\ \text{m}^{-1}$, and $\kappa = 1.7\ \text{W m}^{-1}\ \text{K}^{-1}$. Substituting:

$$\textbf{FOM} = \frac{\sigma \Pi^2}{\kappa} = \frac{\left(10^5\ \Omega^{-1}\ \text{m}^{-1}\right)(0.06\ \text{V})^2}{\left(1.7\ \text{W m}^{-1}\ \text{K}^{-1}\right)} = 212\ \text{K}$$

The values for all materials are listed in Table 5Q16-2:

Table 5Q16-2 Summarized values for FOM.

Material	FOM K
n-Bi_2Te_3	**212**
p-Bi_2Te_3	**338**
Cu	**0.0456**
W	**0.0119**

Obviously metals have the worst FOM and semiconductors have the best FOM. Metal - metal junctions would obviously not make practical Peltier coolers!

*5.17 Seebeck coefficient of semiconductors and thermal drift in semiconductor devices

Consider an *n*-type semiconductor that has a temperature gradient across it. The right end is hot and the left end is cold, as depicted in Figure 5Q17-1. There are more energetic electrons in the hot region than in the cold region. Consequently, electron diffusion occurs from hot to cold regions, which immediately exposes negatively charged donors in the hot region and therefore builds up an internal field and a built-in voltage, as shown in the Figure 5Q17-1. Eventually an equilibrium is reached when the diffusion of electrons is balanced by their drift driven by the built-in field. The net current must be zero. The Seebeck coefficient (or thermoelectric power) *S* measures this effect in terms of the voltage developed as a result of an applied temperature gradient as

$$S = \frac{dV}{dT} \qquad [1]$$

a. How is the Seebeck effect in a *p*-type semiconductor different than that for an *n*-type semiconductor when both are placed in the same temperature gradient in Figure 5Q17-1? Recall that the sign of the Seebeck coefficient is the polarity of the voltage at the cold end with respect to the hot end (see Section 4.8.2 in the textbook)

b. Given that for an *n*-type semiconductor,

$$S_n = -\frac{k}{e}\left[2 + \frac{(E_c - E_F)}{kT}\right] \qquad \begin{matrix}\textit{Seebeck coefficient for} \\ \textit{n – type semiconductor}\end{matrix} \qquad [2]$$

what are typical magnitudes for S_n in Si doped with 10^{14} and 10^{16} donors cm^{-3}? What is the significance of S_n at the semiconductor device level?

c. Consider a *pn* junction Si device that has the *p*-side doped with 10^{18} acceptors cm^{-3} and the *n*-side doped 10^{14} donors cm^{-3}. Suppose that this *pn* junction forms the input stage of an op amp with a large gain, say 100. What will be the output signal if a small thermal fluctuation gives rise to a 1 ˚C temperature difference across the *pn* junction?

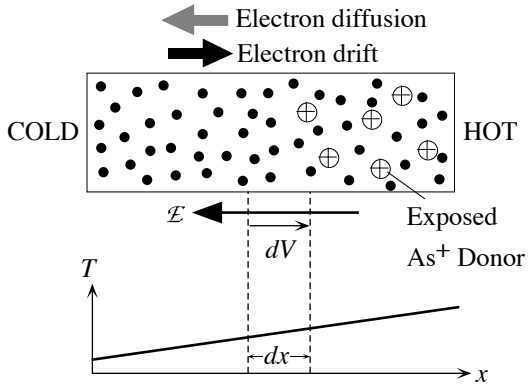


Figure 5Q17-1 In the presence of a temperature gradient, there is an internal field and a voltage difference. The Seebeck coefficient is defined as dV/dT, the potential difference per unit temperature.

Solution

a In a *p*-type semiconductor (SC), we can assume that we only have holes as the mobile charge carriers. The acceptors are negatively charged. The same temperature gradient as in Figure 5Q17-1 results in the diffusion of holes (instead of electrons) from the hot to cold end. This exposes negative acceptors in the hot region (instead of positive charge as in the *n*-type SC). Thus in a *p*-type semiconductor, the Seebeck effect has the reverse sign, or the polarity of the Seebeck voltage is reversed with respect to that for an *n*-type SC for the same temperature gradient. This provides a convenient way to identify whether a semiconductor is doped *n*-type or *p*-type. If the hot side is positive with respect to the cold side then the semiconductor is *n*-type. If the hot side is negative then it is *p*-type. The simplest test is to touch the test leads of a voltmeter (1-10 mV range) to a semiconductor with one lead made hot. In reality, the semiconductor and the copper lead form a thermocouple but the Seebeck coefficient of the semiconductor is much greater than that of the metal lead.

Note: The potential of the cold side with respect to the hot side is taken as the sign of the Seebeck voltage and hence the Seebeck coefficient *S*. Thus, if electrons diffuse from hot to cold, then the Seebeck voltage (and hence *S*) is negative, and if holes diffuse from hot to cold then the Seebeck voltage is positive. Equation 2 only gives the magnitude of *S*.

b The following is a proof of Equation 2 and is given as reference. We consider a small distance, dx, at x where the temperature is T as shown Figure 5Q17-1. The temperature difference across dx is dT. Suppose that we take a single electron from a hot to a cold region across dx. Electrical work done is simply $-edV$. Suppose that we are measuring the energy of the electron from E_F. Then the energy of the electron at x is $(E_c - E_F) + {}^3/_2 kT$ (or, $PE + KE$) so that when we take it across dT, its energy changes by

$$d(E_c - E_F) + {}^3/_2 kdT$$

We should remember that as the temperature increases so does $(E_c - E_F)$. The above energy change must come from the electrical work done ($-edV$) so that

$$-edV = d(E_c - E_F) + {}^3/_2 kdT$$

We can divide through by dT and then use the definition of S_n in Eqn. 1 to obtain

$$S_n = \pm\frac{1}{e}\left[\frac{3k}{2} + \frac{d(E_c - E_F)}{dT}\right]$$

From $$n = N_c \exp\left[-\frac{(E_c - E_F)}{kT}\right]$$

we have $$(E_c - E_F) = kT\ln(N_c/n)$$

The temperature dependence of the term in parenthesis is slower than the pre-exponential term kT so that we can differentiate this quite simply, to obtain,

$$\frac{d(E_c - E_F)}{dT} = \frac{(E_c - E_F)}{T}$$

Substituting in the expression for S_n we get,

$$S_n = \pm\frac{1}{e}\left[\frac{3k}{2} + \frac{E_c - E_F}{T}\right]$$

A more rigorous analysis makes the numerical factor 2 instead of $^3/_2$ so that

$$S_n = \pm\frac{k}{e}\left[2 + \frac{E_c - E_F}{kT}\right] \qquad \textit{(Seebeck Coefficient)} \qquad [2]$$

S_n in Equation 2 depends on $(E_c - E_F)$ which depends on the doping concentration as shown in the table below. $E_c - E_F = \Delta E$ can be found as follows (sample calculation is for $N_d = 10^{14}$ cm^{-3}):

$$N_d = N_c \exp\left(-\frac{\Delta E}{kT}\right)$$

$$\therefore \quad \Delta E = -kT \ln\left(\frac{N_d}{N_c}\right)$$

$$\therefore \quad \Delta E = -\left(1.381\times10^{-23}\ \text{J/K}\right)(300\ \text{K})\ln\left(\frac{10^{20}\ \text{m}^{-3}}{2.8\times10^{25}\ \text{m}^{-3}}\right)\left(\frac{1}{1.602\times10^{-19}\ \text{J/eV}}\right)$$

$$\therefore \quad \Delta E = 0.3244\ \text{eV}$$

Substitute to find S_n:

$$S_n = \pm\left(\frac{1.381\times10^{-23}\ \text{J/K}}{1.602\times10^{-19}\ \text{C}}\right)\left[2 + \frac{(0.3244\ \text{eV})\left(1.602\times10^{-19}\ \text{J/eV}\right)}{\left(1.381\times10^{-23}\ \text{J/K}\right)(300\ \text{K})}\right]$$

$$\boldsymbol{S_n = -1.25\ \text{mV K}^{-1}}$$

Table 5Q17-1 Summarized values for S_n.

N_d (cm^{-3})	$(E_c - E_F)$ (eV)	S_n (mV K^{-1})
1×10^{14}	0.3244	**−1.25**
1×10^{16}	0.2053	**−0.857**
1×10^{18}	0.08618	**−0.460**

c In a typical semiconductor device such as an IC there will be many junctions between differently doped materials. A small local temperature fluctuation can give rise to a temperature induced signal of the order of few millivolts. In a dc op amp of large gain this will appear as a spurious signal in the output. One design cure is to use two different input devices at similar temperatures and use the output signal difference between them. From Table 5Q17-1 above, the p-side with $N_a = 10^{18}$ cm^{-3} has S_p = +0.46 mV/K; and n-side with $N_d = 10^{14}$ cm^{-3} has S_n = -1.25 mV/K.

For ΔT = 1 °C = 1 K, the magnitudes of the individual Seebeck voltages are

$$|V_p| = S_p\,\Delta T = (0.46\ \text{mV/K})(1\ \text{K}) = 0.46\ \text{mV}$$

$$|V_n| = S_n\,\Delta T = (1.25\ \text{mV/K})(1\ \text{K}) = 1.25\ \text{mV}$$

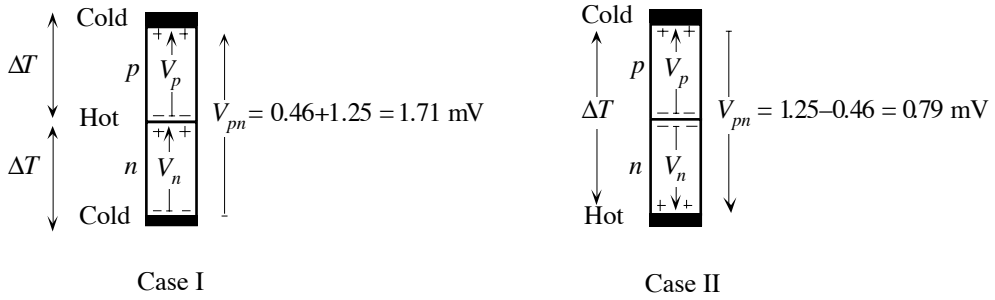


Figure 5Q17-2

There are two possibilities for the temperature variation ΔT. In Case I, ΔT occurs between the junction and the electrodes. A practical example would be the thermal lag between the electrodes and the junction. If the electroded ends were to be cooled, the junction temperature will lag behind by ΔT due to the finite thermal conductivity of the semiconductor. Case II involves a temperature fluctuation across the whole *pn* junction. Thus

Case I $\quad V_{pn} = |V_p| + |V_n| = 0.46 + 1.25 = 1.71$ mV

Case II $\quad V_{pn} = |V_p| + |V_n| = 0.46 - 1.25 = 0.79$ mV

The output drift in the **worst case (Case I)** from the op amp is 100×1.71 mV = **171 mV** or **0.171 V** (substantial drift). The output drift in the **best case (Case II)** from the op amp is 100×0.79 mV = **79 mV** or **0.079 V**.

Before I came here I was confused about this subject. Having listened to your lecture I am still confused. But on a higher level.
Enrico Fermi (1901-1954; Nobel Laureate, 1938)

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Chapter 6: Solutions

If current trends endure, future computers will consist of a single chip. No one will have the foggiest idea what is on it. Somewhere in the basement of Intel or its successor will be a huge computer file with chip's listing. The last electrical engineer will sit nearby, handcuffed to the disk drive in a scene out of *Ben Hur*. That engineer will be extremely well paid, and his or her every demand will be immediately satisfied. That engineer will be last keeper of the secret of the universe: $E = IR$.

Robert Lucky (*Spectrum*, *IEEE,* May 1998 Issue, pg. 21)

6.1 The *pn* junction

Consider an abrupt Si pn^+ junction that has 10^{15} acceptors cm^{-3} on the *p*-side and 10^{19} donors on the *n*-side. The minority carrier recombination times are $\tau_e = 490$ ns for electrons in the *p*-side and $\tau_h = 2.5$ ns for holes in the *n* -side. The cross-sectional area is 1 mm^2. Assuming a long diode, calculate the current, *I*, through the diode at room temperature when the voltage, *V*, across it is 0.6 V. What are *V*/*I* and the incremental resistance (r_d) of the diode and why are they different?

Solution

Consider temperature, T = 300 K. Then kT/e = 0.02586 V.

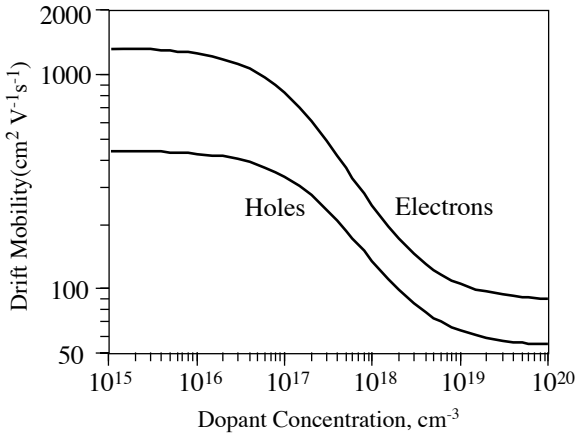


Figure 6Q1-1 The variation of the drift mobility with dopant concentration in Si for electrons and holes at 300 K.

The general expression for the diffusion length is $L = \sqrt{[D\tau]}$ where D is the diffusion coefficient and τ is the carrier lifetime. D is related to the mobility of the carrier, μ, via the Einstein relationship, $D/\mu = kT/e$. We therefore need to know μ to calculate D and hence L. Electrons diffuse in the *p*-region and holes in the *n*-region so that we need μ_e in the presence of N_a acceptors and μ_h in the presence of N_d donors. From the drift mobility, μ vs. dopant concentration graph for silicon (see Figure 6Q1-1) we have the following: with $N_a = 10^{15}$ cm^{-3}, $\mu_e \approx 1350$ cm^2 V^{-1} s^{-1}, and with $N_d = 10^{19}$ cm^{-3},and $\mu_h \approx 65$ cm^2 V^{-1} s^{-1}. Thus,

$$D_e = kT\mu_e/e \approx (0.02586 \text{ V})(1350 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}) = 34.91 \text{ cm}^2 \text{ s}^{-1}$$

and $D_h = kT\mu_h/e \approx (0.02586\text{ V})(65\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}) = 1.681\text{ cm}^2\text{ s}^{-1}$

The diffusion lengths are

$$L_e = \sqrt{[D_e\tau_e]} = \sqrt{[(34.91\text{ cm}^2\text{ s}^{-1})(490\times10^{-9}\text{ s})]} = 4.136\times10^{-3}\text{ cm or }41.36\ \mu\text{m}$$

and $L_h = \sqrt{[D_h\tau_h]} = \sqrt{[(1.681\text{ cm}^2\text{ s}^{-1})(2.5\times10^{-9}\text{ s})]} = 6.483\times10^{-5}\text{ cm or }0.6483\ \mu\text{m}$

To calculate the forward current when $V = 0.6$ V we need to evaluate both the diffusion and recombination components of the current. It is likely that the diffusion component will exceed the recombination component at this forward bias (this can be easily verified). Assuming that the forward current is due to minority carrier diffusion in neutral regions,

$$I = I_{so}[\exp(eV/kT) - 1] \approx I_{so}\exp(eV/kT) \text{ for } V >> kT/e\ (\approx 0.02586\text{ V})$$

where $I_{so} = AJ_{so} = Aen_i^2[(D_h/(L_hN_d)) + (D_e/(L_eN_a))] \approx Aen_i^2D_e/(L_eN_d)$

as $N_d >> N_a$. In other words, the current is mainly due to the diffusion of electrons in the *p*-region. Thus ($n_i = 1.45\times10^{10}$ cm^{-3} from Table 5.1 in the textbook):

$$I_{soe} = \frac{(1\times10^{-2}\text{ cm}^2)(1.602\times10^{-19}\text{ C})(1.45\times10^{10}\text{ cm}^{-3})^2(34.91\text{ cm}^2\text{ s}^{-1})}{(4.14\times10^{-3}\text{ cm})(10^{15}\text{ cm}^{-3})}$$

$\therefore$ $I_{soe} = 2.840\times10^{-12}$ A or 2.840 pA

and $$I_{soh} = \frac{(1\times10^{-2}\text{ cm}^2)(1.602\times10^{-19}\text{ C})(1.45\times10^{10}\text{ cm}^{-3})^2(1.681\text{ cm}^2\text{ s}^{-1})}{(6.48\times10^{-5}\text{ cm})(10^{19}\text{ cm}^{-3})}$$

$\therefore$ $I_{soh} = 8.738\times10^{-16}$ A or 0.0008738 pA

Clearly, as expected, $I_{soe} >> I_{soh}$ and therefore $I_{so} = I_{soe}$.

The forward current is then

$$I = I_{so}\exp[eV/(kT)] = (2.840\times10^{-12}\text{ A})\exp[(0.6\text{ V})/(0.02586\text{ V})]$$

$\therefore$ $\boldsymbol{I \approx}$ **0.0339 A** or **33.9 mA**

$\therefore$ $\boldsymbol{R} = V/I = (0.6\text{ V}) / (0.0339\text{ A}) \approx$ **17.7 Ω**

Author's Note: This is not a true resistance but V/I for a nonlinear device.

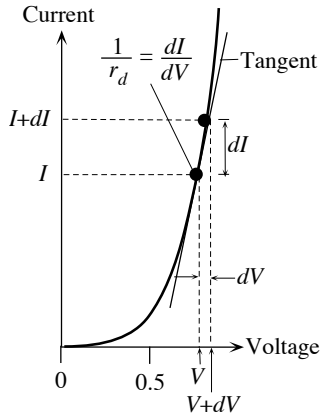


Figure 6Q1-2 The dynamic resistance of the diode is defined as dV/dI, which is the inverse tangent at I.

The incremental diode resistance is (Equation. 6.27 in the textbook):

$$\boldsymbol{r_d} = dV/dI = (kT/e)/I = (0.02586 \text{ V}) / (0.0339 \text{ A}) \approx \mathbf{0.763\ \Omega}$$

See Figure 6Q1-2, for the meaning for r_d ($r_d = \delta V/\delta I$) which is very different than V/I.

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*6.2 The Si *pn* junction

Consider a long *pn* junction diode with an acceptor doping, N_a, of 10^{18} cm^{-3} on the *p*-side and donor concentration of N_d on the *n*-side. The diode is forward biased and has a voltage of 0.6 V across it. The diode cross-sectional area is 1 mm^2. The minority carrier recombination time, τ, depends on the dopant concentration, N_{dopant} (cm^{-3}), through the following approximate relation

$$\tau = \frac{5\times10^{-7}}{\left(1+2\times10^{-17}N_{\text{dopant}}\right)}$$

a. Suppose that $N_d = 10^{15}$ cm^{-3}. Then the depletion layer extends essentially into the *n*-side and we have to consider minority carrier recombination time, τ_h, in this region. Calculate the diffusion and recombination contributions to the total diode current. What is your conclusion?

b. Suppose that $N_d = N_a = 10^{18}$ cm^{-3}. Then W extends equally to both sides and, further, $\tau_e = \tau_h$. Calculate the diffusion and recombination contributions to the diode current. What is your conclusion?

Solution

Consider temperature, T = 300 K. $kT/e = kT/e$ = 0.02586 V.

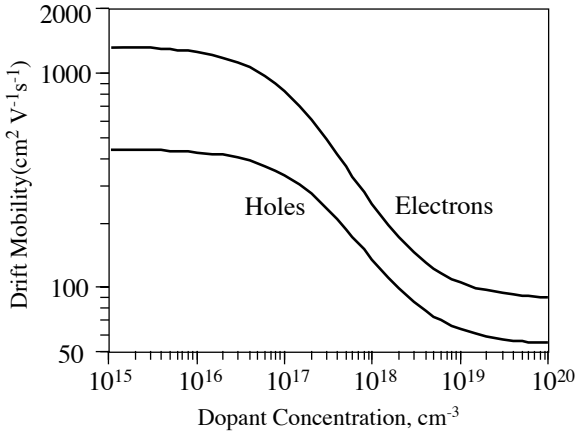


Figure 6Q2-1 The variation of the drift mobility with dopant concentration in Si for electrons and holes at 300 K.

a This is a p^+n diode: $N_d = 10^{15}$ cm^{-3}. Hole lifetime τ_h in the *n*-side is

$$\tau_h = \frac{5\times10^{-7}}{\left(1+2\times10^{-17}N_{\text{dopant}}\right)} = \frac{5\times10^{-7}}{\left(1+2\times10^{-17}\times10^{15}\ \text{cm}^{-3}\right)} = 490.2 \text{ ns}$$

and, using the same equation with $N_{\text{dopant}} = 10^{18}$ cm^{-3}, electron lifetime in the *p*-side is $\tau_e = 23.81$ ns.

I. Diffusion component of diode current

From Figure 6Q2-1, with $N_a = 10^{18}$ cm^{-3}, $\mu_e \approx 250$ cm^2 V^{-1} s^{-1}, and with $N_d = 10^{15}$ cm^{-3}, $\mu_h \approx 450$ cm^2 V^{-1} s^{-1}. Thus:

$$D_e = kT\mu_e/e \approx (0.02586 \text{ V})(250 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}) = 6.465 \text{ cm}^2 \text{ s}^{-1}$$

and $$D_h = kT\mu_h/e \approx (0.02586 \text{ V})(450 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}) = 11.64 \text{ cm}^2 \text{ s}^{-1}$$

The diffusion lengths are then:

$$L_e = \sqrt{[D_e\tau_e]} = \sqrt{[6.465 \text{ cm}^2 \text{ s}^{-1})(23.81 \times 10^{-9} \text{ s})]} = 3.923 \times 10^{-4} \text{ cm, or } 3.923 \ \mu\text{m}$$

and $$L_h = \sqrt{[D_h\tau_h]} = \sqrt{[(11.64 \text{ cm}^2 \text{ s}^{-1})(490.2 \times 10^{-9} \text{ s})]} = 2.389 \times 10^{-3} \text{ cm, or } 23.89 \ \mu\text{m}$$

The diffusion component of the current is

$$I = I_{diff} = I_{so}[\exp(eV/(kT)) - 1] \approx I_{so}\exp(eV/(kT)) \text{ for } V >> kT/e \ (\approx 0.02586 \text{ V})$$

where $$I_{so} = AJ_{so} = Aen_i^2[(D_h/(L_hN_d)) + (D_e/(L_eN_a))] \approx Aen_i^2D_h/(L_hN_d)$$

as $N_a >> N_d$. In other words, the current is mainly due to the diffusion of holes in the *n*-region. Thus:

$$I_{so} = \frac{(1\times10^{-2} \text{ cm}^2)(1.602\times10^{-19} \text{ C})(1.45\times10^{10} \text{ cm}^{-3})^2(11.64 \text{ cm}^2 \text{ s}^{-1})}{(2.389\times10^{-3} \text{ cm})(1\times10^{15} \text{ cm}^{-3})}$$

$\therefore$ $$I_{so} = 1.641 \times 10^{-12} \text{ A or } 1.641 \text{ pA}$$

The forward current due to diffusion is

$$I_{diff} = I_{so}\exp[(eV/(kT)] = (1.641 \times 10^{-12} \text{ A})\exp[(0.6 \text{ V})/(0.02586 \text{ V})]$$

$\therefore$ $$\boldsymbol{I_{diff} = 0.0196 \text{ A}} \text{ or } \boldsymbol{19.6 \text{ mA}}$$

II. Recombination component

The built-in potential is (where n_i is the intrinsic concentration, found in Table 5.1 in the textbook):

$$V_o = (kT/e)\ln(N_dN_a/n_i^2) = (0.02586 \text{ V})\ln[(10^{18} \text{ cm}^{-3} \times 10^{15} \text{ cm}^{-3})/(1.45 \times 10^{10} \text{ cm}^{-3})^2]$$

$\therefore$ $$V_o = 0.7549 \text{ V}$$

The depletion region width *W* is mainly on the *n*-side (ε_r of Si is 11.9 from Table 5.1 in the textbook).

$$W = \left[\frac{2\varepsilon(N_a + N_d)(V_o - V)}{eN_aN_d}\right]^{1/2} \approx \left[\frac{2\varepsilon(V_o - V)}{eN_d}\right]^{1/2}$$

$\therefore$ $$W = \left[\frac{2(11.9)(8.854\times10^{-12} \text{ F m}^{-1})(0.7549 \text{ V} - 0.6 \text{ V})}{(1.602\times10^{-19} \text{ C})(10^{21} \text{ m}^{-3})}\right]^{1/2}$$

i.e. $$W \approx 0.4514 \times 10^{-6} \text{ m or } 0.4514 \ \mu\text{m}$$

The recombination time τ_r here is not necessarily τ_h but let us assume that it is roughly. Then $\tau_r = \tau_h$ which leads to:

$$I_{ro} = \frac{AeWn_i}{2\tau_r} = \frac{(1\times10^{-2} \text{ cm}^2)(1.602\times10^{-19} \text{ C})(0.4514\times10^{-4} \text{ cm})(1.45\times10^{10} \text{ cm}^{-3})}{2(490\times10^{-9} \text{ s})}$$

$\therefore$ $$I_{ro} = 1.070 \times 10^{-9} \text{ A}$$

The forward current due to recombination is

$$I_{recom} = I_{ro}\exp(eV/2kT) = (1.070 \times 10^{-9}\text{ A})\exp[(0.6\text{ V})/(2 \times 0.02586\text{ V})]$$

$\therefore$ $\boldsymbol{I_{recom} = 1.17 \times 10^{-4}\text{ A}}$ or $\mathbf{0.117\ mA}$

Clearly, the diffusion component dominates the recombination component.

b This is a symmetrical *pn* diode: $N_d = N_a = 10^{18}\text{ cm}^{-3}$. Hole lifetime τ_h in the *n*-side is

$$\tau_h = \tau_e = \frac{5\times10^{-7}}{\left(1+2\times10^{-17}N_{\text{dopant}}\right)} = \frac{5\times10^{-7}}{\left(1+2\times10^{-17}\times10^{18}\ \text{cm}^{-3}\right)} = 23.81 \times 10^{-9}\text{ s}$$

I. Diffusion component

From Figure 6Q2-1, with $N_a = 10^{18}\text{ cm}^{-3}$, $\mu_e \approx 250\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$, and with $N_d = 10^{18}\text{ cm}^{-3}$, $\mu_h \approx 130\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$. Thus:

Thus $D_e = kT\mu_e/e \approx (0.02586\text{ V})(250\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}) = 6.465\text{ cm}^2\text{ s}^{-1}$

and $D_h = kT\mu_h/e \approx (0.02586\text{ V})(130\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}) = 3.362\text{ cm}^2\text{ s}^{-1}$

Diffusion lengths are

$$L_e = \sqrt{[D_e\tau_e]} = \sqrt{[6.465\text{ cm}^2\text{ s}^{-1})(23.81 \times 10^{-9}\text{ s})]} = 3.923 \times 10^{-4}\text{ cm or } 3.923\ \mu\text{m}$$

and $L_h = \sqrt{[D_h\tau_h]} = \sqrt{[(3.362\text{ cm}^2\text{ s}^{-1})(23.81 \times 10^{-9}\text{ s})]} = 2.829 \times 10^{-4}\text{ cm or } 2.829\ \mu\text{m}$

The diffusion component of the current is

$$I_{diff} = I = I_{so}[\exp(eV/kT) - 1] \approx I_{so}\exp(eV/kT) \text{ for } V >> kT/e\ (\approx 0.02586\text{ V})$$

where $I_{so} = AJ_{so} = Aen_i^2[(D_h/(L_hN_d)) + (D_e/(L_eN_a))]$

$$I_{so} = \left(1\times10^{-2}\ \text{cm}^2\right)\left(1.602\times10^{-19}\ \text{C}\right)\left(1.45\times10^{10}\ \text{cm}^{-3}\right)^2 \times\left[\frac{\left(3.362\ \text{cm}^2\ \text{s}^{-1}\right)}{\left(2.829\times10^{-4}\ \text{cm}\right)\left(10^{18}\ \text{cm}^{-3}\right)} + \frac{\left(6.465\ \text{cm}^2\ \text{s}^{-1}\right)}{\left(3.923\times10^{-4}\ \text{cm}\right)\left(10^{18}\ \text{cm}^{-3}\right)}\right]$$

$\therefore$ $I_{so} = 9.554 \times 10^{-15}\text{ A}$

The forward current due to diffusion is

$$I_{diff} = I_{so}\exp(eV/kT) = (9.554 \times 10^{-15}\text{ A})\exp(0.6\text{ V}/0.02586\text{ V})$$

$\therefore$ $\boldsymbol{I_{diff} = 1.14 \times 10^{-4}\text{ A}}$ or $\mathbf{0.114\ mA}$

II. Recombination component

The built-in potential is

$$V_o = (kT/e)\ln(N_dN_a/n_i^2) = (0.02586\text{ V})\ln[(10^{18}\text{ cm}^{-3} \times 10^{18}\text{ cm}^{-3})/(1.45 \times 10^{10}\text{ cm}^{-3})^2]$$

$\therefore$ $V_o = 0.9335\text{ V}$

The depletion region width is symmetrical about the junction as $N_a = N_d$.

$$W = \left[\frac{2\varepsilon\left(N_a + N_d\right)\left(V_o - V\right)}{eN_aN_d}\right]^{1/2} = \left[\frac{4\varepsilon\left(V_o - V\right)}{eN_d}\right]^{1/2}$$

$$\therefore \qquad W = \left[\frac{4(11.9)\left(8.854\times10^{-12}\ \text{F m}^{-1}\right)(0.9335\ \text{V} - 0.6\ \text{V})}{\left(1.602\times10^{-19}\ \text{C}\right)\left(10^{24}\ \text{cm}^{-3}\right)}\right]^{1/2}$$

i.e. $W \approx 2.962 \times 10^{-8}$ m or 0.02962 μm

The recombination time τ_r here is not necessarily τ_h but let us assume that it is roughly. Then $\tau_r = \tau_h = \tau_e$ which leads to:

$$I_{ro} = \frac{AeWn_i}{2\tau_r} = \frac{\left(1\times10^{-2}\ \text{cm}^2\right)\left(1.602\times10^{-19}\ \text{C}\right)\left(0.02962\times10^{-4}\ \text{cm}\right)\left(1.45\times10^{10}\ \text{cm}^{-3}\right)}{2\left(23.81\times10^{-9}\ \text{s}\right)}$$

$\therefore \qquad I_{ro} = 1.457 \times 10^{-9}$ A

The forward current due to recombination is

$$I_{recom} = I_{ro}\exp(eV/2kT) = (1.457 \times 10^{-9}\ \text{A})\exp[(0.6\ \text{V})/(2 \times 0.02586\ \text{V})]$$

$\therefore \qquad \boldsymbol{I_{recom} = 1.59 \times 10^{-4}\ \text{A}}$ or **0.159 mA**

Clearly, the recombination component dominates the diffusion component.

6.3 Junction capacitance of a *pn* junction

The capacitance (C) of a reverse-biased abrupt Si p^+n junction has been measured as a function of the reverse bias voltage, V_r, as listed in Table 6Q3-1. The *pn* junction cross-sectional area is 500 μm × 500 μm. By plotting $1/C^2$ versus V_r, obtain the built-in potential, V_o, and the donor concentration, N_d, in the *n*-region. What is N_a?

Table 6Q3-1 Capacitance at various values of reverse bias (V_r)

V_r (V)	1	2	3	5	10	15	20
C (pF)	38.3	30.7	26.4	21.3	15.6	12.9	11.3

Solution

The depletion capacitance of an abrupt *pn* junction is given by Equation. 6.24 (in the textbook),

$$C_{dep} = \frac{\varepsilon A}{W} = \frac{A}{\left(V_o - V\right)^{1/2}}\left[\frac{e\varepsilon\left(N_aN_d\right)}{2\left(N_a + N_d\right)}\right]^{1/2}$$

Taking reverse bias, $V = -V_r$ and $N_a >> N_d$, we get

$$C_{dep} = \frac{A}{\left(V_o + V_r\right)^{1/2}}\left[\frac{e\varepsilon N_d}{2}\right]^{1/2}$$

Rearranging we get: $$\frac{1}{C_{dep}^2} = \left(\frac{2}{e\varepsilon N_d A^2}\right)V_r + \left(\frac{2}{e\varepsilon N_d A^2}\right)V_o$$

which is like: $y = mx + c$

where $y = 1/C_{dep}^2$, $x = V_r$, m is the slope and c is the intercept on the y-axis.

Thus a plot of $y = 1/C_{dep}^2$ vs. V_r should be a straight line $y = mx + c$, with an intercept $V_r = -V_o$. The data is plotted below. The best straight line gives a slope of $m = 3.776 \times 10^{20}$ and a y-intercept of b =

3.119×10^{20}, or $1/C^2 = (3.776 \times 10^{20})V_r + 3.119 \times 10^{20}$. **This gives an *x*-intercept $-V_o$ so that V_o = 0.826 V.**

Thus: $$\text{Slope} = m = \left(\frac{2}{e\varepsilon N_d A^2}\right) = 3.776 \times 10^{20} \ \text{F}^{-2}\ \text{V}^{-1}$$

Substituting $\varepsilon = \varepsilon_o e_r$, $\varepsilon_r = 11.9$ (Table 5.1, pg. 334 in the textbook), $e = 1.602 \times 10^{-19}$ C, $A = (500 \times 10^{-6}\ \text{m})^2$,

i.e. $$N_d = \frac{2}{me\varepsilon A^2}$$

$\therefore$ $$N_d = \frac{2}{(3.776 \times 10^{20}\ \text{F}^{-2}\ \text{V}^{-1})(1.602 \times 10^{-19}\ \text{C})(11.9)(8.854 \times 10^{-12}\ \text{F/m})(500 \times 10^{-6}\ \text{m})^4}$$

we can calculate: $\boldsymbol{N_d = 5.02 \times 10^{21}\ \text{m}^{-3}}$ or $\boldsymbol{5.02 \times 10^{15}\ \text{cm}^{-3}}$

The built-in potential is, $V_o = (kT/e)\ln(N_d N_a / n_i^2)$ thus,

$$N_a = \frac{n_i^2}{N_d}\exp\left(\frac{eV_o}{kT}\right) = \frac{(1.45 \times 10^{10}\ \text{cm}^{-3})^2}{(5.02 \times 10^{15}\ \text{cm}^{-3})}\exp\left(\frac{0.826\ \text{V}}{0.02586\ \text{V}}\right) \approx \boldsymbol{3.12 \times 10^{18}\ \text{cm}^{-3}}$$

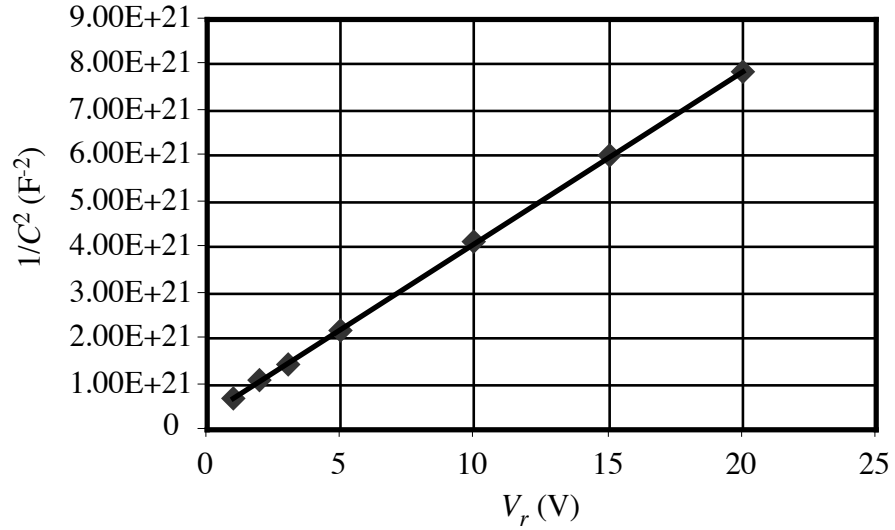


Figure 6Q3-1 Plot of $1/C^2$ versus reverse voltage.

6.4 Temperature dependence of diode properties

a. Consider the reverse current in a *pn* junction. Show that

$$\frac{\delta I_{rev}}{I_{rev}} \approx \left(\frac{E_g}{\eta kT}\right)\frac{\delta T}{T}$$

where $\eta = 2$ for Si and GaAs, in which thermal generation in the depletion layer dominates the reverse current, and $\eta = 1$ for Ge, in which the reverse current is due to minority carrier diffusion to the depletion layer. It is assumed that $E_g >> kT$ at room temperature. Order the semiconductors Ge, Si, and GaAs according to the sensitivity of the reverse current to temperature.

b. Consider a forward-biased *pn* junction carrying a *constant* current *I*. Show that the change in the voltage across the *pn* junction per unit change in the temperature is given by

$$\frac{dV}{dT} = -\frac{V_g - V}{T}$$

where $V_g = E_g/e$ is the energy gap expressed in volts. Calculate typical values for dV/dT for Ge, Si and GaAs assuming that, typically, V = 0.2 V for Ge, 0.6 V for Si, and 0.9 V for GaAs. What is your conclusion? Can one assume that, typically, $dV/dT \approx -2$ mV °C^{-1} for these diodes?

Solution

a The total reverse current is the sum of the diffusion and thermal generation components:

$$I_{rev} = A\left(\frac{eD_h}{L_h N_d} + \frac{eD_e}{L_e N_a}\right)n_i^2 + \frac{eAWn_i}{\tau_g} \tag{1}$$

Neglecting the temperature dependences of parameters other than n_i, the reverse current is,

$$I_{rev} = B_1 n_i^2 + B_2 n_i$$

where B_1 and B_2 are constants.

The intrinsic concentration is given by

$$n_i = \left(N_c N_v\right)^{1/2} \exp\left(-\frac{E_g}{2kT}\right) \tag{2}$$

Thus the reverse current can be written as:

$$I_{rev} = C_1 \exp\left(-\frac{E_g}{kT}\right) + C_2 \exp\left(-\frac{E_g}{2kT}\right)$$

where C_1 and C_2 are constants. The two exponential terms in the right hand side can be combined into one convenient expression that can be written as

$$I_{rev} = C \exp\left(-\frac{E_g}{\eta kT}\right) \tag{3}$$

where η may be 1 or 2 depending on whether the diffusion or the thermal generation contribution respectively dominates the reverse current. Taking the derivative of the reverse current with respect to temperature

$$\frac{dI_{rev}}{dT} = \frac{d}{dT} C \exp\left(-\frac{E_g}{\eta kT}\right) = \left(\frac{E_g}{\eta kT^2}\right) C \exp\left(-\frac{E_g}{\eta kT}\right)$$

i.e.

$$\frac{dI_{rev}}{dT} = \left(\frac{E_g}{\eta kT^2}\right) I_{rev}$$

so that

$$\frac{\delta I_{rev}}{I_{rev}} = \left(\frac{E_g}{\eta kT}\right)\frac{\delta T}{T} \tag{4}$$

This is the fractional change in the current ($\delta I_{rev}/I_{rev}$) due to a fractional change in temperature ($\delta T/T$). Small changes have been represented by differentials i.e. $dT \approx \delta T$ etc.

The ratio of E_g/η for Ge, Si and GaAs are 0.66, 0.55 and 0.71 respectively (E_g from Table 5.1, η = 1 for Ge and η = 2 for Si and GaAs). Therefore the fractional change in I_{rev} for Ge and GaAs will be slightly higher than that for Si. If we consider the actual change of current with temperature, it will be a function of I_{rev} as well as E_g and η. Since the reverse current for Ge is on the order of μA, for Si on the

order of nA and for GaAs on the order of pA, the sensitivity of I_{rev} to temperature can be written in the same order: Ge, Si and GaAs.

b Consider the forward bias current equation with $V >> kT/e$ (Total diode current equation, page 428 of the textbook):

$$I = A\left(\frac{eD_h}{L_h N_d} + \frac{eD_e}{L_e N_a}\right) n_i^2 \exp\left(\frac{eV}{kT}\right) + \left(\frac{eAW}{2\tau_r}\right) n_i \exp\left(\frac{eV}{2kT}\right) \qquad (5)$$

Defining $V_g = E_g/e$, Equation (2) can be written as

$$n_i^2 = \left(N_c N_v\right) \exp\left(-\frac{eV_g}{kT}\right)$$

then Eqn. (5) is: $$I = I_1 \exp\left[\frac{e\left(V - V_g\right)}{kT}\right] + I_2 \exp\left[\frac{e\left(V - V_g\right)}{2kT}\right]$$

which for convenience can be written as

$$I = I_1 \exp\left[\frac{e\left(V - V_g\right)}{\eta kT}\right]$$

where η may be 1 or 2 depending on whether the diffusion or the recombination contribution to the forward current is more dominant respectively.

Since the current is constant, the term in the exponential is constant as well.

$$\frac{e\left(V - V_g\right)}{\eta kT} = \text{Constant}$$

or $$\frac{\left(V - V_g\right)}{T} = C = \text{Constant}$$

Taking the derivative of the both side with respect to temperature:

$$\frac{-\left(V - V_g\right)}{T^2} + \left(\frac{1}{T}\right)\frac{dV}{dT} = 0$$

Rearranging: $$\frac{dV}{dT} = -\frac{V_g - V}{T} \qquad \text{(Constant current)} \qquad (6)$$

Typical values of dV/dT can be calculated at 300 K using values for $E_g = V_g/e$ from Table 5.1 (in the textbook):

For Ge: $$\frac{\boldsymbol{dV}}{\boldsymbol{dT}} = -\frac{0.66\text{ V} - 0.2\text{ V}}{300\text{ K}} = \textbf{-0.00153 V/K or V/°C}$$

For Si: $$\frac{\boldsymbol{dV}}{\boldsymbol{dT}} = -\frac{1.10\text{ V} - 0.6\text{ V}}{300\text{ K}} = \textbf{-0.00167 V/K or V/°C}$$

For GaAs: $$\frac{\boldsymbol{dV}}{\boldsymbol{dT}} = -\frac{1.42\text{ V} - 0.9\text{ V}}{300\text{ K}} = \textbf{-0.00173 V/K or V/°C}$$

This shows that the rate of change of voltage with temperature is very similar for all three semiconductors. It should also be noted that the actual values are slightly higher than those calculated above. Therefore we can simply assume that it is typically of the order of -2 mV/°C.

6.5 Avalanche breakdown

Consider a Si p^+n junction diode that is required to have an avalanche breakdown voltage of 25 V. Given the breakdown field $\mathcal{E}_{br}$ in Figure 6Q5-1, what should be the donor doping concentration?

Solution

The reverse breakdown voltage requirement is $V_{br} = 25$ V. From Table 5.1 (in the textbook), the relative permittivity of Si is $\varepsilon_r = 11.9$, and from Table 5.1 (in the textbook), the intrinsic concentration is $n_i = 1.45 \times 10^{16}$ m^{-3}. Figure 6Q5-1 below shows the relationship between the breakdown field $\mathcal{E}_{br}$ and the doping N_d.

$\mathcal{E}_{br}$ and V_{br} are related by (See Example 6.7 in the textbook):

$$V_{br} = \frac{\varepsilon_r \varepsilon_o \mathcal{E}_{br}^{\,2}}{2eN_d}$$

$$\therefore \quad \mathcal{E}_{br} = \sqrt{\frac{2eN_d V_{br}}{\varepsilon_r \varepsilon_o}} \qquad (1)$$

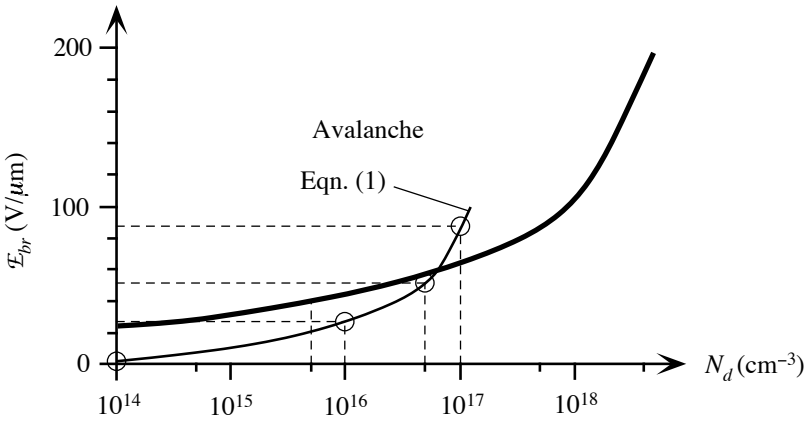


Figure 6Q5-1 The breakdown field, $\mathcal{E}_{br}$, in the depletion layer for the onset of reverse breakdown vs. doping concentration, N_d, in the lightly doped region in a one-sided (p^+n or pn^+) abrupt pn junction. Three points from Eqn. (1) are also shown. The intersection is roughly at $N_d = 6 \times 10^{16}$ cm^{-3}.

We should plot $\mathcal{E}_{br}$ vs. N_d from Eqn. (1) on the same graph and find the intersection point for operation. Given the small figure, this will not be terribly accurate. For four values of N_d calculated below, the plot is shown in Figure 6Q5-1 above. The intersection is roughly at $N_d = 6 \times 10^{16}$ cm^{-3}. An acceptable alternative is to calculate the various $\mathcal{E}_{br}$ by entering a series of N_d values into Eqn. (1) until by trial and error one finds a value of $\mathcal{E}_{br}$ that matches $\mathcal{E}_{br}$ in Figure 6Q5-1.

$N_d = 10^{20}$ m^{-3}:
$$\mathcal{E}_{br} = \sqrt{\frac{2\left(1.602 \times 10^{-19}\ \text{C}\right)\left(10^{20}\ \text{m}^{-3}\right)(25\ \text{V})}{(11.9)\left(8.854 \times 10^{-12}\ \text{F/m}\right)}} = 2.757 \times 10^6\ \text{V/m}$$

Other values for $\mathcal{E}_{br}$ at different values for N_d can be calculated similarly.

Table 6Q5-1 Values of $\mathcal{E}_{br}$ at different N_d.

N_d (m^{-3})	$\mathcal{E}_{br}$ (V/m)
10^{20}	2.757×10^6
10^{22}	2.757×10^7
5×10^{22}	6.165×10^7
10^{23}	8.719×10^7

Since the two graphs intersect at $\boldsymbol{N_d = 6 \times 10^{16}\ cm^{-3}}$, this is the required donor doping for the allowed breakdown voltage.

6.6 Design of a pn^+ junction diode

Design an abrupt Si pn^+ junction that has a reverse breakdown voltage of 100 V and provides a current of 10 mA when the voltage across it is 0.6 V. Assume that, if N_{dopant} is in cm^{-3}, the minority carrier recombination time is given by

$$\tau = \frac{5 \times 10^{-7}}{\left(1 + 2 \times 10^{-17} N_{\text{dopant}}\right)}$$

Mention any assumptions made.

Solution

For this design, we are required to have a reverse breakdown voltage of $V_{br} = 100$ V, and at $V = 0.6$ V the current provided should be $I = 10$ mA. We will assume a room temperature of $T = 300$ K. The relative permittivity of Si is $\varepsilon_r = 11.9$ and the intrinsic concentration is $n_i = 1.45 \times 10^{16}\ m^{-3}$ (Table 5.1 in the textbook).

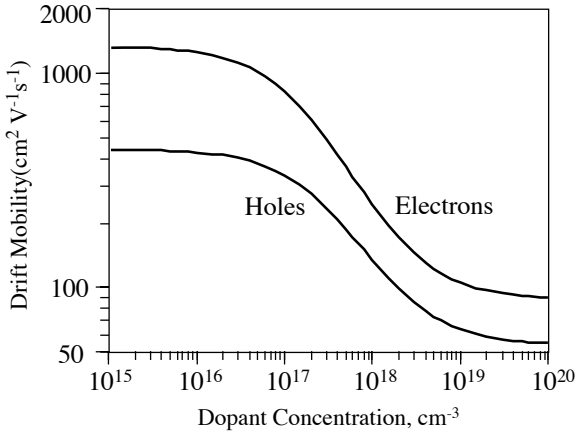


Figure 6Q6-1 The variation of the drift mobility with dopant concentration in Si for electrons and holes at 300 K.

Figure 6Q6-2 shows the relation between $\mathcal{E}_{br}$ and the doping N_d. N_d, the dopant concentration, is equal to the acceptor concentration because we have a pn^+ junction and the depletion region is in the p-side.

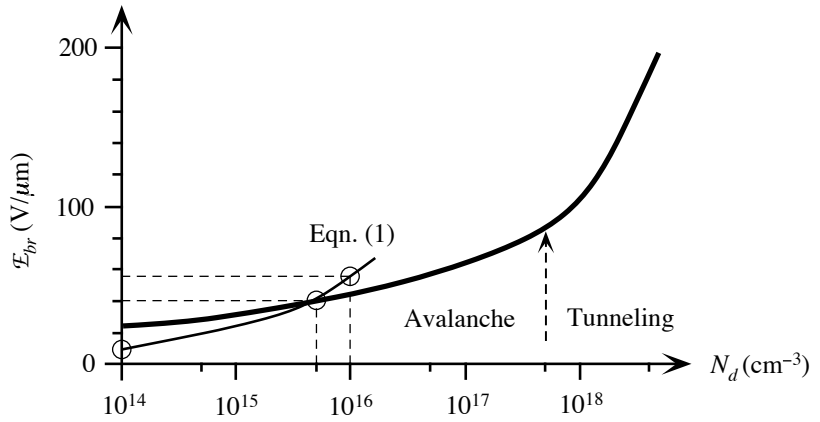


Figure 6Q6-2 The breakdown field, $\mathcal{E}_{br}$, in the depletion layer for the onset of reverse breakdown vs. doping concentration, N_d, in the lightly doped region in a one-sided (p^+n or pn^+) abrupt *pn* junction. Three points from Eqn. (1) are also shown. The intersection is roughly at $N_d = 5 \times 10^{15}$ cm^{-3}.

$\mathcal{E}_{br}$ and V_{br} are related by (see Example 6.7 in the textbook):

$$V_{br} = \frac{\varepsilon_r \varepsilon_o \mathcal{E}_{br}^{\,2}}{2eN_d}$$

$$\therefore \qquad \mathcal{E}_{br} = \sqrt{\frac{2eN_d V_{br}}{\varepsilon_r \varepsilon_o}} \qquad (1)$$

We should plot $\mathcal{E}_{br}$ vs. N_d from Eqn. (1) on the same graph and find the intersection point for operation. Given the small figure, this will not be terribly accurate. For three values of N_d calculated below, the plot is shown in Figure 6Q6-2 above. The intersection is roughly at $N_d = 5 \times 10^{15}$ cm^{-3}. An acceptable alternative is to calculate the various $\mathcal{E}_{br}$ by entering a series of N_d values into Eqn. (1) until by trial and error one finds a value of $\mathcal{E}_{br}$ that matches $\mathcal{E}_{br}$ in Figure 6Q6-2.

$N_d = 10^{20}$ m^{-3}: $$\mathcal{E}_{br} = \sqrt{\frac{2\left(1.602\times10^{-19}\text{ C}\right)\left(10^{20}\text{ m}^{-3}\right)(100\text{ V})}{(11.9)\left(8.854\times10^{-12}\text{ F/m}\right)}} = 5.5147 \times 10^6 \text{ V/m}$$

Other values for $\mathcal{E}_{br}$ at different values for N_d can be calculated similarly.

Table 6Q6-1 Values of $\mathcal{E}_{br}$ at different N_d.

N_d (m^{-3})	$\mathcal{E}_{br}$ (V/m)
10^{20}	5.514×10^6
5×10^{21}	3.899×10^7
10^{22}	5.514×10^7

Since the two graphs intersect at $\mathbf{N_d = 5 \times 10^{15}\ cm^{-3} = 5 \times 10^{21}\ m^{-3}}$, this is the required dopant (acceptor) concentration for the allowed breakdown voltage.

In a pn^+ junction, electron diffusion in the *p*-side gives the diode current. We need the electron lifetime in the *p*-side. Using the given equation:

$$\tau = \frac{5\times10^{-7}}{\left(1+2\times10^{-17}N_d\right)} = \frac{5\times10^{-7}}{\left(1+2\times10^{-17}\left[5\times10^{15}\ \text{cm}^{-3}\right]\right)} = 4.545\times10^{-7}\ \text{s}$$

At N_d (acceptor) = 5×10^{15} cm^{-3} doping, the electron drift mobility (Figure 6Q6-1) is almost the same as at room temperature. Therefore $\mu_e = 1350 \times 10^{-4}$ m^2 V^{-1} s^{-1}.

The diffusion coefficient of electrons (D_e) in the *p*-side is given by:

$$D_e = \frac{kT\mu_e}{e} = \frac{\left(1.381\times10^{-23}\ \text{J/K}\right)(300\ \text{K})\left(0.135\ \text{m}^2\ \text{V}^{-1}\ \text{s}^{-1}\right)}{1.602\times10^{-19}\ \text{C}} = 0.003491\ \text{m}^2/\text{s}$$

The diffusion length of the electrons is then:

$$L_e = \sqrt{D_e\tau} = \sqrt{\left(0.003491\ \text{m}^2/\text{s}\right)\left(4.545\times10^{-7}\ \text{s}\right)} = 3.983\times10^{-5}\ \text{m}$$

Note that $N_a = N_d$ (doping concentration, not "donor").

The forward current is (remember $eV/(kT) >> 1$)

$$I = \left(\frac{AeD_e n_i^{\,2}}{L_e N_a}\right)\exp\left(\frac{eV}{kT}\right)$$

Everything but *A* is given and therefore we can find the required area *A*:

$$A = \frac{IL_eN_a}{eD_e n_i^{\,2}\exp\left(\frac{eV}{kT}\right)}$$

$$\therefore \qquad A = \frac{(0.010\ \text{A})\left(3.983\times10^{-5}\ \text{m}\right)\left(5\times10^{21}\ \text{m}^{-3}\right)}{\left(1.602\times10^{-19}\ \text{C}\right)\left(0.003491\ \text{m}^2/\text{s}\right)\left(1.45\times10^{16}\ \text{m}^{-3}\right)^2\exp\left(\frac{\left(1.602\times10^{-19}\ \text{C}\right)(0.6\ \text{V})}{\left(1.381\times10^{-23}\ \text{J/K}\right)(300\ \text{K})}\right)}$$

$$\therefore \qquad \boldsymbol{A = 1.42\times10^{-6}\ \text{m}^2}$$

If we make a circular device with a radius *r*, the area is πr^2.

$$A = \pi r^2$$

$$\therefore \qquad \boldsymbol{r = \sqrt{A/\pi} = \sqrt{1.42\times10^{-6}\ \text{m}^2/\pi} = 0.000672\ \text{m}\ \text{ or }\ 0.672\ \text{mm}}$$

6.7 Minority carrier profiles (the hyperbolic functions)

Consider a *pnp* BJT under normal operating conditions in which the EB junction is forward biased and the BC junction is reverse biased. The field in the neutral base region outside the depletion layers can be assumed to be negligibly small. The continuity equation for holes, $p_n(x)$, in the *n*-type base region is

$$D_h\frac{d^2p_n}{dx^2} - \frac{p_n - p_{no}}{\tau_h} = 0 \qquad \textbf{[1]}$$

where $p_n(x)$ is the hole concentration at x from just outside the depletion region and p_{no} and τ_h are the equilibrium hole concentration and hole recombination lifetime in the base.

a. What are the boundary conditions at $x = 0$ and $x = W_B$, just outside the collector region depletion layer? (Consider the law of the junction).

b. Show that the following expression for $p_n(x)$ is a solution of the continuity equation

$$p_n(x) = p_{no}\left[\exp\left(\frac{eV}{kT}\right)-1\right]\left[\frac{\sinh\left(\frac{W_B - x}{L_h}\right)}{\sinh\left(\frac{W_B}{L_h}\right)}\right] + p_{no}\left[1 - \frac{\sinh\left(\frac{x}{L_h}\right)}{\sinh\left(\frac{W_B}{L_h}\right)}\right] \qquad [2]$$

where $V = V_{EB}$ and $L_h = \sqrt{[D_h \tau_h]}$.

c. Show that Equation 2 satisfies the boundary conditions.

Solution

The continuity equation for the holes in the *n*-type base region is

$$D_h \frac{d^2 p_n}{dx^2} - \frac{p_n - p_{no}}{\tau_h} = 0 \qquad \text{Eqn. 1}$$

a The boundary conditions (hole concentrations) at $x = 0$ (in the base just outside the EB depletion region) and $x = W_B$ (in the base just outside the BC depletion region) are:

At $x = 0$, $\quad p_n(0) = p_{no}\exp\left(\frac{eV}{kT}\right)$ *Law of the junction*

At $x = W_B$, $\quad p_n(W_B) = 0$

b Substituting Eqn. 2 into Eqn. 1 will show that Eqn. 1 is satisfied. We, of course, have to differentiate $\sinh(x)$ and $\cosh(x)$ with respect to x: $d[\sinh(x)]/dx = \cosh(x)$ and $d[\cosh(x)]/dx = \sinh(x)$.

$$p_n = p_{no}\left[\exp\left(\frac{eV}{kT}\right)-1\right]\frac{\sinh\left(\frac{W_B - x}{L_h}\right)}{\sinh\left(\frac{W_B}{L_h}\right)} + p_{no}\left[1 - \frac{\sinh\left(\frac{x}{L_h}\right)}{\sinh\left(\frac{W_B}{L_h}\right)}\right] \qquad \text{Eqn. 2}$$

$$\therefore \quad \frac{dp_n}{dx} = p_{no}\left[\exp\left(\frac{eV}{kT}\right)-1\right]\frac{\left(\frac{-1}{L_h}\right)\cosh\left(\frac{W_B - x}{L_h}\right)}{\sinh\left(\frac{W_B}{L_h}\right)} - p_{no}\frac{\left(\frac{1}{L_h}\right)\cosh\left(\frac{x}{L_h}\right)}{\sinh\left(\frac{W_B}{L_h}\right)}$$

$$\therefore \quad \frac{d^2 p_n}{dx^2} = p_{no}\left[\exp\left(\frac{eV}{kT}\right)-1\right]\frac{\left(\frac{1}{L_h^2}\right)\sinh\left(\frac{W_B - x}{L_h}\right)}{\sinh\left(\frac{W_B}{L_h}\right)} - p_{no}\frac{\left(\frac{1}{L_h^2}\right)\sinh\left(\frac{x}{L_h}\right)}{\sinh\left(\frac{W_B}{L_h}\right)}$$

Substituting into the left hand side (LHS) of Eqn. 1 and using $L_h^2 = D_h \tau_h$ we get,

$$LHS = D_h \frac{d^2 p_n}{dx^2} - \frac{p_n - p_{no}}{\tau_h}$$

$$LHS = \frac{L_h^{\,2}}{\tau_h}\left[p_{no}\left[\exp\left(\frac{eV}{kT}\right)-1\right]\frac{\left(\frac{1}{L_h^2}\right)\sinh\left(\frac{W_B-x}{L_h}\right)}{\sinh\left(\frac{W_B}{L_h}\right)} - p_{no}\frac{\left(\frac{1}{L_h^2}\right)\sinh\left(\frac{x}{L_h}\right)}{\sinh\left(\frac{W_B}{L_h}\right)}\right]$$

$\therefore$

$$-\frac{p_{no}\left[\exp\left(\frac{eV}{kT}\right)-1\right]\frac{\sinh\left(\frac{W_B-x}{L_h}\right)}{\sinh\left(\frac{W_B}{L_h}\right)} + p_{no}\left[1-\frac{\sinh\left(\frac{x}{L_h}\right)}{\sinh\left(\frac{W_B}{L_h}\right)}\right] - p_{no}}{\tau_h}$$

Simplifying, e.g. multiplying through by τ_h, and canceling terms we get,

$$LHS = 0$$

which verifies Eqn. 1. Thus Eqn. 2 must be a solution of Eqn. 1. Mathematicians would prefer to solve the differential equation in Eqn. 1 directly and hence find Eqn. 2 which is a more rigorous approach.

c Given sinh(0) = 0 then at $x = 0$, and at $x = W_B$ we have

At $x = 0$,
$$p_n(0) = p_{no}\left[\exp\left(\frac{eV}{kT}\right)-1\right]\frac{\sinh\left(\frac{W_B}{L_h}\right)}{\sinh\left(\frac{W_B}{L_h}\right)} + p_{no}[1-0] = p_{no}\exp\left(\frac{eV}{kT}\right)$$

At $x = W_B$,
$$p_n = p_{no}\left[\exp\left(\frac{eV}{kT}\right)-1\right]\frac{\sinh\left(\frac{0}{L_h}\right)}{\sinh\left(\frac{W_B}{L_h}\right)} + p_{no}\left[1-\frac{\sinh\left(\frac{W_B}{L_h}\right)}{\sinh\left(\frac{W_B}{L_h}\right)}\right] = 0 + p_{no}[1-1] = 0$$

Both are the boundary conditions stated in **a**.

*6.8 The *pnp* bipolar transistor

Consider a *pnp* transistor in a common base configuration and under normal operating conditions. The emitter-base junction is forward biased and the base-collector junction is reverse biased. The emitter, base, and collector dopant concentrations are $N_{a(E)}$, $N_{d(B)}$, and $N_{a(C)}$ respectively where $N_{a(E)} >> N_{d(B)} \geq N_{a(C)}$. For simplicity, assume uniform doping in all the regions. The base and emitter widths are W_B and W_E ,respectively, both much shorter than the minority carrier diffusion lengths, L_h and L_e. The minority carrier lifetime in the base is the hole recombination time τ_h. The minority carrier mobility in the base and emitter are denoted by μ_h and μ_e respectively.

The minority carrier concentration profile in the base can be represented by Equation 6.68 (in the textbook).

a. Assuming that the emitter injection efficiency is unity show that

1. $I_E \approx \dfrac{eAD_h n_i^2 \coth(W_B/L_h)}{L_h N_{d(B)}}\exp\left(\dfrac{eV_{EB}}{kT}\right)$

2. $I_C \approx \dfrac{eAD_h n_i^2 \operatorname{cosech}(W_B/L_h)}{L_h N_{d(B)}} \exp\left(\dfrac{eV_{EB}}{kT}\right)$

3. $\alpha \approx \operatorname{sech}\left(\dfrac{W_B}{L_h}\right)$

4. $\beta \approx \dfrac{\tau_h}{\tau_t}$ where $\tau_t = \dfrac{W_B^2}{2D_h}$ is the base transit time.

b. Consider the total emitter current, I_E, through the EB junction, which has diffusion and recombination components as follows:

$$I_E = I_{E(so)} \exp\left(\frac{eV_{EB}}{kT}\right) + I_{E(ro)} \exp\left(\frac{eV_{EB}}{2kT}\right)$$

Only the hole component of the diffusion current (first term) can contribute to the collector current. Show that when $N_{a(E)} >> N_{d(B)}$, the emitter injection efficiency, γ, is given by

$$\gamma \approx \left[1 + \frac{I_{E(ro)}}{I_{E(so)}} \exp\left(-\frac{eV_{EB}}{2kT}\right)\right]^{-1}$$

How does $\gamma < 1$ modify the expressions derived in part (*a*)? What is your conclusion (consider small and large emitter currents, or $V_{EB} = 0.4$ and 0.7 V)?

Solution

a We are given the minority carrier concentration profile in the base:

$$p_n = p_{no}\left[\exp\left(\frac{eV_{EB}}{kT}\right) - 1\right]\frac{\sinh\left(\dfrac{W_B - x}{L_h}\right)}{\sinh\left(\dfrac{W_B}{L_h}\right)} + p_{no}\left[1 - \frac{\sinh\left(\dfrac{x}{L_h}\right)}{\sinh\left(\dfrac{W_B}{L_h}\right)}\right] \qquad \text{Eqn. 1}$$

Under normal operating conditions, EB is forward biased and $eV_{EB}/(kT) >> 1$ and $\exp(eV_{EB}/(kT)) >> 1$.

1. The emitter current I_E is the diffusion current at $x = 0$, i.e. in the base just outside the EB depletion region.

$$I_E = \pm eAD_h\left[\frac{dp_n}{dx}\right]_{x=0} = \pm eAD_h\left\{p_{no}\left[\exp\left(\frac{eV_{EB}}{kT}\right) - 1\right]\frac{\left(\dfrac{-1}{L_h}\right)\cosh\left(\dfrac{W_B}{L_h}\right)}{\sinh\left(\dfrac{W_B}{L_h}\right)} - p_{no}\frac{\left(\dfrac{1}{L_h}\right)(1)}{\sinh\left(\dfrac{W_B}{L_h}\right)}\right\}$$

Using $p_{no} = n_i^2/N_{d(B)}$, $eV_{EB}/kT >> 1$ (forward bias) and $L_h >> W_B$ (small recombination in the base) so that $\cosh(W_B/L_h) \approx 1$, we get,

$$I_E \approx \frac{eAD_h n_i^2}{L_h N_{d(B)}}\frac{\cosh\left(\dfrac{W_B}{L_h}\right)}{\sinh\left(\dfrac{W_B}{L_h}\right)}\exp\left(\frac{eV_{EB}}{kT}\right) = \frac{eAD_h n_i^2}{L_h N_{d(B)}}\coth\left(\frac{W_B}{L_h}\right)\exp\left(\frac{eV_{EB}}{kT}\right)$$

2. The collector current I_C is determined by the concentration gradient at $x = W_B$,

$$I_E = \pm eAD_h\left[\frac{dp_n}{dx}\right]_{x=W_B}$$

$$\therefore \quad I_E = \pm eAD_h\left\{p_{no}\left[\exp\left(\frac{eV_{EB}}{kT}\right)-1\right]\frac{\left(\frac{-1}{L_h}\right)\cosh(0)}{\sinh\left(\frac{W_B}{L_h}\right)} - p_{no}\frac{\left(\frac{1}{L_h}\right)\cosh\left(\frac{W_B}{L_h}\right)}{\sinh\left(\frac{W_B}{L_h}\right)}\right\}$$

Using $p_{no} = n_i^2/N_{d(B)}$, $eV/kT >> 1$ (forward bias) and $L_h >> W_B$ (small recombination in the base) so that $\cosh(W_B/L_h) \approx 1$ and $\cosh(W_B/L_h)/\sinh(W_B/L_h) \approx 0$ and we get,

$$I_C \approx \frac{eAD_h n_i^2}{L_h N_{d(B)}}\frac{1}{\sinh\left(\frac{W_B}{L_h}\right)}\exp\left(\frac{eV_{EB}}{kT}\right) = \frac{eAD_h n_i^2}{L_h N_{d(B)}}\operatorname{cosech}\left(\frac{W_B}{L_h}\right)\exp\left(\frac{eV_{EB}}{kT}\right)$$

3. The current gain, $\alpha = I_C/I_E$. Substituting for I_C and I_E and simplifying by using $\operatorname{sech}(\theta) = 1/\cosh(\theta)$ we obtain:

$$\alpha = \operatorname{cosech}(W_B/L_h) \,/\, \coth(W_B/L_h) = [1/\sinh(W_B/L_h)][\sinh(W_B/L_h)/\cosh(W_B/L_h)]$$

$$\therefore \quad \alpha = 1/[\cosh(W_B/L_h)] = \operatorname{sech}(W_B/L_h)$$

4. Assume (W_B/L_h) is small. For small θ, $\operatorname{sech}(\theta) \approx 1 - (1/2)\theta^2$ (by Taylor expansion). Then:

$$\beta \approx 1/(1-\alpha) = 1/[1 - \operatorname{sech}(W_B/L_h)] \approx 1/[(1/2)(W_B/L_h)^2] = 2L_h^2/W_B^2$$

Given that $L_h^2 = D_h\tau_h$ and $W_B^2 = 2D_h\tau_t$, substituting we find,

$$\beta \approx \tau_h/\tau_t$$

b If $N_a >> N_d$ then we can forget about the electron injection component of the emitter diffusion current (electron injection from the base into the emitter) and consider only the hole diffusion component. The emitter current has two components, diffusion (Shockley) and recombination as stated in total diode current equation on page 428 (in the texttbok):

Diffusion component: $I_{Es} = I_{E(so)}\exp\left(\frac{eV_{EB}}{kT}\right)$

Recombination: $I_{Er} = I_{E(ro)}\exp\left(\frac{eV_{EB}}{2kT}\right)$

The emitter injection efficiency (γ) is given by:

$$\gamma = \frac{I_{Es}}{I_{Es} + I_{Er}}$$

γ is less than unity. Only I_{Es} is useful in the transistor action. The equations in (a) assumed that $\gamma = 1$, i.e. full injection of holes via the diode action (law of the junction). When $\gamma < 1$, α and hence β are reduced. The new current gain is $\alpha' = \gamma\alpha$. Now consider γ:

$$\gamma = \frac{1}{1+\dfrac{I_{Er}}{I_{Es}}} = \frac{1}{1+\dfrac{I_{E(ro)}\exp\left(\dfrac{eV_{EB}}{2kT}\right)}{I_{E(so)}\exp\left(\dfrac{eV_{EB}}{kT}\right)}} = \frac{1}{1+\dfrac{I_{E(ro)}\exp\left(-\dfrac{eV_{EB}}{2kT}\right)}{I_{E(so)}}}$$

Typically, $I_{E(ro)} >> I_{E(so)}$. Assume an order of magnitude for $I_{E(so)} = 10^{-13}$ A and $I_{E(ro)} = 10^{-11}$ A (100 times bigger than $I_{E(so)}$) (see Example 6.4, pg. 432 in the textbook). Assume that temperature $T = 300$ K. We can then calculate γ at two V_{EB} voltages to observe the effect of V_{EB}.

$V_{EB} = 0.4$ V:

$$\gamma = \frac{1}{1+\dfrac{I_{E(ro)}\exp\left(-\dfrac{eV_{EB}}{2kT}\right)}{I_{E(so)}}} = \frac{1}{1+\dfrac{\left(10^{-11}\text{ A}\right)\exp\left(-\dfrac{\left(1.602\times10^{-19}\text{ C}\right)\left(0.4\text{ V}\right)}{2\left(1.381\times10^{-23}\text{ J/K}\right)\left(300\text{ K}\right)}\right)}{\left(10^{-13}\text{ A}\right)}}$$

$\therefore$ γ **= 0.9580**

$V_{EB} = 0.7$ V: γ **= 0.9999**

Clearly, γ is lower when V_{EB} is lower. Injection is more efficient at higher V_{EB} when the diffusion component of I_E dominates over the recombination component. By the way, changing the value of $I_{E(so)}$ but keeping $I_{E(ro)} > I_{E(so)}$ does not change the conclusion.

6.9 Characteristics of an *npn* Si BJT

Consider an idealized silicon *npn* bipolar transistor with the properties listed in Table 6Q9-1. Assume uniform doping in each region. The emitter and base widths are between metallurgical junctions (not neutral regions). The cross-sectional area is 100 μm × 100 μm. The transistor is biased to operate in the normal active mode. The base-emitter forward bias voltage is 0.6 V and the reverse bias base-collector voltage is 18 V.

Table 6Q9-1 Properties of an *npn* BJT

Emitter Width	Emitter Doping	Hole Lifetime in Emitter	Base Width	Base Doping	Electron Lifetime in Base	Collector Doping
10 μm	2×10^{18} cm^{-3}	10 ns	4 μm	1×10^{16} cm^{-3}	400 ns	1×10^{16} cm^{-3}

a. Calculate the depletion layer width extending from the collector into the base and also from the emitter into the base. What is the width of the neutral base region?

b. Calculate α and hence β for this transistor, assuming unity emitter injection efficiency. How do α and β change with V_{CB}?

c. What is the emitter injection efficiency and what are α and β, taking into account that the emitter injection efficiency is not unity?

d. What are the emitter, collector, and base currents?

e. What is the collector current when $V_{CB} = 19$ V but $V_{EB} = 0.6$ V? What is the incremental collector output resistance defined as $\Delta V_{CB}/\Delta I_C$?

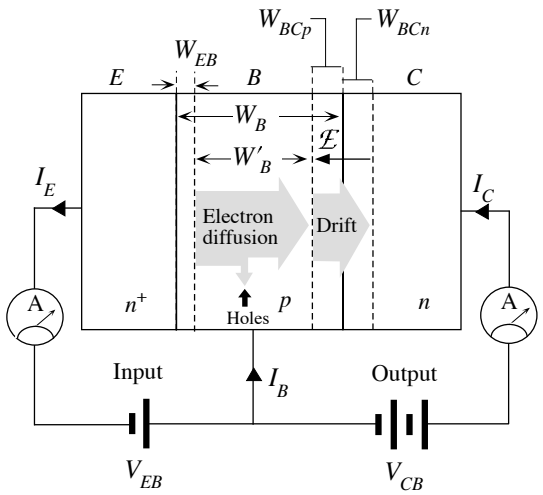
Solution

a Figure 6Q9-1 shows the principle of operation of the *npn* BJT and also defines various device characteristics such as the depletion widths and the base widths. With $V_{BC} >> V_o$, the reverse bias across the base-collector junction $V_r = V_{BC} + V_o \approx V_{BC}$. Thus, the depletion layer W_{BC} at the base-collector junction is given by;

$$W_{BC} \approx \left[\frac{2\varepsilon(N_a + N_d)V_{BC}}{eN_aN_d}\right]^{1/2}$$

i.e.
$$W_{BC} = \left[\frac{2(8.854\times10^{-12}\ \text{F m}^{-1})(11.9)(10^{22}+10^{22}\ \text{m}^{-3})(18\ \text{V})}{(1.602\times10^{-19}\ \text{C})(10^{22}\ \text{m}^{-3})(10^{22}\ \text{m}^{-3})}\right]^{1/2}$$

$$\boldsymbol{W_{BC} = 2.18 \times 10^{-6}\ \text{m}} \text{ or } \boldsymbol{2.18\ \mu\text{m}}$$



An *npn* transistor operated in the normal mode, in the active region, in the common base (CB) configuration (Notation: W = width, $\mathcal{E}$ = electric field).

Figure 6Q9-1

Only a portion of W_{BC} is in the base side. Suppose that W_{BCp} and W_{BCn} are the depletion widths in the base and collector sides of the SCL respectively. Since the total charge on the *p* and *n*-sides of the SCL must be the same,

$$N_aW_{BCp} = N_dW_{BCn}$$

and since
$$W_{BC} = W_{BCp} + W_{BCn}$$

we can find W_{BCp},

$$W_{BCp} = \frac{N_d}{N_a + N_d}W_{BC} = \frac{10^{16}}{10^{16}+10^{16}}(2.17\ \ \mu\text{m}) = 1.09\ \mu\text{m}$$

Since $N_d(E) >> N_a(B)$, the depletion layer width W_{EB} is almost totally in the *p*-side (in the base). With forward bias, $V_{EB} = 0.6$ V across the emitter-base junction, W_{EB} is given by

$$W_{EB} = \left[\frac{2\varepsilon(V_o - V_{EB})}{eN_a}\right]^{1/2}$$

We first need to calculate the built-in voltage V_o between the emitter and base,

$$V_o = \frac{kT}{e}\ln\left(\frac{N_a N_d}{n_i^2}\right) = (0.02586\ \text{V})\ln\left[\frac{(2\times10^{18})(1\times10^{16})}{(1.5\times10^{10})^2}\right]$$

i.e. $V_o = 0.830$ V

Then, $$W_{EB} = \left[\frac{2\varepsilon(V_o - V_{EB})}{eN_a}\right]^{1/2}$$

i.e. $$W_{EB} = \left[\frac{2(8.854\times10^{-12}\ \text{F m}^{-1})(11.9)(0.830 \pm 0.6\ \text{V})}{(1.602\times10^{-19}\ \text{C})(10^{22}\ \text{m}^{-3})}\right]^{1/2}$$

or $$\boldsymbol{W_{EB} = 1.74 \times 10^{-7}\ \text{m}}\ \text{or}\ \boldsymbol{0.174\ \mu\text{m}}$$

Notice that due to the forward bias across the *EB* junction, W_{EB} is an order of magnitude smaller than W_{BCp}.

If W_B is the base width between emitter and collector metallurgical junctions, then the width W'_B of the neutral region in the base between the depletion regions is given by,

$$W'_B = W_B - W_{BCp} - W_{EB}$$

so that $$\boldsymbol{W'_B} = 4\ \mu\text{m} - 1.09\ \mu\text{m} - 0.174\ \mu\text{m} = \mathbf{2.74\ \mu m}$$

b The electron drift mobility μ_e in the base is determined by the dopant (acceptor) concentration here. For $N_a = 1 \times 10^{16}$ cm^{-3}, $\mu_e = 1250$ cm^2 V^{-1} s^{-1} and the diffusion coefficient D_e from the Einstein relationship is,

$$D_e = kT\mu_e/e = (0.02586\ \text{V})(1250 \times 10^{-4}\ \text{m}^2\ \text{V}^{-1}\ \text{s}^{-1}) = 3.23 \times 10^{-3}\ \text{m}^2\ \text{s}^{-1}$$

The electron diffusion length L_e in the base is

$$L_e = (D_e\tau_e)^{1/2} = [(3.23 \times 10^{-3}\ \text{m}^2\ \text{s}^{-1})(400 \times 10^{-9}\ \text{s})]^{1/2}$$

i.e. $L_e = 36.0 \times 10^{-6}$ m or 36.0 μm

In order to calculate α, first we need to find the transit (diffusion) time τ_t through the base

$$\tau_t = \frac{W_B'^2}{2D_e} = \frac{(2.74\times10^{-6}\ \text{m})^2}{2(3.23\times10^{-3}\ \text{m}^2\ \text{s}^{-1})} = 1.161 \times 10^{-9}\ \text{s}$$

If we assume unity injection ($\gamma = 1$), then $\alpha = \alpha_B$, base transport factor:

$$\alpha = \alpha_B = 1 - \frac{\text{Transit (diffusion) time across base}}{\text{Minority carrier recombination time in base}} = 1 - \frac{\tau_t}{\tau_e}$$

i.e. $\alpha = 1 - (1.161\ \text{ns})/(400\ \text{ns}) = \mathbf{0.9971}$

The current gain β is

$$\beta = \frac{\alpha}{1-\alpha} = \frac{0.9971}{1-0.9971} = \mathbf{343}$$

c The hole drift mobility μ_h in the emitter is determined by the dopant (donor) concentration here. For $N_d = 2 \times 10^{18}$ cm^{-3}, $\mu_h \approx 100$ cm^2 V^{-1} s^{-1} and the diffusion coefficient D_h from the Einstein relationship is,

$$D_h = kT\mu_h/e = (0.02586\ \text{V})(100 \times 10^{-4}\ \text{m}^2\ \text{V}^{-1}\ \text{s}^{-1}) = 2.59 \times 10^{-4}\ \text{m}^2\ \text{s}^{-1}$$

The hole diffusion length L_h in the base is

$$L_h = (D_h\tau_h)^{1/2} = [(2.59 \times 10^{-4}\ \text{m}^2\ \text{s}^{-1})(10 \times 10^{-9}\ \text{s})]^{1/2}$$

$$L_h = 1.61 \times 10^{-6} \text{ or } 1.61\ \mu\text{m}$$

Thus the hole diffusion length is much shorter than the emitter width.

The emitter current is given by electron diffusion in the base and hole diffusion in the emitter so that

$$I_E = I_{E(\text{electron})} + I_{E(\text{hole})}$$

where for electrons diffusing in the base,

$$I_{E(\text{electron})} = I_{soe}\exp\left(\frac{eV_{EB}}{kT}\right); \qquad I_{soe} = \frac{eAD_e n_i^2}{N_a W_B}$$

and holes diffusing in the emitter,

$$I_{E(\text{hole})} = I_{soh}\exp\left(\frac{eV_{EB}}{kT}\right); \qquad I_{soh}\frac{eAD_h n_i^2}{N_d L_h}$$

where we used L_h instead of W_E because $W_E >> L_h$ (emitter width is 10 μm and hole diffusion length is 1.62 μm). Substituting the values we find,

$$I_{soe} = \frac{(1.602\times10^{-19}\ \text{C})(1\times10^{-8}\ \text{m}^2)(3.23\times10^{-3}\ \text{m}^2\ \text{s}^{-1})(1.5\times10^{16}\ \text{m}^{-3})^2}{(1\times10^{22}\ \text{m}^{-3})(2.74\times10^{-6}\ \text{m})}$$

i.e. $I_{soe} = 4.267 \times 10^{-14}$ A or 42.67 fA

and $$I_{soh} = \frac{(1.602\times10^{-19}\ \text{C})(1\times10^{-8}\ \text{m}^2)(2.59\times10^{-4}\ \text{m}^2\ \text{s}^{-1})(1.5\times10^{16}\ \text{m}^{-3})^2}{(2\times10^{24}\ \text{m}^{-3})(1.61\times10^{-6}\ \text{m})}$$

i.e. $I_{soh} = 2.93 \times 10^{-17}$ A or 0.0293 fA

The emitter injection efficiency is the fraction of the emitter current that is due to minority carriers injected into the base; those that diffuse across the base towards the collector.

$$\gamma = \frac{I_{E(\text{electron})}}{I_{E(\text{electron})} + I_{E(\text{hole})}} = \frac{I_{soe}}{I_{soe} + I_{soh}}$$

i.e. $$\gamma = \frac{4.267\times10^{-14}}{4.267\times10^{-14} + 2.93\times10^{-17}} = \mathbf{0.99931}$$

The current gains, taking into account the emitter injection efficiency, are

$$\alpha = \gamma\alpha_B = (0.99931)(0.9971) = \mathbf{0.99641}$$

and $$\beta = \alpha/(1-\alpha) = 0.99641/(1-0.99641) = \mathbf{278}$$

d The emitter current with $V_{EB} = 0.6$ V is

$$I_E = (I_{soe} + I_{soh})\exp(eV_{EB}/kT)$$

$$I_E = (4.267 \times 10^{-14}\ \text{A} + 2.93 \times 10^{-17}\ \text{A})\exp(0.6\ \text{V}/0.2586\ \text{V})$$

$$\mathbf{I_E = 5.13 \times 10^{-4}\ A} \text{ or } \mathbf{0.513\ mA}$$

The collector current is determined by those minority carriers in the base that reach the collector junction. Only γI_E of I_E is injected into the base as minority carriers and only a fraction α_B make it to the collector,

$$\boldsymbol{I_C = \alpha_B \gamma I_E = \alpha I_E} = (0.99641)(0.513 \text{ mA}) = \mathbf{0.511\ mA}$$

The base current is given by,

$$\boldsymbol{I_B} = I_C/\beta = (0.511 \text{ mA})/278 = \mathbf{1.83 \times 10^{-3}\ mA} \text{ or } \mathbf{1.83\ \mu A}$$

Table 6Q9-2 Characteristics of BJT with V_{CB} = 18 and 19 V.

V_{CB}	W'_B	α	β	I_E	I_C
18 V	2.74 μm	0.99641	278	0.513 mA	0.511 mA
19 V	2.71 μm	0.99649	283	0.517 mA	0.515 mA

e Suppose that we increase V_{CB} to 19 V and repeat all the calculations above. We then find results tabulated in Table 6Q9-2. We can calculate the small signal collector incremental resistance from,

$$\boldsymbol{r_c} = \frac{\Delta V_{CB}}{\Delta I_C} = \frac{19 \text{ V} - 18 \text{ V}}{0.515 \text{ mA} - 0.511 \text{ mA}} = \mathbf{250\ k\Omega}$$

We can also calculate the BJT characteristics using the hyperbolic expressions given in Question 6.8 in the textbook.

The base transport factor α_B is given by,

$$\alpha_B \approx \text{sech}\left(\frac{W_B}{L_e}\right) = \text{sech}\left(\frac{2.74\ \mu\text{m}}{36.0\ \mu\text{m}}\right) = 0.99711$$

which is, within calculation errors, almost identical to the simplified theory.

The emitter current is given by

$$I_{E(\text{electron})} = I_{soe} \exp\left(\frac{eV_{EB}}{kT}\right);\ I_{soe} = \frac{eAD_e n_i^2}{N_a L_e} \coth\left(\frac{W_B}{L_e}\right)$$

$$I_{soe} = \frac{(1.602 \times 10^{-19}\ \text{C})(1 \times 10^{-8}\ \text{m}^2)(3.23 \times 10^{-3}\ \text{m}^2\ \text{s}^{-1})(1.5 \times 10^{16}\ \text{m}^{-3})^2}{(36 \times 10^{-6}\ \text{m})(1 \times 10^{22}\ \text{m}^{-3}) \tanh\left(\frac{2.74\ \mu\text{m}}{36\ \mu\text{m}}\right)}$$

i.e. $I_{soe} = 4.260 \times 10^{-14}$ A or 42.60 fA

which, for all practical purposes, is identical to the simplified theory.

NOTE: We have ignored any bandgap narrowing in the emitter due to heavy doping in this region.

6.10 The JFET Pinch-off Voltage

Consider the symmetric *n*-channel JFET shown in Figure 6Q10-1. The width of each depletion region extending into the *n*-channel is W. The thickness, or depth, of the channel, defined between the two metallurgical junctions, is $2a$. Assuming an abrupt *pn* junction and $V_{DS} = 0$, show that when the gate to source voltage is $-V_P$ the channel is pinched off where

$$V_P = \frac{a^2 e N_d}{2\varepsilon} - V_o$$

where V_o is the built-in potential between p^+n junction and N_d is the donor concentration of the channel.

Calculate the pinch-off voltage of a JFET that has an acceptor concentration of 10^{19} cm^{-3} in the p^+ gate, a channel donor doping of 10^{16} cm^{-3}, and a channel thickness (depth), $2a$, of 2μm.

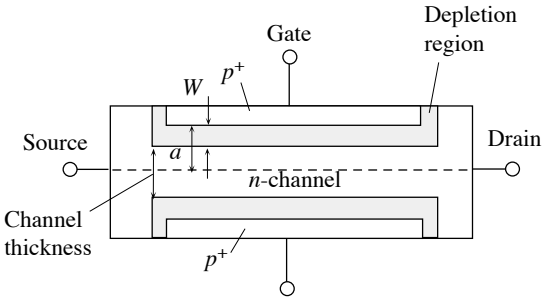


Figure 6Q10-1 A symmetric JFET.

Solution

Assume we have a Si JFET. The depletion region of a p^+n junction extends essentially into n-side. W depends on the reverse bias $V_r = -V_{GS}$ between the p^+ and n-channel,

$$W = \left[\frac{2\varepsilon(V_o - V_{GS})}{eN_d}\right]^{1/2}$$

As V_{GS} is negative $V_o - V_{GS}$ is greater than V_o and W extends more into the channel than the $V_{GS} = 0$ case. As V_{GS} is made more negative, W extends more into the channel. Typically the magnitude of V_{GS} is greater than the built-in potential V_o. The channel is totally pinched off when W extends halfway into the channel thickness, when $W = a$. Then

$$a^2 = \frac{2\varepsilon(V_o - V_{GS})}{eN_d}$$

Rearranging we obtain,

$$V_{GS} = V_o - \frac{a^2 e N_d}{2\varepsilon}$$

With $V_{DS} = 0$, the channel is pinched off when, by definition, $V_{GS} = -V_P$. Thus

$$V_P = \frac{a^2 e N_d}{2\varepsilon} - V_o$$

The built-in potential is

$$V_o = (kT/e)\ln(N_d N_a / n_i^2) = (0.02586\text{ V})\ln[(10^{19}\text{ cm}^{-3})(10^{16}\text{ cm}^{-3})/(1.45\times10^{10}\text{ cm}^{-3})^2]$$

$$V_o = 0.8740\text{ V}$$

The pinch-off voltage is,

$$\boldsymbol{V_P} = \frac{(1\times10^{-6}\text{ m})^2(1.602\times10^{-19}\text{ C})(10^{22}\text{ m}^{-3})}{2(11.9)(8.854\times10^{-12}\text{ F m}^{-1})} - 0.8740 = \mathbf{6.73\text{ V}}$$

It is also apparent that V_P is highly sensitive to the channel doping. We would expect device to device variation between different batches of fabrication.

6.11 The JFET

Consider an *n*-channel JFET that has a symmetric p^+n gate-channel structure as shown in Figures 6Q11-1a and 6Q11-2. Let L be the gate length, Z the gate width, and $2a$ the channel thickness. The pinch-off voltage is given by Question 6.10. The drain saturation current, I_{DSS}, is the drain current when $V_{GS} = 0$. This occurs when $V_{DS} = V_{DS(sat)} = V_P$ (Figure 6Q11-3) so $I_{DSS} = V_P G_{ch}$, where G_{ch} is the conductance of the channel between the source and the pinched-off point (Figure 6Q11-4). Taking into account the shape of the channel at pinch-off, if G_{ch} is about $^1/_3$ of the conductance of the free or unmodulated (rectangular) channel, show that

$$I_{DSS} = V_P\left[\frac{1}{3}\frac{(e\mu_e N_d)(2a)Z}{L}\right]$$

A particular *n*-channel JFET with a symmetric p^+n gate-channel structure has a pinch-off voltage of 3.9 V and an I_{DSS} of 5.5 mA. If the gate and channel dopant concentrations are $N_a = 10^{19}$ cm^{-3} and $N_d = 10^{15}$ cm^{-3} respectively, find the channel thickness $2a$ and Z/L. If $L = 10$ μm, what is Z? What is the gate-source capacitance when the JFET has no voltage supplies connected to it?

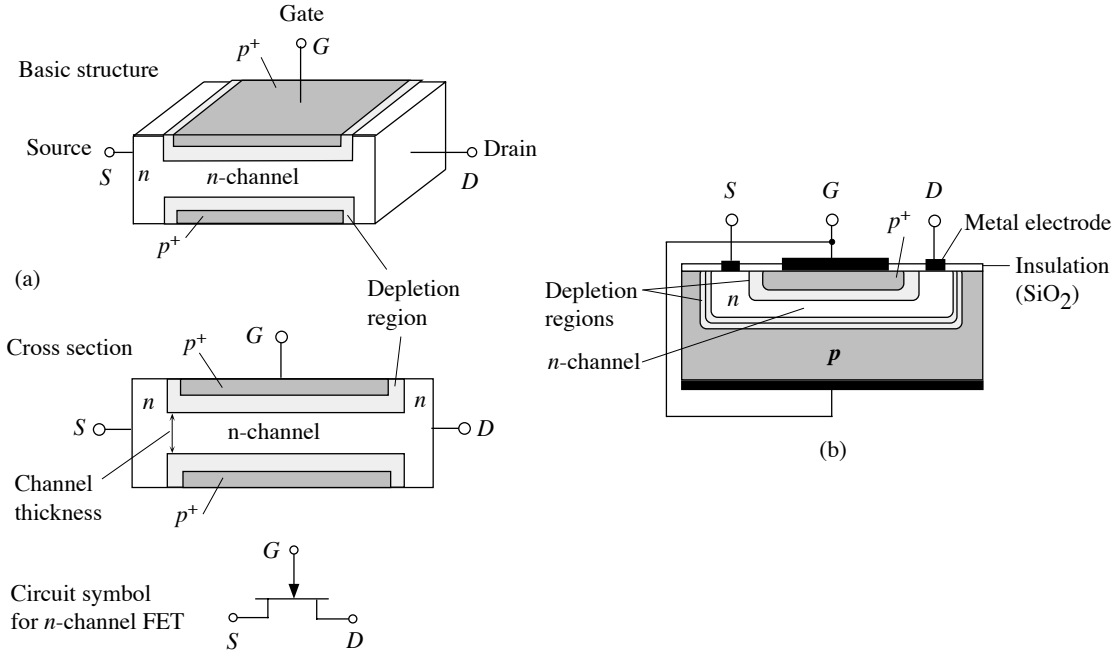


Figure 6Q11-1

(a) The basic structure of the junction field transistor (JFET) with an *n*-channel. The two p^+ regions are electrically connected and form the gate.

(b) A simplified sketch of the cross section of a more practical *n*-channel JFET.

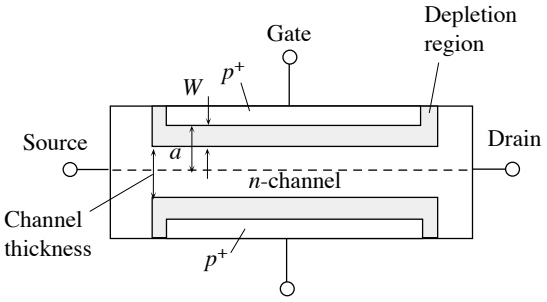


Figure 6Q11-2 A symmetric JFET.

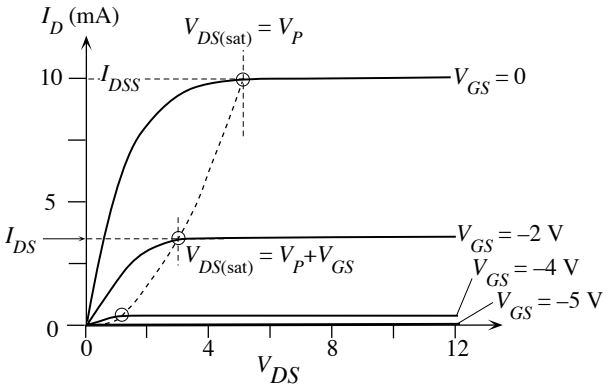


Figure 6Q11-3 Typical I_D versus V_{DS} characteristics of a JFET for various fixed gate voltages V_{GS}.

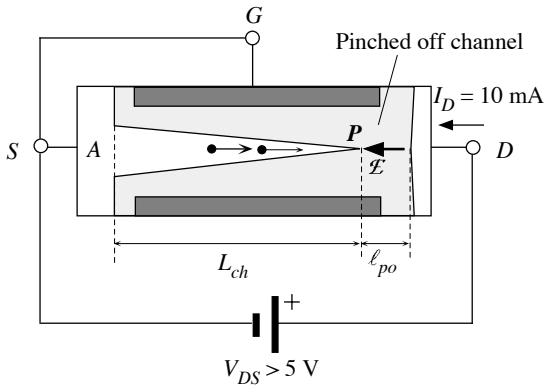


Figure 6Q11-4 The pinched-off channel and conduction for $V_{DS} > V_P$ (= 5 V)

Solution

The conductivity of the channel is:

$$\sigma = eN_d\mu_e$$

The channel width is Z and the depth is $2a$. Therefore the area A is $A = 2aZ$. The conductance of the channel is given as 1/3 of the conductance of the free channel, therefore:

$$G_{ch} = \frac{1}{3}\frac{\sigma A}{L}$$

Substitute:
$$G_{ch} = \frac{2}{3}\frac{eN_d\mu_e aZ}{L}$$

The voltage across the channel is the pinch-off voltage V_P (see Figure 6Q11-5). At pinch-off the drain current is I_{DSS}, given as:

$$I_{DSS} = V_P G_{ch}$$

$$\therefore \qquad I_{DSS} = V_P\left(\frac{2}{3}\frac{eN_d\mu_e aZ}{L}\right) \qquad (1)$$

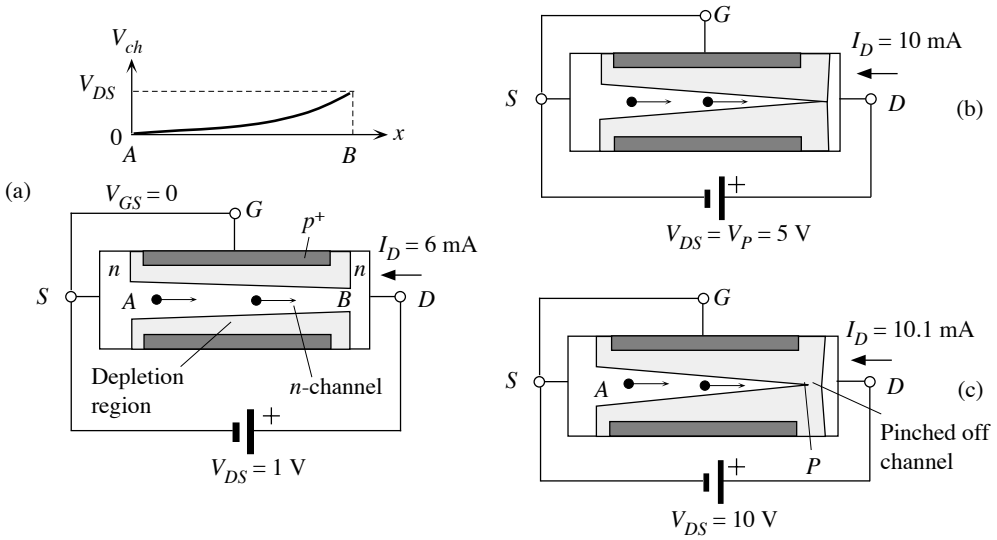


Figure 6Q11-5

(a) The gate and source are shorted ($V_{GS} = 0$) and V_{DS} is small.

(b) V_{DS} has increased to a value that allows the two depletion layers to just touch, when $V_{DS} = V_P$ (= 5 V) and the p^+n junction voltage at the drain end, $V_{GD} = -V_{DS} = -V_P = -5$ V.

(c) V_{DS} is large ($V_{DS} > V_P$), so a short length of the channel is pinched off.

Assume temperature $T = 300$ K and that the JFET is Si. The relative permittivity of Si is $\varepsilon_r = 11.9$ (Table 5.1 in the textbook) and the intrinsic concentration is $n_i = 1.45 \times 10^{10}$ cm^{-3}. The channel donor concentration N_d is given as 10^{15} cm^{-3} and the gate acceptor concentration $N_a = 10^{19}$ cm^{-3}. The electron drift mobility with $N_d = 10^{15}$ cm^{-3} is approximately $\mu_e = 1350$ cm^2 V^{-1} s^{-1} (see Table 5.1 in the textbook, the dopant concentration is too low to affect μ_e). V_o can be calculated as follows:

$$V_o = \frac{kT}{e}\ln\left(\frac{N_d N_a}{N_i^2}\right) = \frac{(1.381\times10^{-23}\ \text{J/K})(300\ \text{K})}{(1.602\times10^{-19}\ \text{C})}\ln\left(\frac{(10^{21}\ \text{m}^{-3})(10^{25}\ \text{m}^{-3})}{(1.45\times10^{16}\ \text{m}^{-3})^2}\right)$$

$\therefore$ $V_o = 0.8145$ V

We can calculate a from the given pinch-off voltage, $V_P = 3.9$ V:

$$V_P = \frac{a^2 e N_d}{2\varepsilon} - V_o$$

$\therefore$ $$a = \sqrt{\frac{2\varepsilon(V_P + V_o)}{eN_d}} = \sqrt{\frac{2(11.9)(8.854\times10^{-12}\ \text{F/m})(3.9\ \text{V} + 0.8145\ \text{V})}{(1.602\times10^{-19}\ \text{C})(10^{21}\ \text{m}^{-3})}}$$

$\therefore$ $a = 2.490 \times 10^{-6}$ m or 2.490 μm

Therefore the **channel width ($2a$) is 4.98 μm**. I_{DSS} is given as 5.5 mA. The Z/L ratio can then be found from Eqn. (1) above:

$$I_{DSS} = V_P\left(\frac{2}{3}\frac{eN_d\mu_e aZ}{L}\right)$$

$\therefore$ $$\frac{Z}{L} = \frac{3I_{DSS}}{2V_P eN_d\mu_e a}$$

$$\therefore \qquad \frac{Z}{L} = \frac{3\left(5.5\times10^{-3}\ \text{A}\right)}{2(3.9\ \text{V})\left(1.602\times10^{-19}\ \text{C}\right)\left(10^{21}\ \text{m}^{-3}\right)\left(0.135\ \text{m}^2\ \text{V}^{-1}\ \text{s}^{-1}\right)\left(2.49\times10^{-6}\ \text{m}\right)}$$

$$\therefore \qquad \boldsymbol{\frac{Z}{L} = 39.28}$$

The channel length L is given as 10 μm, therefore:

$$\boldsymbol{Z} = 39.28(10 \times 10^{-6}\ \text{m}) = \mathbf{3.93 \times 10^{-4}\ m}\ \text{or}\ \mathbf{393\ \mu m}$$

The gate area is A_{gate} and is equal to $Z \times L = (3.93 \times 10^{-4}\ \text{m})(10 \times 10^{-6}\ \text{m}) = 3.93 \times 10^{-9}\ \text{m}^2$. The depletion capacitance per unit area is:

$$C_{dep} = A_{gate}\left(\frac{e\varepsilon N_d N_a}{2\left[N_d + N_a\right]V_o}\right)^{\frac{1}{2}}$$

and the gate capacitance C_{gate} is twice C_{dep} as the JFET is symmetric and the two gates are connected in parallel as in Figure 6Q11-1.

$$C_{gate} = 2\left(3.93\times10^{-9}\ \text{m}^2\right)\left(\frac{\left(1.602\times10^{-19}\ \text{C}\right)(11.9)\left(8.854\times10^{-12}\ \text{F/m}\right)\left(10^{21}\ \text{m}^{-3}\right)\left(10^{25}\ \text{m}^{-3}\right)}{2\left[10^{21}\ \text{m}^{-3} + 10^{25}\ \text{m}^{-3}\right](0.81486\ \text{V})}\right)^{\frac{1}{2}}$$

$$\therefore \qquad \boldsymbol{C_{gate} = 8.00 \times 10^{-13}\ F}$$

This neglects stray capacitances (e.g. between gate and source leads, gate and drain leads etc.).

6.12 The JFET amplifier

Consider an *n*-channel JFET that has a pinch-off voltage (V_P) of 5 V and I_{DSS} = 10 mA. It is used in a common source configuration as in Figure 6Q12-1a in which the gate to source bias voltage (V_{GS}) is –1.5 V. Suppose that V_{DD} = 25 V.

a. If a small signal voltage gain of 10 is needed, what should be the drain resistance (R_D)? What is V_{DS}?

b. If an ac signal of 3 V peak-to-peak is applied to the gate in series with the dc bias voltage, what will be the ac output voltage peak-to peak? What is the voltage gain for positive and negative input signals? What is your conclusion?

Solution

Given, V_P = 5 V, I_{DSS} = 10 mA, $V_{GS} = V_{GG}$ = -1.5 V.

a If a small signal voltage gain of 10 is needed, what should be the drain resistance (R_D)? See Example 6.11 (in the textbook):

$$I_{DS} = I_{DSS}\left[1 - \left(\frac{V_{GS}}{-V_P}\right)\right]^2 = (10\ \text{mA})\left[1 - \left(\frac{-1.5\ \text{V}}{-5\ \text{V}}\right)\right]^2 = 4.90\ \text{mA}$$

$$\therefore \qquad g_m = \frac{dI_{DS}}{V_{GS}} = \frac{2I_{DSS}}{V_P}\left[1 - \left(\frac{V_{GS}}{-V_P}\right)\right] = \frac{2(10\ \text{mA})}{5\ \text{V}}\left[1 - \left(\frac{-1.5\ \text{V}}{-5\ \text{V}}\right)\right] = 2.80 \times 10^{-3}\ \text{A/V}$$

From Equation 6.56 (in the textbook), the small signal voltage gain $A_V = g_m R_D$, so that

$$\boldsymbol{R_D} = A_V / g_m = (10)/(2.80 \times 10^{-3}\ \text{A/V}) = \mathbf{3571\ \Omega}$$

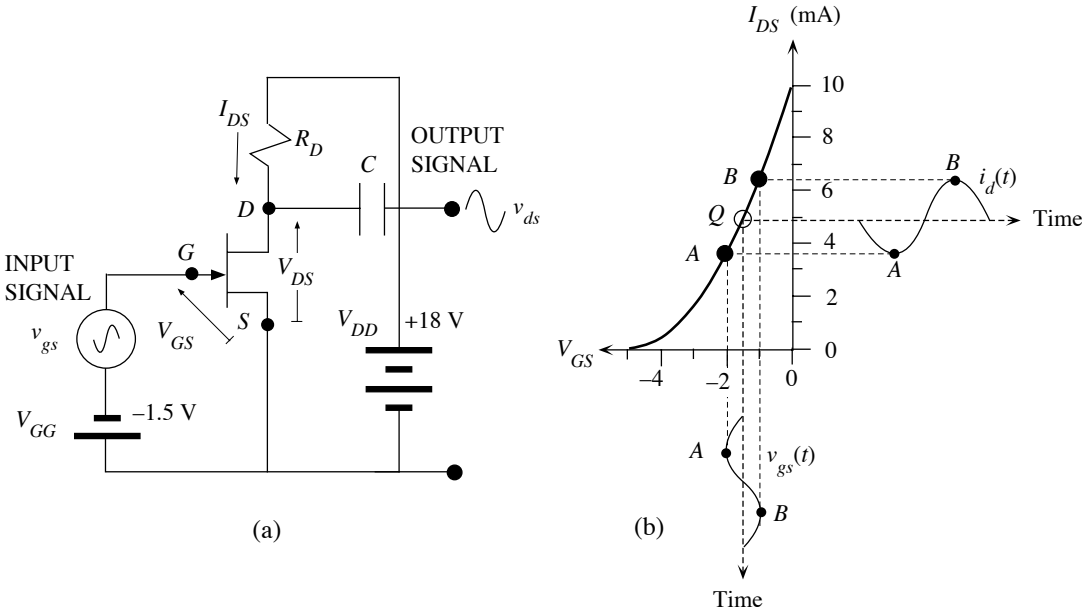


Figure 6Q12-1

(a) Common source (CS) ac amplifier using a JFET.

(b) Explanation of how I_D is modulated by the signal, v_{gs} in series with the dc bias voltage V_{GG}.

V_{DS} is given by Eqn. 6.53 (in the textbook) (I_{DS} = 4.9 mA from Table 6.1 in the textbook):

$$V_{DS} = V_{DD} - I_{DS}R_D$$

$$\boldsymbol{V_{DS}} = 25\text{ V} - (4.9 \times 10^{-3}\text{ A})(3571\ \Omega) = \mathbf{7.50\ V}$$

b V_{DD} is given as 25 V. Let R_D = 3571 Ω.

Negative input signal:

When v_{signal} = -1.5 V at the input then,

$$V_{GS\min} = V_{GG} + v_{\text{signal}} = -1.5\text{ V} - 1.5\text{ V} = -3\text{ V}$$

$$\therefore \quad I_{DS\min} = I_{DSS}\left[1 - \left(\frac{V_{GS}}{-V_P}\right)\right]^2 = (10\ \text{mA})\left[1 - \left(\frac{-3\text{ V}}{-5\text{ V}}\right)\right]^2 = 1.60\text{ mA}$$

$$\therefore \quad \boldsymbol{V_{DS\max}} = V_{DD} - I_{DS\min}R_D = (25\text{ V}) - (1.60\text{ mA})(3.571\text{ k}\Omega) = \mathbf{19.29\ V}$$

For negative going signals, the gain is,

$$A_{V-} = \frac{\text{Change in output voltage}}{\text{Change in input voltage}} = \frac{V_{DS\max} - V_{DS}}{v_{\text{signal}}} = \frac{19.29\text{ V} - 7.51\text{ V}}{-1.5\text{ V}}$$

$$\therefore \quad \boldsymbol{A_{V-}} = \mathbf{-7.85}$$

Positive input signal:

When v_{signal} = +1.5 V at the input then,

$$V_{GS\max} = V_{GG} + v_{\text{signal}} = -1.5\text{ V} + 1.5\text{ V} = 0\text{ V}$$

$$\therefore \quad I_{DS\max} = I_{DSS}\left[1 - \left(\frac{V_{GS}}{-V_P}\right)\right]^2 = (10\ \text{mA})\left[1 - \left(\frac{0\text{ V}}{-5\text{ V}}\right)\right]^2 = 10.0\text{ mA}$$

$\therefore$ $$\boldsymbol{V_{DS\min}} = V_{DD} - I_{DS\max}R_D = 25\text{ V} - (10.0\text{ mA})(3.571\text{ k}\Omega) = \mathbf{-10.7\text{ V}}$$

This is nonsense as $V_{DS\min}$ can not be below zero. The peak to peak voltage is then 19.3 V - 0 V = 19.3 V. Therefore the JFET has saturated with $V_{DS} \approx 0$ and the voltage across R_D being V_{DD}. This occurs when $I_{DS} = I_{DS\text{sat}}$:

The JFET amplifier saturates when

$$I_{DS} = I_{DS\text{sat}} \approx V_{DD}/R_D = 25\text{ V} / (3.571\text{ k}\Omega) = 7.00\text{ mA}$$

For positive going signals the output eventually becomes saturated and the gain is,

$$A_{V+} = \frac{\text{Change in output voltage}}{\text{Change in input voltage}} = \frac{V_{DS\,\min} - V_{DS}}{v_{\text{signal}}} \approx \frac{0 - 7.5}{+1.5}$$

$\therefore$ $$\boldsymbol{A_{V+} = -5.00}$$

The JFET amplifier is operating non-linearly. The negative going *output* signal will be clipped for the positive going *input* signal. The negative sign in both cases represents a phase shift of 180°.

Can we find the signal v_{signal} for this saturation (clipping) condition in the positive going input signal? Normally this would be done using a *load line* with the JFET characteristics (as in electronics circuits courses). It may be thought that we can at least estimate $V_{GS\min}$ from

$$I_{DS} = I_{DSS}\left[1 - \left(\frac{V_{GS}}{-V_P}\right)\right]^2$$

with $I_{DS} = I_{DS\max}$ to find the required $V_{GS\min}$. However this equation is not valid when $V_{DS} < V_{DS(\text{sat})}$, see Figure 6Q12-2, which will be the case when the drain current drops V_{DD} across R_D. As a very rough estimate, using the above equation with $I_{DS\max} = 10$ mA, gives $V_{GS\min} = -0.82$ V which can be interpreted as a rough condition for *approaching* saturation (clipping). Thus when $V_{GG} + v_{\text{signal}} \approx -0.82$ V, the output should be approaching saturation, or when $v_{\text{signal}} \approx +0.68$ V.

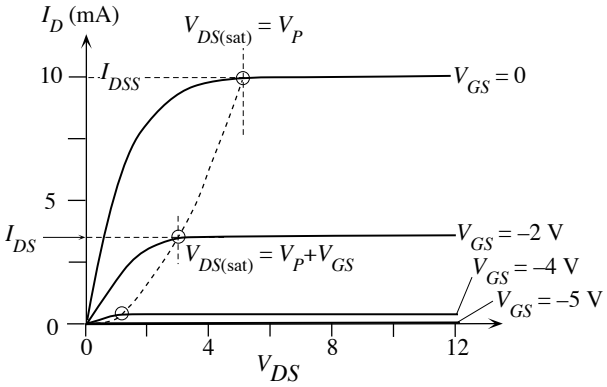


Figure 6Q12-2 Typical I_D versus V_{DS} characteristics of a JFET for various fixed gate voltages V_{GS}.

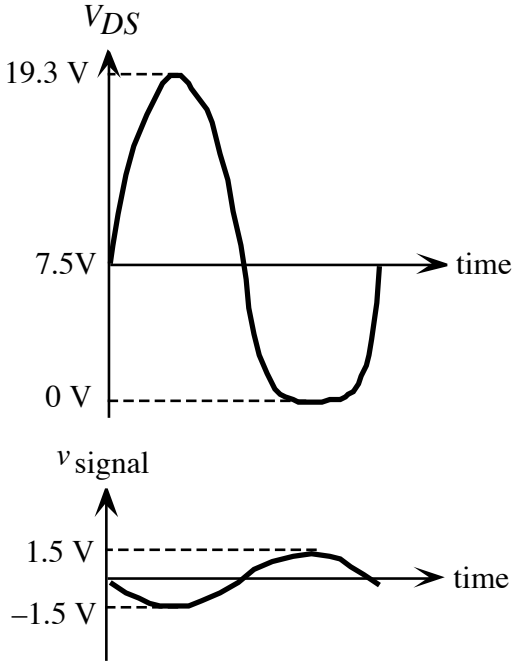


Figure 6Q12-3

6.13 The Enhancement NMOSFET amplifier

Consider an *n*-channel Si enhancement NMOS that has a gate width (Z) of 150 μm, channel length (L) of 10 μm, and oxide thickness (t_{ox}) of 500 Å. The channel has μ_e = 700 cm^2 V^{-1} s^{-1} and the threshold voltage (V_{th}) is 2 V (ε_r = 3.9 for SiO_2).

a. Calculate the drain current when V_{GS} = 5 V and V_{DS} = 5 V and assuming $\lambda = 0.01$.

b. What is the small-signal voltage gain if the NMOSFET is connected as a common source amplifier as shown in Figure 6Q13-1, with a drain resistance R_D of 2.2 kΩ, the gate biased at 5 V with respect to source (V_{GG} = 5 V) and V_{DD} is such that V_{DS} = 5 V? What is V_{DD}? What will happen if the drain supply is smaller?

c. Estimate the most positive and negative input signal voltages that can be amplified if V_{DD} is fixed at the above value in b.

d. What factors will lead to a higher voltage amplification?

Solution

a Given the device geometry and fabrication characteristics we can calculate K for this device. Using Z = 150 μm, L = 10 μm, $\mu_e \approx$ 700 cm^2 V^{-1} s^{-1}, $\varepsilon = \varepsilon_o\varepsilon_r$ and ε_r = 3.9 for SiO_2, and t_{ox} = 500 Å = 500 × 10^{-10} m, then,

$$K = \frac{Z\mu_e\varepsilon}{2Lt_{ox}}$$

$$K = \frac{(150\times10^{-6}\text{ m})(700\times10^{-4}\text{ m}^2\text{ V}^{-1}\text{ s}^{-1})(3.9\times8.854\times10^{-12}\text{ F m}^{-1})}{2(10\times10^{-6}\text{ m})(5\times10^{-8}\text{ m})}$$

giving, $K = 0.363 \times 10^{-3}$ A V^{-2}

Given $V_{th} = 2$ V, we note that $V_{DS(sat)}$ at $V_{GS} = 5$ V is determined by

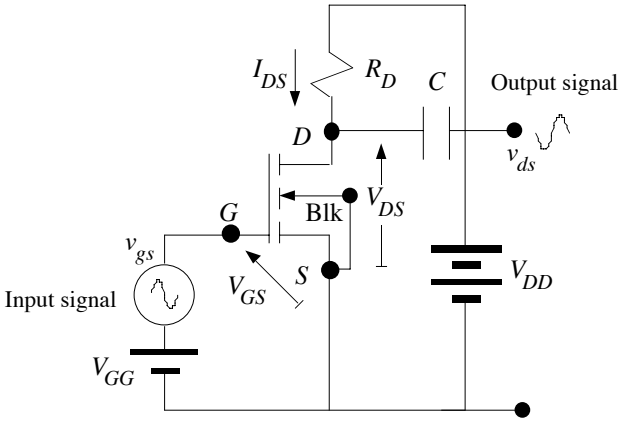
i.e. $$V_{DS(sat)} = V_{GS} - V_{th} = 5 \text{ V} - 2 \text{ V} = 3 \text{ V}$$

Thus when $V_{GS} = 5$ V and $V_{DS} = 5 \text{ V} > V_{DS(sat)}$, we have

$$I_{DS} = K\left(V_{GS} - V_{th}\right)^2\left(1 + \lambda V_{DS}\right)$$

$$I_{DS} = \left(3.63 \times 10^{-4} \text{ A V}^{-2}\right)(5 \text{ V} - 2 \text{ V})^2\left(1 + 0.01 \text{ V}^{-1} \times 5 \text{ V}\right)$$

i.e. $$\boldsymbol{I_{DS} = 3.42 \times 10^{-3} \text{ A}} \text{ or } \boldsymbol{3.4 \text{ mA}}$$



An NMOSFET Amplifier

Figure 6Q13-1

b The mutual conductance (or transconductance) is

$$g_m = 2K(V_{GS} - V_{th})(1 + \lambda V_{DS})$$

i.e. $$g_m = 2(0.363 \times 10^{-3} \text{ A V}^{-2})(5 \text{ V} - 2 \text{ V})(1 + 0.01 \text{ V}^{-1} \times 5 \text{ V}) = 2.28 \times 10^{-3} \text{ AV}^{-1}$$

The small signal voltage gain is,

$$A_V = \frac{v_{ds}}{v_{gs}} = \frac{-R_D i_d}{v_{gs}} = -\frac{R_D \delta I_{DS}}{\delta V_{GS}} = \pm g_m R_D$$

i.e. $$\boldsymbol{A_V = -g_m R_D} = -(2.28 \times 10^{-3} \text{ AV}^{-1})(2200 \ \Omega) = \boldsymbol{-5.05}$$

Since $V_{DS} = V_{DD} - I_{DS}R_D$ we can find V_{DD}.

$$\boldsymbol{V_{DD}} = V_{DS} + I_{DS}R_D = 5 \text{ V} + (3.4 \text{ mA})(2.2 \text{ k}\Omega) = \boldsymbol{12.5 \text{ V}}$$

A smaller drain supply will reduce the gain and also limit the voltage swing.

c The output V_{DS} can only change between 0 and V_{DD}. Obviously, $V_{DS} = V_{DD}$ when $I_D = 0$, which corresponds to $V_{GSmin} = V_{th}$. This means that the most negative input voltage v_{gs} has to be

$$\boldsymbol{v_{gsmin}} = -(V_{GS} - V_{th}) = -(5 \text{ V} - 2 \text{ V}) = \boldsymbol{-3 \text{ V}}$$

As the input voltage v_{gs} and hence V_{GS} increase, the drain current I_{DS} also increases and V_{DS} decreases. As soon as V_{DS} reaches $V_{DS(sat)}$, any further increase in V_{GS} will lead $V_{DS} < V_{DS(sat)}$ which means $I_D \neq I_{DS}$, or I_D is no longer the saturated drain current and therefore no longer given by $I_{DS} = \text{K}(V_{GS} - V_{th})^2$. One can assume that once $V_{DS} < V_{D(sat)}$, the MOSFET amplifier will become highly non-linear and will be operating in the non-desirable range, that is, further increases in V_{GS} will cause the output V_{DS} to change by small amounts. Suppose that V_{GSmax} is the GS voltage that takes V_{DS} to $V_{DS(sat)}$ so that the channel is just pinched off. Then just at pinch-off,

$$V_{GSmax} - V_{DS(sat)} = V_{th}$$

and
$$I_{DS} = \frac{V_{DD} - V_{DS(sat)}}{R_D} \approx K(V_{GSmax} - V_{th})^2$$

Combining the two equations we find a quadratic equation in V_{GSmax},

$$(KR_D)V_{GSmax}^{\ 2} + (1 - 2KR_DV_{th})V_{GSmax} + (KR_DV_{th}^{\ 2} - V_{DD} - V_{th}) = 0$$

Substituting for $K = 0.363 \times 10^{-3}$ A V^{-2}, $R_D = 22000\ \Omega$, $V_{DD} = 12.5$ V, $V_{th} = 2$ V,

$$(0.7977)\ V_{GSmax}^{\ 2} + (-2.19)V_{GSmax} + (-11.31) = 0$$

and solving, $V_{GSmax} \approx 5.38$ V

This corresponds to an the input signal is

$$v_{gsmax} = V_{GSmax} - V_{GG} = 5.38\ \text{V} - 5\ \text{V} = \mathbf{0.38\ V}$$

d If we set $V_{DS} = {}^1/_2 V_{DD}$ (typical design for a symmetrical swing about the dc value), then $I_{DS} = V_{DD}/(2R_D)$ and the *magnitude* of voltage gain is,

$$A_V = g_m R_D \propto [KI_{DS}]^{1/2} R_D = \left[K\frac{V_{DD}}{2R_D}\right]^{1/2} R_D = \left[\frac{KR_DV_{DD}}{2}\right]^{1/2}$$

or
$$A_V \propto [KR_DV_{DD}]^{1/2}$$

Thus we can increase the gain by increasing R_D, V_{DD} or the device constant K. For the latter, we can use a wider gate width (greater Z), narrower channel (smaller L), or a thinner oxide thickness (smaller t_{ox}).

*6.14 Ultimate limits to device performance

a. Consider the speed of operation of an *n*-channel FET-type device. The time required for an electron to transit from the source to the drain is $\tau_t = L/v_d$, where L is the channel length and v_d is the drift velocity. This transit time can be shortened by shortening L and increasing v_d. As the field increases, the drift velocity eventually saturates at about $v_{dsat} = 10^5$ m s^{-1} when the field in the channel is equal to $\mathcal{E}_c \approx 10^6$ V m^{-1}. A short τ_t requires a field that is at least $\mathcal{E}_c$.

1. What is the change in the *PE* of an electron when it traverses the channel length L from source to drain if the voltage difference is V_{DS}?
2. This energy must be greater than the energy due to thermal fluctuations, which is of the order of kT. Otherwise, electrons would be brought in and out of the drain due to thermal fluctuations. Given the minimum field and V_{DS}, what is the minimum channel length and hence the minimum transit time?

b. Heisenberg's uncertainty principle relates the energy and the time duration in which that energy is possessed through a relationship of the form (Chapter 3) $\Delta E \Delta t > \hbar$. Given that during the transit of the electron from the source to the drain its energy changes by eV_{DS}, what is the shortest transit time, τ, satisfying Heisenberg's uncertainty principle? How does it compare with your calculation in part (*a*)?

c. How does electron tunneling limit the thickness of the gate oxide and the channel length in a MOSFET? What would be typical distances for tunneling to be effective? (Consider Example 3.10 in the textbook).

Solution

Assume temperature $T = 300$ K. The saturation velocity is given as $v_{dsat} = 10^5$ m/s and the saturation field is given as $\mathcal{E}_c = 10^6$ V/m.

a(1) The change in the PE is ΔPE. This is the charge times the voltage, i.e. $\Delta PE = eV_{DS}$.

(2) The lower limit to ΔPE due to thermal fluctuations is kT. Therefore, substituting into the equation above:

$$eV_{DS} = kT$$

$$\therefore \quad V_{DS} = kT/e = (1.381 \times 10^{-23} \text{ J/K})(300 \text{ K})/(1.602 \times 10^{-19} \text{ C}) = 0.02586 \text{ V}$$

This is the lower limit to V_{DS}. The minimum channel length L can now be found from the minimum electric field, given by $\mathcal{E}_c = V_{DS}/L$:

$$\therefore \quad \boldsymbol{L = V_{DS}/\mathcal{E}_c = (0.02586 \text{ V})/(10^6 \text{ V/m}) = 2.586 \times 10^{-8} \text{ m}}$$

The minimum transit time τ_t is then:

$$\tau_t = L/v_{dsat} = (2.586 \times 10^{-8} \text{ m})/(10^5 \text{ m/s}) = \mathbf{2.586 \times 10^{-13} \text{ s}}$$

The above limit is the thermal fluctuation limit (thermal noise limit).

b Consider the Heisenberg uncertainty principle. Let $\tau = \Delta t$ be the transit time. During this time the energy changes by $\Delta E = eV_{DS}$. We are given $\Delta E \Delta t > \hbar$, therefore, substituting for the shortest transit time:

$$eV_{DS}\tau = \hbar$$

$$\therefore \quad \tau = \frac{\hbar}{eV_{DS}} = \frac{1.055 \times 10^{-34} \text{ J s}}{(1.602 \times 10^{-19} \text{ C})(0.02586 \text{ V})} = \mathbf{2.55 \times 10^{-14} \text{ s}}$$

The uncertainty limit allows a shorter transit time down to 0.0255 ps. Thus thermal fluctuation limit will operate at room temperature.

c If the oxide becomes too thin then the electron tunneling will allow gate charge to tunnel into the channel. This will prevent the gate from holding the necessary charge and will lead to a gate current. The field effect will fail. From Example 3.10, one can guess that the thickness should be less than 1 nm or 10 Å depending on various material properties.

6.15 LED Output Spectrum

Given that the width of relative light intensity vs. photon energy spectrum of a LED is typically around ~3kT, what is the linewidth $\Delta\lambda$ in the output spectrum in terms of wavelength?

Solution

The emitted wavelength λ is related to the photon energy E_{ph} by

$$\lambda = \frac{c}{\upsilon} = \frac{hc}{E_{ph}}$$

If we differentiate λ with respect to the photon energy E_{ph} we get

$$\frac{d\lambda}{dE_{ph}} = -\frac{hc}{E_{ph}^2}$$

We can represent small changes or intervals (or Δ) by differentials, e.g. $\frac{\Delta\lambda}{\Delta E_{ph}} \approx \left|\frac{d\lambda}{dE_{ph}}\right|$,

then

$$\Delta\lambda \approx \frac{hc}{E_{ph}^2}\Delta E_{ph}$$

We are given the energy width of the output spectrum, $\Delta E_{ph} = h\Delta\nu \approx 3k_B T$

Then using the later and substituting for E_{ph} in terms of λ we find

$$\Delta\lambda \approx \lambda^2 \frac{3kT}{hc}$$

6.16 LED Output Wavelength Variations

Show that the change in the emitted wavelength λ with temperature T from a LED is given by

$$\frac{d\lambda}{dT} = -\frac{hc}{E_g^2}\left(\frac{dE_g}{dT}\right)$$

where E_g is the band gap. Consider a GaAs LED. The band gap of GaAs at 300 K is 1.42 eV which changes (decreases) with temperature as $dE_g/dT = -4.5 \times 10^{-4}$ eV K^{-1}. What is the change in the emitted wavelength if the temperature change is 10 °C?

Solution

The emitted photon energy is approximately equal to the band gap energy E_g. The emitted wavelength is then

$$\lambda = \frac{c}{\upsilon} = \frac{hc}{E_g}$$

Differentiating λ with respect to temperature (note that $E_g = f(T)$) we receive

$$\frac{d\lambda}{dT} = -\frac{hc}{E_g^2}\left(\frac{dE_g}{dT}\right) = \frac{\left(6.626\times10^{-34}\ \text{J s}\right)\left(3\times10^{8}\ \text{m s}^{-1}\right)}{\left(1.42\times1.602\times10^{-19}\ \text{J}\right)}\left(-4.5\times10^{-4}\times1.602\ \text{J K}^{-1}\right)$$

so that

$$\frac{d\lambda}{dT} = \mathbf{2.77 \times 10^{-10}}\ \text{m K}^{-1} = \mathbf{0.277\ nm\ K^{-1}}$$

The change in the wavelength for ΔT = 10 °C is

$$\Delta\lambda = \left(\frac{d\lambda}{dT}\right)\Delta T = (0.277\ \text{nm K}^{-1})(10\ \text{K}) \approx \mathbf{2.8\ nm}$$

Since E_g decreases with temperature, the wavelength increases with temperature. This calculated change is within 10% of typical values for GaAs LEDs quoted in the data books.

6.17 Solar Cell Driving a Load

a. A Si solar cell of area 1 cm^2 is connected to drive a load *R* as in Figure 6Q17-1a. It has the I-V characteristics in Figure 6Q17-1b under illumination of 500 W m^{-2}. Suppose that the load is 20 Ω and it is used under a light intensity of 1 kW m^{-2}. What are the current and the voltage in the circuit? What is the power delivered to the load? What is the efficiency of the solar cell in this circuit?

b. What should be the load to obtain maximum power transfer from the solar cell to the load at 1 kW m^{-2} illumination? What is this load at 500 W m^{-2}?

c. Consider using a number of such solar cells to drive a calculator that needs a minimum of 3 V and draws 3.0 mA at 3-4 V. It is to be used at a light intensity of about 500 W m^{-2}. How many solar cells would you need and how would you connect them?

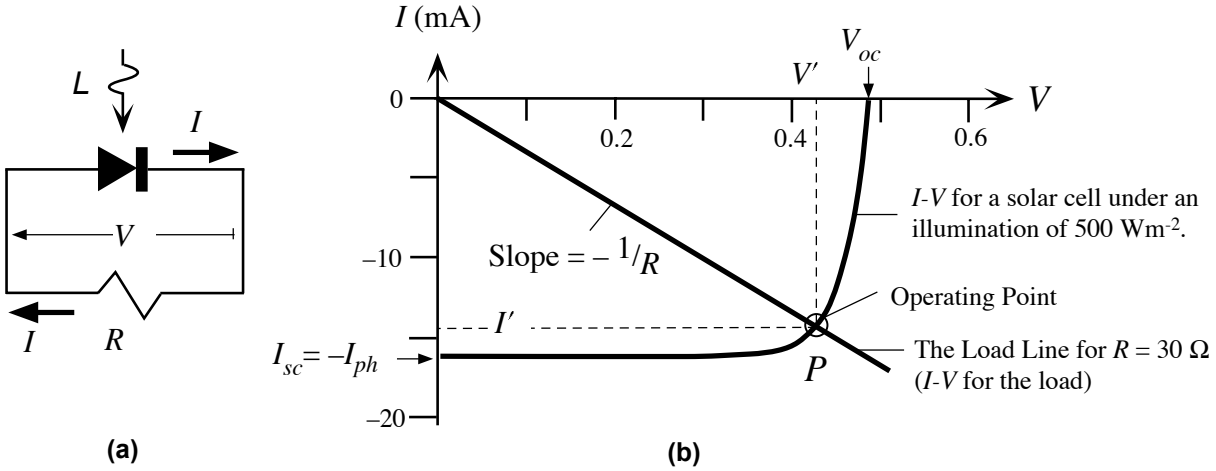


Figure 5Q17-1

(a) When a solar cell drives a load *R*, *R* has the same voltage as the solar cell but the current through it is in the opposite direction to the convention that current flows from high to low potential.

(b) The current *I′* and voltage *V′* in the circuit of (a) can be found from a load line construction. Point *P* is the operating point (*I′*, *V′*). The load line is for *R* = 30 Ω.

Solution

a The total current through the solar cell is given by Equation 6.64 (in the textbook) which can be written as

$$I_{500} = -I_{500\,ph} + I_0\left[\exp\left(\frac{V}{\eta V_t}\right) - 1\right], \qquad (1)$$

where $V_t = \dfrac{kT}{e}$. There are three unknowns in Equation (1) - $I_{500\,ph}$, I_0 and ηV_t. From Figure 5Q17-1b we can determine that:

V, [V]	*I*, [mA]
0	-16.2
.45	-13.1
.49	0

It is clear that I_{ph} = 16.2 mA. In order to determine I_0 and ηV_t we have to solve the following system of equations:

$$\begin{cases} I_0\left[\exp\left(\dfrac{0.49}{\eta V_t}\right)-1\right]=16.2\times10^{-3} \\ I_0\left[\exp\left(\dfrac{0.45}{\eta V_t}\right)-1\right]=(16.2-13.1)\times10^{-3} \end{cases}$$

The solution of this system is:

$$I_0 = 2.58 \times 10^{-11} \text{ A} \quad \text{and} \quad \eta V_t = 0.024$$

Now we are ready to scale I-V characteristics of the solar cell for illumination of 1000 W m^{-1}. Using Equations 6.63 and 6.64 (in the textbook), it is easy to show that for this light intensity the current through the solar cell will be

$$I_{1000} = -2\,I_{500\,ph} + I_0\left[\exp\left(\frac{V}{\eta V_t}\right)-1\right] \qquad (2)$$

Figure 6Q17-2 shows the I-V curves for both illuminations together with the load lines for R = 30Ω and R = 20 Ω. Now we can find the interception of the solar cell I-V curve for 1000 W m^{-2} with the load line for R = 20 Ω solving the equation

$$-2\,I_{500\,ph} + I_0\left[\exp\left(\frac{V}{\eta V_t}\right)-1\right] = -\frac{V}{R}$$

The solution of this equation can be obtained by several iterations or graphically using Figure 6Q17-2. The result is V' = **0.475 V.**

At this voltage the current trough the circuit will be I' = 0.475/20 = **0.024 A** and the power delivered to the load is P' =0.475 × 0.024 = **0.011 W.** *This is not the maximum power available from the solar cell***.** The input sun-light power is

$$P_{in} = (\text{Light intensity})\ (\text{Surface area}) = \left(1000\ \text{W m}^{-2}\right)\left(1\times10^{-4}\ \text{m}^2\right) = \mathbf{0.1\ W}$$

The efficiency of the solar cell is then

$$\eta_{\text{solar}} = \frac{P©}{P_{in}}\times100\% = \frac{(0.011)}{(0.1)}\times100\% = \mathbf{11\ \%}$$

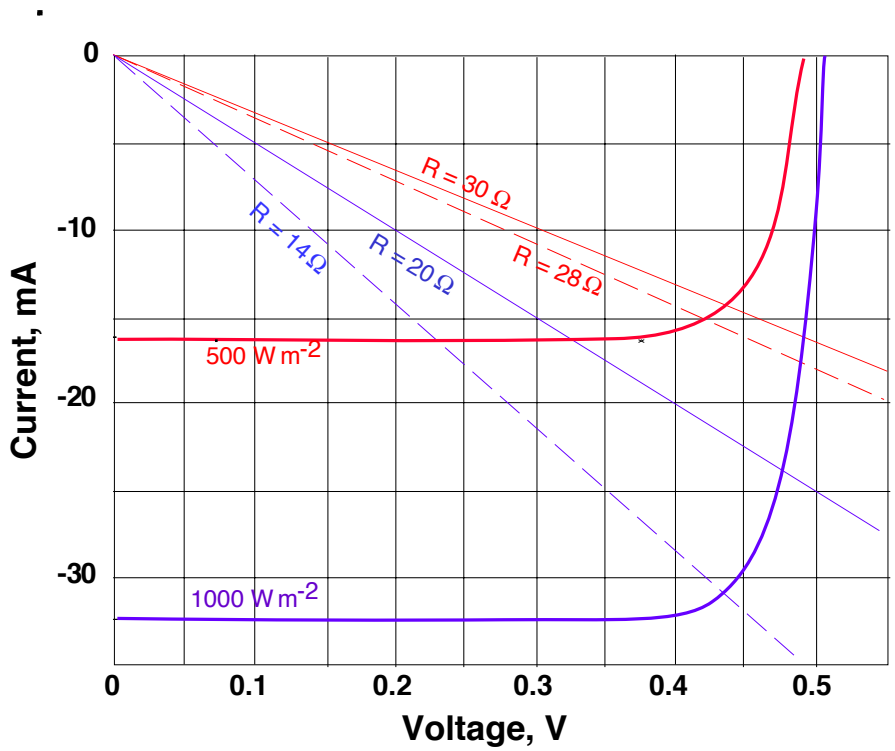


Figure 6Q17-2

b The power delivered to the load for the case of 1000 W illumination as a function of the voltage across the load is

$$P(V) = IV = V\left\{-2I_{500\ ph}\left[\exp\left(\frac{V}{\eta V_t}\right)-1\right]\right\} \qquad (3)$$

The maximum efficiency point corresponds to the extrema in expression (3). Differentiating Eqn. 3 and setting the derivative equal to zero we receive

$$\frac{I_0}{\eta V_t}V\exp\left(\frac{V}{\eta V_t}\right)-2I_{500\,ph}+I_0\left[\exp\left(\frac{V}{\eta V_t}\right)-1\right]=0$$

and the solution of that equation is V_m = **0.436 V.** Substituting this value in Eqn. (2) we receive the current at that operating point: I_m = **0.031 mA.** To achieve this operation point we must connect to the solar cell load with resistance of

$$R_{m\ 1000}=\frac{V_m}{I_m}=\frac{(0.436\ \text{V})}{(0.031\ \text{mA})}=\mathbf{14.07\ \Omega}$$

Now it is easy to find the fill factor of the solar cell for illumination of 1000 W m.$^{-2}$**.** Setting Eqn. (2) equal to zero we can calculate the open circuit voltage V_{oc} = **0.507 V.** The fill factor (Equation 6.66 in the textbook) is then

$$FF_{1000}=\frac{I_m V_m}{I_{sc}V_{oc}}=\frac{(0.031\ \text{A})(0.436\ \text{V})}{(0.0324\ \text{A})(0.507\ \text{V})}=\mathbf{0.823}$$

Performing similar calculations we can find that when the illumination is 500 W m^{-2} the most efficient operating point is V_m = **0.42 V** and I_m = **0.015 A,** which corresponds to load $R_{m\,500}$ = **28 Ω**

(very close to 30 Ω). The fill factor for this case is $FF_{500} = \mathbf{0.794}$. The most efficient load lines are also plotted in Figure 6Q17-2.

c We need to connect the solar cells in series to obtain the necessary voltage. Assume that all the solar cells are identical and in series. Each has the same current. The short circuit current from Figure 6Q17-2 is much higher than the calculator current. From Figure 6Q17-2, when I = 3 mA, V is around 0.48 V. We need at least 3/0.48 = 6.25 or **7 solar cells**.

6.18 Open Circuit Voltage

A solar cell under an illumination of 100 W m^{-2} has a short circuit current I_{sc} of 50 mA and an open circuit voltage V_{oc}, of 0.55 V. What are the short circuit current and open circuit voltage when the light intensity is halved?

Note: For more examples and problems on LEDs and solar cells, see Optoelectronics and Photonics: Principles and Practices. S. O. Kasap (Prentice Hall, 2001) chs 3 and 6.

Solution

The general I-V characteristics under illumination are given by Equation 6.64 (in the textbook)

$$I = -I_{ph} + I_0\left[\exp\left(\frac{eV}{\eta kT}\right) - 1\right]$$

Setting $I = 0$ for open circuit we have

$$-I_{ph} + I_0\left[\exp\left(\frac{eV}{\eta kT}\right) - 1\right] = 0$$

Assuming that $V_{oc} >> \eta kT$ and rearranging the above equation we can find V_{oc}

$$V_{oc} = \frac{\eta kT}{e}\ln\left(\frac{I_{ph}}{I_o}\right) \qquad (1)$$

In Eqn. (1), the photocurrent I_{ph} , depends on the light intensity I , via $I_{ph} = K I$

At a given temperature, then, the change in V_{oc} is

$$V_{oc\,2} - V_{oc\,1} = \frac{\eta kT}{e}\ln\left(\frac{I_{ph\,2}}{I_{ph\,1}}\right) = \frac{\eta kT}{e}\ln\left(\frac{I_2}{I_1}\right)$$

The short circuit current is the photocurrent, so that at halved intensity this is

$$I_{sc\,2} = I_{sc\,1}\frac{I_2}{I_1} = (50\ \text{mA})\left(\frac{1}{2}\right) = \mathbf{25\ mA}$$

Assuming $\eta = 1$ and $T = 300$ K the new open circuit voltage is

$$V_{oc\,2} = V_{oc\,1} + \frac{\eta kT}{e}\ln\left(\frac{I_2}{I_1}\right) = (0.55\ \text{V}) + (0.02585\ \text{V})\ln\left(\frac{1}{2}\right) = \mathbf{0.532\ V}$$

"Al Asuli writing in Bukhara some 900 years ago divided his pharmacopoeia into two parts, 'Diseases of the rich' and 'Diseases of the poor'."

Abdus Salam (1926-1997; Nobel Laureate, 1979)

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Chapter 7

7.1 Relative permittivity and polarizability

a. Show that the local field is given by

$$\mathcal{E}_{\text{loc}} = \mathcal{E}\left(\frac{\varepsilon_r + 2}{3}\right) \qquad \textit{Local field}$$

b. Amorphous selenium (a-Se) is a high resistivity semiconductor that has a density of approximately 4.3 g cm^{-3} and an atomic number and mass of 34 and 78.96 g/mol respectively. Its relative permittivity at 1 kHz has been measured to be 6.7. Calculate the relative magnitude of the local field in a-Se. Calculate the polarizability per Se atom in the structure. What type of polarization is this? How will ε_r depend on the frequency?

c. If the electronic polarizability of an isolated atom is given by

$$\alpha_e \approx 4\pi\varepsilon_o r_o^{\,3}$$

where r_o is the radius of the atom, then calculate the electronic polarizability of an isolated Se atom, which has r_0 = 0.12 nm, and compare your result with that for an atom in a-Se. Why is there a difference?

Solution

a The polarization, *P*, is given by:

$$P = \left(\varepsilon_o[\varepsilon_r - 1]\right)\mathcal{E}$$

where $\mathcal{E}$ is the electric field.

The local field $\mathcal{E}_{\text{loc}}$ is given by:

$$\mathcal{E}_{loc} = \mathcal{E} + \frac{P}{3\varepsilon_o}$$

Substitute for *P*: $$\mathcal{E}_{loc} = \mathcal{E} + \frac{\left(\varepsilon_o[\varepsilon_r - 1]\right)\mathcal{E}}{3\varepsilon_o} = \left(1 + \frac{(\varepsilon_r - 1)}{3}\right)\mathcal{E} = \left(\frac{3 + \varepsilon_r - 1}{3}\right)\mathcal{E}$$

$\therefore$ $$\mathcal{E}_{loc} = \left(\frac{\varepsilon_r + 2}{3}\right)\mathcal{E}$$

b The relative magnitude of the local field refers to the local field compared to the applied field, i.e.: $\mathcal{E}_{\text{loc}} / \mathcal{E}$. Therefore, with ε_r = 6.7:

$$\frac{\mathcal{E}_{\text{loc}}}{\mathcal{E}} = \left(\frac{\varepsilon_r + 2}{3}\right) = \left(\frac{6.7 + 2}{3}\right) = \mathbf{2.9}$$

If *D* is the density then the concentration of Se atoms *N* is

$$N = \frac{DN_A}{M_{at}} = \frac{\left(4.3\times10^{3}\ \text{kg/m}^3\right)\left(6.022\times10^{23}\ \text{mol}^{-1}\right)}{\left(78.96\times10^{-3}\ \text{kg/mol}\right)}$$

$$= 3.279 \times 10^{28}\ \text{m}^{-3}$$

The Clausius-Mossotti equation relates the relative permittivity to the electronic polarizability, α_e:

$$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = \frac{N}{3\varepsilon_o}\alpha_e$$

$\therefore$ $$\alpha_e = \frac{3\varepsilon_o(\varepsilon_r - 1)}{N(\varepsilon_r + 2)} = \frac{3\left(8.854\times10^{-12}\ \text{F/m}\right)(6.7-1)}{\left(3.279\times10^{28}\ \text{m}^{-3}\right)(6.7+2)}$$

$\therefore$ $$\alpha_e = \mathbf{5.31 \times 10^{-40}\ F\ m^2}$$

This would be a type of electronic polarization, as Se is a covalent solid. ε_r is flat up to optical frequencies.

c The Se atom has a radius of about $r_o = 0.12$ nm. Substituting into the given equation:

$$\alpha_e' \approx 4\pi\varepsilon_o r_o^{\,3} = 4\pi(8.854 \times 10^{-9}\ \text{F/m})(0.12 \times 10^{-9}\ \text{m})^3$$

$\therefore$ $$\alpha_e' \approx \mathbf{1.92 \times 10^{-40}\ F\ m^2}$$

Comparing this value and our previous value:

$$\frac{\alpha_e}{\alpha_e'} = \frac{5.30\times10^{-40}\ \text{F m}^2}{1.92\times10^{-40}\ \text{F m}^2} = 2.76$$

The observed polarizability per Se atom in the solid is 2.8 times greater than the polarizability of the isolated Se atom. In the solid, valence electrons are involved in bonding and these electrons contribute to electronic bond polarization (the field can displace these electrons).

7.2 Relative permittivity, bond strength, bandgap and refractive index

Diamond, silicon, and germanium are covalent solids with the same crystal structure. Their relative permittivities are shown in Table 7Q2-1.

a. Explain why ε_r increases from diamond to germanium.
b. Calculate the polarizability per atom in each crystal and then plot polarizability against the elastic modulus *Y* (Young's modulus). Should there be a correlation?
c. Plot the polarizability from part *b* against the bandgap energy, E_g. Is there a relationship?
d. Show that the refractive index *n* is $\sqrt{\varepsilon_r}$. When does this relationship hold and when does it fail?
e. Would your conclusions apply to ionic crystals such as NaCl?

Table 7Q2-1 Properties of diamond, Si, and Ge

	ε_r	M_{at}	Density (g cm^{-3})	α_e	Y (GPa)	E_g (eV)	n
Diamond	5.8	12	3.52		827	5.5	2.42
Si	11.9	28.09	2.33		190	1.12	3.45
Ge	16	72.61	5.32		75.8	0.67	4.09

Solution

a In diamond, Si, and Ge, the polarization mechanism is electronic (bond). There are two factors that increase the polarization. First is the number of electrons available for displacement and the ease with which the field can displace the electrons. The number of electrons in the core shells increases from diamond to Ge. Secondly, and most importantly, the bond strength per atom decreases from diamond to Ge, making it easier for valence electrons in the bonds to be displaced.

b For diamond, atomic concentration N is:

$$N = \frac{DN_A}{M_{at}} = \frac{\left(3.52\times10^{3}\ \text{kg/m}^3\right)\left(6.022\times10^{23}\ \text{mol}^{-1}\right)}{\left(12\times10^{-3}\ \text{kg/mol}\right)} = 1.766\times10^{29}\ \text{m}^{-3}$$

The polarizability can then be found from the Clausius-Mossotti equation:

$$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = \frac{N}{3\varepsilon_o}\alpha_e$$

$$\therefore \qquad \alpha_e = \frac{3\varepsilon_o(\varepsilon_r - 1)}{N(\varepsilon_r + 2)} = \frac{3\left(8.854\times10^{-12}\ \text{F/m}\right)(5.8-1)}{\left(1.766\times10^{29}\ \text{m}^{-3}\right)(5.8+2)}$$

$$\therefore \qquad \alpha_e = \mathbf{9.256\times10^{-41}\ F\ m^2}$$

The polarizability for Si and Ge can be found similarly, and are summarized in Table 7Q2-2:

Table 7Q2-2 Polarizability values for diamond, Si and Ge

	N (m^{-3})	α_e (**F m^2**)
Diamond	$1.766\times10^{29}\ \text{m}^{-3}$	$\mathbf{9.256\times10^{-41}\ F\ m^2}$
Si	$4.995\times10^{28}\ \text{m}^{-3}$	$\mathbf{4.170\times10^{-40}\ F\ m^2}$
Ge	$4.412\times10^{28}\ \text{m}^{-3}$	$\mathbf{5.017\times10^{-40}\ F\ m^2}$

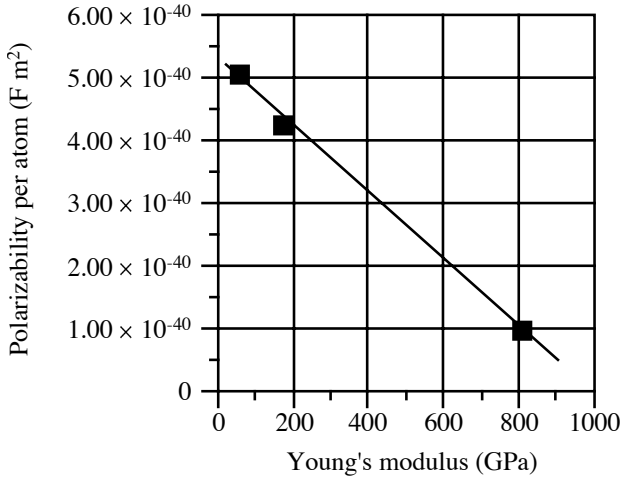


Figure 7Q2-1 Plot of polarizability per atom versus Young's modulus.

As the polarization mechanism in these crystals is due to electronic bond polarization, the displacement of electrons in the covalent bonds depends on the flexibility or elasticity of these bonds and hence also depends on the elastic modulus.

c

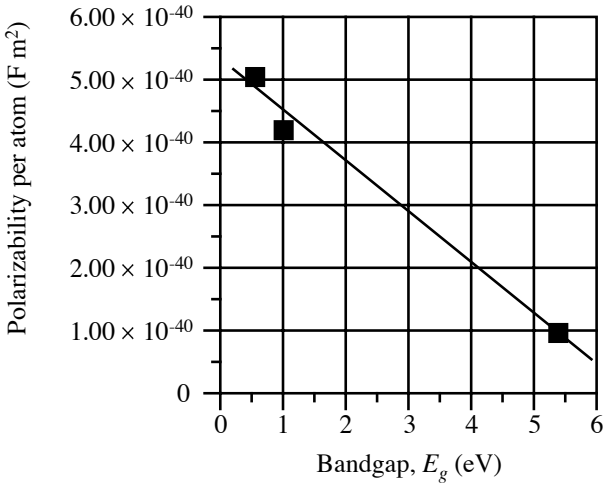


Figure 7Q2-2 Plot of polarizability versus bandgap energy.

There indeed seems to be a linear relationship between polarizability and bandgap energy.

d To facilitate this proof, we can plot a graph of refractive index, n, versus relative permittivity, ε_r.

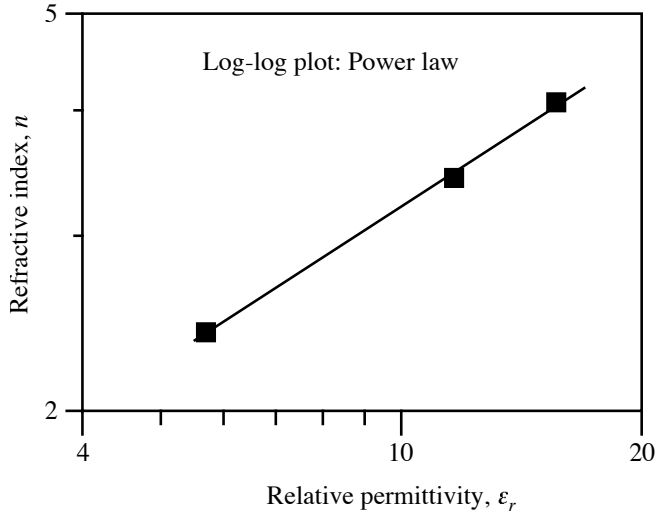


Figure 7Q2-3 Logarithmic plot of refractive index versus relative permittivity.

The log-log plot exhibits a straight line through the three points. The best fit line is $n = A\varepsilon_r^x$ (Correlation coefficient is 0.9987) where $x = 0.513 \approx 1/2$ and $A = \exp(-0.02070) \approx 1$. Thus $n = \sqrt{(\varepsilon_r)}$.

The refractive index n is an optical property that represents the speed of a light wave, or an electromagnetic wave, through the material ($v = c/n$). The light wave is a high frequency electromagnetic wave where the frequency is of the order of $10^{14} - 10^{15}$ Hz ($f_{optical}$). n and polarizability (or ε_r) will be related if the polarization can follow the field oscillations at this frequency ($f_{optical}$). This will be the case in electronic polarization because electrons are light and rapidly respond to the fast oscillations of the field.

The relationship between n and ε_r will not hold if we take ε_r at a low frequency (<< $f_{optical}$) where other slow polarization contributions (such as ionic polarization, dipolar polarization, interfacial polarization) also contribute to ε_r.

e $n = \sqrt{\varepsilon_r}$ would apply to ionic crystals if ε_r is taken at the corresponding optical frequency rather than at frequencies below $f_{optical}$. Tabulated data for ionic crystals typically quote ε_r that includes ionic polarization and hence this data does NOT conform to $n = \sqrt{\varepsilon_r}$.

7.3 Dipolar liquids

Given the static dielectric constant of water as 80, its high frequency dielectric constant (due to electronic polarization) as 4, its density as 1 g cm^{-3} calculate the permanent dipole moment p_o per water molecule assuming that it is the orientational and electronic polarization of individual molecules that gives rise to the dielectric constant. Use both the simple relationship in Equation 7.14 (in the textbook) where the local field is the same as the macroscopic field and also the Clausius-Mossotti equation and compare your results with the permanent dipole moment of the water molecule which is 6.1×10^{-30} C m. What is your conclusion? What is ε_r calculated from the Clasius-Mossotti equation taking the true p_o (6.1×10^{-30} C m) of a water molecule? (Note: Static dielectric constant is due to both orientational and electronic polarization. The Clausius-Mossotti equation does not apply to dipolar materials because the local field is not described by the Lorentz field.)

Solution

We first need the number H_2O molecules per unit volume. The molecular mass of water M_{mol} is 18×10^{-3} kg mol^{-1}, its density is $d = 10^3$ kg m^{-3}. The number of H_2O molecules per unit volume is

$$N = \frac{d\,N_A}{M_{mol}} = \frac{\left(10^3 \text{ kg m}^{-3}\right)\left(6.022 \times 10^{23} \text{ mol}^{-1}\right)}{\left(18 \times 10^{-3} \text{ kg mol}^{-1}\right)} = \mathbf{3.35 \times 10^{28}\ m^{-3}}$$

Using the high frequency dielectric constant ε_{r_HV} which is due only to electronic polarization and the Clausius-Mossotti equation (Equation 7.15 in the textbook), we can calculate the electronic polarizability α_e of water molecules

$$\alpha_e = \frac{3\varepsilon_0\left(\varepsilon_{r\,HF} - 1\right)}{\left(\varepsilon_{r\,HF} + 2\right)N} = \frac{3\left(8.85 \times 10^{-12} \text{ F m}^{-1}\right)(4-1)}{(4+2)\left(3.35 \times 10^{28} \text{ m}^{-3}\right)} = \mathbf{3.964 \times 10^{-40}\ F\ m^2}$$

Now using this result and the static dielectric constant $\varepsilon_{r\,Stat}$ which is due both to electronic and dipolar polarization, we can solve the Clausius-Mossotti equation for the dipolar polarizability α_d of water

$$\alpha_d = \frac{\left[3\varepsilon_0\left(\varepsilon_{r\,Stat} - 1\right) - N\alpha_e\left(\varepsilon_{r\,Sat} + 2\right)\right]}{\left(\varepsilon_{r\,Sat} + 2\right)N} =$$

$$= \frac{3\left(8.85 \times 10^{-12} \text{ F m}^{-1}\right)(80-1) - \left(3.35 \times 10^{28} \text{ m}^{-3}\right)\left(3.96 \times 10^{-40} \text{ F m}^2\right)(80+2)}{(80+2)\left(3.35 \times 10^{28} \text{ m}^{-3}\right)} = \mathbf{3.674 \times 10^{-40}\ F\ m^2}$$

The permanent dipolar moment per water p_0 molecule can be calculated from Equation 7.19 (in the textbook)

$$p_0 = \sqrt{3kT\alpha_d} = \sqrt{3\left(1.38\times10^{-23}\ \text{J K}^{-1}\right)(300\ \text{K})\left(3.964\times10^{-40}\ \text{F m}^2\right)} = \mathbf{2.137\times10^{-30}\ C\ m}$$

This result is around 3 times smaller than the real permanent dipolar moment of the water molecule.

The same calculations performed on the base of the simple relationship in Equation 7.14 (in the textbook) will result in

$$\alpha_e = \frac{\varepsilon_0\left(\varepsilon_{r\,HF}-1\right)}{N} = \frac{\left(8.85\times10^{-12}\ \text{F m}^{-1}\right)(4-1)}{\left(3.35\times10^{28}\ \text{m}^{-3}\right)} = \mathbf{7.928\times10^{-40}\ F\ m^2}$$

$$\alpha_d = \frac{\varepsilon_0\left(\varepsilon_{r\,Stat}-1\right)}{N} - \alpha_e = \frac{\left(8.85\times10^{-12}\ \text{F m}^{-1}\right)(4-1)}{\left(3.35\times10^{28}\ \text{m}^{-3}\right)} - \left(7.298\times10^{-40}\ \text{F m}^2\right) = \mathbf{2.009\times10^{-38}\ F\ m^2}$$

and

$$p_0 = \sqrt{3kT\alpha_d} = \sqrt{3\left(1.38\times10^{-23}\ \text{J K}^{-1}\right)(300\ \text{K})\left(2.009\times10^{-38}\ \text{F m}^2\right)} = \mathbf{1.58\times10^{-29}\ C\ m}$$

which is about 3 times bigger than the actual value.

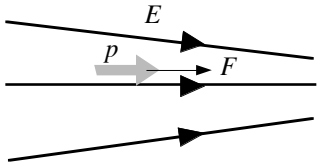
Both are unsatisfactory calculations. The reasons for the differences are two fold. First is that the individual H_2O molecules are not totally free to rotate. In the liquid, H_2O molecules cluster together through hydrogen bonding so that the rotation of individual molecules is then limited by this bonding. Secondly, the local field can neither be totally neglected nor taken as the Lorentz field. A better theory for dipolar liquids is based on the Onsager theory which is beyond the scope of this book. Interestingly, if one uses the actual $p_o = 6.1\times10^{-30}$ C m in the Clasius-Mossotti equation, then ε_r turns out to be negative, which is nonsense.

7.4 Dipole moment in a nonuniform electric field

Figure 7Q4-1 shows an electric dipole moment p in a nonuniform electric field. Suppose the gradient of the field is dE/dx at the dipole p, and the dipole is oriented to be along the direction of increasing E as in figure 7Q4-1. Show that the *net force* acting on this dipole is given by

$$F = p\frac{dE}{dx} \qquad \textit{Net force on a dipole}$$

Which direction is the force? What happens to this net force when the dipole moment is facing the direction of increasing field? Given that a dipole normally also experiences a torque (as described in Section 7.3.2 of the textbook), explain qualitatively what happens to a randomly placed dipole in a nonuniform electric field. Explain the experimental observation of bending a flow of water by a nonuniform field from a charged comb as shown in the photograph in Figure 7Q4-1? (Remember that a dielectric medium placed in a field develops polarization P directed along the field.)



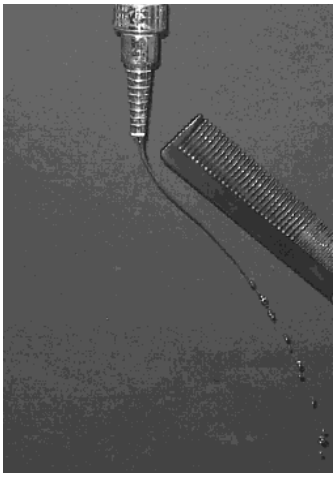


Figure 7Q4-1
Left: A dipole moment in a nonuniform field experiences a net force F that depends on the dipole moment p and the field gradient dE/dx. Right: When a charged comb (by combing hair) is brought close to a water jet, the field from the comb polarizes the liquid by orientational polarization. The induced polarization vector P and hence the liquid is attracted to the comb where the field is higher.

Solution

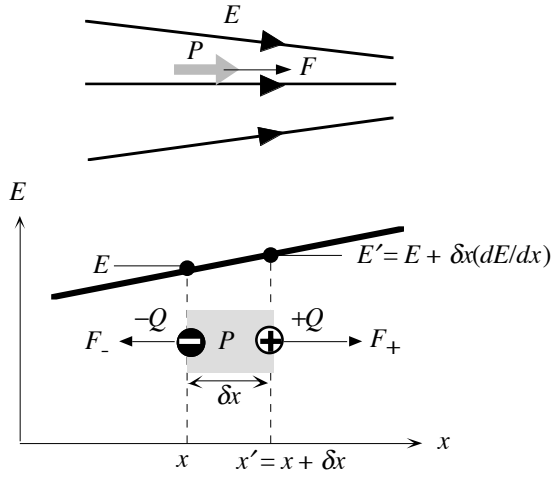


Figure 7Q4-2
When a dielectric is placed in a nonuniform field it develop a polarization P *along* the direction of the field. We can represent this polarization by two oppositle charges $+Q$ and $-Q$ separaed by δx.

We can represent the dipole p as two opposite charges $+Q$ and $-Q$ separated by a distance δx as in Figure 7Q4-2. Thus $p = Q\delta x$.

Let $-Q$ be at x, then $+Q$ is at $x' = x + \delta x$. The field is nonuniform. If the field is E at x and then it is E' at x', that is

$$E' = E + \delta x(dE/dx)$$

The force F_- acting on $-Q$ is along the $-x$ direction and its magnitude is given by

$$F_- = QE$$

and that on $+Q$ is along $+x$ and is given by

$$F_+ = QE' = Q[E + \delta x(dE/dx)].$$

The net force along the $+x$ direction is

$$F = F_+ - F_- = Q[E + \delta x(dE/dx)] - QE$$

or

$$F = Q\delta x(dE/dx) = p(dE/dx).$$

The force is along the $+x$ direction. The force changes direction if p is pointing in the opposite direction.

When a dipole moment is placed in a random direction in an electric field, it will always rotate to align with the electric field. Thus it will rotate until p is along E. Since the field is nonuniform, it will then experience a net force towards higher electric fields. Thus, p rotates and moves towards higher fields.

When a dielectric is placed in a nonuniform electric field, it develops a polarization P along the field as shown in Figure 7Q4-2. We can of course represent the polarized dielectric by the surface charges $+Q$ and $-Q$ as in Figure 7Q4-2. Following the above lines of argument, since induced P is along the field direction, the dielectric experiences a net force F towards higher fields along $+x$ direction,

$$F = P(dE/dx)$$

The water jet near a charged comb is attracted to the comb because water is a dielectric, develops a polarization P along the field and becomes attracted towards the higher field region which is near the comb.

7.5 Ionic and electronic polarization

Consider a CsBr crystal that has the CsCl unit cell crystal structure (one Cs^+–Br^- pair per unit cell) with a lattice parameter (a) of 0.430 nm. The electronic polarizability of Cs^+ and Cl^- ions are 3.35×10^{-40} F m^2 and 4.5×10^{-40} F m^2 respectively, and the mean ionic polarizability per ion pair is 5.8×10^{-40} F m^2. What is the low frequency dielectric constant and that at optical frequencies?

Solution

The CsBr structure has a lattice parameter given by $a = 0.430$ nm, and there is one CsBr ion pair per unit cell. If n is the number of ion pairs in the unit cell, the number of ion pairs, or individual ions, per unit volume (N) is

$$N = \frac{n}{a^3} = \frac{1}{\left(0.430 \times 10^{-9}\ \text{m}\right)^3} = 1.258 \times 10^{28}\ \text{m}^{-3}$$

At low frequencies both ionic and electronic polarizability contribute to the relative permittivity. Thus, from Equation 7.20 (in the textbook), (where α_i is the mean ionic polarizability per ion pair, $\alpha_{e\text{Cs}}$ is the electronic polarizability of Cs^+ and $\alpha_{e\text{Br}}$ is the electronic polarizability of Br^-):

$$\frac{\varepsilon_{r(low)} - 1}{\varepsilon_{r(low)} + 2} = \frac{1}{3\varepsilon_o}\left(N\alpha_i + N\alpha_{e\text{Cs}} + N\alpha_{e\text{Br}}\right)$$

Remember that ($N\alpha_i + N\alpha_{e\text{Cs}} + N\alpha_{e\text{Br}}$) should be written as ($N_i\alpha_i + N_{\text{Cs}}\alpha_{e\text{Cs}} + N_{\text{Br}}\alpha_{e\text{Cl}}$), but since there is a one-to-one ratio between the number of molecules and ions in CsCl, we can take all the N's to be the same.

$\therefore$

$$\varepsilon_{r(low)} = \frac{1}{3\varepsilon_o}\left(N\alpha_i + N\alpha_{e\text{Cs}} + N\alpha_{e\text{Br}}\right)\left(\varepsilon_{r(low)} + 2\right) + 1$$

$$\therefore \quad \varepsilon_{r(low)} - \frac{1}{3\varepsilon_o}\left(N\alpha_i + N\alpha_{e\mathrm{Cs}} + N\alpha_{e\mathrm{Br}}\right)\varepsilon_{r(low)} = \frac{2}{3\varepsilon_o}\left(N\alpha_i + N\alpha_{e\mathrm{Cs}} + N\alpha_{e\mathrm{Br}}\right) + 1$$

Isolate and simplify: $\varepsilon_{r(low)} = \dfrac{2N\left(\alpha_i + \alpha_{e\mathrm{Cs}} + \alpha_{e\mathrm{Br}}\right) + 3\varepsilon_o}{3\varepsilon_o - N\left(\alpha_i + \alpha_{e\mathrm{Cs}} + \alpha_{e\mathrm{Br}}\right)}$

$$\varepsilon_{r(low)} = \frac{2\left(1.258\times10^{28}\ \mathrm{m^{-3}}\right)\left(5.8\times10^{-40}\ \mathrm{F\,m^2} + 3.35\times10^{-40}\ \mathrm{F\,m^2} + 4.50\times10^{-40}\ \mathrm{F\,m^2}\right) + 3\left(8.854\times10^{-12}\ \mathrm{F/m}\right)}{3\left(8.854\times10^{-12}\ \mathrm{F/m}\right) - \left(1.258\times10^{28}\ \mathrm{m^{-3}}\right)\left(5.8\times10^{-40}\ \mathrm{F\,m^2} + 3.35\times10^{-40}\ \mathrm{F\,m^2} + 4.50\times10^{-40}\ \mathrm{F\,m^2}\right)}$$

$$\therefore \quad \boldsymbol{\varepsilon_{r(low)} = 6.48}$$

At optical frequencies there is no contribution from ionic polarization. We only consider electronic polarization of individual ions and therefore the relative permittivity at optical frequencies, $\varepsilon_{r(op)}$, is:

$$\varepsilon_{r(op)} = \frac{2\left(1.258\times10^{28}\ \mathrm{m^{-3}}\right)\left(3.35\times10^{-40}\ \mathrm{F\,m^2} + 4.50\times10^{-40}\ \mathrm{F\,m^2}\right) + 3\left(8.854\times10^{-12}\ \mathrm{F/m}\right)}{3\left(8.854\times10^{-12}\ \mathrm{F/m}\right) - \left(1.258\times10^{28}\ \mathrm{m^{-3}}\right)\left(3.35\times10^{-40}\ \mathrm{F\,m^2} + 4.50\times10^{-40}\ \mathrm{F\,m^2}\right)}$$

$$\therefore \quad \boldsymbol{\varepsilon_{r(op)} = 2.77}$$

7.6 Electronic polarizability and KCl

KCl has the NaCl crystal structure with a lattice parameter of 0.629 nm. Calculate the relative permittivity of a KCl crystal at optical frequencies given that the electronic polarizability of K^+ is 1.264×10^{-40} F m^2 and that of Cl^- is 3.408×10^{-40} F m^2. How does this compare with the measured value of 2.19?

Solution

The CsCl structure has a lattice parameter given by $a = 0.629$ nm, and there are 4 KCl ion pairs per unit cell (see Table 1.3, pg. 48 in the textbook). The number of ion pairs, or individual ions, per unit volume (N) is therefore:

$$N = \frac{4}{a^3} = \frac{4}{\left(0.629\times10^{-9}\ \mathrm{m}\right)^3} = 1.607\times10^{28}\ \mathrm{m^{-3}}$$

The electronic polarizability of the K^+ ion is given as $\alpha_{e\mathrm{K}} = 1.264\times10^{-40}$ F m^2, and polarizability of the Cl^- ion is given as $\alpha_{e\mathrm{Cl}} = 3.408\times10^{-40}$ F m^2. From Equation 7.20 (in the textbook), the relative permittivity at optical frequencies, $\varepsilon_{r(op)}$, can be found (see solution for question 7.5 for derivation):

$$\varepsilon_{r(op)} = \frac{2N\left(\alpha_{e\mathrm{K}} + \alpha_{e\mathrm{Cl}}\right) + 3\varepsilon_o}{3\varepsilon_o - N\left(\alpha_{e\mathrm{K}} + \alpha_{e\mathrm{Cl}}\right)}$$

$$\therefore \quad \varepsilon_{r(op)} = \frac{2\left(1.607\times10^{28}\ \mathrm{m^{-3}}\right)\left(1.264\times10^{-40}\ \mathrm{F\,m^2} + 3.408\times10^{-40}\ \mathrm{F\,m^2}\right) + 3\left(8.854\times10^{-12}\ \mathrm{F/m}\right)}{3\left(8.854\times10^{-12}\ \mathrm{F/m}\right) - \left(1.607\times10^{28}\ \mathrm{m^{-3}}\right)\left(1.264\times10^{-40}\ \mathrm{F\,m^2} + 3.408\times10^{-40}\ \mathrm{F\,m^2}\right)}$$

$$\therefore \quad \boldsymbol{\varepsilon_{r(op)} = 2.18}$$

This value is very close to the experimental value of 2.19.

7.7 Equivalent circuit of a polyester capacitor

Consider a 1 nF polyester capacitor that has a polymer (PET) film thickness of 1 μm. Calculate the equivalent circuit of this capacitor at 50 °C and at 120 °C for operation at 1 kHz. What is your conclusion?

Solution

The capacitance is given as 1 nF at room temperature and also at 50 °C where ε_r' is constant as shown in Figure 7Q7-1. The relative permittivities at 50 °C and 120 °C can be found approximately from Figure 7Q7-1. Upon inspection, at 50 °C, $\varepsilon_r' = 2.6$ and at 120 °C $\varepsilon_r' = 2.8$.

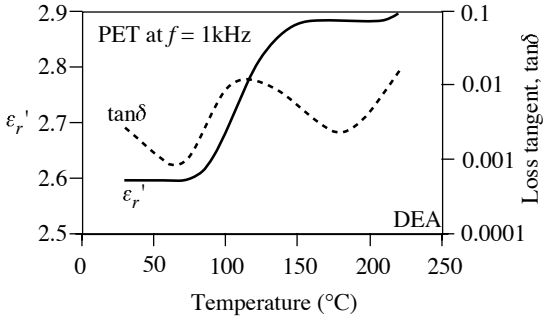


Figure 7Q7-1 Real part of the dielectric constant ε_r' and loss tangent, tan δ, at 1 kHz versus temperature for PET.

SOURCE: Data obtained by Kasap and Maeda (1955) using a dielectric analyzer (DEA).

From the equation for capacitance, the area of the dielectric can be found (where d is the thickness of the film):

$$C = \frac{\varepsilon_r' \varepsilon_o A}{d}$$

$\therefore$

$$A = \frac{dC}{\varepsilon_r' \varepsilon_o} = \frac{(1\times10^{-6}\ \text{m})(1\times10^{-9}\ \text{F})}{(2.6)(8.854\times10^{-12}\ \text{F/m})} = 4.344\times10^{-5}\ \text{m}^2$$

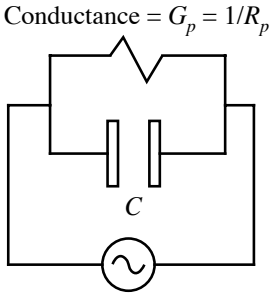


Figure 7Q7-2 Equivalent circuit of a capacitor at one given frequency.

Let $\tan\delta$ equal the loss tangent or loss factor. From Figure 7Q7-1, at 50 ˚C, $\tan\delta = 0.001$ and at 120 ˚C $\tan\delta = 0.01$.

We need the equivalent parallel conductance, G_p, (or resistance R_p) at 50 ˚C and 120 ˚C. For temperature T = 50 ˚C:

$$\boldsymbol{G_p} = \omega C \tan\delta = 2\pi(1000\ \text{Hz})\left(1\times10^{-9}\ \text{F}\right)(0.001) = \mathbf{6.28\ \times\ 10^{-9}\ \Omega^{-1}}$$

$$\boldsymbol{R_p} = 1\ /\ G_p = \mathbf{1.59\ \times\ 10^{8}\ \Omega}\ \text{or}\ \mathbf{159\ M\Omega}$$

At 120 ˚C, the capacitance is different (C') due to the change in ε_r'. The loss is higher which means a lower equivalent parallel resistance R_p. Assuming no change in area A:

$$C' = \frac{(2.8)\left(8.854\times10^{-12}\ \text{F/m}\right)\left(4.344\times10^{-5}\ \text{m}^2\right)}{\left(1\times10^{-6}\ \text{m}\right)} = 1.077\times10^{-9}\ \text{F}$$

A value 7.7% higher than at the lower temperature. The new conductance and resistance are:

$$\boldsymbol{G_p} = \omega C \tan\delta = 2\pi(1000\ \text{Hz})\left(1.077\times10^{-9}\ \text{F}\right)(0.01) = \mathbf{6.77\ \times\ 10^{-8}\ \Omega^{-1}}$$

$$\boldsymbol{R_p} = 1\ /\ G_p = \mathbf{1.48\ \times\ 10^{7}\ \Omega}\ \text{or}\ \mathbf{14.8\ M\Omega}$$

7.8 Dielectric loss per unit capacitance

Consider the three dielectric materials listed in Table 7Q8-1 with the real and imaginary dielectric constants, ε_r' and ε_r''. At a given voltage, which dielectric will have the lowest power dissipation per unit capacitance at 1 kHz and at an operating temperature of 50 °C? Is this also true at 120 °C?

Table 7Q8-1 Dielectric properties of three insulators at 1 kHz

Material	T =50 °C		T = 120 °C	
	ε_r'	ε_r''	ε_r'	ε_r''
Polycarbonate	2.47	0.003	2.535	0.003
PET	2.58	0.003	2.75	0.027
PEEK	2.24	0.003	2.25	0.003

Solution

Since we are merely comparing values, assume voltage V = 1 V for calculation purposes. From example 7.5 (in the textbook), the power dissipated per unit capacitance (W_{cap}) is given by:

$$W_{cap} = V^2\omega\frac{\varepsilon_r''}{\varepsilon_r'}$$

where ω is the angular frequency ($2\pi f$) and ε_r' and ε_r'' represent the real and imaginary components of the relative permittivity ε_r, respectively. As a sample calculation, the power dissipated in polycarbonate is:

$$\boldsymbol{W_{cap}} = (1\ \text{V})^2\left[2\pi(1000\ \text{Hz})\right]\frac{(0.003)}{(2.47)} = \mathbf{7.63\ \ W/F}$$

Therefore, 7.63 W per F are dissipated at 50 ˚C at 1 V. The values for the other materials at both 50 ˚C and 120 ˚C are listed below in Table 7Q8-2:

Table 7Q8-2 Power dissipated at different temperatures for the given materials.

	50 ˚C	120 ˚C
Material	**Power Dissipated (W per F)**	**Power Dissipated (W per F)**
Polycarbonate	**7.63**	**7.44**
PET	**7.31**	**61.7**
PEEK	**8.41**	**8.38**

At 50 ˚C, all three are comparable in magnitude but PET has the lowest power dissipation.

At 120 ˚C, polycarbonate has the lowest dissipation, while PET is almost ten times worse.

*7.9 TCC of a polyester capacitor

Consider the parallel plate capacitor equation

$$C = \frac{\varepsilon_o \varepsilon_r xy}{z}$$

where ε_r is the relative permittivity (or ε_r'), x and y are the side lengths of the dielectric so that xy is the area A, and z is the thickness of the dielectric. The quantities ε_r, x, y and z change with temperature. By differentiating this equation with respect to temperature, show that the temperature coefficient of capacitance (TCC) is

$$\text{TCC} = \frac{1}{C}\frac{dC}{dT} = \frac{1}{\varepsilon_r}\frac{d\varepsilon_r}{dT} + \lambda \qquad \textit{Temperature coefficient of capacitance}$$

where λ is the linear expansion coefficient defined by

$$\lambda = \frac{1}{L}\frac{dL}{dT}$$

where L stands for any length of the material (x, y or z). Assume that the dielectric is isotropic, λ is the same in all directions. Using ε_r' versus T behavior in Figure 7Q9-1 and taking $\lambda = 50 \times 10^{-6}$ K^{-1} as a typical value for polymers, predict the TCC at room temperature and at 10 kHz.

Solution

The real part of the relative permittivity, ε_r', is usually simply written as ε_r. We are given xy = area (A) and z = thickness. From the definition of the linear expansion coefficient λ:

$$\frac{dx}{dT} = \lambda x \qquad \frac{dy}{dT} = \lambda y \qquad \frac{dz}{dT} = \lambda z$$

Now differentiate C with respect to T (remember that ε_r depends on temperature):

$$C = \frac{\varepsilon_o \varepsilon_r xy}{z}$$

$$\therefore \quad \frac{dC}{dT} = \frac{\left(\varepsilon_o xy\frac{d\varepsilon_r}{dT} + \varepsilon_o \varepsilon_r y\frac{dx}{dT} + \varepsilon_o \varepsilon_r x\frac{dy}{dT}\right)z - \varepsilon_o \varepsilon_r xy\frac{dz}{dT}}{z^2}$$

Substitute for the derivatives of x, y and z as according to the definition of λ and simplify:

$$\frac{dC}{dT} = \frac{\varepsilon_o xy\frac{d\varepsilon_r}{dT} + \lambda\varepsilon_o \varepsilon_r xy}{z}$$

The temperature coefficient of capacitance is defined as:

$$\text{TCC} = \frac{1}{C}\frac{dC}{dT}$$

substitute for dC/dT: $$\text{TCC} = \frac{\varepsilon_o xy\frac{d\varepsilon_r}{dT} + \lambda\varepsilon_o \varepsilon_r xy}{Cz}$$

substitute for C: $$\text{TCC} = \frac{\varepsilon_o xy\frac{d\varepsilon_r}{dT} + \lambda\varepsilon_o \varepsilon_r xy}{\left(\frac{\varepsilon_o \varepsilon_r xy}{z}\right)z}$$

simplify: $$\text{TCC} = \frac{1}{\varepsilon_r}\frac{d\varepsilon_r}{dT} + \lambda$$

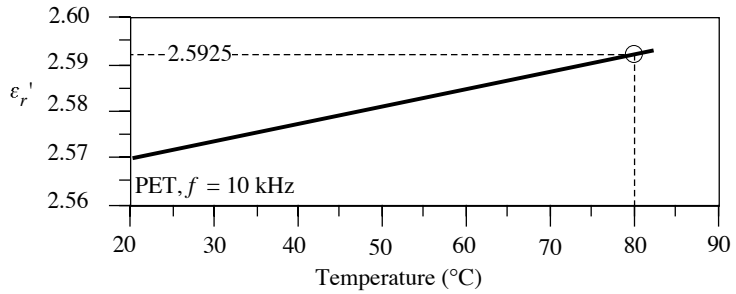


Figure 7Q9-1 Temperature dependence of ε_r' at 10 kHz.

From the slope of the straight line in Figure 7Q9-1, we can estimate the value of $d\varepsilon_r/dt$:

$$\frac{d\varepsilon_r}{dT} = \frac{2.5925 - 2.57}{80\ \text{C} - 20\ \text{C}} = 0.000375\ \ \text{C}^{-1} \text{ or } \text{K}^{-1}$$

Using the given value of $\lambda = 50 \times 10^{-6}$ K^{-1} and $\varepsilon_r = 2.57$ (ε_r at room temperature from inspecting Figure 7Q9-1):

$$\text{TCC} = \frac{1}{\varepsilon_r}\frac{d\varepsilon_r}{dT} + \lambda = \frac{1}{(2.57)}\left(0.000375\ \ \text{K}^{-1}\right) + 50 \times 10^{-6}\ \ \text{K}^{-1}$$

$\therefore$ **TCC = 0.000196 K^{-1} or °C^{-1}**

This is 196 ppm per °C. PET capacitors are quoted to have typically 200 ppm/°C.

7.10 Dielectric breakdown of gases and Paschen curves

Dielectric breakdown in gases typically involves the avalanche ionization of the gas molecules by energetic electrons accelerated by the applied field. The mean free path between collisions must be sufficiently long to allow the electrons to gain sufficient energy from the field to impact-ionize the gas molecules. The breakdown voltage, V_{br}, between two electrodes depends on the distance, d, between the electrodes as well as the gas pressure, P, as shown in Figure 7Q10-1. V_{br} versus Pd plots are called **Paschen curves**. We consider gaseous insulation, air and SF_6, in an HV switch.

a. What is the breakdown voltage between two electrodes of a switch separated by a 5 mm gap with air at 1 atm when the gaseous insulation is air and when it is SF_6?
b. What are the breakdown voltages in the two cases when the pressure is 10 times greater? What is your conclusion?
c. At what pressure is the breakdown voltage minimum?
d. What air gap spacing, d, at 1 atm gives the minimum breakdown voltage?
e. What would be the reasons for preferring gaseous insulation over liquid or solid insulation?

Solution

a At pressure P = 1 atm = 1.013×10^5 Pa and air gap d = 5 mm, $P \times d = (1.013 \times 10^5 \text{ Pa})(0.005 \text{ m})$ = 506.5 Pa m. From Figure 7Q10-1, the corresponding values of breakdown voltage for air (V_{air}) and for SF_6 (V_{SF6}) are:

$$\boldsymbol{V_{air} = 21000 \text{ V}} \text{ or } \boldsymbol{21.0 \text{ kV}}$$

$$\boldsymbol{V_{SF6} = 50000 \text{ V}} \text{ or } \boldsymbol{50.0 \text{ kV}}$$

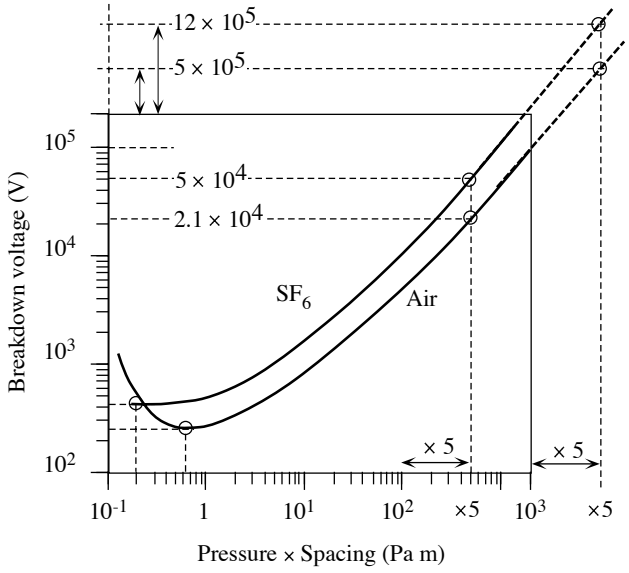


Figure 7Q10-1 Breakdown voltage versus (pressure × electrode spacing) (Paschen curves)

b At P = 10 atm = 1.013×10^6 Pa and d = 5 mm, $P \times d = (1.013 \times 10^6 \text{ Pa})(0.005 \text{ m})$ = 5065 Pa m. Using linear extrapolation on Figure 7Q10-1:

$$\boldsymbol{V_{air} = 500000 \text{ V}} \text{ or } \boldsymbol{500 \text{ kV}}$$

$$\boldsymbol{V_{SF6} = 1200000 \text{ V}} \text{ or } \boldsymbol{1200 \text{ kV}}$$

Pressure increases by 10 times but the breakdown voltages increase by a factor of about 25 times - this is a good improvement.

c With a gap length of 5 mm, we need to know the pressure at which the breakdown voltage is a minimum. From the graph, the minimum breakdown voltage of air is about $V_{air} = 250$ V, and the minimum for SF_6 is about $V_{SF6} = 420$ V. The corresponding values for $P \times d$ are $(P \times d)_{air} = 0.62$ Pa m and $(P \times d)_{SF6} = 0.2$ Pa m. From these we can determine the values of pressure needed for minimum breakdown voltage:

$$P_{air} d = 0.62 \text{ Pa m} \qquad P_{SF6} d = 0.2 \text{ Pa m}$$

$$\therefore \quad P_{air} = \frac{0.62 \text{ Pa m}}{0.005 \text{ m}} \qquad P_{SF6} = \frac{0.2 \text{ Pa m}}{0.005 \text{ m}}$$

$$\therefore \quad \boldsymbol{P_{air}} = \mathbf{124\ Pa} \text{ or } \mathbf{0.00122\ atm} \qquad \boldsymbol{P_{SF6}} = \mathbf{40.0\ Pa} \text{ or } \mathbf{0.000395\ atm}$$

A low pressure is needed for minimum breakdown which explains why discharge tubes operate at a low pressure.

d At a set pressure $P = 1$ atm $= 1.013 \times 10^5$ Pa, the air gap spacing d for minimum breakdown voltage can be found in a similar manner to the one above, using the same values for $P \times d$:

$$Pd_{air} = 0.62 \text{ Pa m}$$

$$\therefore \quad d_{air} = \frac{0.62 \text{ Pa m}}{1.013 \times 10^5 \text{ Pa}}$$

$$\therefore \quad \boldsymbol{d_{air}} = \mathbf{6.12 \times 10^{-6}\ m}$$

This value corresponds to a breakdown voltage of 250 V. Therefore a gap of about 6 μm will only need 250 V for breakdown.

e HV and high current switches or relays that have moving parts cannot be practically insulated using solid dielectrics. Liquid dielectrics are not as efficient as gaseous dielectrics because some undergo chemical changes under partial discharges. Further, they have a higher viscosity than gases that may affect the efficiency of the moving parts. Gas naturally permeates all the necessary space or locations where insulation is critical.

*7.11 Capacitor design

Consider a nonpolarized 100 nF capacitor design at 60 Hz operation. Note that there are three candidate dielectrics, as listed in Table 7Q11-1.

a. Calculate the volume of the 100 nF capacitor for each dielectric, given that they are to be used under low voltages and each dielectric has its minimum fabrication thickness. Which one has the smallest volume?

b. How is the volume affected if the capacitor is to be used at a 500 V application and the maximum field in the dielectric must be a factor of 2 less than the dielectric strength? Which one has the smallest volume?

c. At a 500 V application, what is the power dissipated in each capacitor at 60 Hz operation? Which one has the lowest dissipation?

Table 7Q11-1 Comparison of dielectric properties at 60 Hz (typical values)

	Polymer Film PET	**Ceramic TiO_2**	**High-K Ceramic ($BaTiO_3$ based)**
Name	Polyester	Polycrystalline titania	X7R
ε_r'	3.2	90	1800
tan δ	5×10^{-3}	4×10^{-4}	5×10^{-2}
$\mathcal{E}_{br}$ (kV cm^{-1})	150	50	100
Typical minimum thickness	1-2 μm	10 μm	10 μm

Solution

Note: All sample calculations are for Polymer film (PET). All methods of calculation for the other materials are identical, and the obtained values are summarized in Table 7Q11-2.

a To find the volume needed for C = 100 nF given that the dielectric has the minimum practical thickness, d (Table 7Q11-1), find the capacitance per unit volume (C_{vol}):

$$C_{\text{vol}} = \frac{\varepsilon_o \varepsilon_r}{d^2} = \frac{(8.854 \times 10^{-12} \text{ F/m})(3.2)}{(1 \times 10^{-6} \text{ m})^2} = 28.33 \text{ F/m}^3$$

The volume V can now be found as follows:

$$\boldsymbol{V} = C / C_{\text{vol}} = (100 \times 10^{-9} \text{ F}) / (28.33 \text{ F/m}^3) = \mathbf{3.53 \times 10^{-9} \ m^3}$$

This is the volume at low voltage operation based on the minimum practical thickness.

b Suppose that d is the minimum thickness (in m) which gives a maximum field of half of $\mathcal{E}_{br}$ at 500 V. Then:

$$\frac{1}{2}\mathcal{E}_{br} = \frac{V_{\max}}{d}$$

$\therefore$

$$d = 2\frac{V_{\max}}{\mathcal{E}_{br}} = 2\left(\frac{500 \text{ V}}{1.50 \times 10^7 \text{ V/m}}\right) = 6.667 \times 10^{-5} \text{ m}$$

Now the capacitance per unit volume can be found:

$$C_{\text{vol}} = \frac{\varepsilon_o \varepsilon_r}{d^2} = \frac{(8.854 \times 10^{-12} \text{ F/m})(3.2)}{(6.667 \times 10^{-5} \text{ m})^2} = 0.006374 \text{ F/m}^3$$

$\therefore$

$$\boldsymbol{V} = C / C_{\text{vol}} = (100 \times 10^{-9} \text{ F}) / (0.006374 \text{ F/m}^3) = \mathbf{1.57 \times 10^{-5} \ m^3}$$

This is the dielectric volume at 500 V.

c The power dissipation in the capacitor at 500 V (60 Hz operation) can be found by first obtaining the power lost per unit volume W_{vol}. It is given by:

$$W_{\text{vol}} = \frac{\mathcal{E}_{br}^{\,2}}{\eta^2} \omega \varepsilon_o \varepsilon_r' \tan\delta$$

where η is the safety factor (assumed to be equal to 2) and $\omega = 2\pi f$ is the angular frequency. Evaluating:

$$W_{vol} = \frac{(1.5\times10^{7}\ \text{V/m})^2}{(2)^2}(2\pi(60\ \text{Hz}))(8.854\times10^{-12}\ \text{F/m})(3.2)(5\times10^{-3})$$

$\therefore$ $W_{vol} = 3004\ \text{W/m}^3$

The power dissipated (W) is therefore:

$$W = W_{vol}V = (3004\ \text{W/m}^3)(1.57\times10^{-5}\ \text{m}^3)$$

$\therefore$ $\boldsymbol{W = 0.0472\ W}$

Table 7Q11-2 Summarized values for volume and power of given capacitors.

		Polymer Film PET	Ceramic TiO_2	High-K Ceramic ($BaTiO_3$ based)
a	Low voltage volume (m^3)	$\mathbf{3.53\times10^{-9}}$	$\mathbf{1.25\times10^{-8}}$	$\mathbf{6.27\times10^{-10}}$
b	High voltage volume (m^3)	$\mathbf{1.57\times10^{-5}}$	$\mathbf{5.02\times10^{-6}}$	$\mathbf{6.27\times10^{-8}}$
c	Power dissipated (W)	**0.0472**	**0.00377**	**0.471**

Upon inspection we see that for part **a** and part **b**, high-K ceramic has the smallest volumes, and for part **c**, ceramic has the lowest power dissipation.

*7.12 Dielectric breakdown in a coaxial cable

Consider a coaxial underwater high-voltage cable as in Figure 7Q12-1a. The current flowing through the inner conductor generates heat, which has to flow through the dielectric insulation to the outer conductor where it will be carried away by conduction and convection. We will assume that steady state has been reached and the inner conductor is carrying a dc current I. Heat generated per unit second, $Q' = dQ/dt$, by joule heating of the inner conductor is

$$Q' = RI^2 = \frac{\rho L I^2}{\pi a^2} \qquad \textit{Rate of heat generation} \qquad [1]$$

where ρ is the resistivity, a the radius of the conductor, and L the cable length.

This heat flows radially out from the inner conductor through the dielectric insulator to the outer conductor, then to the ambient. This heat flow is by thermal conduction through the dielectric. The rate of heat flow Q' depends on the temperature difference $T_i - T_o$, between the inner and outer conductors; on the sample geometry (a, b and L); and on the thermal conductivity κ of the dielectric. From elementary thermal conduction theory, this is given by

$$Q' = (T_i - T_o)\frac{2\pi\kappa L}{\ln\left(\frac{b}{a}\right)} \qquad \textit{Rate of heat conduction} \qquad [2]$$

The inner core temperature, T_i, rises until, in the steady state, the rate of joule heat generation by the electric current in Equation 1 is just removed by the rate of thermal conduction through the dielectric insulation, given by Equation 2.

a. Show that the inner conductor temperature is

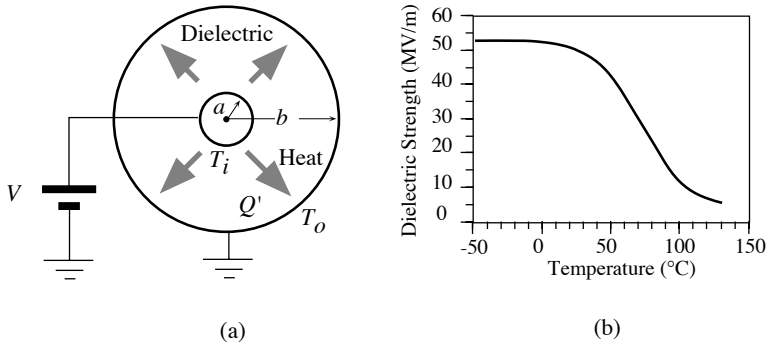
$$T_i = T_o + \frac{\rho I^2}{2\pi^2 a^2 \kappa} \ln\left(\frac{b}{a}\right) \qquad \textit{Steady state inner conductor temperature} \qquad [3]$$

b. The breakdown occurs at the maximum field point, which is at $r = a$, just outside the inner conductor, and is given by (see Example 7.10 in the textbook)

$$\mathcal{E}_{\max} = \frac{V}{a \ln\left(\frac{b}{a}\right)} \qquad \textit{Maximum field in a coaxial cable} \qquad [4]$$

The dielectric breakdown occurs when $\mathcal{E}_{\max}$ reaches the dielectric strength $\mathcal{E}_{br}$. However the dielectric strength $\mathcal{E}_{br}$ for many polymeric insulation materials depends on the temperature, and generally it decreases with temperature, as shown for a typical example in Figure 7Q12-1b. If the load current, I, increases, then more heat, Q', is generated per second and this leads to a higher inner core temperature, T_i, by virtue of Equation 3. The increase in T_i with I eventually lowers $\mathcal{E}_{br}$ so much that it becomes equal to $\mathcal{E}_{\max}$ and the insulation breaks down (thermal breakdown). Suppose that a certain coaxial cable has an aluminum inner conductor of diameter 10 mm and resistivity 27 nΩ m. The insulation is 3 mm thick and is a polyethylene-based polymer whose long-term dc dielectric strength is shown in Figure 7Q12-1b. Suppose that the cable is carrying a voltage of 40 kV and the outer shield temperature is the ambient temperature, 25 °C. Given that the thermal conductivity of the polymer is about 0.3 W K^{-1} m^{-1}, at what dc current will the cable fail?

c. Rederive T_i in Equation 3 by considering that r depends on the temperature as $\rho = \rho_0\left[1 + \alpha_0\left(T - T_0\right)\right]$ (Chapter 2 of the textbook). Recalculate the maximum current in *b* given that $\alpha_o = 3.9 \times 10^{-3}$ °C^{-1} at 25 °C.



(a) (b)

Figure 7Q12-1

(a) The joule heat generated in the core conductor flows outward radially through the dielectric material.

(b) Typical temperature dependence of the dielectric strength of a polyethylene-based polymeric insulation.

Solution

a The resistance R of the inner conductor is:

$$R = \frac{\rho L}{A} = \frac{\rho L}{\pi a^2}$$

where ρ is the resistivity of the conductor, L is the length and a is the radius.

Joule heating power generated by the current I through the conductor (P) is given by:

$$P = I^2R = I^2\frac{\rho L}{\pi a^2} \qquad \text{Eqn. [1]}$$

The rate of heat conduction (Q') from inner conductor to outer conductor is:

$$Q' = \left(T_i - T_o\right)\frac{2\pi\kappa L}{\ln\left(\frac{b}{a}\right)} \qquad \text{Eqn. [2]}$$

In the steady state, the rate of Joule heating of the inner conductor, P, is equal to the rate of heat flow, Q', through the insulator. Therefore:

$$I^2\frac{\rho L}{\pi a^2} = \left(T_i - T_o\right)\frac{2\pi\kappa L}{\ln\left(\frac{b}{a}\right)}$$

$$\therefore \qquad \left(T_i - T_o\right) = I^2\frac{\rho L}{2\kappa\pi^2 a^2 L}\ln\left(\frac{b}{a}\right)$$

$$\therefore \qquad T_i = T_o + \frac{\rho I^2}{2\kappa\pi^2 a^2}\ln\left(\frac{b}{a}\right)$$

Note: A major assumption is the inner conductor resistivity ρ and thermal conductivity κ are constant (do not change with temperature).

b We are given the cable's characteristics: resistivity $\rho = 27 \times 10^{-9}$ Ω, thermal conductivity $\kappa = 0.3$ W K^{-1} m^{-1} (polyethylene material at 25 °C), inner conductor radius $a = 0.005$ m, and total radius $b = 0.005$ m + 0.003 m = 0.008 m. The outer temperature is given as $T_o = 25$ °C + 273 = 298 K, and the applied voltage is 40 kV. The maximum field $\mathcal{E}_{max}$ depends on the voltage V by Equation 4:

$$\mathcal{E}_{max} = \frac{V}{a\ln\left(\frac{b}{a}\right)} = \frac{40\times10^3 \text{ V}}{(0.005 \text{ m})\ln\left(\frac{0.008 \text{ m}}{0.005 \text{ m}}\right)} = 1.702\times10^7 \text{ V/m or } 17.02 \text{ MV/m}$$

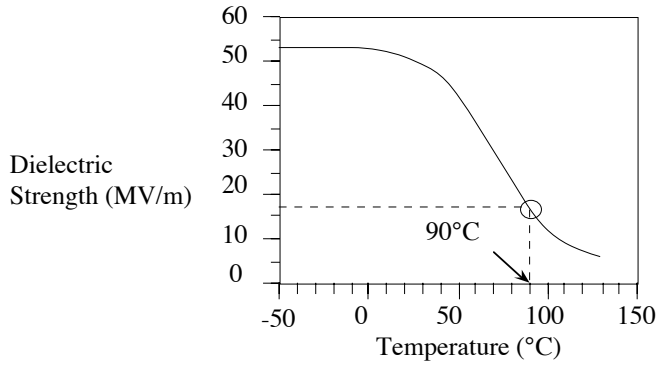


Figure 7Q12-2 Temperature dependence of the dielectric strength of a polyethylene-based polymeric insulation.

The breakdown field $\mathcal{E}_{br}$ is equal to this maximum field $\mathcal{E}_{max}$ when the inner temperature $T_i = 90$ °C = 363 K (see Figure 7Q12-2 above).

Now, from Equation 3:

$$T_i = T_o + \frac{\rho I^2}{2\kappa\pi^2 a^2}\ln\left(\frac{b}{a}\right)$$

$$\therefore \qquad I = \sqrt{\frac{2\kappa\pi^2 a^2 (T_i - T_o)}{\rho \ln\left(\frac{b}{a}\right)}} = \sqrt{\frac{2\left(0.3 \text{ W K}^{-1} \text{ m}^{-1}\right)\pi^2 (0.005 \text{ m})^2 (363 \text{ K} - 298 \text{ K})}{\left(27\times10^{-9} \ \Omega \text{ m}\right)\ln\left(\frac{0.008 \text{ m}}{0.005 \text{ m}}\right)}}$$

$$\therefore \qquad \boldsymbol{I = 871 \text{ A}}$$

c As in **a** for the steady state, the rate of Joule heating of the inner conductor, P, is equal to the rate of heat flow, Q', through the insulator but now ρ is temperature dependent. Therefore:

$$I^2 \frac{\rho_0\left[1 + \alpha_0 (T_i - T_o)\right]L}{\pi a^2} = (T_i - T_o)\frac{2\pi\kappa L}{\ln\left(\frac{b}{a}\right)} \qquad (5)$$

Solving for T_i and simplifying we receive

$$T_i = T_o + \frac{I^2}{\frac{2\pi^2 a^2 \kappa}{\rho_0 \ln\left(\frac{b}{a}\right)} - I^2\alpha_0}$$

Now we have to calculate at what current T_i will reach 90 °C and hence thermal breakdown will occur.

Solving for I Equation (5) we have

$$I = \sqrt{\frac{2\pi^2 a^2 \kappa (T_i - T_o)}{\rho_0\left[1 + \alpha_0 (T_i - T_o)\right]\ln\left(\frac{b}{a}\right)}}$$

So that

$$I = \sqrt{\frac{2\pi^2 \left(5\times10^{-3} \text{ m}\right)^2 \left(0.3 \text{ W K}^{-1} \text{ m}^{-1}\right)\left((90 - 25) \text{ K}\right)}{\left(27\times10^{-9} \ \Omega \text{ m}\right)\left[1 + \left(3.9\times10^{-3}\right)\left((90 - 25) \text{ K}\right)\right]\ln\left(\frac{8}{5}\right)}} = \boldsymbol{777.8 \text{ A}}$$

7.13 Piezoelectricity

Consider a quartz crystal and a PZT ceramic filter both designed for operation at f_s = 1 MHz. What is the bandwidth of each? Given Young's modulus (Y), density (ρ) of each, and that the filter is a disk with electrodes and is oscillating radially, what is the diameter of the disk for each material? For quartz, Y = 80 GPa and ρ = 2.65 g cm^{-3}. For PZT, Y = 70 GPa and ρ = 7.7 g cm^{-3}. Assume that the velocity of mechanical oscillations in the crystal is $v = \sqrt{Y/\rho}$ and the wavelength $\lambda = v/f_s$. Consider only the fundamental mode (n = 1).

Solution

We are given the first resonance frequency, f_s = 1 MHz. There is a second resonance frequency, f_a, related to the first by (see example 7.13, pg. 565 in the textbook)):

$$f_a = \frac{f_s}{\sqrt{1-K^2}}$$

where K is the electromechanical conversion factor. From Table 7.7 (pg. 561 in the textbook), K_{quartz} = 0.1 and K_{PZT} = 0.72. The second resonance frequencies can then be found:

Quartz:

$$f_a = \frac{f_s}{\sqrt{1-K^2}} = \frac{\left(1\times10^6\ \text{Hz}\right)}{\sqrt{1-(0.1)^2}} = 1.005\times10^6\ \text{Hz}$$

The bandwidth B_{quartz} is defined as the difference between these two frequencies:

$\therefore$ $$\boldsymbol{B_{\text{quartz}}} = f_a - f_s = 1.005\times10^6\ \text{Hz} - 1\times10^6\ \text{Hz} = \mathbf{5000\ Hz}$$

PZT:

$$f_a = \frac{f_s}{\sqrt{1-K^2}} = \frac{\left(1\times10^6\ \text{Hz}\right)}{\sqrt{1-(0.72)^2}} = 1.441\times10^6\ \text{Hz}$$

$\therefore$ $$\boldsymbol{B_{\text{PZT}}} = f_a - f_s = 1.441\times10^6\ \text{Hz} - 1\times10^6\ \text{Hz} = \mathbf{4.41\times10^5\ Hz}$$

A much wider bandwidth than the quartz crystal.

Next, we need to find the diameter of the disks of each material. We are given Young's modulus ($Y_{\text{quartz}} = 80\times10^9$ Pa, $Y_{\text{PZT}} = 70\times10^9$ Pa) and the density (ρ_{quartz} = 2650 kg/m^3, ρ_{PZT} = 7700 kg/m^3). From these we can determine the velocity and wavelength of the mechanical oscillations in the crystal, and then the diameter (L) from Equation 7.50 (in the textbook).

Quartz:

$$v_{\text{quartz}} = \sqrt{\frac{Y_{\text{quartz}}}{\rho_{\text{quartz}}}} = \sqrt{\frac{\left(80\times10^9\ \text{Pa}\right)}{\left(2650\ \text{kg/m}^3\right)}} = 5494\ \text{m/s}$$

$\therefore$ $$\lambda_{\text{quartz}} = \frac{v_{\text{quartz}}}{f_s} = \frac{5494\ \text{m/s}}{1\times10^6\ \text{Hz}} = 0.005494\ \text{m}$$

$\therefore$ $$\boldsymbol{L_{\text{quartz}}} = n\left(\frac{1}{2}\lambda_{\text{quartz}}\right) = 1\left(\frac{1}{2}(0.005494\ \text{m})\right) = \mathbf{0.00275\ m}\ \text{or}\ \mathbf{2.75\ mm}$$

PZT:

$$v_{\text{PZT}} = \sqrt{\frac{Y_{\text{PZT}}}{\rho_{\text{PZT}}}} = \sqrt{\frac{\left(70\times10^9\ \text{Pa}\right)}{\left(7700\ \text{kg/m}^3\right)}} = 3015\ \text{m/s}$$

$\therefore$ $$\lambda_{\text{PZT}} = \frac{v_{\text{PZT}}}{f_s} = \frac{3015\ \text{m/s}}{1\times10^6\ \text{Hz}} = 0.003015\ \text{m}$$

$\therefore$ $$\boldsymbol{L_{\text{PZT}}} = n\left(\frac{1}{2}\lambda_{\text{PZT}}\right) = 1\left(\frac{1}{2}(0.003015\ \text{m})\right) = \mathbf{0.00151\ m}\ \text{or}\ \mathbf{1.51\ mm}$$

7.14 Piezoelectric voltage coefficient

The application of a stress T to a piezoelectric crystal leads to a polarization P and hence to an electric field E in the crystal such that

$$E = gT \qquad \textit{Piezoelectric voltage coefficient}$$

where g is the *piezoelectric voltage coefficient*. If $\varepsilon_o\varepsilon_r$ is the permittivity of the crystal, show that

$$g = \frac{d}{\varepsilon_o\varepsilon_r}$$

A $BaTiO_3$ sample, along a certain direction (called 3), has d = 190 pC N^{-1}, and its $\varepsilon_r \approx 1900$ along this direction. What do you expect for its g coefficient for this direction and how does this compare with the measured value of approximately 0.013 C^{-1} m^2?

Solution

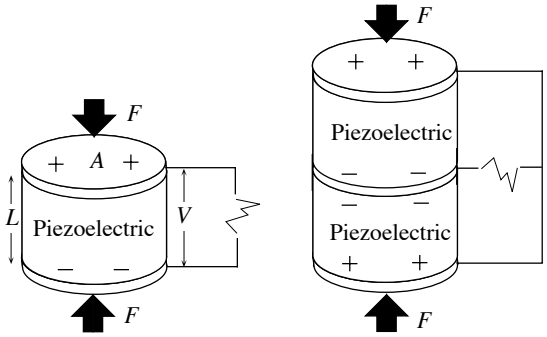


Figure 7Q14-1 The piezoelectric spark generator.

The electric field E in the piezoelectric sample due to induced voltage V over the sample (see Figure 7Q14-1a) is

$$E = \frac{V}{L}$$

The induced voltage is proportional to the induced charge Q by

$$Q = \frac{V}{C} \ ,$$

where C is the capacitance of the sample.

The induced charge in his turn is related to the polarization P by

$$Q = AP$$

And finally, the capacitance of parallel plate capacitor is given by

$$C = \frac{\varepsilon_0\varepsilon_r A}{L}$$

Combining these expressions we receive

$$E = \frac{V}{L} = \frac{Q}{C\,L} = \frac{A\,P}{\varepsilon_0\,\varepsilon_r\,\frac{A}{L}L} = \frac{P}{\varepsilon_0\,\varepsilon_r} \qquad (1)$$

The applied stress T and the induced polarization P are related through the piezoelectric voltage coefficient d

$$P = d\,T$$

Thus substituting in (1) we can obtain the relation between the electric field E and T

$$E = \frac{d}{\varepsilon_0\varepsilon_r}T$$

A direct comparison between this expression and the piezoelectric voltage coefficient definition shows that

$$g = \frac{d}{\varepsilon_0\varepsilon_r}$$

The piezoelectric voltage coefficient for the $BaTiO_3$ sample described in the problem is

$$g = \frac{d}{\varepsilon_o\varepsilon_r} = \frac{\left(190\times10^{-12}\ \text{C N}^{-1}\right)}{\left(8.8534\times10^{-12}\ \text{F m}^{-1}\right)(1900)} = \mathbf{0.0113\ C^{-1}\ m^2}$$

The received value is in pretty good agreement with the experimental one.

7.15 Piezoelectricity and the piezoelectric bender

a. Consider using a piezoelectric material in an application as a mechanical positioner where the displacements are expected to be small (as in a scanning tunneling microscope). For the piezoelectric plate shown in Figure 7Q15-1a, we will take L = 20 mm, W = 10 mm, and D (thickness) = 0.25 mm. Under an applied voltage of V, the plate changes length, width, and thickness according to the piezoelectric coefficients d_{ij}, relating the applied field along i to the resulting strain along j. Suppose we define direction 3 along the thickness D and direction 1 along the length L, as shown in the figure. Show that the changes in the thickness and length are

$$\delta D = d_{33}\,V \qquad \textit{Piezoelectric effects}$$

$$\delta L = \left(\frac{L}{D}\right)d_{31}V$$

Given $d_{33} \approx 500\times10^{-12}$ m V^{-1} and $d_{31} \approx -250\times10^{-12}$ m V^{-1}, calculate the changes in the length and thickness for an applied voltage of 100 V. What is your conclusion?

b. Consider two oppositely poled and joined ceramic plates, A and B, forming a bimorph, as shown in Figure 7Q15-1b. This piezoelectric bimorph is mounted as a cantilever; one end is fixed and the other end is free to move. Oppositely poled means that the electric field elongates A and contracts B, and the two relative motions *bend* the plate. The displacement h of the tip of the cantilever is given by

$$h = \frac{3}{2}d_{31}\left(\frac{L}{D}\right)^2 V \qquad \textit{Piezoelectric bending}$$

What is the deflection of the cantilever for an applied voltage of 100 V? What is your conclusion?

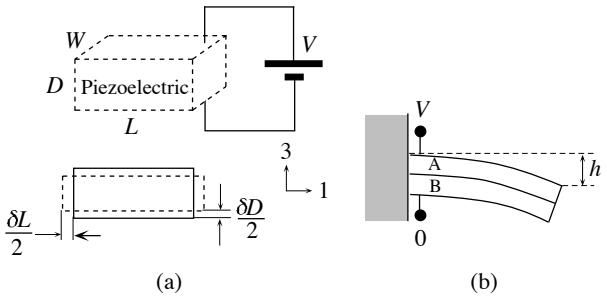


Figure 7Q15-1

(a) A mechanical positioner using a piezoelectric plate under an applied voltage of V.

(b) A cantilever-type piezoelectric bender. An applied voltage bends the cantilever.

Solution

a The strain along direction 3 is given by the change in thickness divided by the thickness, $S_3 = \delta D/D$, and the electric field along 3 is given by $E_3 = V/D$ where V is the voltage. The piezoelectric effect relates the strain S and the field E via Equation 7.47 (in the textbook), such that:

$$S_3 = d_{33}E_3$$

$$\therefore \quad \frac{\delta D}{D} = d_{33}\frac{V}{D}$$

$$\therefore \quad \delta D = d_{33}V$$

Along direction 1, the strain is the change in length divided by the length, $S_1 = \delta L/L$. By the piezoelectric effect (Equation 7.47 in the textbook):

$$S_1 = d_{31}E_3$$

$$\therefore \quad \frac{\delta L}{L} = d_{31}\frac{V}{D}$$

$$\therefore \quad \delta L = \left(\frac{L}{D}\right)d_{31}V$$

Now, using the given values, we can find the change in thickness and length for 100 V:

$$\delta \boldsymbol{D} = d_{33}V = \left(500\times10^{-12}\ \text{m/V}\right)(100\ \text{V}) = \mathbf{5.00\ \times\ 10^{-8}\ m}$$

$$\delta \boldsymbol{L} = \left(\frac{L}{D}\right)d_{31}V = \left(\frac{0.02\ \text{m}}{0.00025\ \text{m}}\right)\left(-250\times10^{-12}\ \text{m/V}\right)(100\ \text{V}) = \mathbf{-2.00\ \times\ 10^{-6}\ m}$$

There is a much greater change in the length than in the thickness of the plate.

b The cantilever tip is displaced by h, whose magnitude is:

$$h = \frac{3}{2}\left|d_{31}\right|\left(\frac{L}{D}\right)^2 V = \frac{3}{2}\left|\left(-250\times10^{-12}\ \text{m/V}\right)\right|\left(\frac{0.02\ \text{m}}{0.00025\ \text{m}}\right)^2(100\ \text{V})$$

$$\therefore \quad \boldsymbol{h\ =\ 0.000240\ m}\ \text{or}\ \mathbf{0.240\ mm}$$

The cantilever configuration provides a greater displacement than the above cases.

7.16 Piezoelectricity

The wavelength, λ, of mechanical oscillations in a piezoelectric slab satisfies

$$n\left(\frac{1}{2}\lambda\right) = L$$

where n is an integer, L is the length of the slab along which mechanical oscillations are set up, and the wavelength λ is determined by the frequency f and velocity υ of the waves. The ultrasonic wave velocity υ depends on Young's modulus Y as

$$\upsilon = \left[\frac{Y}{\rho}\right]^{1/2}$$

where ρ is the density. For quartz, Y = 80 GPa and ρ = 2.65 g cm^{-3}. Considering the fundamental mode (n = 1), what are practical dimensions for crystal oscillators operating at 1 kHz and 1 MHz?

Solution

We are given the characteristics of quartz, $Y = 80 \times 10^9$ Pa and $\rho = 2650$ kg/m^3. We can then calculate the ultrasonic wave velocity, υ:

$$\upsilon = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{80\times10^9 \text{ Pa}}{2650 \text{ kg/m}^3}} = 5494 \text{ m/s}$$

First, consider f = 1 kHz:

$$\lambda = \frac{\upsilon}{f} = \frac{5494 \text{ m/s}}{1000 \text{ Hz}} = 5.494 \text{ m}$$

The length L of the quartz crystal at 1 kHz at the fundamental mode (n = 1) is therefore:

$$\boldsymbol{L} = n\left(\frac{1}{2}\lambda\right) = (1)\left(\frac{1}{2}\times 5.494 \text{ m}\right) = \mathbf{2.75\ m}$$

A huge crystal, very impractical.

Now, consider f' = 1 MHz:

$$\lambda' = \frac{\upsilon}{f'} = \frac{5494 \text{ m/s}}{1\times10^6 \text{ Hz}} = 0.005494 \text{ m}$$

$\therefore$ $$\boldsymbol{L'} = n\left(\frac{1}{2}\lambda\right) = (1)\left(\frac{1}{2}\times 0.005494 \text{ m}\right) = \mathbf{0.00275\ m} \text{ or } \mathbf{2.75\ mm}$$

This is a much more practical value.

Note: Using $n > 1$ increases the size of the quartz crystal.

7.17 Pyroelectric detectors

Consider two different radiation detectors using PZT and PVDF as pyroelectric materials whose properties are summarized in Table 7Q17-1. The receiving area is 4 mm^2. The thicknesses of the PZT ceramic and the PVDF polymer film are 0.1 mm and 0.005 mm respectively. In both cases the incident radiation is chopped periodically to allow the radiation to pass for a duration of 0.05 seconds.

a. Calculate the magnitude of the output voltage for each detector if both receive a radiation of intensity 10 μW cm^{-2}. What is the corresponding current in the circuit? In practice, what would limit the magnitude of the output voltage?

b. What is the minimum detectable radiation intensity if the minimum detectable signal voltage is 10 nV?

Table 7Q17-1 Properties of PZT and PVDF

	ε_r'	**Pyroelectric Coefficient ($\times$ 10^{-6} C m^{-2} K^{-1})**	**Density (g cm^{-3})**	**Heat Capacity (J K^{-1} g^{-1})**
PZT	290	380	7.7	0.3
PVDF	12	27	1.76	1.3

Solution

a The change in voltage is given by Equation 7.54 (in the textbook):

$$\Delta V = \frac{p I \Delta t}{\rho c \varepsilon_r \varepsilon_o}$$

where p is the pyroelectric coefficient, ρ is the density, c is the specific heat capacity, I is the light intensity, and Δt is the duration of radiation. The output voltages for the two materials can then be calculated:

$$\Delta V_{PZT} = \frac{\left(380\times10^{-6}\ \text{C m}^{-2}\ \text{K}^{-1}\right)\left(0.1\ \text{W/m}^2\right)(0.05\ \text{s})}{\left(7700\ \text{kg/m}^3\right)\left(300\ \text{J K}^{-1}\ \text{kg}^{-1}\right)(290)\left(8.854\times10^{-12}\ \text{F/m}\right)}$$

$\therefore$ ΔV_{PZT} **= 0.000320 V**

$$\Delta V_{PVDF} = \frac{\left(27\times10^{-6}\ \text{C m}^{-2}\ \text{K}^{-1}\right)\left(0.1\ \text{W/m}^2\right)(0.05\ \text{s})}{\left(1760\ \text{kg/m}^3\right)\left(1300\ \text{J K}^{-1}\ \text{kg}^{-1}\right)(12)\left(8.854\times10^{-12}\ \text{F/m}\right)}$$

$\therefore$ ΔV_{PVDF} **= 0.000555 V**

The corresponding currents can be found in two ways:

(1) Neglecting thermal conduction, the current is equal to the change in polarization P over time, or:

$$i = \frac{dP}{dt} = p\frac{dT}{dt} = \frac{pdH}{mc_s dt}$$

where p is the pyroelectric coefficient, m is the mass, c_s is the specific heat capacity and dH is the change in heat. The change in heat can be expressed as $dH = IAdt$, and the mass can be found from the density, $\rho = m/V$, i.e. $m = \rho Ad$. Substituting into the previous equation, we obtain:

$$i = \frac{p\mathcal{I}}{\rho d c_s}$$

$$\boldsymbol{i_{PZT}} = \frac{\left(380\times10^{-6}\ \text{C m}^{-2}\ \text{K}^{-1}\right)\left(0.1\ \text{W m}^{-2}\right)}{\left(7700\ \text{kg m}^{-3}\right)\left(0.1\times10^{-3}\ \text{m}\right)\left(300\ \text{J K}^{-1}\ \text{kg}^{-1}\right)} = \mathbf{1.65\ \times\ 10^{-7}\ A}$$

similarly, $\boldsymbol{i_{PVDF}} = \mathbf{2.36\ \times\ 10^{-7}\ A}$

(2) We can also estimate through the equation for current in a capacitor (where d is the thickness):

$$i = C\frac{dV}{dt} \approx C\frac{\Delta V}{\Delta t} \approx \frac{\varepsilon_o\varepsilon_r A\Delta V}{d\Delta t}$$

$$\boldsymbol{i_{PZT}} \approx \frac{\left(8.854\times10^{-12}\ \text{F/m}\right)(290)\left(4\times10^{-6}\ \text{m}^2\right)(0.000320\ \text{V})}{\left(0.1\times10^{-3}\ \text{m}\right)(0.05\ \text{s})} = \mathbf{6.57\ \times\ 10^{-13}\ A}$$

similarly, $\boldsymbol{i_{PVDF}} \approx \mathbf{9.43\ \times\ 10^{-13}\ A}$

Remember that these are only approximations and that the first method is more accurate.

Heat losses, e.g. thermal conduction, would prevent the absorbed heat from raising the temperature indefinitely. The calculation assumes that the rise in the temperature is small so that the heat losses, proportional to the temperature difference, can be neglected.

b Rearrange the equation for change in voltage and isolate $\mathcal{I}$:

$$\mathcal{I} = \frac{\Delta V\rho c\varepsilon_r\varepsilon_o}{\Delta t p}$$

The minimum signal is given as $\Delta V' = 10\times10^{-9}$ V. Substituting to find the detectable light intensity:

$$\mathcal{I}_{PZT} = \frac{\left(10\times10^{-9}\ \text{V}\right)\left(7700\ \text{kg/m}^3\right)\left(300\ \text{J K}^{-1}\ \text{kg}^{-1}\right)(290)\left(8.854\times10^{-12}\ \text{F/m}\right)}{(0.05\ \text{s})\left(380\times10^{-6}\ \text{C m}^{-2}\ \text{K}^{-1}\right)}$$

$\therefore$ $\boldsymbol{\mathcal{I}_{PZT}} = \mathbf{3.12\ \times\ 10^{-6}\ W/m^2}$ (Small - advantage of pyroelectric detectors)

$$\mathcal{I}_{PVDF} = \frac{\left(10\times10^{-9}\ \text{V}\right)\left(1760\ \text{kg/m}^3\right)\left(1300\ \text{J K}^{-1}\ \text{kg}^{-1}\right)(12)\left(8.854\times10^{-12}\ \text{F/m}\right)}{(0.05\ \text{s})\left(27\times10^{-6}\ \text{C m}^{-2}\ \text{K}^{-1}\right)}$$

$\therefore$ $\boldsymbol{\mathcal{I}_{PVDF}} = \mathbf{1.80\ \times\ 10^{-6}\ W/m^2}$ (about the same order of magnitude as PZT)

Note: Notice that we did not need the pyroelectric material thickness (Example 7.15 in the textbook). This is because we neglected the thermal conduction loss. If we were to include heat losses via thermal conduction then we would indeed need the thickness - this matter is treated in advanced texts.

*7.18 Pyroelectric detectors

Consider a typical pyroelectric radiation detector circuit as shown in Figure 7Q18-1. The FET circuit acts as a voltage follower (source follower). The resistance R_1 represents the input resistance of the FET in parallel with a bias resistance that is usually inserted between the gate and source. C_1 is the overall input capacitance of the FET including any stray capacitance but excluding the capacitance of the pyroelectric detector. Suppose that the incident radiation intensity is constant and equal to $\mathcal{I}$. Emissivity

η of a surface characterizes what fraction of the incident radiation that is absorbed. ηI is the energy absorbed per unit area per unit time. Some of the absorbed energy will increase the temperature of the detector and some of it will be lost to surroundings by thermal conduction and convection. Let the detector receiving area be A, thickness be L, density be ρ, and specific heat capacity (heat capacity per unit mass) be c. The heat losses will be proportional to the temperature difference between the detector temperature, T, and the ambient temperature, T_o, as well as the surface area A (much greater than L). Energy balance requires that

Rate of increase in the internal energy (heat content) of the detector

= Rate of energy absorption – Rate of heat losses

that is, $$(AL\rho)c\frac{dT}{dt} = A\eta I - KA(T - T_o)$$

where K is a constant of proportionality that represents the heat losses and hence depends on the thermal conductivity κ. If the heat loss involves pure thermal conduction from the detector surface to the detector base (detector mount), then $K = \kappa/L$. In practice, this is generally not the case and $K = \kappa/L$ is an oversimplification.

a. Show that the temperature of the detector rises exponentially as

$$T = T_o + \frac{\eta I}{K}\left[1 - \exp\left(-\frac{t}{\tau_{th}}\right)\right] \qquad \textit{Detector temperature}$$

where τ_{th} is a **thermal time constant** defined by $\tau_{th} = L\rho c/K$. Further show that for very small K, the above equation simplifies to

$$T = T_o + \frac{\eta I}{L\rho c}t$$

b. Show that temperature change dT in time dt leads to a pyroelectric current, i_p, given by

$$i_p = Ap\frac{dT}{dt} = \frac{Ap\eta I}{L\rho c}\exp\left(-\frac{t}{\tau_{th}}\right) \qquad \textit{Pyroelectric current}$$

where p is the pyroelectric coefficient. What is the initial current?

c. The voltage across the FET and hence the output voltage $v(t)$ is given by

$$v(t) = V_o\left[\exp\left(-\frac{t}{\tau_{th}}\right) - \exp\left(-\frac{t}{\tau_{el}}\right)\right] \qquad \textit{Pyroelectric detector output voltage}$$

where V_0 is a constant and τ_{el} is the **electrical time constant** given by R_1C_t , where C_t , total capacitance, is $(C_1 + C_{det})$, where C_{det} is the capacitance of the detector. Consider a particular PZT pyroelectric detector with an area 1 mm^2, and thickness 0.05 mm. Suppose that this PZT has ε_r = 250, ρ = 7.7 g cm^{-3}, c = 0.3 J K^{-1} g^{-1}, and κ = 1.5 W K^{-1} m^{-1}. The detector is connected to an FET circuit that has R_1 = 10 MΩ and C_1 = 3 pF. Taking the thermal conduction loss constant K as κ/L, and η = 1, calculate τ_{th} and τ_{el}. Sketch schematically the output voltage. What is your conclusion?

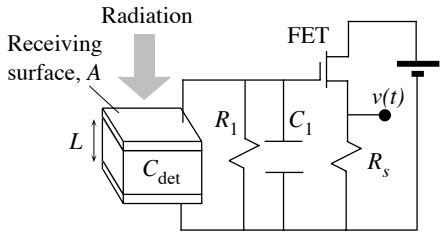


Figure 7Q18-1 A pyroelectric detector with an FET voltage follower circuit.

Solution

a Energy balance requires the following:

$$AL\rho c\frac{dT}{dt} = A\eta I - KA\left(T - T_o\right) \tag{1}$$

Consider the given expression for the temperature with thermal time constant $\tau_{th} = L\rho c/K$:

$$T = T_o + \frac{\eta I}{K}\left[1 - \exp\left(-\frac{t}{\tau_{th}}\right)\right] \tag{2}$$

We can show that Eqn. (2) is a solution by substituting Eqn. (2) into Eqn. (1) and verifying that is indeed a solution. Mathematicians prefer solving Eqn. (1) directly and this is how Eqn. (2) was actually obtained in the first place. However, given the expression in Eqn. (2), the simplest method is to substitute Eqn. (2) into Eqn. (1) and show that it is satisfied.

Substitute for T in Eqn. (1):

$$AL\rho c\frac{d}{dt}\left(T_o + \frac{\eta I}{K}\left[1 - \exp\left(-\frac{t}{\tau_{th}}\right)\right]\right) = A\eta I - KA\left(T_o + \frac{\eta I}{K}\left[1 - \exp\left(-\frac{t}{\tau_{th}}\right)\right] - T_o\right)$$

$\therefore$ $$L\rho c\frac{\eta I}{K}\left(-\frac{1}{\tau_{th}}\right)\left[-\exp\left(-\frac{t}{\tau_{th}}\right)\right] = \eta I - \left(\eta I\left[1 - \exp\left(-\frac{t}{\tau_{th}}\right)\right]\right)$$

$\therefore$ $$L\rho c\frac{\eta I}{K}\left(\frac{1}{\tau_{th}}\right)\left[\exp\left(-\frac{t}{\tau_{th}}\right)\right] = \eta I\exp\left(-\frac{t}{\tau_{th}}\right)$$

substitute for τ_{th}: $$L\rho c\frac{\eta I}{K}\left(\frac{1}{L\rho c/K}\right)\left[\exp\left(-\frac{t}{\tau_{th}}\right)\right] = \eta I\exp\left(-\frac{t}{\tau_{th}}\right)$$

simplify: $$\eta I = \eta I$$

As the right hand side equals the left hand side, Eqn. (1) is satisfied and Eqn. (2) is a solution.

For small K, we can expand the exponential of Eqn. (2) ($\exp(-t / \tau_{th}) = \exp(-Kt / L\rho C)$), i.e.: Assuming x is small and neglecting second order and higher order terms (x^2 and higher), $\exp(-x) = 1 - x$.

$$T = T_o + \frac{\eta I}{K}\left[1 - \exp\left(-\frac{Kt}{L\rho c}\right)\right] = T_o + \frac{\eta I}{K}\left[1 - \left(1 - \frac{Kt}{L\rho c}\right)\right]$$

$$\therefore \qquad T = T_o + \frac{\eta I}{K}\left(\frac{Kt}{L\rho c}\right)$$

$$\therefore \qquad T = T_o + \frac{\eta I}{L\rho c}t$$

Therefore, for small loss (K small) or for sufficiently short periods of time (t small), the temperature rise is linear with time.

b The pyroelectric current i_p is given by:

$$i_p = Ap\frac{dT}{dt}$$

This is so because in time dt, the temperature changes by dT. The change in the polarization is $p = dP/dT$. This changes the charge on the pyroelectric crystal by an amount $dQ = AdP = A(pdT)$. The current i_p is dQ/dt, hence the above equation.

Substitute Eqn. (2) into the above:

$$i_p = Ap\frac{d}{dt}\left(T_o + \frac{\eta I}{K}\left[1 - \exp\left(-\frac{t}{\tau_{th}}\right)\right]\right)$$

$$\therefore \qquad i_p = Ap\frac{\eta I}{K}\left(-\frac{1}{\tau_{th}}\right)\left[-\exp\left(-\frac{t}{\tau_{th}}\right)\right] = Ap\frac{\eta I}{K}\left(-\frac{1}{L\rho c/K}\right)\left[-\exp\left(-\frac{t}{\tau_{th}}\right)\right]$$

$$\therefore \qquad i_p = \frac{Ap\eta I}{L\rho c}\exp\left(-\frac{t}{\tau_{th}}\right)$$

To find the initial current, substitute $t = 0$.

$$i_p(0) = \frac{Ap\eta I}{L\rho c}\exp(0) = \frac{Ap\eta I}{L\rho c}$$

c To find the electrical time constant τ_{el}, we need the capacitance of the detector C_{det}, which can be found as follows (relative permittivity $\varepsilon_r = 250$, area $A = 1$ mm^2, thickness $L = 0.05$ mm):

$$C_{\text{det}} = \frac{\varepsilon_o \varepsilon_r A}{L} = \frac{\left(8.854\times10^{-12}\ \text{F/m}\right)(250)\left(1\times10^{-6}\ \text{m}^2\right)}{0.05\times10^{-3}\ \text{m}} = 4.427\times10^{-11}\ \text{F}$$

Now the total capacitance C_t can be found (FET circuit capacitance C_1 is given as 3 pF):

$$C_t = C_1 + C_{\text{det}} = 3\times10^{-12}\ \text{F} + 4.427\times10^{-11}\ \text{F} = 4.727\times10^{-11}\ \text{F}$$

and the electrical time constant is then (where FET circuit resistance $R_1 = 10$ MΩ):

$$\tau_{el} = R_1 C_t = (10\times10^6\ \Omega)(4.727\times10^{-11}\ \text{F}) = \mathbf{0.000473\ s}$$

To find the thermal time constant τ_{th}, we need the constant of proportionality K (where κ is the thermal conductivity and L is the thickness):

$$K = \kappa / L = (1.5\ \text{W K}^{-1}\ \text{m}^{-1}) / (0.05\times10^{-3}\ \text{m}) = 30000\ \text{W K}^{-1}$$

τ_{th} is therefore:

$$\tau_{th} = \frac{L\rho c}{K} = \frac{\left(0.05\times10^{-3}\ \text{m}\right)\left(7700\ \text{kg/m}^3\right)\left(300\ \text{J K}^{-1}\ \text{kg}^{-1}\right)}{30000\ \text{W K}^{-1}} = \mathbf{0.00385\ \ s}$$

For the sketch, we can set V_o to 1 for the purpose of simplicity. Note: Intensity is not required as the output is a schematic sketch. The magnitude of the output voltage is:

$$v(t) = \left| V_o\left[\exp\left(-\frac{t}{\tau_{th}}\right) - \exp\left(\frac{t}{\tau_{el}}\right)\right]\right|$$

The output peaks and drops. This is because R_1 leaks the induced charge on the pyroelectric sample to ground at a rate determined by the electrical time constant τ_{el}. If we were to set $\tau_{el} = \infty$ ($R_1 = \infty$) then there would be no decay in the output voltage v, at least in theory. However, in practice there is always some leakage or finite R_1.

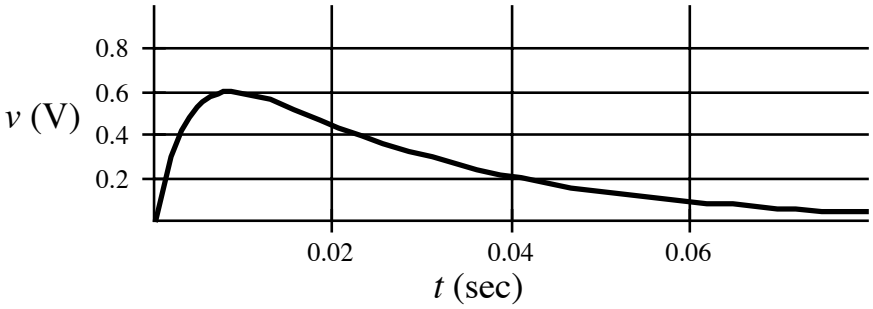


Figure 7Q18-2 Schematic plot of output voltage.

Note that the light intensity is switched on suddenly at $t = 0$, i.e. it is a step excitation.

7.19 Spark generator design

Design a PLZT piezoelectric spark generator using two back-to-back PLZT crystals that provide a 60 μJ spark in an air gap of 0.5 mm from a force of 50 N. At 1 atmosphere in an air gap of 0.5 mm, the breakdown voltage is about 3000 V. The design will need to specify the dimensions of the crystal and the dielectric constant. Assume that the piezoelectric voltage coefficient is 0.023 V m N^{-1}.

Solution

Given, F = applied force = 50 N to be applied through a mechanical lever arrangement.

Two ceramics are used to obtain a greater energy in the spark. Energy stored in each ceramic just before breakdown has to be 30 μJ.

The stress T induces an electric field E and hence a voltage V given by,

$$E = gT \tag{1}$$

where g is the piezoelectric voltage coefficient. Thus, at breakdown when $V = V_{br}$,

$$\frac{V_{\text{br}}}{L} = g\frac{F}{A}$$

$$\therefore \qquad \frac{L}{A} = \frac{V_{\text{br}}}{gF}$$

$$\therefore \qquad \frac{L}{A} = \frac{(3000\text{ V})}{(0.023\text{ V m N}^{-1})(50\text{ N})} = 2608\text{ m}^{-1}.$$

When V reaches V_{br} and the air breaks down, the energy stored on the piezoelectric sample capacitance is very *roughly* the energy in the spark, Equation (2),

$$E = \frac{1}{2}CV^2 \approx \frac{\varepsilon_o \varepsilon_r A}{2L} V_{br}^2 \qquad (2)$$

$$\varepsilon_r \approx \frac{2E}{\varepsilon_o V_{br}^2}\left(\frac{L}{A}\right) = \frac{2(30\times10^{-6}\text{ J})}{(8.85\times10^{-12}\text{ F m}^{-1})(3000\text{ V})^2}\left(2.6\times10^{3}\text{ m}^{-1}\right) = \mathbf{1958}$$

This is the minimum dielectric constant needed. Recall the units relationship: [J] = [F][V]^2.

Taking a crystal length ***L* of 5 mm** gives, ***A* = 0.0055/2.6 × 10^3 m^2** or **1.92 mm^2**.

This means a sample **diameter of 1.56 mm**.

Note: g determines the force needed to spark the gap and hence determines the crystal dimensions. ε_r determines the amount of energy in the gap. This the reason for giving g and requesting L/A and ε_r. Alternatively one can give ε_r and request g. The problem can also be approached in terms of d and ε_r but it is more usual to use g when a voltage needs to be generated; of course $g = d/(\varepsilon_o\varepsilon_r)$

"In Zipf's law the quantity under study is inversely proportional to the rank, that is, proportional to 1, 1/2, 1/3, 1/4, etc."

Murray Gell-Mann (California Institute of Technology; Nobel Laureate 1969)

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Chapter 8

8.1 Inductance of a long solenoid

Consider the very long (ideally infinitely long) solenoid shown in Figure 8Q1-1. If r is the radius of the core and ℓ is the length of the solenoid, then $\ell >> r$. The total number of turns is N and the number of turns per unit length is $n = N/\ell$. The current through the coil wires is I. Apply Ampere's law around C, which is the rectangular circuit $PQRS$, and show that

$$B \approx \mu_o \mu_r n I$$

Further, show that the inductance is

$$L \approx \mu_o \mu_r n^2 V_{\text{core}} \qquad \textit{Inductance of long solenoid}$$

where V_{core} is the volume of the core. How would you increase the inductance of a long solenoid?

What is the approximate inductance of an air-cored solenoid with a diameter of 1 cm, length of 20 cm, and 500 turns? What is the magnetic field inside the solenoid and the energy stored in the whole solenoid when the current is 1 A? What happens to these values if the core medium has a relative permeability μ_r of 600?

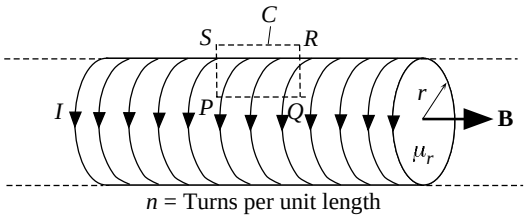


Figure 8Q1-1

Solution

We use Ampere's law in Equation 8.15 (in the textbook). Consider Figure 8Q1-2. If H is the field along a small length $d\ell$ along a closed path C, then around C, $\int H d\ell$ = total threaded current $= I_{\text{total}} = NI$.

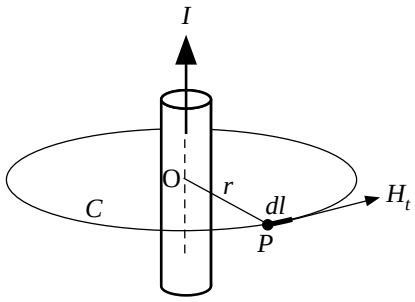


Figure 8Q1-2 Ampere's circuital law

Assume that the solenoid is infinitely long. The rectangular loop $PQRS$ has $n(PQ)$ number of turns where n is the number of turns per unit length or $n = N/\ell$ (See Figure 8Q1-1). The field is only inside the solenoid and only along the PQ direction (long solenoid assumption) and therefore the field along QR, RS and SP is zero. Assume that the field H is uniform across the solenoid core cross section. Then the path

integral of the magnetic field intensity H around $PQRS$ is simply is $H\ell = H(PQ)$. Ampere's law $\int H d\ell = I_{total}$ is then

$$H(PQ) = I(nPQ)$$

i.e. $$H = nI$$

The dimensions of the solenoid are such that length >> diameter. We can assume that H field is relatively uniform at all points inside the solenoid. Note: The approximate equality sign in the text (equation for B) is due to the fact that we assumed H is uniform across the core and, further, along the whole length of the solenoid from one end to the other. The ends of the solenoid will have different fields (lower). Let A be the cross-sectional area of the solenoid. The magnetic field B, the flux Φ and hence the inductance L are

$$B = \mu_o\mu_r H \approx \mu_o\mu_r nI$$

$\therefore$ $$\Phi = BA \approx \mu_o\mu_r nAI = \mu_o\mu_r\,(N/\ell)AI$$

and $$L = (N\Phi)/I = N[\mu_o\mu_r(N/\ell)AI]/I = \mu_o\mu_r(N^2/\ell)A = \mu_o\mu_r n^2(\ell A)$$

$\therefore$ $$L = \mu_o\mu_r n^2 V_{core}$$

where V_{core} is the volume of the core. Inductance depends on n^2, where n is the number of turns per unit length, on the relative permeability μ_r and on the volume of the core containing the magnetic flux. For a given volume inductor, L can be increased by using a higher μ_r material or increasing n, e.g. thinner wire to get more turns per unit length (not so thin that the skin effect diminishes the Q-factor, quality factor; see §2.8 in the textbook). The theoretical inductance of the coil is

$$L = (4\pi\times 10^{-7}\text{ H/m})(1)[(500)/(0.2\text{ m})]^2(0.2\text{ m})(\pi)[(0.01\text{ m})/(2)]^2$$

$\therefore$ $$\boldsymbol{L = 1.23\times 10^{-4}\text{ H}}\text{ or }\mathbf{0.123\text{ mH}}$$

and $$\boldsymbol{B} \approx (4\pi\times 10^{-7}\text{ Wb A}^{-1}\text{ m}^{-1})(1)[(500)/(0.2\text{ m})](1\text{ A}) = \mathbf{3.14\times 10^{-3}\text{ T}}$$

The energy per unit volume is,

$$E_{vol} = B^2/(2\mu_o) = (3.14\times 10^{-3}\text{ T})^2/[2(4\pi\times 10^{-7}\text{ Wb A}^{-1}\text{ m}^{-1})]$$

$\therefore$ $$E_{vol} = 3.92\text{ J / m}^3$$

The total energy stored is then,

$$\boldsymbol{E}_{\mathbf{tot}} = E_{vol}(\text{Length}\times\text{Area}) = (3.92\ \text{J/m}^3)\pi\left(\frac{0.01\ \text{m}}{2}\right)^2(0.2\ \text{m}) = \mathbf{61.6\ \mu J}$$

Suppose that $\mu_r = 600$ and suppose that the core does not saturate (an ideal ferromagnetic material) then,

$$\boldsymbol{L} \approx (1.23\times 10^{-4}\ \text{H})(600) = \mathbf{0.0738\text{ H}}$$

$$\boldsymbol{B} \approx (3.14\times 10^{-3}\ \text{T})(600) = \mathbf{1.88\text{ T}}$$

and $$E_{vol} \approx \frac{(1.88\text{ T})^2}{2(4\pi\times 10^{-7}\ \text{Wb/A}\cdot\text{m})(600)} = 2344\ \text{J/m}^3$$

so that $$\boldsymbol{E}_{\mathbf{tot}} = (E_{vol})(\text{Volume}) = \mathbf{36.8\text{ mJ}}$$

This is a dramatic increase and shows the virtue of using a magnetic core material for increasing the inductance and the stored magnetic energy.

8.2 Magnetization

Consider a long solenoid with a core that is an iron alloy (see Problem 8.1 for the relevant formulas). Suppose that the diameter of the solenoid is 2 cm and the length of the solenoid is 20 cm. The number of turns on the solenoid is 200. The current is increased until the core is magnetized to saturation at about $I = 2$ A and the saturated magnetic field is 1.5 T.

a. What is the magnetic field intensity at the center of the solenoid and the applied magnetic field, $\mu_o H$, for saturation?

b. What is the saturation magnetization M_{sat} of this iron alloy?

c. What is the total magnetization current on the surface of the magnetized iron alloy specimen?

d. If we were to remove the iron-alloy core and attempt to obtain the same magnetic field of 1.5 T inside the solenoid, how much current would we need? Is there a practical way of doing this?

Solution

a Applying Ampere's law or $H\ell = NI$ we have,

$$H = \frac{NI}{\ell} = \frac{(200)(2\ \text{A})}{0.2\ \text{m}}$$

Since $I = 2$ A gives saturation, corresponding magnetizing field is

$$\boldsymbol{H_{sat} \approx 2000\ \text{A/m}}$$

Suppose the applied magnetic field is the magnetic field in the toroid core in the absence of material. Then

$$B_{app} = \mu_o H_{sat} = \left(4\pi \times 10^{-7}\ \text{Wb A}^{-1}\ \text{m}^{-1}\right)(2000\ \text{A/m})$$

$\therefore$ $\boldsymbol{B_{app} = 2.51 \times 10^{-3}\ \text{T}}$

b Apply

$$B_{sat} = \mu_o\left(M_{sat} + H_{sat}\right)$$

$\therefore$ $$M_{sat} = \frac{B_{sat}}{\mu_o} - H_{sat} = \frac{1.5\ \text{T}}{4\pi \times 10^{-7}\ \text{Wb A}^{-1}\ \text{m}^{-1}} - 2000\ \text{A/m}$$

$\therefore$ $\boldsymbol{M_{sat} \approx 1.19 \times 10^{6}\ \text{A/m}}$

c Since M is the magnetization current per unit length,

$$I_m = M_{sat} \approx 1.19 \times 10^{6}\ \text{A/m}$$

Then $I_{surface}$ = Total circulating surface current:

$\therefore$ $\boldsymbol{I_{surface}} = I_m \ell = \left(1.19 \times 10^{6}\ \text{A/m}\right)(0.2\ \text{m}) = \boldsymbol{2.38 \times 10^{5}\ \text{A}}$

Note that the actual current in the wires, 2 A is negligible compared with $I_{surface}$.

d Apply, $B \approx \mu_o n I$ (for air)

$$\boldsymbol{I} \approx \frac{1.5\ \text{T}}{\left(4\pi \times 10^{-7}\ \text{Wb A}^{-1}\ \text{m}^{-1}\right)\left(\frac{200}{0.2\ \text{m}}\right)} = \boldsymbol{1194\ \text{A}}$$

Not very practical in every day life! Perhaps this current (thus field $B = 1.5$ T) could be achieved by using a superconducting solenoid.

8.3 Pauli spin paramagnetism

Paramagnetism in metals depends on the number of conduction electrons that can flip their spins and align with the applied magnetic field. These electrons are near the Fermi level E_F, and their number is determined by the density of states $g(E_F)$ at E_F. Since each electron has a spin magnetic moment of β, paramagnetic susceptibility can be shown to be given by

$$\chi_{\text{para}} \approx \mu_o \beta^2 g(E_F) \qquad \textit{Pauli spin paramagnetism}$$

where the density of states is given by Equation 4.10 (in the textbook). The Fermi energy of calcium, E_F, is 4.68 eV. Evaluate the paramagnetic susceptibility of calcium and compare with the experimental value of 1.9×10^{-5}.

Solution

Apply, $$g(E) = \left(8\pi\sqrt{2}\right)\left(\frac{m_e}{h^2}\right)^{3/2}\sqrt{E} \qquad \text{(Equation 4.10)}$$

so that $$g(E_F) = \left(8\pi\sqrt{2}\right)\left(\frac{9.109\times10^{-31}\ \text{kg}}{\left(6.626\times10^{-34}\ \text{J}\cdot\text{s}\right)^2}\right)^{3/2}\sqrt{(4.68\ \text{eV})\left(1.602\times10^{-19}\ \text{J/eV}\right)}$$

$\therefore$ $$g(E_F) = 9.197\times10^{46}\ \text{J}^{-1}\,\text{m}^{-3}$$

Then, $$\chi_{\text{para}} \approx \mu_o \beta^2 g(E_F)$$

$$\chi_{\text{para}} \approx \left(4\pi\times10^{-7}\ \text{Wb A}^{-1}\,\text{m}^{-1}\right)\left(9.273\times10^{-24}\,\text{A m}^2\right)^2\left(9.199\times10^{46}\ \text{J}^{-1}\,\text{m}^{-3}\right)$$

$\therefore$ $$\boldsymbol{\chi_{\text{para}} \approx 0.994\times10^{-5}}$$

This is in reasonable agreement within an order of magnitude with the experimental value of 1.9×10^{-5}.

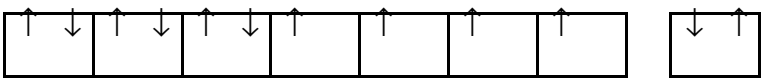
8.4 Ferromagnetism and the exchange interaction

Consider dysprosium (Dy), which is a rare earth metal with a density of 8.54 g cm^{-3} and atomic mass of 162.50 g mol^{-1}. The isolated atom has the electron structure [Xe] $4f^{10}6s^2$. What is the spin magnetic moment in the isolated atom in terms of number of Bohr magnetons? If the saturation magnetization of Dy near absolute zero of temperature is 2.4 MA m^{-1}, what is the effective number of spins per atom in the ferromagnetic state? How does this compare with the number of spins in the isolated atom? What is the order of magnitude for the exchange interaction in eV per atom in Dy if the Curie temperature is 85 K?

Solution

In an isolated Dy atom, the valence shells will fill in accordance with the exchange interaction:

$4f^{10}$ $6s^2$



Obviously, there are 4 unpaired electrons. Therefore for an isolated Dy atom, **the spin magnetic moment = 4β**.

Atomic concentration in dysprosium (Dy) solid is (where ρ is the density, N_A is Avogadro's number and M_{at} is the atomic mass):

$$n_{at} = \frac{\rho N_A}{M_{at}} = \frac{\left(8.54\times10^{3}\ \text{kg/m}^3\right)\left(6.022\times10^{23}\ \text{mol}^{-1}\right)}{162.50\times10^{-3}\ \text{kg/mol}} = 3.165\times10^{28}\ \text{m}^{-3}$$

Suppose that each atom contributes x Bohr magnetons, then

$$M_{\text{sat}} = n_{at} x\beta$$

$$\boldsymbol{x} = \frac{M_{\text{sat}}}{n_{at}\beta} = \frac{2.4\times10^{6}\ \text{A/m}}{\left(3.165\times10^{28}\ \text{m}^{-3}\right)\left(9.273\times10^{-24}\ \text{A m}^2\right)} = \mathbf{8.18}$$

This is almost twice the net magnetic moment in the isolated atom. Suppose that the Dy atom in the solid loses all the 4 electrons that are paired into the "electron gas" in the solid. This would make Dy^{+4} have 8 unpaired electrons and a net spin magnetic moment of 8β (this is an oversimplified view).

$$\textbf{Exchange interaction} \sim kT_C = \left(8.617\times10^{-5}\ \text{eV/K}\right)(85\ \text{K}) = \mathbf{0.00732\ eV}$$

The order of magnitude of exchange interaction ~ 10^{-2} eV/atom for Dy (small).

*8.5 Toroidal inductor and radio engineers toroidal inductance equation

a. Consider a toroidal coil (Figure 8Q5-1) whose mean circumference is ℓ and has N tightly wound turns around it. Suppose that the diameter of the core is $2a$ and $\ell >> a$. By applying Ampere's law, show that if the current through the coil is I, then the magnetic field in the core is

$$B = \frac{\mu_o \mu_r N I}{\ell} \qquad \textbf{[8.30]}$$

where μ_r is the relative permeability of the medium. Why do you need $\ell >> a$ for this to be valid? Does this equation remain valid if the core cross section is not circular but rectangular, $a \times b$, and $\ell >> a$ and b?

b. Show that the inductance of the toroidal coil is

$$L = \frac{\mu_o \mu_r N^2 A}{\ell} \qquad \textit{Toroidal coil inductance} \qquad \textbf{[8.31]}$$

where A is the cross-sectional area of the core.

c. Consider a toroidal inductor used in electronics that has a ferrite core size FT-37, that is, round but with a rectangular cross section. The outer diameter is 0.375 in (9.52 mm), the inner diameter is 0.187 in (4.75 mm), and the height of the core is 0.125 in (3.175 mm). The initial relative permeability of the ferrite core is 2000, which corresponds to a ferrite called the 77 Mix. If the inductor has 50 turns, then using Equation 8.31, calculate the approximate inductance of the coil.

d. Radio engineers use the following equation to calculate the inductances of toroidal coils,

$$L(\text{mH}) = \frac{A_L N^2}{10^6} \qquad \textit{Radio engineers inductance equation} \qquad \textbf{[8.32]}$$

where L is the inductance in millihenries (mH) and A_L is an inductance parameter, called an **inductance index**, that characterizes the core of the inductor. A_L is supplied by the manufacturers of ferrite cores and is typically quoted as millihenries (mH) per 1000 turns. In using Equation 8.32, one simply substitutes the numerical value of A_L to find L in millihenries. For the FT-37 ferrite toroid with the 77 Mix as the ferrite core, A_L is specified as 884 mH/1000 turns. What is the inductance of the toroidal inductor in part (*c*) from the radio engineers equation in Equation 8.32? What is the

percentage difference in values calculated by Equations 8.32 and 8.31? What is your conclusion? (*Comment*: The agreement is not always this close).

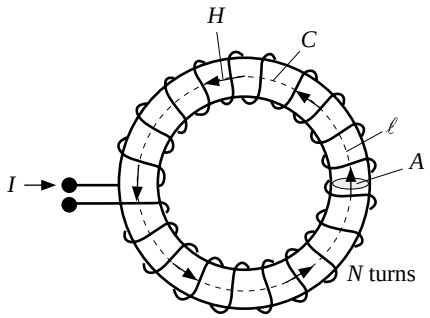


Figure 8Q5-1 A toroidal coil with *N* turns.

Solution

a As in Figure 8Q5-1, if we choose a closed path C that runs along the geometric center of the core and if *I* is the current, *N* is the number of turns and ℓ = mean circumference, then:

$$\oint_C H_t d\ell = H\ell = NI \qquad \text{or} \qquad H = (NI)/\ell$$

$$\therefore \qquad B = \mu_o \mu_r H = \frac{\mu_o \mu_r NI}{\ell}$$

This particular derivation only applies in the special case of $\ell >>$ radius (a); that is, for a very long, narrow core. If the core is very long and narrow, it may be safely assumed that the magnetic flux density B is uniform across the entire width ($2a$) of the core. If B was not uniform, then applying Ampere's Law to different (concentric) closed paths would yield different results.

This above derivation for B is valid for a rectangular cross-sectioned core of area $a \times b$, provided that $\ell >> a$ and $\ell >> b$. The magnetic field is then,

$$B = \frac{\mu_o \mu_r NI}{\ell}$$

b The inductance by definition is given by,

$$L = \frac{N\Phi}{I} = \frac{N(BA)}{I} = \frac{N\left(\dfrac{\mu_o \mu_r NAI}{\ell}\right)}{I} = \frac{\mu_o \mu_r N^2 A}{\ell}$$

c

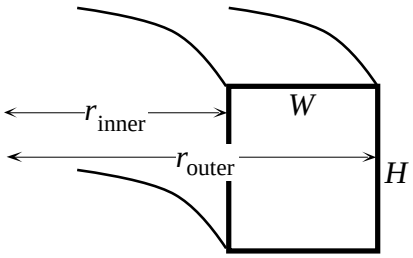


Figure 8Q5-2 Toroidal core with rectangular cross section.

Given outer radius = r_{outer} = 0.00476 m, inner radius = r_{inner} = 0.002375 m, height = H = 0.003175 m. We take the mean circumference ℓ through the geometric center of the core so that the mean radius is:

$$r_\ell = \frac{r_{\text{outer}} - r_{\text{inner}}}{2} + r_{\text{inner}} = 3.5675 \text{ mm}$$

$$\therefore \quad \ell = 2\pi r_\ell = (2\pi)\left(3.5675 \times 10^{-3} \text{ m}\right) = 22.42 \times 10^{-3} \text{ m}$$

Width = $W = r_{\text{outer}} - r_{\text{inner}} = 0.002385$ m

A = Cross sectional area = $W \times H$ = (0.002385 m)(0.003175 m) = 7.572×10^{-6} m^2.

Since $\ell >> a$ and $\ell >> b$ (at least approximately) we can calculate L as follows:

$$L = \frac{\mu_o \mu_r N^2 A}{\ell}$$

$$\therefore \quad L = \frac{\left(4\pi \times 10^{-7} \text{ H/m}\right)(2000)(50)^2\left(7.572 \times 10^{-6} \text{ m}^2\right)}{22.42 \times 10^{-3} \text{ m}}$$

$$\therefore \quad \boldsymbol{L = 0.00212 \text{ H}} \text{ or } \mathbf{2.12 \text{ mH}}$$

d Using the radio engineer's equation,

$$\boldsymbol{L(\text{mH}) = \frac{A_L N^2}{10^6} = \frac{(884 \text{ mH})(50)^2}{10^6} = 2.21 \text{ mH}}$$

so that $$\textbf{\% } \boldsymbol{difference} = \frac{2.21 \text{ mH} - 2.12 \text{ mH}}{2.21 \text{ mH}} \times 100\% = \mathbf{4.07\%}$$

Equations 2 and 3 differ only by 4.1% in this case. This is a good agreement.

*8.6 A toroidal inductor

a. Equations 8.31 and 8.32 (in the textbook) allow the inductance of a toroidal coil in electronics to be calculated. Equation 8.32 (in the textbook) is the equation that is used in practice. Consider a toroidal inductor used in electronics that has a ferrite core of size FT-23 that is round but with a rectangular cross section. The outer diameter is 0.230 in (5.842 mm), the inner diameter is 0.120 in (3.05 mm), and the height of the core is 0.06 in (1.5 mm). The ferrite core is a 43-Mix that has an initial relative permeability of 850 and a maximum relative permeability of 3000. The inductance index for this 43-Mix ferrite core of size FT-23 is A_L =188 (mH/1000 turns). If the inductor has 25 turns, then using Equations 8.31 and 8.32 (in the textbook), calculate the inductance of the coil under small-signal conditions and comment on the two values.

b. The saturation field, B_{sat}, of the 43-mix ferrite is 0.2750 T. What will be typical dc currents that will saturate the ferrite core (an estimate calculation is required)? It is not unusual to find such an inductor in an electronic circuit also carrying a dc current? Will your calculation of the inductance remain valid in these circumstances?

c. Suppose that the above toroidal inductor is in the vicinity of a very strong magnet that saturates the magnetic field inside the ferrite core. What will be the inductance of the coil?

Solution

a Provided that the mean circumference is much greater than any long cross sectional dimension (e.g. $\ell >>$ diameter or $\ell >> a$ and $\ell >> b$), then we can use,

$$L \approx \frac{\mu_o \mu_r N^2 A}{\ell}$$

We need the mean circumference ℓ which can be calculated from the mean radius r_ℓ,

$$r_\ell = \frac{r_{\text{outer}} - r_{\text{inner}}}{2} + r_{\text{inner}} = 2.223 \text{ mm so that } \ell = 2\pi r_\ell = 13.97 \text{ mm}$$

Width = $W = r_{\text{outer}} - r_{\text{inner}} = 0.001396$ m and Height = $H = 0.0015$ m.

A = Cross sectional area = $W \times H$

Since $\ell >> a$ and $\ell >> b$ (at least approximately) we can calculate L as follows:

$$L \approx \frac{\mu_o \mu_r N^2 (WH)}{\ell}$$

$$L \approx \frac{\left(4\pi \times 10^{-7} \text{ H/m}\right)(850)(25)^2\left(1.396 \times 10^{-3} \text{ m}\right)\left(1.5 \times 10^{-3} \text{ m}\right)}{13.97 \times 10^{-3} \text{ m}}$$

$$\boldsymbol{L \approx 0.10 \text{ mH}}$$

And, the radio engineer's equation, Equation 8.32 (in the textbook), (radio engineers toroidal inductance equation, pg. 6.25 and data pg. 24.7 in The ARRL Handbook 1995)

$$\boldsymbol{L} = \frac{A_L N^2}{10^6} = \frac{(188 \text{ mH})(25)^2}{10^6} = \mathbf{0.118 \text{ mH}}$$

There is a 15% difference between the two inductances; limitations of the approximation are apparent.

b To estimate H_{sat}, we'll take the maximum relative permeability $\mu_{r\,\max} = 3000$ as an estimate in order to find H_{sat} (see Figure 8Q6-1).

We know that

$$H = \frac{NI}{\ell} \text{ and } B_{\text{sat}} = \mu_{r\,\max} \mu_o H_{\text{sat}}$$

$\therefore$ $$B_{\text{sat}} = \frac{\mu_o \mu_{r\,\max} N I_{\text{sat}}}{\ell}$$

$\therefore$ $$0.2750 \text{ T} = \frac{\left(4\pi \times 10^{-7} \text{ Wb A}^{-1} \text{ m}^{-1}\right)(3000)(25) I_{\text{sat}}}{13.97 \times 10^{-3} \text{ m}}$$

$\therefore$ $\boldsymbol{I_{\text{sat}} = 40.8 \text{ mA}}$

will saturate the core.

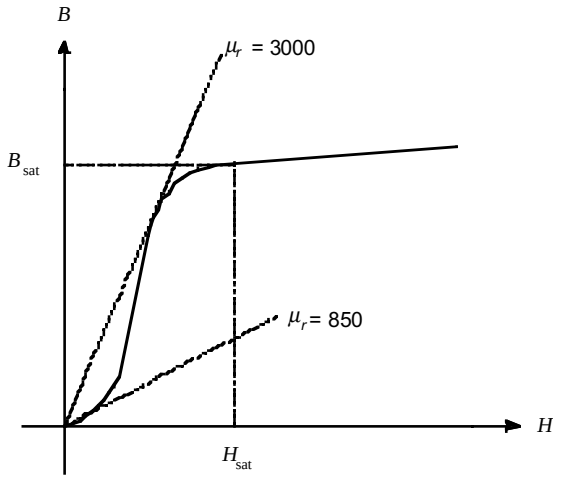


Figure 8Q6-1 B versus H curve for 43 - Mix ferrite.

If the core is saturated, the calculation of inductance is of course no longer valid as it used the initial permeability. The inductance now will be much reduced. Since under saturation $\Delta B = \mu_0 \Delta H$, effectively $\mu_r = 1$. The use of an initial permeability such as $\mu_r = 850$ implies that we have small changes (and *also reversible changes)* near around $H = 0$ or $I = 0$.

When an inductor carries a dc current in addition to an ac current, the core will operate "centered" at a different part of the material's *B*-*H* curve, depending on the magnitude of the dc current inasmuch as $H \propto I$. Thus, a dc current I_1 imposes a constant H_1 and shifts the operation of the inductor to around H_1. μ_r will depend on the magnitude of the dc current.

c The same effect as passing a large dc current and saturating the core. When the core is saturated, the increase in the magnetic field *B* in the core is that due to free space, i.e. $\Delta B = \mu_o \Delta H$. This means we can use $\mu_r = 1$ in Equation 8.31 (in the textbook) to find **the inductance** under saturation. It **will be about 0.10 mH / 850** or **0.12 μH**, very small.

Author's Note: Static inductance can be defined as *L* = Flux linked per unit dc current or $L = N\Phi/I$. It applies under dc conditions and *I* = dc current. For ac signals, the current *i* will be changing harmonically (or following some other time dependence) and we use the definition

$$L = \frac{N\delta\Phi}{\delta i} = \frac{N\left(\frac{d\Phi}{dt}\right)}{\left(\frac{di}{dt}\right)}$$

Further, by Faraday's law, since *v* is the induced voltage across the inductor by the changing total flux $Nd\Phi/dt$, we can also define *L* by,

$$v = L\left(\frac{di}{dt}\right)$$

Moreover, since $L = N\delta\Phi/\delta i$ we see that the ac *L* is proportional to the slope of the *B* vs. *H* behavior.

*8.7 The transformer

a. Consider the transformer shown in Figure 8Q7-1a whose primary is excited by an ac (sinusoidal) voltage of frequency f. The current flowing into the primary coil sets up a magnetic flux in the transformer core. By virtue of Faraday's law of induction and Lenz's law, the flux generated in the core is the flux necessary to induce a voltage nearly equal and opposite to the applied voltage. Thus,

$$\upsilon = \frac{d(\text{Total flux linked})}{dt} = \frac{NAdB}{dt}$$

where *A* is the cross-sectional area, assumed constant, and *N* is the number of turns in the primary. Show that if V_{rms} is the rms voltage at the primary ($V_{max} = V_{rms}\sqrt{2}$) and B_m is the maximum magnetic field in the core, then

$$V_{rms} = 4.44\, NAfB_m \qquad \textit{Transformer equation} \qquad \textbf{[8.33]}$$

Transformers are typically operated with B_m at the "knee" of the *B*-*H* curve, which corresponds roughly to maximum permeability. For transformer irons, $B_m \approx 1.2$ T. Taking $V_{rms} = 120$ V and a transformer core with $A = 10\text{ cm} \times 10\text{ cm}$, what should *N* be for the primary winding? If the secondary winding is to generate 240 V, what should be the number of turns for the secondary coil?

b. The transformer core will exhibit hysteresis and eddy current losses. The hysteresis loss per unit second, as power loss in watts, is given by

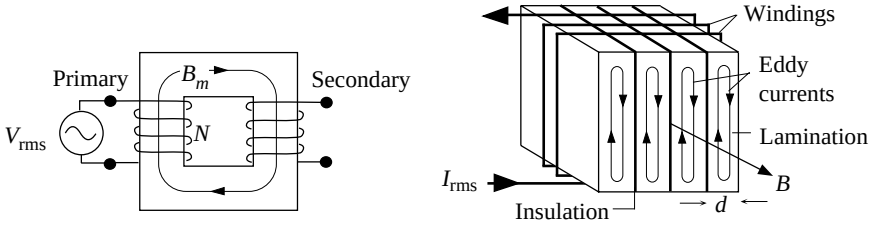
$$P_h = KfB_m{}^nV_{\text{core}} \qquad \textit{Hysteresis loss} \qquad \textbf{[8.34]}$$

where K = 150.7, f is the ac frequency (Hz), B_m is the maximum magnetic field (T) in the core (assumed to be in the range 0.2 – 1.5 T), n = 1.6 and, V_{core} is the volume of the core. The eddy current losses are reduced by laminating the transformer core as shown in Figure 8Q7-1b. The eddy current loss is given by

$$P_e = 1.65 f^2 B_m^2 \left(\frac{d^2}{\rho}\right) V_{\text{core}} \qquad \textit{Eddy current loss} \qquad \textbf{[8.35]}$$

where d is the thickness of the laminated iron sheet in meters (8Q7-1b) and ρ is its resistivity (Ω m).

Suppose that the transformer core has a volume of 0.0108 m^3 (corresponds to a mean circumference of 1.08 m). If the core is laminated into sheets of thickness 1 mm and the resistivity of the transformer iron is 6×10^{-7} Ω m, calculate both the hysteresis and eddy current losses at f = 60 Hz, and comment on their relative magnitudes. How would you achieve this?



(a) (b)

Figure 8Q7-1

(a) A transformer with N turns in the primary.

(b) Laminated core reduces eddy current losses.

Solution

a The induced voltage either at the primary or the secondary is given by Faraday's law of induction (the negative sign indicates an induced voltage opposite to the applied voltage), that is,

$$\upsilon = -NA\frac{dB}{dt}$$

Suppose we apply this to the primary winding. Then $\upsilon = V_m \sin(2\pi ft)$ where f = 60 Hz and V_m is the maximum voltage, that is $V_m = V_{\text{rms}}(\sqrt{2})$. Then

$$V_m \sin(2\pi ft) = -NA\frac{dB}{dt}$$

It is clear that B is also a sinusoidal waveform. Integrating this equation we find,

$$B = \frac{V_m}{NA(2\pi f)}\cos(2\pi ft)$$

which can be written as

$$B = B_m \cos(2\pi ft)$$

so that the maximum B is B_m given by

$$B_m = \frac{V_m}{2\pi fNA} = \frac{V_{\text{rms}}\sqrt{2}}{2\pi fNA}$$

where we used $V_m = V_{\text{rms}}(\sqrt{2})$. Thus,

$$\boldsymbol{V_{\text{rms}} = 4.44NAfB_m}$$

Let N_P = Primary winding turns and assume f = 60 Hz, then

$$\boldsymbol{N_P} = \frac{V_{\text{rms}}}{4.44AfB_m} = \frac{120\ \text{V}}{(4.44)(0.1\ \text{m})^2(60\ \text{Hz})(1.2\ \text{T})} = \textbf{38 turns on primary}$$

Secondary winding turns are

$$\boldsymbol{N_S} = \frac{240\ \text{V}}{(4.44)(0.1\ \text{m})^2(60\ \text{Hz})(1.2\ \text{T})} = \textbf{75 turns on secondary}$$

b Part **a** gives B_m = 1.2 T. Then the hysteresis loss is

$$P_h = KfB_m^n V_{\text{core}}$$

B_m = 1.2 T

$$\boldsymbol{P_h} = (150.7)(60\ \text{Hz})(1.2\ \text{T})^{1.6}\left(0.0108\ \text{m}^3\right) = \textbf{131 W}$$

Eddy current loss is

$$P_e = 1.65 f^2 B_m^2 \left(\frac{d^2}{\rho}\right) V_{\text{core}}$$

B_m = 1.2 T

$$\boldsymbol{P_e} = 1.65(60\ \text{Hz})^2(1.2\ \text{T})^2\left(\frac{(0.001\ \text{m})^2}{6\times10^{-7}\ \Omega\ \text{m}}\right)\left(0.0108\ \text{m}^3\right) = \textbf{154 W}$$

With 1 mm lamination and at this low frequency (60 Hz) hysteresis loss seems to dominate the eddy current loss. Eddy current loss can be reduced with thinner laminations or higher resistivity core materials (e.g. ferrites at the expense of B_{max}). Hysteresis loss can be reduced by using different core material but that changes B. Eqn. 8.34 is valid for only silicon steel cores only which have the required typical B_m for power applications.

8.8 Losses in a magnetic recording head

Consider eddy current losses in a permalloy magnetic head for audio recording up to 10 kHz. We will use the following equation for the eddy current losses,

$$P_e = 1.65 f^2 B_m^2 \left(\frac{d^2}{\rho}\right) V_{\text{core}}$$

where V_{core} is the volume of the core. Consider a magnetic head weighing 30 grams and made from a permalloy with density 8.8 g cm^{-3} and resistivity 6×10^{-7} Ω m. The head is to operate at B_m of 0.5 T. If the eddy current losses are not to exceed 1 mW, estimate the thickness of laminations needed. How would you achieve this?

Solution

We will apply the eddy current loss equation in this problem though this equation has a number of assumptions so that answer is only an *estimate*. The eddy current loss is

$$P_e = 1.65 f^2 B_m^2 \left(\frac{d^2}{\rho}\right) V_{\text{core}}$$

where the core volume is

$$V_{\text{core}} = \frac{\text{Mass}}{\text{Density}} = \frac{30\ \text{g}}{8.8\ \text{g/cm}^3} = 3.409\ \text{cm}^3\ \text{ or }\ 3.409\times10^{-6}\ \text{m}^3$$

Then, $$d^2 = \frac{P_e \rho}{1.65 f^2 B_m^2 V_{\text{core}}}$$

$\therefore$ $$d = \sqrt{\frac{(1\times10^{-3}\ \text{W})(6\times10^{-7}\ \text{W m})}{(1.65)(10{,}000\ \text{Hz})^2(0.5\ \text{T})^2(3.409\times10^{-6}\ \text{m}^3)}}$$

$\therefore$ $\boldsymbol{d = 2.07\ \mu\text{m}}$

Very thin (a page of a textbook is typically about 50 - 100 μm).

*8.9 Design of a ferrite antenna for an AM receiver

We consider an AM radio receiver that is to operate over the frequency range 530 - 1600 kHz. Suppose that the receiving antenna is to be a coil with a ferrite rod as core, as depicted in Figure 8Q9-1. The coil has N turns, its length is ℓ, and the cross-sectional area is A. The inductance, L, of this coil is tuned with a variable capacitor C. The maximum value of C is 265 pF, which with L should correspond to tuning in the lowest frequency at 530 kHz. The coil with the ferrite core receives the EM waves, and the magnetic field of the EM wave permeates the ferrite core and induces a voltage across the coil. This voltage is detected by a sensitive amplifier, and in subsequent electronics it is suitably demodulated. The coil with the ferrite core therefore acts as the antenna of the receiver (ferrite antenna). We will try to find a suitable design for the ferrite coil by carrying out approximate calculations - in practice some trial and error experimentation would also be necessary. We will assume that the inductance of a finite solenoid is

$$L = \frac{\gamma \mu_{ri} \mu_o A N^2}{\ell} \qquad \textit{Inductance of a solenoid} \qquad \textbf{[8.36]}$$

where A is the cross-sectional area of the core, ℓ is the coil length, N is the number of turns, and γ is a geometric factor that accounts for the solenoid coil being of finite length. Assume $\gamma \approx 0.75$. The resonant frequency f of an LC circuit is given by

$$f = \frac{1}{2\pi (LC)^{1/2}} \qquad \textbf{[8.37]}$$

a. If d is the diameter of the enameled wire to be used as the coil winding, then the length $\ell \approx Nd$. If we use an enameled wire of diameter 1 mm, what is the number of coil turns, N, we need for a ferrite rod given its diameter is 1 cm and its initial relative permeability is 100?

b. Suppose that the magnetic field intensity H of the signal in free space is varying sinusoidally, that is

$$H = H_m \sin(2\pi f t) \qquad \textbf{[8.38]}$$

where H_m is the maximum magnetic field intensity. H is related to the electric field $\mathcal{E}$ at a point by $H = \mathcal{E}/Z_{\text{space}}$ where Z_{space} is the impedance of free space given by 377 Ω. Show that the induced voltage at the antenna coil is

$$V_m = \frac{\mathcal{E}_m d}{2\pi 377 C f \gamma} \qquad \textit{Induced voltage across a ferrite antenna} \qquad \textbf{[8.39]}$$

where f is the frequency of the AM wave and $\mathcal{E}_m$ is the electric field intensity of the AM station at the receiver point. Suppose that the electric field of a local AM station at the receiver is 10 mV m^{-1}. What is the voltage induced across the ferrite antenna and can this voltage be detected by an amplifier? Would you use a ferrite rod antenna at short wave frequencies, given the same C but less N?

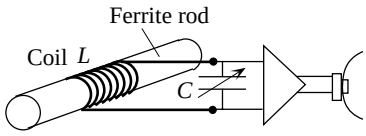


Figure 8Q9-1 A ferrite antenna of an AM receiver.

Solution

a Given C_{max} = 265 pF and resonant frequency f_o = 530 kHz (the lower frequency will need the largest inductance).

From $$f_o = \frac{1}{2\pi\sqrt{LC}}$$

$$L = \frac{1}{C(2\pi f_o)^2} = \frac{1}{(265\times10^{-12}\ \text{F})(2\pi 530\times10^{3}\ \text{Hz})^2} = 3.403\times10^{-4}\ \text{H}$$

Given $$L = \frac{\gamma\mu_{ri}\mu_o AN^2}{\ell_{\text{coil}}} = \frac{\gamma\mu_{ri}\mu_o AN}{d} \quad \text{since } \ell_{\text{coil}} \approx Nd$$

$\therefore$ $$\boldsymbol{N} = \frac{Ld}{\gamma\mu_{ri}\mu_o A} = \frac{(3.40\times10^{-4}\ \text{H})(0.001\ \text{m})}{(0.75)(100)(4\pi\times10^{-7}\ \text{H/m})(\pi)\left(\frac{0.01\ \text{m}}{2}\right)^2} = \textbf{46 turns}$$

b H-field is continuous everywhere. Let $\omega = 2\pi f$ where f is the frequency.

$H = H_m \sin\omega t \ \ \therefore \ \ B \approx \mu_r\mu_o H_m \sin\omega t$. This is the flux density inside the ferrite rod.

The induced voltage is

$$v(t) = NA\frac{dB}{dt} = \mu_r\mu_o NA\omega H_m \cos\omega t$$

$\therefore$ $$V_m = \mu_r\mu_o\omega NAH_m = \frac{\mu_r\mu_o\omega^2 NAH_m}{\omega}; \quad \omega^2 = \frac{1}{LC}$$

Substitute in the final equation for N in part **a**:

$\therefore$ $$V_m = \frac{\mu_r\mu_o NAH_m}{LC\omega} = \frac{\mu_r\mu_o\left(\frac{Ld}{\gamma\mu_r\mu_o A}\right)AH_m}{LC\omega} = \frac{H_m d}{2\pi f\gamma C};\ H_m = \frac{\mathcal{E}_m}{377}$$

$\therefore$ $$V_m = \frac{\mathcal{E}_m d}{2\pi 377 Cf\gamma}$$

Assume f_o = 530 kHz, corresponding to C = 265 pF:

$$V_m = \frac{(10\times10^{-3}\ \text{V/m})(1\times10^{-3}\ \text{m})}{(2\pi)(377\ \Omega)(265\times10^{-12}\ \text{F})(530\times10^{3}\ \text{Hz})(0.75)}$$

$\therefore$ $$\boldsymbol{V_m} = \textbf{40.1 } \boldsymbol{\mu}\textbf{V}$$

40 μV can be detected without corruption by noise (see Example 1.8, pg. 39 where the noise voltage is at most typically a few microvolts).

Author's Note: Suppose that the radio engineer uses the same tuning capacitor *C* for different bands. Then, since V_m depends inversely on *f*, the ferrite antennas will not be efficient at high frequencies. Changing the tuning capacitor *C* will obviously affect V_m. Note also that thicker *d* is better though there is another reason for using a thick *d* (Skin effect).

*8.10 A permanent magnet with an air gap

The magnetic field energy in the gap of a permanent magnet is available to do work. Suppose that B_m and B_g are the magnetic field in the magnet and the gap, H_m and H_g are the field intensities in the magnet and the gap and V_m and V_g are the volumes of the magnet and gap; show that, in terms of magnitudes,

$$B_g H_g V_g \approx B_m H_m V_m$$

What is the significance of this result?

Solution

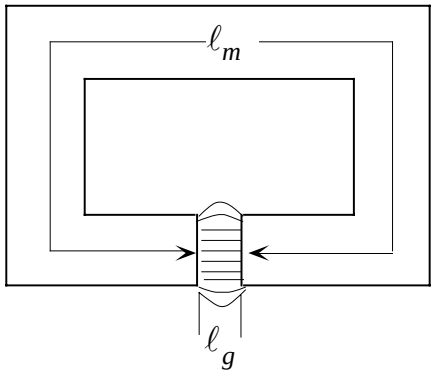


Figure 8Q10-1 A permanent magnet with a small air gap.

Assume uniform cross-sectional area *A* in the magnet.

Apply Ampere's Law to the closed path $\ell_m + \ell_g$ (refer to Figure 8Q10-1):

$$\oint_{\ell_m+\ell_g} H d\ell = H_m \ell_m + H_g \ell_g = \text{Total Current} = 0$$

because there are no windings on the magnet.

$$\therefore \qquad H_g = -\frac{\ell_m}{\ell_g} H_m \text{ which means } B_g = -\mu_o \frac{\ell_m}{\ell_g} H_m$$

The magnetic flux, $\Phi = BA$, must be continuous through the magnet and the air gap (a fundamental law in magnetism). Since the flux inside the magnet Φ_m also flows through the air gap, $\Phi_g = \Phi_m$. Also, if the gap length ℓ_g is sufficiently small, then fringing will be negligible. Then, since both the gap and the magnet have the same cross sectional area *A*, the flux density in the gap $B_g \approx B_m$.

Magnetic energy stored in the air gap is

$$E_g = V_g\left(\frac{1}{2} B_g H_g\right) = \left(A\ell_g\right)\left(\frac{1}{2} B_g H_g\right)$$

$$E_g \approx \frac{1}{2}\left(A\ell_g\right) B_m H_m \left(\frac{\ell_m}{\ell_g}\right) = \frac{1}{2}\left(A\ell_m\right) B_m H_m = \frac{1}{2} V_m B_m H_m$$

Thus, $V_g B_g H_g \approx V_m B_m H_m$

This is the required result. If we only consider magnitudes without reference to a direction then $B_g H_g V_g = B_m H_m V_m$. The greater the magnet volume (V_m large) or smaller the air gap volume (V_g small), the greater is the magnetic field B_g in the gap. For an efficient permanent magnet, i.e. high B_g, we need a large magnet volume, small air gap volume, and a high magnetic field in the magnet which implies a high remanent magnetic field (B_r) and a high coercive field intensity (H_c).

8.11 A permanent magnet with an air gap

a. Show that the maximum energy stored in the air gap of a permanent magnet can be written very roughly as

$$E_{\text{gap}} \approx {}^1/_8 B_r H_c V_m$$

where V_m is the volume of the magnet, which is much greater than that of the gap, B_r is the remanent magnetic field, and H_c is the coercivity of the magnet.

b. Using Table 8.5, compare the $(BH)_{\text{max}}$ with the product $[(1/2H_c)][(1/2B_r)]$ and comment on the closeness of agreement.

c. Calculate the energy in the gap of a rare earth cobalt magnet that has a volume of 0.1 m^3. Give an example of typical work (e.g., raising so many apples, each 100 grams, by so many meters) that could be done if all this energy could be converted to mechanical work.

Solution

a For most magnetic materials, to a fairly close approximation, $(BH)_{\text{max}}$ from the *B*-*H* characteristics would be (see Figure 8Q11-1 below):

$$(BH)_{\text{max}} = B'_m H'_m \approx \left(\frac{1}{2}B_r\right)\left(\frac{1}{2}H_c\right) = \frac{1}{4}B_r H_c$$

Since $E_{\text{gap}} \approx E_m$ (from Question 8.10),

$$E_{\text{gap}} \approx \frac{1}{2}(BH)_{\text{max}} V_m$$

$\therefore$ $$E_{\text{gap}} \approx \frac{1}{8}B_r H_c V_m$$

where V_m is the volume of the magnet. This is the required result and it is an approximation. It is useful, obviously, when the *B* vs. *H* behavior of the magnet is not available but we know B_r and H_c as below.

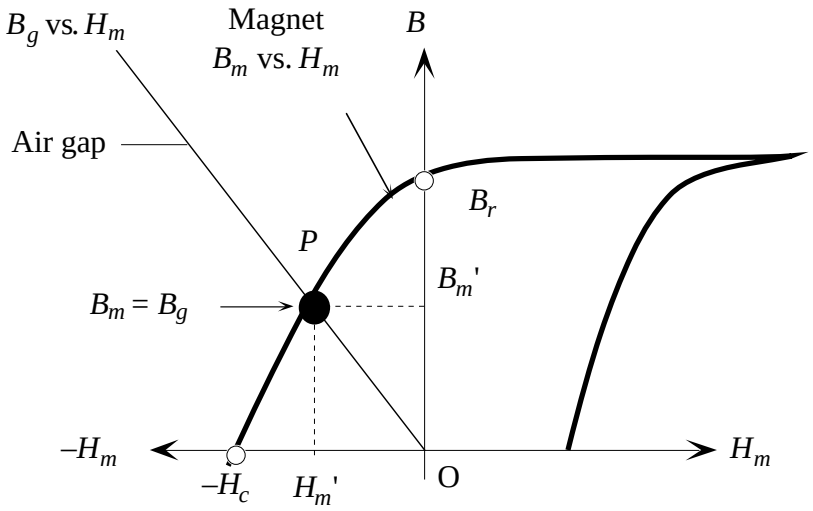


Figure 8Q11-1 Magnet with air gap. Point *P* is the operating point of the magnet and determines the field inside the magnet and in the air gap.

b $(H_c/2)(B_r/2)$ calculations are summarized in the table below.

Table 8Q11-1 Summarized values and comments for different magnetic materials.

Material	Tabulated $(BH)_{max}$ kJ/m^3	$(H_c/2)(B_r/2)$ kJ/m^3	Comment
Alnico (Fe-Al-Ni-Co-Cu)	50	34.0	30% difference
Alnico (Columnar)	60	20.1	Substantial difference
Strontium Ferrite (anisotropic)	24 - 34	21.5 - 34.2	Very close
Rare Earth Cobalt	150 - 240	135.7 - 240.7	Very close
NdFeB Magnets	200 - 275	179.0 - 238.7	Close

c For rare-earth cobalt magnet, from Table 8.5 (in the textbook), $(BH)_{max}$ = 150 - 240 kJ/m^3. Taking an average,

$$\boldsymbol{E}_{\mathbf{gap}} \approx \frac{1}{2}(BH)_{\max} V_m = \frac{1}{2}\left(\frac{150+240 \ \text{kJ/m}^3}{2}\right)\left(0.1 \ \text{m}^3\right) = \mathbf{9750\ J}$$

This corresponds to a mechanical work of raising 1000 apples (each 100 g) by 10 m.

8.12 Weight, cost, and energy of a permanent magnet with an air gap

For a certain application, an energy of 1 kJ is required in the gap of a permanent magnet. There are three candidates, as shown in Table 8Q12-1. Which material will give the lightest magnet? Which will give the cheapest magnet?

Table 8Q12-1 Three permanent magnet candidates.

Magnet	$(BH)_{max}$ kJ m^{-3}	Density g cm^{-3}	Yesterday's Relative Price (per unit mass)
Alnico	50	7.3	1
Rare earth	200	8.2	2
Ferrite	30	4.8	0.5

Solution

The energy in the magnet gap is,

$$E_{\text{gap}} \approx \frac{1}{2}(BH)_{\max} V_m$$

$$\therefore \quad V_m \approx \frac{2E_{\text{gap}}}{(BH)_{\max}} = \frac{2(1000\ \text{J})}{(BH)_{\max}}$$

If ρ is the density, then the corresponding magnet mass is

$$M_m = \text{Volume} \times \text{Density} = V_m \rho = \frac{2(1000\ \text{J})\rho}{(BH)_{\max}} \qquad (1)$$

The magnet material cost is

$$\text{Cost} = (\text{Price per unit mass}) M_m \qquad (2)$$

The table below lists the results of the calculations of magnet mass and cost using Eqns. (1) and (2) given $(BH)_{max}$.

Table 8Q12-2 Summarized results for magnet candidates.

Magnet	$(BH)_{max}$ kJ/m^3	Density g/cm^3	Yesterday's Relative Price (per unit mass)	Mass, M_m (kg)	Relative cost
Alnico	50	7.3	1	292	292
Rare Earth	200	8.2	2	**82**	164
Ferrite	30	4.8	0.5	320	**160**

Conclusions: (1) Rare earth magnet is lightest. (2) Ferrite magnet is cheapest.

8.13 Superconductivity and critical current density

Consider two superconducting wires, tin (Sn; Type I) and Nb_3Sn (Type II), each 1 mm in thickness. The magnetic field on the surface of a current-carrying conductor is given by

$$B = \frac{\mu_o I}{2\pi r}$$

a. Assuming that Sn wire loses its superconductivity when the field at the surface reaches the critical field (0.2 T), calculate the maximum current and hence the critical current density that can be passed through the Sn wire near absolute zero of temperature.

b. Calculate the maximum current and critical current density for the Nb_3Sn wire using the same assumption as in part (*a*) but taking the critical field to be the upper critical field, B_{c2}, which is 24.5 at 0 K. How does your calculation of J_c compare with the critical density of about 10^{11} A m^{-2} for Nb_3Sn at 0 K?

Solution

a We assume that when the current through the wire generates a magnetic field at the wire surface that is equal to the critical field, then superconductivity is extinguished.

$$I_c = \frac{B_c 2\pi r}{\mu_o} = \frac{(0.2\ \text{T})(2\pi)\left(\frac{0.001\ \text{m}}{2}\right)}{4\pi\times10^{-7}\ \text{Wb A}^{-1}\,\text{m}^{-1}}$$

$\therefore$ $\boldsymbol{I_c = 500\ \text{A}}$

$\therefore$ $$\boldsymbol{J_c = \frac{I_c}{A} = \frac{I_c}{\pi r^2} = \frac{500\ \text{A}}{\pi\left(\frac{0.001\ \text{m}}{2}\right)^2} = 6.37\times10^{8}\ \text{A/m}^2}$$

It turns out that this is indeed about the order of magnitude for the critical current density for Type I superconductors. The current that generates a field on the conductor surface that is equal to B_c extinguishes the superconductivity.

b Type II superconductor, Nb_3Sn. The critical field is the upper critical field, B_{c2}

$$\boldsymbol{I_c = \frac{(24.5\ \text{T})(2\pi)\left(\frac{0.001\ \text{m}}{2}\right)}{4\pi\times10^{-7}\ \text{Wb A}^{-1}\,\text{m}^{-1}} = 61250\ \text{A}}$$

and $$\boldsymbol{J_c = \frac{61250\ \text{A}}{\pi\left(\frac{0.001\ \text{m}}{2}\right)^2} = 7.80\times10^{10}\ \text{A/m}^2}$$

This value is close to experimental value of ~10^{11} A/m^2.

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*8.14 Enterprising engineers in the high arctic building a superconducting inductor

A current-carrying inductor has energy stored in its magnetic field that can be converted to electrical work. A group of enterprising engineers and scientists living in Resolute in Nunavut (Canada) have decided to build a toroidal inductor to store energy so that this energy can be used to supply a small community of 10 houses each consuming on average 3 kW of energy during the night (6 months). They have discovered a superconductor (Type II) that has a B_{c2} = 100 T and a critical current density of J_c = 5 $\times$ 10^{10} A m^{-2} at night temperatures (it is obviously a novel high-T_c superconductor of some sort). Their superconducting wire has a diameter of 5 mm and is available in any desirable length. All the wiring in the community is done by superconductors except where energy needs to be converted to other forms (mechanical, heat, etc.). They have decided on the following design specification for their toroid:

The mean diameter, D_{toroid}, of toroid, (1/2)(Outside diameter + Inside diameter), is 10 times longer than the core diameter, D_{core}. The field inside the toroid is therefore reasonably uniform to within 10%.

The maximum operating magnetic field in the core is 35 T. Fields larger than this can result in mechanical fracture and failure.

Assume that J_c decreases linearly with the magnetic field and that the mechanical engineers in the group can take care of the forces trying to blow open the toroid by building a proper support structure.

Find the size of the toroid (mean diameter and circumference), the number of turns and the length of the superconducting wire they need, the current in the coil, and whether this current is sufficiently below the critical current at that field. Is it feasible?

Solution

As a design question, there are several solutions.

Design I (Author's design)

Critical field B_c = 100 T and the critical current density $J_c = 5 \times 10^{10}$ A/m^2.

Engineers choose B = 35 T based on the mechanical support technology for the solenoid.

Assume that J_c decreases linearly with B_c so that when the actual field is B, J_c is J'_c.

$$J'_c = J_c(B_c - B)/B_c$$

$$J'_c = (5 \times 10^{10} \text{ A/m}^2)(100 \text{ T} - 35 \text{ T})/(100 \text{ T}) = 3.250 \times 10^{10} \text{ A/m}^2$$

Power per house = 3000 W

1 night is 6 months and in seconds this is

Time = (1 Nordic night)(6 months/night)(30 days/month)(24 hours/day)
(60 minutes/hour)(60 seconds/minute)

Time = 1.555×10^7 s

Total energy requirement is

$$E_{\text{total}} = (\text{Houses})(\text{Power per house})(\text{Time})$$

$$E_{\text{total}} = (10 \text{ houses})(3000 \text{ J/s per house})(1.555 \times 10^7 \text{ s}) = 4.665 \times 10^{11} \text{ J}$$

The available energy density, E_{volume}, in the toroid is

$$E_{\text{volume}} = \frac{B^2}{2\mu_o} = \frac{(35 \text{ T})^2}{2\left(4\pi \times 10^{-7} \text{ Wb A}^{-1} \text{ m}^{-1}\right)} = 4.874 \times 10^8 \text{ J/m}^3$$

This provides the energy per unit volume in the toroid.

Let V = volume of the toroidal core.

By the definition of energy density, E_{volume},

$$V = \text{Volume} = \frac{E_{\text{total}}}{E_{\text{volume}}} = \frac{4.665 \times 10^{11} \text{ J}}{4.874 \times 10^8 \text{ J m}^{-3}} = 957.1 \text{ m}^3$$

which is the necessary toroid volume.

We now need the maximum current I_{max} for destroying superconductivity.

Let d = diameter of superconducting wire; d = 5 mm.

I_{max} corresponding to J'_c is

$$I_{\text{max}} = J'_c\left[\pi\left(\frac{d}{2}\right)^2\right] = \left(3.250 \times 10^{10} \text{ A/m}^2\right)\left[\pi\left(\frac{0.005 \text{ m}}{2}\right)^2\right] = 6.381 \times 10^5 \text{ A}$$

The operating current must be less than this maximum current.

Let L = mean circumference of the toroid and D = diameter of cross section of toroid core. Let the mean diameter of the toroid be D_{toroid}, whereas D is the diameter of the core. We need $D_{\text{toroid}} >> D$ or by *design choice*:

$$D_{\text{toroid}} = 10\ D$$

Mean circumference is

$$L = \pi(D_{\text{toroid}}) = 10\pi D \text{ or } D = L/(10\pi)$$

Now consider the volume of the core,

$$V = \pi\left(\frac{D}{2}\right)^2 L = \pi\left(\frac{(L/10\pi)}{2}\right)^2 L = \frac{L^3}{400\pi}$$

$\therefore$ $$\boldsymbol{L} = (400\pi V)^{1/3} = \left(400\pi\left[957.1\ \text{m}^{-3}\right]\right)^{1/3} = \mathbf{106.3\ m}$$

which is the mean circumference of the inductor. And:

$$\boldsymbol{D}_{\textbf{toroid}} = L/\pi = 106.4\ \text{m}/\pi = \mathbf{33.9\ m}$$

Let N = number of turns. If d is the diameter of the superconducting wire, then

$Nd \approx$ Length (mean circumference) of the inductor = L

$$\boldsymbol{N} = \frac{L}{d} = \frac{106.3}{0.005} = \mathbf{21260\ turns}$$

Suppose that I_{op} is the operating current. Then the magnetic field inside the core of the toroid is

$$B = \frac{\mu_o N I_{op}}{L}$$

so that $$\boldsymbol{I_{op}} = \frac{BL}{\mu_o N} = \frac{(35\ \text{T})(106.3\ \text{m})}{(4\pi\times10^{-7}\ \text{Wb A}^{-1}\text{ m}^{-1})(21260)} = \mathbf{1.39\times10^5\ A}$$

The operating current is some 4.5 times less than the maximum allowed current, I_{max}, that extinguishes superconductivity.

Length of superconducting wire needed = NL = (21260)(106.3 m)= 2.26×10^6 m = **2260 km**

Author's Note: There are various assumptions in the above design. In addition, there are various technological problems such as setting up the current and tapping out the current as needed etc. Furthermore, there has to be some energy margin in the design (perhaps 50%). Feasible? The currents are not trivial. Any vanishing of superconductivity at any location would have disastrous consequences. Though not feasible today, why not in 50-70 years given the present rate of development in superconducting technologies?

The students find the following analogy helpful in appreciating why we considered an inductive storage device rather than a capacitor. Suppose that we build a huge capacitor that is charged to the required voltage and hence to the required energy during the summer and then the energy is tapped as needed during the winter. If the voltage is 10 kV, then we need a capacitor of 10^4 F! (And, it must not leak.)

<u>Design II (Submitted by an instructor using the textbook)</u>

Consider the energy need of the village.

E_{tot} = Total energy needed

E_{tot} = (10 houses)(3000 W/house)(3600 s/hour)(24 hours/day)(180 days)

$E_{\text{tot}} = 4.665\times10^{11}$ J

If E_{vol} is the energy density, then

$$E_{\text{tot}} = 4.665\times10^{11}\ \text{J} = E_{\text{vol}}\left(\pi D_{\text{toroid}}\right)\pi\left(\frac{D_{\text{core}}}{2}\right)^2$$

where D_{core} and D_{toroid} are the diameters of the core and the toroid. Assume $D_{\text{toroid}} = 10D_{\text{core}}$. Then

$$E_{tot} = E_{vol} 10\pi^2 D_{core} \frac{D_{core}^{\ 2}}{4}$$

$$\therefore \qquad E_{tot} = \frac{5}{2}\pi^2 D_{core}^3 E_{vol}$$

$$\therefore \qquad E_{vol} = \frac{E_{tot}}{\frac{5}{2}\pi^2 D_{core}^3} = \frac{4.665\times10^{11}\ \text{J}}{\frac{5}{2}\pi^2 (10\ \text{m})^3} = 1.891\times10^7\ \text{J/m}^3$$

Now these engineers are guided by the mechanical size of the toroid and choose, D_{core} = 10 m. In Design I, the electrical engineers instead chose *B* and left the mechanical size to mechanical and civil engineers.

Select D_{core} = 10 m

$$\therefore \qquad E_{vol} = 1.891\times10^7\ \text{J/m}^3 = \frac{B^2}{2\mu_o}$$

$$\therefore \qquad B = \sqrt{2\left(4\pi\times10^{-7}\ \text{H/m}\right)\left(1.891\times10^7\ \text{J/m}^3\right)} = 6.894\ \text{T}$$

Since D_{toroid} is 10 times D_{core}, we know $\boldsymbol{D_{toroid}}$ **is 100 m**. The mean circumference *L* is therefore $\boldsymbol{L = (100\ \text{m})\pi = 314\ \text{m}}$. If the turns on the toroid touched each other along the inside edge of the toroid, there would be $(D_{toroid} - D_{core})\pi = (90\pi)$ m of wire side-by-side which leads to

$$\boldsymbol{N} = \frac{(90\pi)\ \text{m}}{(0.005)\ \text{m}} = \mathbf{56549\ turns}$$

Length of wire needed: $\ell_{\mathbf{wire}} = N\pi D_{core} = (56549)(\pi)(10\ \text{m}) =$ **1776.5 km** (just for the toroid itself; more would be needed to wire the houses).

$$\therefore \qquad \boldsymbol{I} = \frac{B\ell_{toroid}}{\mu_o N} = \frac{(6.89\ \text{T})(100\ \text{m})(\pi)}{\left(4\pi\times10^{-7}\ \text{Wb A}^{-1}\ \text{m}^{-1}\right)(56549)} = \mathbf{3.05\times10^4\ A}$$

$$\therefore \qquad \boldsymbol{J_{wire}} = \frac{I}{\pi r^2} = \frac{3.05\times10^4\ \text{A}}{(\pi)\left(\frac{0.005\ \text{m}}{2}\right)^2} = \mathbf{1.55\times10^9\ A/m^2}$$

which is OK.

[*Author's Note*: This design uses a larger toroid than the first - I'm open to other design suggestions.]

8.15 Magnetic storage media

a. Consider the storage of video information (FM signal) on a video tape. Suppose that the maximum signal frequency to be recorded as a spatial magnetic pattern is 10 MHz. The heads helically scan the tape, and the relative velocity of the tape to head is about 10 m s^{-1}. What is the minimum spatial wavelength of the stored magnetic pattern (information) on the tape?

b. Suppose that the speed of an audio cassette tape in a cassette player is 5 cm s^{-1}. If the maximum frequency that needs to be recorded is 20 kHz, what is the minimum spatial wavelength on the tape?

Solution

a Consider a video tape. The maximum frequency in the signal is f_{video} = 10 MHz. The speed of the video tape is 10 m/s. The spatial wavelength is

$$\lambda = v/f_{video} = (10 \text{ m/s}) / (10 \times 10^6 \text{ s}^{-1}) = \mathbf{10^{-6}\ m} \text{ or } \mathbf{1\ \mu m}$$

By comparison, typically, a page of a book is 50 - 100 μm thick and so is a typical human hair.

b Consider an audio tape. The maximum frequency in the signal is f_{audio} = 20 kHz. The speed of the video tape is 0.05 m/s. The spatial wavelength is

$$\lambda = \frac{5 \times 10^{-2} \text{ m/s}}{20 \times 10^{3} \text{ Hz}} = \mathbf{2.5 \times 10^{-6}\ m} \text{ or } \mathbf{2.5\ \mu m}$$

This is a spatial wavelength of 2.5 μm on the tape.

"We have a habit in writing articles published in scientific journals to make the work as finished as possible, to cover up all the tracks, to not worry about the blind alleys or describe how you had the wrong idea first, and so on. So there isn't any place to publish, in a dignified manner, what you actually did in order to get to do the work."

Richard P. Feynmann (Nobel Lecture, 1966)

Second Edition (© 2001 McGraw-Hill)

Chapter 9

9.1 Refractive index and relative permittivity

Using $n = \sqrt{\varepsilon_r}$, calculate the refractive index *n* of the materials in the table given their low frequency relative permittivities ε_r(LF). What is your conclusion?

Material →	a-Se	Ge	NaCl	MgO
ε_r (LF)	6.4	16.2	5.90	9.83
n (~ 1 - 5 μm)	2.45	4.0	1.54	1.71

Solution

The results from the calculations are summarized in Table 9Q1-1.

For a-Se and Ge there is an excellent agreement between $\sqrt{\varepsilon_r(LF)}$ and *n*. Both are covalent solids in which electronic polarization (electronic bond polarization) is the only polarization mechanism at low and high frequencies. Electronic polarization involves the displacement of light electrons with respect to positive ions. This process can readily respond to the field oscillations up to optical frequencies.

For NaCl and MgO $\sqrt{\varepsilon_r(LF)}$ is larger than *n* because at low frequencies both of these solids possess a degree of ionic polarization. The bonding has a substantial degree of ionic character which contributes to polarization at frequencies below far-infrared wavelengths.

Table 9Q1-1

Material	ε_r (LF)	$\sqrt{\varepsilon_r(LF)}$	*n* - optical for (1 - 5) μm range	Comment
a-Se	6.4	2.53	2.45	Electronic bond polarization
Ge	16.2	4.02	4.0	Electronic bond polarization
NaCl	5.9	2.43	1.54	Ionic polarization contributes to ε_r (LF)
MgO	9.83	3.14	1.71	Ionic polarization contributes to ε_r (LF)

9.2 Refractive index and bandgap

Diamond, silicon, and germanium all have the same diamond unit cell. All three are covalently bonded solids. Their refractive indices (*n*) and energy bandgaps (E_g) are shown in the table below. (a) Plot *n* versus E_g and (b) Plot also n^4 versus $1/E_g$. What is your conclusion? According to **Moss's rule**, very roughly,

$$n^4 E_g \approx K = const \qquad \textit{Moss's rule}$$

What is the value of *K*?

Material →	Diamond	Silicon	Germanium
Bandgap, E_g (eV)	5	1.1	0.66
n	2.4	3.46	4.0

Solution

The plots of n versus E_g and of n^4 versus $1/E_g$ are presented in Figures 9Q2-1 and 9Q2-2 respectively. The lines presented on both plots are the best fits of the Moss's rule to the experimental data with K being the only parameter. On the base of the high values of r^2 and $Adj\ r^2$ we can conclude that there is a reasonable agreement between the experimental data and Moss's rule.
Fitting the Moss's rule to the experimental data gives a value for K of about 164.3 eV.

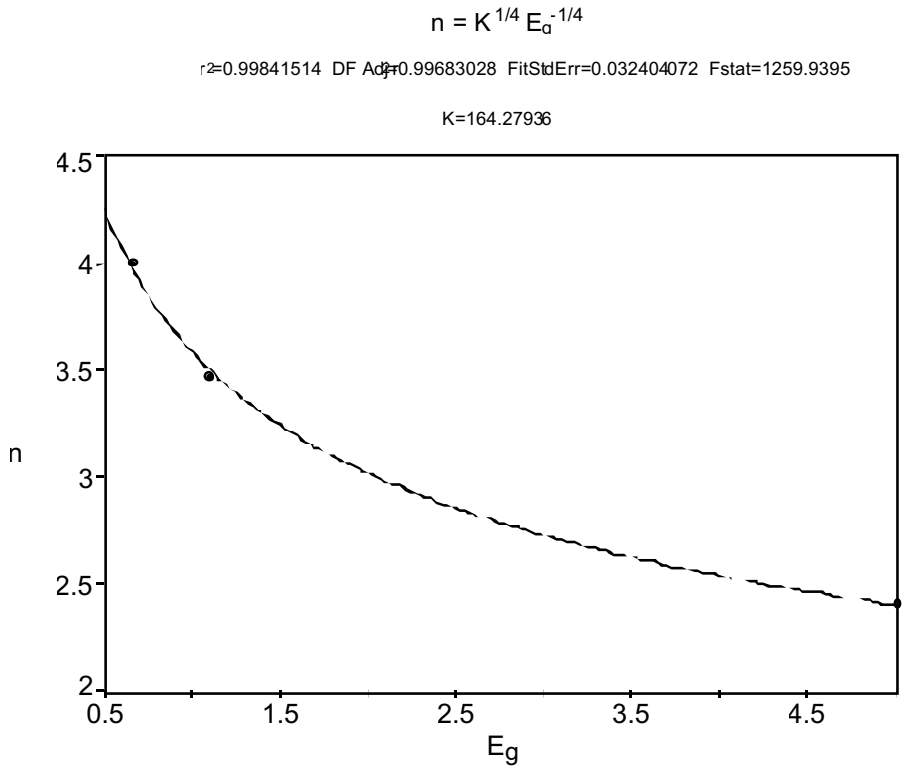


Figure 9Q2-1

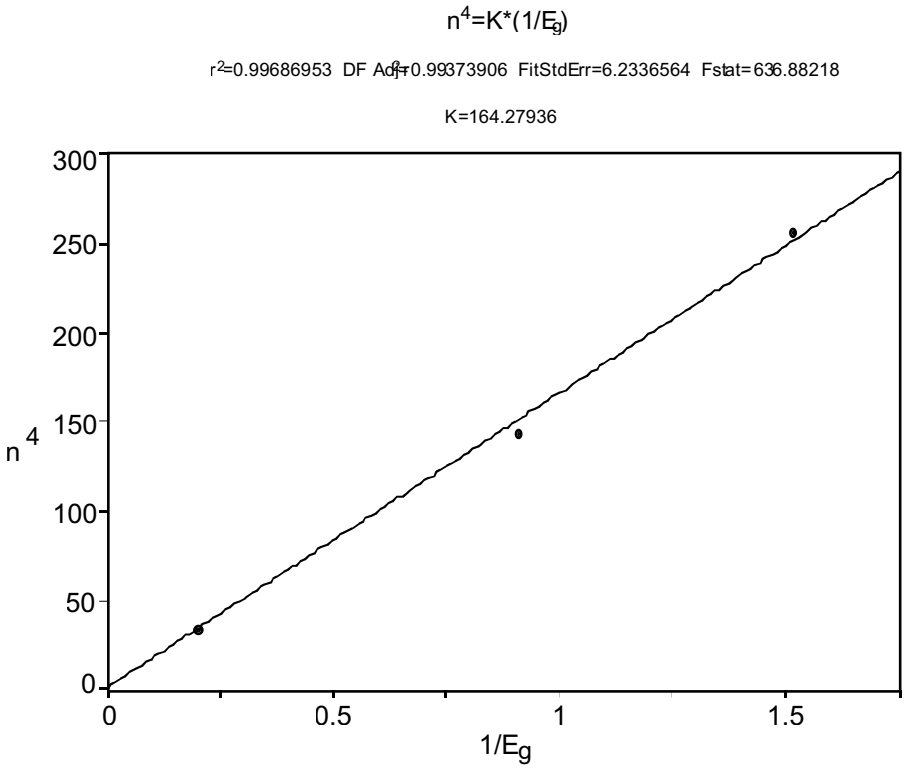


Figure 9Q2-2

*9.3 Temperature coefficient of refractive index

Suppose that we could write the relationship between the refractive index n (at frequencies much less than ultraviolet light) and the band gap E_g of a semiconductor as suggested by Hervé and Vandamme,

$$n^2 = 1 + \left(\frac{A}{E_g + B} \right)^2$$

where E_g is in eV, A = 13.6 eV and B = 3.4 eV. (B depends on the incident photon energy.) Temperature dependence in n results from dE_g/dT and dB/dT. Show that the temperature coefficient of refractive index (TCRI) is given by[1],

$$\text{TCRI} = \frac{1}{n} \cdot \frac{dn}{dT} = -\frac{(n^2 - 1)^{3/2}}{13.6 n^2} \left[\frac{dE_g}{dT} + B' \right] \qquad \textit{Hervé-Vandamme relationship}$$

where B' is dB/dT. Given that $B' = 2.5 \times 10^{-5}$ eV K^{-1}, calculate TCRI for two semiconductors: Si with $n \approx 3.5$, $dE_g/dT \approx -3 \times 10^{-4}$ eV K^{-1} and AlAs with $n \approx 3.2$, $dE_g/dT \approx -4 \times 10^{-4}$ eV K^{-1}.

Solution

We can start the proof of the Hervé-Vandamme relationship for TCRI from the empirical relation for suggested by them, which connects the refractive index and energy band gap

[1] P.J.L. Hervé and L.K.J. Vandamme, *J. Appl. Physics*, **77**, 5476, 1995, and references therein.

$$n^2 = 1 + \left(\frac{A}{E_g + B} \right)^2 \qquad (1)$$

Taking natural logarithm from both sides of (1) and differentiating with respect to temperature we subsequently receive

$$2\ln(n) = \ln\left[1 + \left(\frac{A}{E_g + B} \right)^2 \right] \quad \frac{d}{dT}$$

$$\frac{2}{n}\frac{dn}{dT} = \frac{1}{\left[1 + \left(\frac{A}{E_g + B} \right)^2 \right]} \frac{d}{dT}\left[1 + \left(\frac{A}{E_g + B} \right)^2 \right] = \frac{1}{n^2} 2\left(\frac{A}{E_g + B} \right) \frac{d}{dT}\left(\frac{A}{E_g + B} \right) =$$

$$= -\frac{2}{n^2} \frac{A^2}{\left(E_g + B \right)^3} \frac{d}{dT}\left(E_g + B \right) = -\frac{2}{n^2} \frac{A^2}{\left(E_g + B \right)^3} \left(\frac{dE_g}{dT} + \frac{dB}{dT} \right)$$

or

$$\frac{1}{n}\frac{dn}{dT} = -\frac{1}{n^2} \frac{A^2}{\left(E_g + B \right)^3} \left(\frac{dE_g}{dT} + \frac{dB}{dT} \right) \qquad (2)$$

The left hand side of this equation is exactly the TCRI. Using Equation (1) we can easily show that

$$\frac{A^2}{\left(E_g + B \right)^3} = \frac{(n^2 - 1)^{\frac{3}{2}}}{A}$$

and substituting in (2) we receive exactly Hervé-Vandamme relationship for TCRI

$$\text{TCRI} = -\frac{\left(n^2 - 1 \right)^{\frac{3}{2}}}{A n^2} \left(\frac{dE_g}{dT} + \frac{dB}{dT} \right) = -\frac{\left(n^2 - 1 \right)^{\frac{3}{2}}}{13.6\, n^2} \left(\frac{dE_g}{dT} + B© \right)$$

Applying Hervé-Vandamme relationship for Si and for AlAs we receive:

$$\text{TCRI}_{\text{Si}} = -\frac{\left[(3.5)^2 - 1 \right]^{\frac{3}{2}}}{(13.6\ \text{eV})(3.5)^2} \left[\left(-3 \times 10^{-4}\ \ \text{eV}\ \ \text{K}^{-1} \right) + \left(2.5 \times 10^{-5}\ \ \text{eV}\ \ \text{K}^{-1} \right) \right] = \mathbf{6.23 \times 10^{-5}\ K^{-1}}$$

$$\text{TCRI}_{\text{AlAs}} = -\frac{\left[(3.2)^2 - 1 \right]^{\frac{3}{2}}}{(13.6\ \text{eV})(3.2)^2} \left[\left(-4 \times 10^{-4}\ \ \text{eV}\ \ \text{K}^{-1} \right) + \left(2.5 \times 10^{-5}\ \ \text{eV}\ \ \text{K}^{-1} \right) \right] = \mathbf{7.56 \times 10^{-5}\ K^{-1}}$$

9.4 Dispersion (n vs. λ) in diamond

The dispersion relationship for diamond can be written as

$$n^2 = 1 + \frac{A\lambda^2}{\lambda^2 - \lambda_1^2} + \frac{B\lambda^2}{\lambda^2 - \lambda_2^2} \qquad \textit{Sellmeier equation}$$

where A = 0.3306 and B = 4.3356, λ_1 = 175 nm and λ_2 = 106 nm. This type of $n(\lambda)$ dispersion relation is called the **Sellmeier equation** and is quite common for describing the refractive index of many

materials; in general, there may be more terms on the right hand side. Calculate the refractive index at 500 nm.

Solution

The refractive index for diamond at 500 nm is

$$n = \sqrt{1 + \frac{A\lambda^2}{\lambda^2 - \lambda_1^2} + \frac{B\lambda^2}{\lambda^2 - \lambda_2^2}} = \sqrt{1 + \frac{(0.3306)(500)^2}{(500)^2 - (175)^2} + \frac{(0.4.3356)(500)^2}{(500)^2 - (106)^2}} = \mathbf{2.432}$$

9.5 Dispersion (n vs. λ) in GaAs

By using the dispersion relation for GaAs, calculate the refractive index n and the group index N_g of GaAs at a wavelength of 1300 nm.

Solution

The dispersion relation for GaAs is given by Equation 9.13 (in the textbook)

$$n^2 = 7.10 + \frac{3.78\lambda^2}{\lambda^2 - 0.2767} \qquad (1)$$

where λ must be in μm. At λ =1300 nm = 1.3 μm for the refractive index of GaAs we receive

$$n = \sqrt{7.10 + \frac{3.78(1.3)^2}{(1.3)^2 - 0.2767}} = \mathbf{3.409}$$

The group index is defined as (see Equation 9.18 in the textbook)

$$N_g = n - \lambda\frac{dn}{d\lambda} \qquad (2)$$

To calculate $\frac{dn}{d\lambda}$ we have to differentiate both sides of (1) with respect to λ:

$$2n\frac{dn}{d\lambda} = \frac{3.78(2\lambda)(\lambda^2 - 0.2767) - 3.78\lambda^2(2\lambda)}{(\lambda^2 - 0.2767)^2}$$

and

$$\frac{dn}{d\lambda} = -\frac{1}{n}\frac{1.045926\lambda}{(\lambda^2 - 0.2767)^2} = -\frac{1}{3.409}\frac{1.045926(1.3)}{((1.3)^2 - 0.2767)^2} = \mathbf{-0.1997\ \mu m^{-1}}$$

Substituting this value in (2) we can calculate the group index

$$N_g = 3.409 - (1.3\ \mu\text{m})(-0.1997\ \mu\text{m}^{-1}) = \mathbf{3.669}$$

*9.6 Dispersion (n vs. λ)

Consider an atom in the presence of an oscillating electric field as in Figure 9Q6-1. The applied field oscillates harmonically in the +x and -x directions and is given by $E = E_o\exp(j\omega t)$. The energy losses can be represented by a frictional force whose magnitude is proportional to the velocity, dx/dt. If γ is the proportionality constant per electron and per unit electron mass then Newton's second law for Z electrons in the polarized atom is

$$Zm_e \frac{d^2x}{dt^2} = -ZeE_o \exp(j\omega t) - Zm_e\omega_o^2 x - Zm_e\gamma \frac{dx}{dt}$$

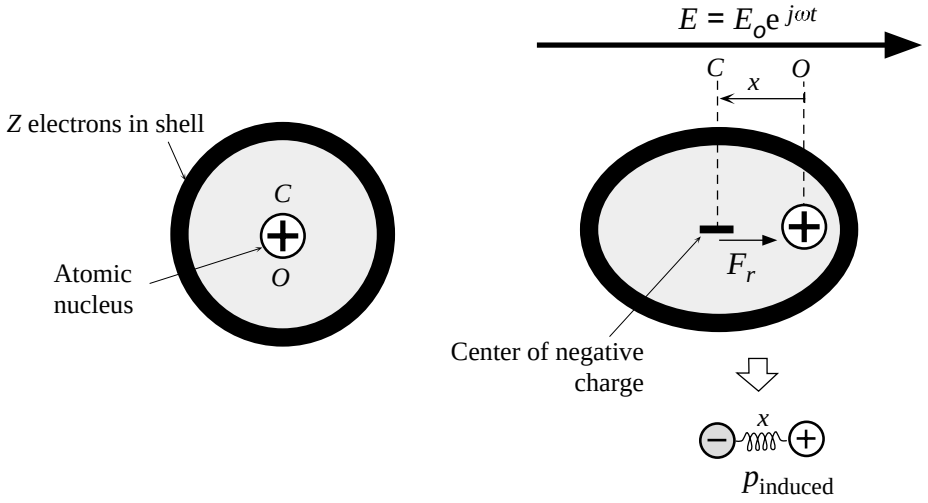
where $\omega_o = (\beta/Zm_e)^{1/2}$ is the **natural frequency** of the system composed of Z electrons and a $+Ze$ nucleus and β is a force constant for the restoring Coulombic force between the electrons and the nucleus. Show that the electronic polarizability α_e is

$$\alpha_e = \frac{p_{\text{induced}}}{E} = \frac{Ze^2}{m_e(\omega_o^2 - \omega^2 + j\gamma\omega)} \qquad \textit{Electronic polarizability}$$

What does a complex polarizability represent? Since α_e is a complex quantity, so is ε_r and hence the refractive index. By writing the complex refractive index $\mathrm{N} = \sqrt{(\varepsilon_r)}$ and ε_r is related to α_e by the Clasius-Mossotti equation, show that,

$$\frac{\mathrm{N}^2 - 1}{\mathrm{N}^2 + 2} = \frac{NZe^2}{3\varepsilon_o m_e(\omega_o^2 - \omega^2 + j\gamma\omega)} \qquad \textit{Complex refractive index}$$

where N is the number of atoms per unit volume. What is your conclusions?



(a) A neutral atom in $E = 0$. (b) Induced dipole moment in a field

Figure 9Q6-1 Electronic polarization of an atom.

Solution

The induced electronic dipole moment is simply given by $p_{induced} = -(Ze)x$. Using this simple relation we can easily prove that if the expression for the electronic polarizability is correct, the magnitude of the displacement of the negative charge with respect to the positive charge in the atom is given by

$$x = -\frac{eE}{m_e\left(\omega_o^2 - \omega^2 + j\gamma\omega\right)}$$

Further taking into account that the electric field is given by $E = E_0 \exp(j\omega t)$ for the magnitude of the displacement we receive

$$x = -\frac{eE_o \exp(j\omega t)}{m_e\left(\omega_o^2 - \omega^2 + j\gamma\omega\right)}$$

This expression has to satisfy the differential equation representing Newton's second law. We can prove that the expression is solution of the differential equation by substituting it in the equation and showing that it turns it into an identity. To do so, we need the first and the second derivative of the displacement with respect to time:

$$\frac{dx}{dt} = -\frac{j\omega eE_o \exp(j\omega t)}{m_e\left(\omega_o^2 - \omega^2 + j\gamma\omega\right)}$$

$$\frac{d^2x}{dt^2} = \frac{\omega^2 eE_o \exp(j\omega t)}{m_e\left(\omega_o^2 - \omega^2 + j\gamma\omega\right)}$$

Substituting the derivatives into the displacement differential equation we receive

$$\frac{Z\omega^2 eE_o \exp(j\omega t)}{\left(\omega_o^2 - \omega^2 + j\gamma\omega\right)} = -ZeE_o \exp(j\omega t) + \frac{Z\omega_o^2 eE_o \exp(j\omega t)}{\left(\omega_o^2 - \omega^2 + j\gamma\omega\right)} + \frac{Z\gamma j\omega eE_o \exp(j\omega t)}{\left(\omega_o^2 - \omega^2 + j\gamma\omega\right)}$$

$$\frac{Z\omega^2 e}{\left(\omega_o^2 - \omega^2 + j\gamma\omega\right)} = -Ze + \frac{Z\omega_o^2 e}{\left(\omega_o^2 - \omega^2 + j\gamma\omega\right)} + \frac{Z\gamma j\omega e}{\left(\omega_o^2 - \omega^2 + j\gamma\omega\right)}$$

$$Z\omega^2 e = -Ze\left(\omega_o^2 - \omega^2 + j\gamma\omega\right) + Z\omega_o^2 e + Z\gamma\, j\omega e$$

$$Z\omega^2 e = -Ze\omega_o^2 + Ze\omega^2 - Zej\gamma\omega + Z\omega_o^2 e + Z\gamma\, j\omega e$$

$$Z\omega^2 e = Z\omega^2 e$$

Thus the substitution turns the equation into an identity and this actually proves that the expression for the electronic polarizability is correct, as the expression for the displacement was derived directly from it.

The Clausius-Mossotti relation (Equation 7.15 in the textbook) relates ε_r and α_e as follows:

$$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = \frac{N\alpha_e}{3\varepsilon_0}$$

Substituting ε_r with N^2 and α_e with $\dfrac{Ze^2}{m_e(\omega_o^2 - \omega^2 + j\gamma\omega)}$ we receive exactly the relation we wanted to prove

$$\frac{N^2 - 1}{N^2 + 2} = \frac{NZe^2}{3\varepsilon_0 m_e(\omega_o^2 - \omega^2 + j\gamma\omega)}$$

The complex polarizability accounts for some energy losses in the media. If γ is zero, there are no losses in the media and we have the usual expression for lossless media.

9.7 Dispersion and diamond

Consider applying the simple electronic polarizability and Clasius-Mossotti equations to diamond. Neglecting losses,

$$\alpha_e = \frac{Ze^2}{m_e(\omega_o^2 - \omega^2)}$$

and

$$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = \frac{NZe^2}{3\varepsilon_o m_e(\omega_o^2 - \omega^2)} \qquad \textit{Dispersion in diamond}$$

For diamond we can take $Z = 4$ (valence electrons only as these are the most responsive), $N = 1.8 \times 10^{29}$ atoms m^{-3}, ε_r(DC) = 5.7. Find ω_o and then find the refractive index at $\lambda = 0.5$ μm and 5 μm.

Solution

We can obtain the value of ω_0 from the dispersion relation in diamond using that for DC $\omega = 0$. The dispersion relation then reduces to

$$\frac{\varepsilon_{r\,DC}-1}{\varepsilon_{r\,DC}+2}=\frac{NZe^2}{3\varepsilon_o m_e\omega_o^2}$$

Solving for ω_0 we have

$$\omega_o=\sqrt{\frac{NZe^2}{3\varepsilon_o m_e}\frac{\varepsilon_{r\,DC}+2}{\varepsilon_{r\,DC}-1}}=\sqrt{\frac{\left(1.8\times10^{29}\ \text{m}^{-3}\right)(4)\left(1.6022\times10^{-19}\ \text{C}\right)^2}{3\left(8.85\times10^{-12}\ \text{F m}^{-1}\right)\left(9.1\times10^{-31}\ \text{kg}\right)}\frac{(5.7)+2}{(5.7)-1}}=\mathbf{3.54\times10^{16}\ s^{-1}}$$

This corresponds to linear frequency of $f_o=\dfrac{\omega_o}{2\pi}=5.63\times10^{15}$ Hz and wavelength of

$$\lambda_o=\frac{c}{f_o}=53.2\ \text{nm}$$

If we neglect the losses the refractive index is simply given by $n=\sqrt{\varepsilon_r}$ and the value of ε_r for given ω can be found from the dispersion relation solving for ε_r

$$\varepsilon_r=\frac{2NZe^2+3\varepsilon_o m_e\left(\omega_o^2-\omega^2\right)}{3\varepsilon_o m_e\left(\omega_o^2-\omega^2\right)-NZe^2}$$

The refractive index is then

$$n=\sqrt{\frac{2NZe^2+3\varepsilon_o m_e\left(\omega_o^2-\omega^2\right)}{3\varepsilon_o m_e\left(\omega_o^2-\omega^2\right)-NZe^2}}$$

The circular frequency corresponding to $\lambda = 0.5\ \mu$m is

$$\omega=\frac{2\pi c}{\lambda}=\frac{2\pi\left(3\times10^8\ \text{m s}^{-1}\right)}{\left(0.5\times10^{-6}\ \text{m}\right)}=3.767\times10^{15}\ \text{s}^{-1}$$

and the value of the refractive index is

$$n=\sqrt{\frac{2\left(1.8\times10^{29}\ \text{m}^{-3}\right)(4)\left(1.6022\times10^{-19}\ \text{C}\right)^2+3\left(8.85\times10^{-12}\ \text{F m}^{-1}\right)\left(9.1\times10^{-31}\ \text{kg}\right)\left[\left(3.54\times10^{16}\ \text{s}^{-1}\right)^2-\left(3.77\times10^{15}\ \text{s}^{-1}\right)^2\right]}{3\left(8.85\times10^{-12}\ \text{F m}^{-1}\right)\left(9.1\times10^{-31}\ \text{kg}\right)\left[\left(3.54\times10^{16}\ \text{s}^{-1}\right)^2-\left(3.77\times10^{15}\ \text{s}^{-1}\right)^2\right]-\left(1.8\times10^{29}\ \text{m}^{-3}\right)(4)\left(1.6022\times10^{-19}\ \text{C}\right)}}=\mathbf{2.417}$$

Analogously for $\lambda = 5\ \mu$m we have

$$\omega=\frac{2\pi c}{\lambda}=\frac{2\pi\left(3\times10^8\ \text{m s}^{-1}\right)}{\left(5\times10^{-6}\ \text{m}\right)}=3.767\times10^{14}\ \text{s}^{-1}$$

and

n = **2.388**.

9.8 Electric and magnetic fields in light

The intensity (irradiance) of the red laser beam from a He-Ne laser in air has been measured to be about 1 mW cm^{-2}. What are the magnitudes of the electric and magnetic fields? What are the magnitudes if this 1 mW cm^{-2} beam were in a glass medium with a refractive index n = 1.45 and still had the same intensity?

Solution

The average irradiance according to Equation 9.24 (in the textbook) is given by

$$I = \frac{1}{2} c\varepsilon_o n E_o^2$$

For vacuum or air $n = 1$ and we can calculate the magnitude of the electric field from the above relation:

$$E_o = \sqrt{\frac{2I}{c\varepsilon_o n}} = \sqrt{\frac{2\left(10\ \text{W m}^{-2}\right)}{\left(3\times10^{8}\ \text{m s}^{-1}\right)\left(8.85\times10^{-12}\ \text{F m}^{-1}\right)(1)}} = \mathbf{86.772\ V\ m^{-1}}$$

The corresponding magnetic field is

$$B_o = \frac{nE_o}{c} = \frac{(1)\left(86.772\ \text{V m}^{-1}\right)}{\left(3\times10^{8}\ \text{m s}^{-1}\right)} = \mathbf{2.892 \times 10^{-7}\ T = 0.2892\ \mu T}$$

If this beam was traveling in a glass medium of $n = 1.45$ and still had the same intensity (1 mW cm^{-2}), then

$$E_o = \sqrt{\frac{2I}{c\varepsilon_o n}} = \sqrt{\frac{2\left(10\ \text{W m}^{-2}\right)}{\left(3\times10^{8}\ \text{m s}^{-1}\right)\left(8.85\times10^{-12}\ \text{F m}^{-1}\right)(1.45)}} = \mathbf{72.06\ V\ m^{-1}}$$

and

$$B_o = \frac{nE_o}{c} = \frac{(1.45)\left(86.772\ \text{V m}^{-1}\right)}{\left(3\times10^{8}\ \text{m s}^{-1}\right)} = \mathbf{3.483 \times 10^{-7}\ T = 0.3483\ \mu T.}$$

9.9 Reflection of light from a less dense medium (internal reflection)

A ray of light which is traveling in a glass medium of refractive index $n_1 = 1.450$ becomes incident on a less dense glass medium of refractive index $n_2 = 1.430$. Suppose that the free space wavelength (λ) of the light ray is 1 μm.

a. What should be the minimum incidence angle for TIR?
b. What is the phase change in the reflected wave when $\theta_i = 85°$ and when $\theta_i = 90°$?
c. What is the penetration depth of the evanescent wave into medium 2 when $\theta_i = 85°$ and when $\theta_i = 90°$?

Solution

a The critical angle q_c for TIR is given by Equation 9.26 (in the textbook)

$\sin\theta_c = n_2/n_1 = 1.430/1.450$ so that $\theta_c = 80.47°$.

b Since the incidence angle $\theta_i > \theta_c$, there is a phase shift in the reflected wave. The phase change in $E_{r,\perp}$ is given by $\phi_\perp$ (Equation 9.39 in the textbook). With $n_1 = 1.450$, $n_2 = 1.430$ and $\theta_i = 85°$, the phase change is

$$\phi_\perp = 2\arctan\left[\frac{\left(\sin^2\theta_i - n^2\right)^{1/2}}{\cos\theta_i}\right] = 2\arctan\left[\frac{\left((\sin 85°)^2 - \left(\frac{1.43}{1.45}\right)^2\right)^{1/2}}{\cos 85°}\right] = \mathbf{116.45°}$$

For the $E_{r,//}$ component (Equation 9.40 in the textbook), the phase change is

$$\phi_{//} = 2\left\{\arctan\left[\frac{\left(\sin^2\theta_i - n^2\right)^{1/2}}{n^2\cos\theta_i}\right] - \frac{1}{2}\pi\right\} =$$

$$= 2\left\{\arctan\left[\frac{\left((\sin 85°)^2 - \left(\frac{1.43}{1.45}\right)^2\right)^{1/2}}{\left(\frac{1.43}{1.45}\right)^2\cos 85°}\right] - \frac{1}{2}\pi\right\} = \mathbf{-62.1°}$$

(Note: If we were to invert the reflected field, this phase change would be 117.9°) .

We can repeat the calculation with $\theta_i = 90°$ to find $\phi_\perp = 180°$ and $\phi_{//} = 0°$.

Note that as long as $\theta_i > \theta_c$, the magnitude of the reflection coefficients are unity. Only the phase changes.

c The amplitude of the evanescent wave as it penetrates into medium 2 is

$$E_{t,\perp}(y,t) \sim E_{to,\perp}\exp(-\alpha_2 y)$$

We ignore the z-dependence, $\exp j(\omega t - k_z z)$, as this only gives a propagating property along z. The field strength drops to e^{-1} when $y = 1/\alpha_2 = d$, which is called the **penetration depth**. The attenuation constant α_2 (Equation 9.42 in the textbook) is

$$\alpha_2 = \frac{2\pi n_2}{\lambda}\left[\left(\frac{n_1}{n_2}\right)^2\sin^2\theta_i - 1\right]^{\frac{1}{2}}$$

i.e.

$$\alpha_2 = \frac{2\pi(1.43)}{\left(1\times10^{-6}\ \text{m}\right)}\left[\left(\frac{(1.45)}{1.43}\right)^2(\sin 85°)^2 - 1\right]^{\frac{1}{2}} = \mathbf{1.28\times10^6\ m^{-1}.}$$

so that the penetration depth is, $d = 1/\alpha_2 = 1/(1.28\times10^6\ \text{m}^{-1}) = \mathbf{7.8\times10^{-7}\ m}$, or **0.78 mm**. For 90°, repeating the calculation we find, $\boldsymbol{\alpha_2 = 1.5\times10^6\ m^{-1}}$, so that $d = 1/\alpha_2 =$ **0.66 mm**. We see that the penetration is greater for smaller incidence angles.

9.10 Internal and external reflection at normal incidence

Consider the reflection of light at normal incidence on a boundary between a GaAs crystal medium of refractive index 3.6 and air of refractive index 1.

a. If light is traveling from air to GaAs, what is the reflection coefficient and the intensity of the reflected light in terms of the incident light?

b. If light is traveling from GaAs to air, what is the reflection coefficient and the intensity of the reflected light in terms of the incident light?

Solution

a The light travels in air and becomes partially reflected at the surface of the GaAs crystal which corresponds to external reflection. Thus $n_1 = 1$ and $n_2 = 3.6$. Then according to Equation 9.37 (in the textbook),

$$r_{//} = r_\perp = \frac{n_1 - n_2}{n_1 + n_2} = \frac{1 - 3.6}{1 + 3.6} = \mathbf{-0.565}$$

This is negative which means that there is a 180° phase shift. The reflectance R (Equation 9.43 in the textbook), which gives the fractional reflected power, is

$$R = \left(r_{\perp}\right)^2 = (-0.565)^2 = \mathbf{0.319} = \mathbf{31.9\ \%}$$

b The light travels in GaAs crystal and becomes partially reflected at the crystal-air interface which corresponds to internal reflection. Thus $n_1 = 3.6$ and $n_2 = 1$. Then,

$$r_{//} = r_{\perp} = \frac{n_1 - n_2}{n_1 + n_2} = \frac{3.6 - 1}{3.6 + 1} = \mathbf{0.565}$$

There is no phase shift. The reflectance is again 0.319 or 31.9%. In both cases, **a** and **b**, the amount of reflected light is the same.

9.11 Antireflection coating

a. Consider three dielectric media with flat and parallel boundaries with refractive indices n_1, n_2 and n_3. Show that for normal incidence the reflection coefficient between layers 1 and 2 is the same as that between layers 2 and 3 if $n_2 = \sqrt{n_1 n_3}$. What is the significance of this?

b. Consider a Si photodiode that is designed for operation at 900 nm. Given a choice of two possible antireflection coatings, SiO_2 with a refractive index of 1.5 and TiO_2 with a refractive index of 2.3 which would you use and what would be the thickness of the antireflection coating you chose? The refractive index of Si is 3.5.

Solution

For light traveling in medium 1 incident on the 1-2 interface at normal incidence,

$$r_{12} = \frac{n_1 - n_2}{n_1 + n_2} = \frac{n_1 - \sqrt{n_1 n_3}}{n_1 + \sqrt{n_1 n_3}} = \frac{1 - \sqrt{\dfrac{n_3}{n_1}}}{1 + \sqrt{\dfrac{n_3}{n_1}}}$$

For light traveling in medium 2 incident on the 2-3 interface at normal incidence,

$$r_{23} = \frac{n_2 - n_3}{n_2 + n_3} = \frac{\sqrt{n_1 n_3} - n_3}{\sqrt{n_1 n_3} + n_3} = \frac{\sqrt{\dfrac{n_1}{n_3}} - 1}{\sqrt{\dfrac{n_1}{n_3}} + 1} = \frac{1 - \sqrt{\dfrac{n_3}{n_1}}}{1 + \sqrt{\dfrac{n_3}{n_1}}}$$

thus, $r_{23} = r_{12}$

Significance? For an efficient antireflection effect, the coefficient the waves A and B (see Figure 9Q11-1) should interfere destructively and at the same time should have comparable magnitudes to cancel each other. This can be achieved by $r_{12} = r_{23}$.

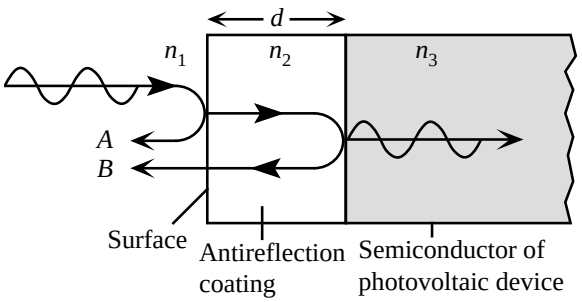


Figure 9Q11-1 Illustration of how an antireflection coating reduces the reflected light intensity.

The best antireflection coating has to have a refractive index n_2 such that $n_2 = (n_1 n_3)^{1/2} = [(1)(3.5)]^{1/2} = 1.87$. Given a choice of two possible antireflection coatings, SiO_2 with a refractive index of 1.5 and TiO_2 with a refractive index of 2.3, SiO_2 both are very close.

To find the thickness of the coating with a 900 nm wavelength (equation from Example 9.7 in the textbook),

$$d = m\left(\frac{\lambda}{4n_2}\right)$$

where m = 1, 3, 5, … is an odd integer.

For SiO_2: $$d = \frac{(900\ \text{nm})}{4(1.5)}$$

$$\boldsymbol{d = 150\ \text{nm}}$$

or odd multiples of d.

For TiO_2 (if chosen): $$d = \frac{(900\ \text{nm})}{4(2.3)}$$

$$\boldsymbol{d = 97.8\ \text{nm}}$$

or odd multiples of d.

9.12 Complex refractive index

Spectroscopic ellipsometry measurements on a silicon crystal at a wavelength of 620 nm show that the real and imaginary parts of the complex relative permittivity are 15.2254 and 0.172, respectively. Find the complex refractive index. What is the reflectance and absorption coefficient at this wavelength? What is the phase velocity?

Solution

We know that $\varepsilon_r' = 15.2254$ and that $\varepsilon_r'' = 0.172$. The real part n and the imaginary part K of the complex refractive index are solutions of the following system of equations (see Equation 9.55 in the textbook)

$$n^2 + K^2 = 15.2254 \qquad \text{and} \qquad 2nK = 0.172$$

We can take K from the second equation and substitute for it in the first equation,

$$n^2 + \left(\frac{0.172}{2n}\right)^2 = 15.2254$$

This is a quadratic equation in n^2 that can be easily solved to find that the four roots are:

$$n_{1,2} = \pm 3.902 \qquad \text{and} \qquad n_{3,4} = \pm 0.022$$

Since $n > 0$, only the positive roots can have significance for us. The wavelength of 620 nm corresponds to photon energy of

$$E = \frac{hc}{\lambda} = \frac{\left(6.62\times10^{-34}\ \text{J s}\right)\left(3\times10^{8}\ \text{m s}^{-1}\right)}{\left(620\times10^{-9}\ \text{m}\right)} = 3.2\times10^{-19}\ \text{J} = 2\ \text{eV}.$$

This photon energy is comparable with the energy bandgap of Si (1.12 eV) and for such photon energies n should greater than one and much greater than K. Thus, we can conclude that n = **3.902.** Once we know n, we can find $K = \frac{1}{2n}$ = **0.022**. If we simply square root the real part of ε_r, we would find still find n = 3.902, because the extinction coefficient K is very small.

The reflectance of the Si crystal is given by Equation 9.57 (in the textbook)

$$R = \frac{(n-1)^2 + K^2}{(n+1)^2 + K^2} = \frac{(3.902-1)^2 + 0.022^2}{(3.902+1)^2 + 0.022^2} = 0.35$$

which is the same as simply using $(n - 1)^2/(n + 1)^2 = 0.35$, because $K << n$.

The absorption coefficient α describes the loss in the light intensity I via $I = I_o\exp(-\alpha z)$ and by virtue of Equation 9.52 (in the textbook),

$$\alpha = 2k'' = 2k_oK = 2\frac{2\pi}{\lambda}K = 2\frac{2\pi}{\left(620\times10^{-9}\ \text{m}\right)}(0.022) = \mathbf{4.459\times10^{5}\ m^{-1}}$$

The phase velocity is given by

$$v = \frac{c}{n} = \frac{\left(3\times10^{8}\ \text{m s}^{-1}\right)}{(3.902)} = \mathbf{7.683\times10^{7}\ m\ s^{-1}}$$

9.13 Complex refractive index

Spectroscopic ellipsometry measurements on a germanium crystal at a photon energy of 1.5 eV show that the real and imaginary parts of the complex relative permittivity are 21.56 and 2.772 respectively. Find the complex refractive index. What is the reflectance and absorption coefficient at this wavelength? How do your calculations match with the experimental values of n = 4.653 and K = 0.298, R = 0.419 and $\alpha = 4.53\times10^{6}\ \text{m}^{-1}$?

a. Show that the attenuation coefficient a due to free carrier absorption is given by

$$\alpha = \left(\frac{\omega}{c}\right)\frac{\varepsilon_r''}{n} = \left(\frac{1}{c\varepsilon_o}\right)\frac{\sigma}{n} \qquad \textit{Free carrier absorption}$$

where ω is the angular frequency of the EM radiation, ε_r'' is the imaginary part of the relative permittivity, n is the refractive index and σ is the conductivity due to free carriers in the sample.

b. Intrinsic germanium has a conductivity of about 2.1 $\Omega^{-1}\ \text{m}^{-1}$. Calculate the imaginary part ε_r'' of the relative permittivity at a wavelength of 20 μm. Find the attenuation coefficient α due to free carrier absorption. The refractive index of germanium at the specified wavelength is n = 4.

Solution

a We know that $\varepsilon_r' = 21.56$ and that $\varepsilon_r'' = 2.772$. Thus, from Equation 9.55 (in the textbook), we have

$$n^2 + K^2 = 21.56 \qquad \text{and} \qquad 2nK = 2.772$$

We can take K from the second equation and substitute for it in the first equation,

$$n^2 + \left(\frac{2.772}{2n}\right)^2 = 21.56$$

This is a quadratic equation in n^2 that can be easily solved to find, that the four roots are:

$$n_{1,2} = \pm 4.634 \qquad \text{and} \qquad n_{3,4} = \pm 0.299 \times 10^{-3}$$

Since n and K should be positive, and for photon energies of about 1.5 eV, n should be greater than one and greater than K, n = **4.634** is the only root having physical significance.

Knowing n, we can find $K = \frac{1}{2n}$ = **0.299**. Both values compare very well with the experimental results.

From Equations 9.55 and 9.52 (in the textbook), we can calculate the reflectance R and the absorption coefficient α, respectively

$$R = \frac{(n-1)^2 + K^2}{(n+1)^2 + K^2} = \frac{(4.634-1)^2 + 0.299^2}{(4.634+1)^2 + 0.299^2} = \mathbf{0.418}$$

$$\alpha = 2k'' = 2k_o K = 2\frac{2\pi}{\lambda}K = 2\frac{2\pi E_{ph}}{hc}K =$$

$$= 2\frac{2\pi(1.5\ \text{eV})}{\left(4.135\times10^{-15}\ \text{eV s}\right)\left(3\times10^{8}\ \text{m s}^{-1}\right)}(0.299) = \mathbf{4.540 \times 10^{6}\ m^{-1}}$$

Since n and K were in good agreement with the experiment, α and R are also very close to their experimental values.

b The absorption coefficient is defined through Equation 9.52 (in the textbook) as $\alpha = 2k''$. Further, using Equation 9.53 (in the textbook), we can easily show that

$$\alpha = 2k_o K \tag{1}$$

By definition $k_o = \frac{2\pi}{\lambda}$, or expressed through the angular frequency ω

$$k_o = \frac{\omega}{c} \tag{2}$$

From Equation 9.55 (in the textbook),

$$K = \frac{\varepsilon_r''}{2n} \tag{3}$$

Combining (1), (2) and (3) we receive

$$\alpha = \left(\frac{\omega}{c}\right)\frac{\varepsilon_r''}{n}, \tag{4}$$

which is exactly the relation we had to prove.

Taking into account the *Conduction loss equation* (Equation 9.49 in the textbook) we can easily transform (4) to

$$\alpha = \left(\frac{1}{c\varepsilon_o}\right)\frac{\sigma}{n} \tag{5}$$

c Equation 9.49 (in the textbook) relates the imaginary part of relative permittivity ε_r'' and the conductivity due to free carriers in the sample:

$$\varepsilon_r'' = \frac{\sigma}{\varepsilon_o \omega}$$

The angular frequency can be trivially calculated from the wavelength λ

$$\omega = \frac{2\pi c}{\lambda}$$

Thus, for ε_r'', we receive

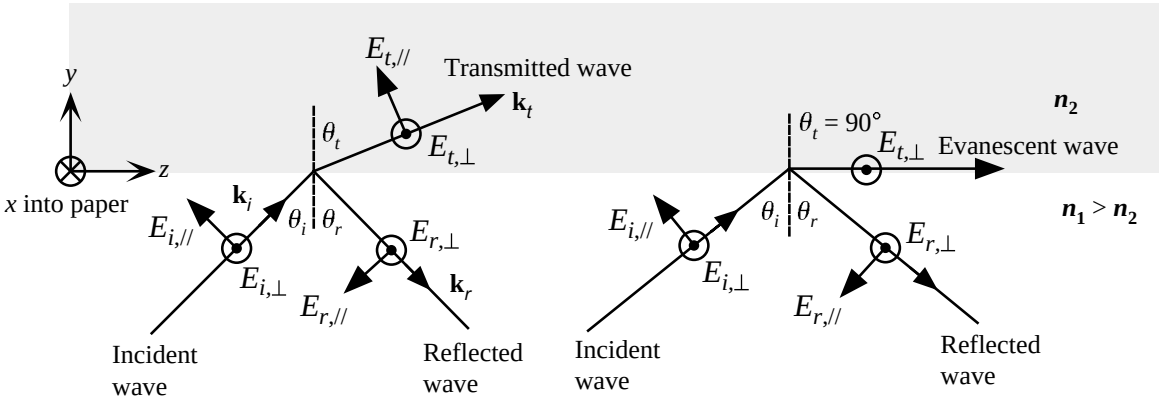
$$\varepsilon_r'' = \frac{\lambda\sigma}{2\pi c\varepsilon_o} = \frac{\left(20\times10^{-6}\ \text{m}\right)\left(2.1\ \Omega^{-1}\ \text{m}^{-1}\right)}{2\pi\left(3\times10^{8}\ \text{m s}^{-1}\right)\left(8.85\times10^{-12}\ \text{F m}^{-1}\right)} = \mathbf{0.002518}$$

Since we are given that for a wavelength of 20 μm, the refraction index of germanium crystal is 4, we can calculate the attenuation coefficient α from (5)

$$\alpha = \left(\frac{1}{c\varepsilon_o}\right)\frac{\sigma}{n} = \frac{1}{\left(3\times10^{8}\ \text{m s}^{-1}\right)\left(8.85\times10^{-12}\ \text{F m}^{-1}\right)}\frac{\left(2.1\ \Omega^{-1}\ \text{m}^{-1}\right)}{(4)} = \mathbf{197.74\ m^{-1}}$$

9.14 Evanescent wave

Total internal reflection (TIR) of light from a boundary between a more dense medium n_1 and a less dense medium n_2 is accompanied by an evanescent wave propagating in medium 2 near the boundary. Find the functional form of this wave for the component normal to the plane of incidence and discuss how its magnitude varies with the distance y into medium 2. (See Figure 9Q14-1)



(a) $\theta_i < \theta_c$ then some of the wave is transmitted into the less dense medium. Some of the wave is reflected.

(b) $\theta_i > \theta_c$ then the incident wave suffers total internal reflection. However, there is an evanescent wave at the surface of the medium

Figure 9Q14-1 Light wave traveling in a more dense medium strikes a less dense medium. The plane of incidence is the plane of the paper and is perpendicular to the flat interface between the two media. The electric field is normal to the direction of propagation. It can be resolved into perpendicular ($\perp$) and parallel ($||$) components.

Solution

The transmitted wave has the general form

$$E_{t,\perp} = t_\perp E_{io,\perp} \exp j(\omega t - \mathbf{k}_t \cdot \mathbf{r})$$

where $t_\perp$ is the transmission coefficient. The dot product, examining Figure 9Q14-2, is

$$\mathbf{k}_t \cdot \mathbf{r} = yk_t \cos\theta_t + zk_t \sin\theta_t.$$

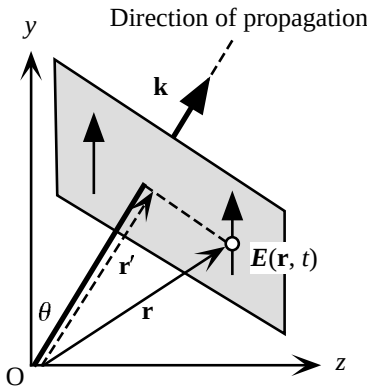


Figure 9Q14-2 A traveling plane EM wave along a direction **k**.

However, from Snell's law, when $\theta_i > \theta_c$, $\sin\theta_t = (n_1/n_2)\sin\theta_i > 1$ and $\cos\theta_t = \sqrt{[1 - \sin^2\theta_t]} = \pm jA_2$ is a purely imaginary number. Thus, taking $\cos\theta_t = -jA_2$

$$E_{t,\perp} = t_\perp E_{io,\perp} \exp j(\omega t - zk_t \sin\theta_t + jyk_t A_2)$$

$$= t_\perp E_{io,\perp} \exp(-yk_t A_2) \exp j(\omega t - zk_t \sin\theta_t)$$

which has an amplitude that decays along y as $\exp(-\alpha_2 y)$ where $\alpha_2 = k_t A_2$. Note that $+jA_2$ is ignored because it implies a light wave in medium 2 whose amplitude and hence intensity grows.

Consider, the traveling wave part, $\exp j(\omega t - zk_t \sin\theta_t)$. Here, $k_t \sin\theta_t = k_i \sin\theta_i$ (by virtue of Snell's law). But $k_i \sin\theta_i = k_{iz}$ which is the wavevector along z, that is, along the boundary. Thus the evanescent wave propagates along z at the same speed as the incident and reflected waves along z.

Furthermore, for TIR we need $\sin\theta_i > n_2/n_1$. This means that the transmission coefficient,

$$t_\perp = \frac{n_i \cos\theta_i}{\cos\theta_i + \left[\left(\frac{n_2}{n_1}\right)^2 - \sin^2\theta_i\right]^{1/2}} = t_{\perp 0} \exp(j\psi_\perp)$$

must be a complex number as indicated by $t_{\perp 0}\exp(j\psi_\perp)$ where $t_{\perp 0}$ is a real number and $\psi_\perp$ is a phase change. Note that $t_\perp$ does not however change the general behavior of propagation along z and the penetration along y.

9.15 Quartz half-wave plate

What are the possible thicknesses of a half-wave quartz plate for a wavelength $\lambda \approx 1.01\ \mu$m given the extraordinary and ordinary refractive indices are $n_o = 1.534$ and $n_e = 1.543$?

Solution

Equation 9.69 (in the textbook) describes the relative phase difference through a retarder plate

$$\phi = \frac{2\pi}{\lambda}(n_e - n_o)L$$

Halfwave retardation is a phase difference of π but the effect will be the same if the phase difference is

$$\phi = (2m+1)\pi, \qquad m = 0,1,2,...$$

All possible thicknesses of halfwave plate retarder are then

$$L = \frac{(2m+1)\lambda}{2(n_e - n_o)}, \qquad m = 0,1,2,.....$$

When m = 0 we receive the smallest possible value

$$L = \frac{\lambda}{2(n_e - n_o)} = \frac{(1.01\times10^{-6})}{2(1.543-1.534)} = \mathbf{56.1\ \mu m}$$

For m = 1 we have L = 168 μm, for m = 2 we have L = 280.6 μm, for m = 3 we have L = 392.8 μm and so on.

9.16 Pockels Cell Modulator

What should be the aspect ratio d/L for the transverse $LiNiO_3$ phase modulator in Figure 9Q16-1 that will operate at a free-space wavelength of 1.3 μm and will provide a phase shift $\Delta\phi$ of π (half wavelength) between the two field components propagating through the crystal for an applied voltage of 20 V? The Pockels coefficient r_{22} is 3.2×10^{-12} m/V and n_o = 2.2.

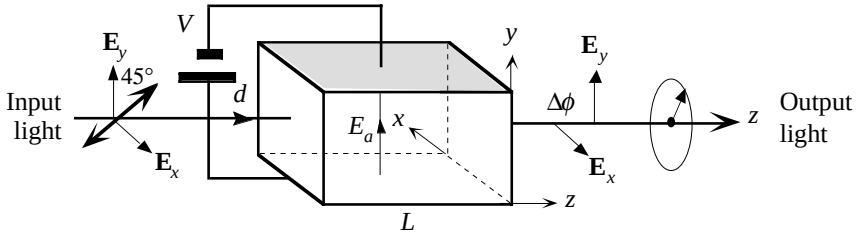


Figure 9Q16-1 Transverse Pockels cell phase modulator. A linearly polarized input light into an electro-optic crystal emerges as a circularly polarized light.

Solution

From Equation 9.75 (in the textbook), putting $\Delta\phi = \pi$ for the phase difference between the field components $\mathbf{E}_x$ and $\mathbf{E}_y$ in Figure 9Q16-1 (in the textbook) gives,

$$\Delta\phi = \frac{2\pi}{\lambda} n_o^3 r_{22} \frac{L}{d} V_{\lambda/2} = \pi$$

or

$$\frac{d}{L} = \frac{1}{\Delta\phi} \cdot \frac{2\pi}{\lambda} n_o^3 r_{22} V_{\lambda/2} = \frac{1}{\pi} \cdot \frac{2\pi}{(1.3\times10^{-6}\ \text{m})} (2.2)^3 (3.2\times10^{-12}\ \text{m V}^{-1})(20\ \text{V})$$

giving $\boldsymbol{d/L = 1.048 \times 10^{-3}}$

This particular transverse phase modulator has the field applied along the *y*-direction and light traveling along the *z*-direction as in Figure 9Q16-1. If we were to use the transverse arrangement in which the field is applied along the *z*-axis, and the light travels along the *y*-axis, the relevant Pockels coefficients would be greater and the corresponding aspect ratio d/L would be $\sim 5 \times 10^{-2}$. We cannot arbitrarily set d/L to any ratio we like for the simple reason that when d becomes too small, the light will suffer diffraction effects that will prevent it from passing through the device. d/L ratios 10^{-3} - 10^{-2} in practice can be implemented by fabricating an integrated optical device.

"I know you believe you understand what you think I said, but I am not sure you realize that what you heard is not what I meant"

Alan Greenspan
Chairman of US Federal Reserve Board
